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Applicability of 30-day hospital readmission risk predictive models in paediatrics: a systematic review

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TITLE

Applicability of 30-day hospital readmission risk predictive models in paediatrics: a systematic review

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ABSTRACT

Objective: To summarize multivariable predictive models for 30-day hospital readmissions (HRs) in paediatrics, describe their performance and completeness in reporting, and to determine their potential for application in practice.

Design: Systematic review.

Data source: CINAHL, EMBASE and PubMed up to 1st January 2021.

Eligibility criteria: English- or German-language studies aiming to develop or validate a multivariable predictive model for 30-day paediatric HRs related to all-cause, surgical conditions or general medical conditions were included.

Data extraction and synthesis: Study characteristics, risk factors significant for predicting readmissions and information about performance measures (e.g., c-statistic) were extracted. Reporting quality was addressed by the 'Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis' (TRIPOD) adherence form. The study quality was assessed by applying six domains of potential biases. Due to expected heterogeneity among the studies, the data were qualitatively synthesized.

Results: Based on 13 studies, 21 predictive models were identified, which could potentially be used for determining individual 30-day HR risk in paediatrics. The number of study participants ranged from 328 patients to 1.4 million encounters. The three most common significant risk factors were comorbidity, (postoperative) length of stay and illness severity. Six models showed a c-statistic above 0.7 and are potentially applicable at discharge. The median TRIPOD adherence of the models was 63% ($P_{25} - P_{75}$ 55% - 69%), ranging from a minimum of 33% to a maximum of 81%. Overall, the quality of the studies was moderate in almost all domains.

Conclusions: Predictive models may be useful in identifying paediatric patients at increased risk for readmission. To support the application of predictive models, more attention should be placed on completeness in reporting, particularly for those items that may be relevant for implementation in practice.

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Strengths and limitations of this study

- This is the first systematic review to summarize multivariable predictive models for 30-day paediatric hospital readmissions.
- The comprehensive presentation of predictive models is based on a well-founded methodological approach and provides information about applicability, performance and reporting quality at model level, differentiated by 30-day all-cause, surgical conditions and general medical conditions related hospital readmissions.
- Due to study heterogeneity, the models were qualitatively synthesized.

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INTRODUCTION

Hospital readmissions (HRs) are becoming increasingly important as a quality indicator for paediatric inpatient care[1, 2]. HR is often defined as a subsequent, unplanned admission within a period of 30 days after the index hospitalization[3]. According to published evidence, all-cause 30-day unplanned HR rates for the paediatric population range from 3.4% to 18.7%[3-5]. In addition, taking 27 US states into account, it has been estimated that paediatric HRs can cost up to $2 billion annually, with approximately 40% of these occurring HRs being potentially preventable[6].

Identifying the reasons for paediatric HRs is a major challenge, as the health of children is also affected by factors aside of inpatient care[7]. Predictive models can be applied as a tool for the identification of patients with an increased risk of HR and for the implementation of preventive interventions to reduce the risk of HR[8]. Especially in the context of the ongoing COVID-19 pandemic, where children and adolescents are also being hospitalized with a variety of symptoms[9-11], the prevention of unplanned HRs can be beneficial, as it would allow hospital resources to be used in a more target-orientated way.

Predictive models with good performance are useful in practice when clinicians and other stakeholders have all the necessary information for their application in clinical practice and critical assessment[12]. Previous systematic reviews discussed the shortcomings in reporting the quality of prediction models[13-15] and also for paediatric clinical prediction rules[16]. A previous systematic review has already identified 36 significant risk factors for unplanned HRs in paediatric patients with different health conditions[3]. The largest number of risk factors were identified for surgical procedure related unplanned HRs. Among others, comorbidity was one of the most common risk factors across the 44 studies included in the review[3]. The review[3] extends the findings of an earlier systematic review that focused on 29 paediatric studies targeting predictors for asthma-related unplanned HRs[17]. Both reviews[3, 17] were primarily addressed to predictor finding studies[14], while to date there is no published review of existing 30-day HR predictive models in paediatrics.

This systematic review aims to address the two identified research gaps. This paper provides an overview of existing multivariable predictive models for individualized risk prediction of 30-day HR in the paediatric population. The objective of this analysis is to determine the potential application of the identified predictive models by evaluating the models´

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discriminative ability, completeness in reporting and the risk factors shown to be significant for prediction of 30-day HR.

METHOD

The 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) statement was adhered to for conducting and reporting of this systematic review[18]. Screening of the titles and abstracts, data extraction, quality assessment and analyses (e.g., completeness in reporting) were performed by two independent reviewers, while disagreements were discussed with a third author. A protocol was pre-specified and is available from the corresponding author.

Data source and search strategy

CINAHL, EMBASE and PubMed were used for an electronic database search to identify studies published up to 1 January 2021. According to previous systematic reviews[3, 19], the key search terms included ("readmission*" OR "rehospitali*") AND ("predict*" OR "AUC" OR "c-statistic*") AND ("adolescen*" OR "child*" OR "teen*") (see supplementary material for a full search strategy – Tables A1-A3).

Inclusion criteria

Studies addressing multivariable predictive models for children and adolescents (except newborns/ preterm newborns, as the index admission is the birth hospitalization) were included if they were published in English or German and available as full texts in peer-reviewed original journal articles. Studies aiming to develop a new model or to validate an existing model were included (i) if the model was potentially appropriate for the individual prediction of 30-day HR from acute healthcare service after discharge or after index procedure in paediatrics; and (ii) if the model provided at least one discrimination measure (e.g., c-statistic). Discriminative ability is a key factor in evaluating predictive models[20] and a necessary information to make well-founded conclusions about the performance of a model.

In addition, (iii) predictive model studies that developed a new model (i.e., development design) or determined the incremental or added value of a predictor for an existing model (i.e., incremental value design) had to be based on a regression modelling approach. This inclusion criterion enables us to identify significant risk factors and to apply the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) adherence form, which was originally developed for regression models[21]. Due to specific

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requirements of mental diseases, studies were only included (iv) if they addressed non-mental health conditions related 30-day HRs[3].

Study selection and data extraction

Potential articles were assessed for meeting the inclusion criteria. The reference lists of the included studies were examined for further potential studies. Just as in previous systematic reviews[3, 19], studies were categorized by health conditions in all tables.

Basic study characteristics were extracted in table 1. To assess the applicability of the predictive models, significant risk factors (i.e., odds ratio (OR) or hazard ratio >1 with a p-value of < 0.05) were assigned to established and revised variable categories[3] in table 2. If all variables of a predictive model are available for a patient at the time of index admission (e.g., previous health service usage before index admission), the model is applicable at admission. Applicability of predictive models at discharge is given if all variables are available at this point for a patient (e.g., length of stay, operative time).

Reporting quality and performance

In order to understand whether the predictive models can be implemented with the given information, the completeness of reporting was assessed by using the TRIPOD adherence form and scoring rules[12, 22, 23]. The TRIPOD adherence form consists of 22 main criteria based on the TRIPOD statement[21], resulting in 37 items that are applicable to varying degrees to the development, validation and incremental value studies [22]. We decided to apply the TRIPOD adherence form at predictive model level. Therefore, publications that report the development and validation of the same predictive model, for example, are assessed separately. According to previous research, our analysis concentrates on items that could be reported in the main text or supplements[24].

TRIPOD adherence at model level was merged with the performance results (i.e., discrimination and calibration measures) and the applicability assignment in table 3. The discrimination of a predictive model is often evaluated by the c-statistic or area under the receiver operating characteristic curve. The c-statistic can take a value between 0.5 and 1. A value of 0.5 indicates that the model is not superior to a random prediction of outcomes, while values between 0.7 and 0.8 indicate that the model is appropriate. A value of 0.8 or greater indicates a strong discrimination of a model.[25]

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Quality assessment

Following previous systematic reviews[3, 19, 26], the refined version of the quality in prognosis studies (QUIPS) tool with its prompting items[27] was used to appraise the studies critically with regard to the included predictive models based on six domains. Each domain was rated with a 'high', 'moderate' or 'low' risk of bias.

The six domains are[27]: 'study participation', 'study attrition', 'prognostic factor measurement', 'outcome measurement', 'study confounding' and 'statistical analysis and reporting'.

Data synthesis

Because a quantitative evaluation in the form of a meta-analysis was not possible due to the high heterogeneity among the studies, the studies were qualitatively synthesized, i.e., the results for performance, completeness in reporting and significant risk factors were presented in a narrative and in a simplified quantitative form.

RESULTS

Search result

From the electronic database search, 9,035 records were obtained. After duplicates had been removed, the titles and abstracts were screened for 6,881 records. Based on the predefined inclusion criteria, 6,568 records were excluded, resulting in 295 records included in the full-text assessment. Among the 282 excluded records, three were predictive model studies with no discrimination metrics[28-30], and four were non-regression-based predictive model studies for 30-day HRs in paediatrics[31-34]. Of the 57 excluded prognostic factor studies for 30-day paediatric HRs, nine studies provided information about discriminative ability, where the c-statistic ranged from 0.61 to 0.79[35-43]. Based on the full-text assessments, 13 studies were included in the systematic review, with two of them[44, 45] already presented in a previous systematic review[3] with a different focus. The results of the review process regarding the database search is provided in figure 1.

[…Placement of figure 1…]

Figure 1: Flowchart for the search and study selection process (PRISMA).

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Quality assessment

Overall, the quality of many studies was moderate for many domains. In the domain 'study participants', for example, nine studies were judged as having a 'moderate' risk of bias due to (i) limited availability of basic information (i.e., age and/or gender)[46-48], (ii) no detailed description for the validation cohort[48-51], (iii) a lack of source description for census data[49, 50, 52] and/or (iv) a potential limitation of generalizability, as just one hospital was included in the study[46, 47, 49, 52-54]. For seven studies, the domain 'study attrition' was judged as 'moderate' because of (i) limited information about patient flow[48, 50, 55, 56], (ii) no detailed description for the validation cohort[48, 50, 51], (iii) limited reporting of outcome events[56] and/or (iv) a high number of missing values for one variable might affect the representation of the study sample[44, 47]. For the domain 'prognostic factor measurement', seven studies[47, 49, 50, 52, 53, 55, 56] were judged as 'moderate', and eight studies were judged as 'moderate' for the domain 'outcome measurement' [44-48, 51, 54, 56]. For one study, the prognostic factor measurement was rated with having a 'high' risk of bias due to a lack of information about the predictors included[51]. All studies were rated as 'low' for the domain 'study confounding'. For the domain 'statistical analysis and reporting', six studies were judged as 'moderate'[45, 48-50, 54, 56], and one as 'high'[51](see table A4 in the supplementary material for the detailed results of the risk of bias assessment).

Study characteristics

All studies except one[53] were conducted in the USA. In addition, all studies were based on retrospective data, with six studies based on tertiary or paediatric hospital data[46, 47, 49, 52-54], and seven studies were based on centralized databases[44, 45, 48, 50, 51, 55, 56]. Five of 13 studies additionally included census data in the analysis[49, 50, 52, 53, 55]. The period of data collection ranged from one year[46, 55, 56] to seven years[48, 51]. The majority of studies included patients up to an age of <18 or ≤18 years. Only one study considered patients up to 21 years of age[47]. The sample size was specified with different units in the individual studies (e.g., encounters, admissions) and varies between 328 patients[54] and 1.4 million encounters[51].

The 13 included studies resulted in 21 predictive models for 30-day HRs in paediatrics. Eight of 13 studies developed or validated more than one predictive model, which were in part excluded due to non-agreement with the inclusion criteria[48-52, 54-56]. The models included were grouped into three health conditions: (i) all-cause HR (n=7)[46, 47, 49, 51, 53, 55]; (ii)

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surgical conditions related HR (n=11)[44, 45, 48, 50, 56]; and (iii) general medical conditions related HR (n=3)[52, 54]. 12 of 21 predictive models explicitly aim to predict 30-day unplanned HRs with respect to the grouped health condition[44-49, 52-55]. 30-day unplanned HR rates vary from 3.0%[54] to 25.3%[46].

Among the 21 predictive models included, nine (43%) used a development design[44-50, 52, 53], ten (47%) an external validation design[49-51, 54, 56] and two (10%) an incremental value design[52, 55]. Nine out of ten external validated models were based on existing predictive models that had been previously used in the adult population[49, 51], for different outcomes[54, 56] or both[56]. Furthermore, four of the 13 studies included did not state the primary aim to develop, validate externally or assess the incremental value of the respective 30-day HR predictive model[48, 49, 51, 55].

Of the predictive models with a development or incremental value design, two employed an apparent validation[53, 55] and nine employed an internal validation[44-50, 52]. The most commonly applied internal validation method was split-sample (n=4)[44, 48-50], followed by bootstrapping (n=3)[45, 52] and cross-validation (n=2)[46, 47]. In order to analyse the data, either a logistic regression[45-50, 52, 53, 55] or a Cox proportional hazard regression[44] was used. Most models present their results by ORs with a 95% confidence interval. With a p-value of < 0.05, the results are considered as statistically significant[3]. A summary of characteristics of all included studies is provided in table 1.

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Table 1: Summary of study characteristics

Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
All-cause related HRs									
Brittan et al. 2018[47], USA	Composite Score	All-cause	30-day unplanned HRs	Retrospective/ 1 children's hospital	29,542 patients	0-21 years	Jan. 2014 - Dez. 2015	4.0%	Development study/ Internal: Cross
Sills et al. 2017[55], USA	PACR + SDH	All-cause	30-day unplanned HRs	Retrospective / PHIS database, US Census's American Community Survey data, 47 hospitals	458,686 index discharges	<18 years	Jan. - Dec. 2014	6.1%	Incremental value study/ Apparent
Ehwerhemuepha et al. 2018[49], USA	Unnamed	All-cause	30-day unplanned HRs	Retrospective / US Census's American Community Survey data, 1 tertiary paediatric hospital	38,143 inpatient clinical encounters (DC: 19,072, VC: 19,071)	between 28-days and 17 years	July 2013 to June 2017	10.4%	Development study/ Internal: Random-Split-Sample
	LACE (validation)				VC: 19,071 inpatient clinical encounters			NR	External validation study
Bradshaw et al. 2020[46], USA	HARRPS-tool	All-cause	30-day unplanned HRs	Retrospective / 1 paediatric hospital	5,306 patients	<18 years	May 2017-June 2018	25.3%	Development study/ Internal: Cross
Zhou et al. 2020[53], Australia	Unnamed	All-cause	30-day unplanned HRs	Retrospective/ Australian Census data, 1 tertiary paediatric hospital	73,132 patients	age limit for admission: 15 years, special permissions by hospital executives possible	Jan. 2010 - Dez. 2014	4.6%	Development study/ Apparent
Ehwerhemuepha et al. 2020[51], USA	LACE (validation)	All-cause	30-day HRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters	< 18 years	Jan. 2000-Dez. 2017	12.6% (DC)	External validation study
Surgical conditions related HRs									
Vo et al. 2018[45], USA	Unnamed	All surgical specialties without cardiac surgery	30-day unplanned post- surgical HRs relating to noncardiac surgery	Retrospective/ ACS NSQIP-P database	182,589 patients	< 18 Years	2012-2014	4.8%	Development study/ Internal: Bootstrap

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Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
Polites et al. 2016[44], USA	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Retrospective/ ACS NSQIP-P database	54,870 patients (DC: 38,397, VC: 16473)	29 days - < 18 years	2012-2014	3.6%	Development study/ Internal: Random-Split-Sample
Smith et al. 2015[50], USA	Developed model	Cardiac surgery	30-day HRs following congenital heart surgery	Retrospective/ PHIS database, US census data, 43 not-for-profit, tertiary care paediatric hospitals	17,871 discharges (DC: 9,104, VC: 8,767)	< 18 years	Jan. - Dec. 2011	11.0%	Development study/ Internal: Split-Sample (Non-Random temporal)
	Validated model						Jan. - Dec. 2012	NR	External validation study /Split-Sample (Non-Random temporal)
Jiang et al. 2018[56], USA	CCI (validation NRD)	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ Nationwide Readmissions Database	8,006 patients	<18 years	2013	NR	External validation study
	VWI (validation NRD)								External validation study
	Rhee index (validation NRD)								External validation study
	CCI (validation SID)	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ State Inpatient Databases	6,236 patients	<18 years	2007-2010	8%[a]	External validation study
	VWI (validation SID)								External validation study
	Rhee index (validation SID)								External validation study
Delaplain et al. 2020[48], USA	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Retrospective / Cerner Health Facts database, 28 hospitals	82,532 patients (DC: 75%, VC: 25%)	< 18 years	2000-2017	8.8%	Development study/ Internal: Random-Split-Sample[b]
General medical conditions related HRs									
Learly et al. 2019[52], USA	Prediction at admission	Complex chronic conditions	30-day unplanned HRs	Retrospective / US Census Bureau data, 1 academic medical centre	2,296 index admissions	6 months to 18 years	Oct. 2010 - July 2016	8.2%	Development study/ Internal: Bootstrap

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Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
	Prediction at discharge								Incremental value study/ Internal: Bootstrap
Ryan et al. 2021[54], USA	PASS (validation)	Asthma	30-day unplanned HRs	Retrospective/ 1 university-affiliated, tertiary paediatric referral centre	328 patients	5-18 years	May 2015-Oct. 2017	3.0%	External validation study

Abbreviations: ACS NSQIP-P, American College of Surgeons' (ACS) National Surgical Quality Improvement Program Pediatric; CCI, Charlson Comorbidity Index; DC, derivation cohort; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; NRD, nationwide readmission database; PACR, Pediatric All-Condition Readmission; PASS, Pediatric Asthma Severity Score; PHIS, Paediatric Health Information Systems; SDH, social determinants of health; SID, state inpatient database; VWI, Van Walraven Index; VC, validation cohort

[a] Own approximation resulting from Table 3 in the study based on readmission per risk score[56].

[b] Assumption for validation method: Odds ratios for 30-day unplanned HRs are displayed in a Table that is part of the DC from the 7-day unplanned HRs predictive model[48].

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Applicability and significant risk factors in predictive models

Based on the eleven predictive models with a development or incremental value design, 23 significant risk factors associated with 30-day HRs were identified (see table 2). The most common risk factors were comorbidity (n=8), (postoperative) length of stay (n=6), illness severity (n=6) and health service usage prior to index admission (n=5). The significant risk factors were inconsistently defined across predictive models, allowing a direct comparison only to a limited extent. ORs for comorbidity ranged from 1.08[55] to 2.41[53] across predictive models. A length of stay of >=15 days (OR=2.39)[53] and a postoperative length of stay of >4 days (Hazard Ratio=3.12)[44] were considered to be a major risk factor. For all-cause HRs, 14 significant risk factors were found, and 15 for HRs related to surgical conditions, respectively. For HRs related to general medical conditions, five significant risk factors were identified, while ORs for health service usage prior to index admission are similar (i.e., any admissions (OR=1.7) or emergency department visits (OR=2.04) in the previous six months[52]).

Most predictive models are potentially applicable at discharge (n=19), while two predictive models can be used at index admission[46, 52], based on the significant and examined variables (see table A5 in the supplementary material for an overview of variables and table 3 for an application description).

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Table 2: Significant risk factors for 30-day HR predictive models with a development or incremental value design

Health condition group	All-cause (n=5)					Surgical conditions related (n=4)				General medical conditions related (n=2)	
Reference	[47]	[55]	[49]	[46]	[53]	[45]	[44]	[50]	[48]	[52][a]	[52][b]
Location of residence*		x			x						
Health insurance				x	x			x			
Type of index hospital						x					
Living environment				x							
Characteristics of primary care provider	x										
Age at admission/operation					x						
Race/Ethnicity		x						x			
Health service usage prior to index admission			x	x					x	x	x
Comorbidity		x	x	x	x	x	x			x	x
Illness severity			x	x			x	x		x	x
LOS/Postop LOS			x		x		x	x	x		x
Principal diagnoses			x					x	x		
Principal procedures							x				
Inpatient complications						x	x				
Specific medication at index admission									x		
Devices at index admission								x			
Length of operation							x				
Wound contamination before operation							x				
The ASA class						x					
Discharge on Friday or Weekend					x						
Discharge disposition											x
Discharge with increased medication/further treatment	x										
Admission on Friday					x						

Abbreviations: ASA class, American Society of Anesthesiologists class; LOS, length of stay; postop LOS, postoperative length of stay
x=risk factor (Odds ratio/Hazard ratio >1)
[a] admission model, [b] discharge model, *social determinants of health are included (e.g., median household income)

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Completeness in reporting and discriminative ability at model level

Information about TRIPOD adherence and performance at model level are provided in table 3. The median TRIPOD adherence of the models was 63% ($P_{25} - P_{75}$ 55% - 69%, average: 62%), ranging from 33%[51] to 81%[52]. Developed predictive models had a more favourable reporting quality in comparison to external validated models (i.e., median adherence of 63% ($P_{25} - P_{75}$ 62% - 73%, average: 66%) compared with 61% ($P_{25} - P_{75}$ 54% - 66%, average: 58%), respectively). The two models with the poorest adherence in reporting were based on an external validation design, and the validation of these models was not the primary aim of the study[49, 51].

Table 3: Performance, application and TRIPOD adherence of 30-day HR predictive models in paediatrics (n=21)

Reference	Model name	Performance		TRIPOD score	Potentially applicable…
		Discrimination (c-statistic)	Calibration		
All-cause related HRs					
Brittan et al. 2018[47]	Composite Score	0.62		73.33%	at discharge
Sills et al. 2017[55]	PACR + SDH	0.708		64.71%	at discharge
Ehwerhemuepha et al. 2018[49]	Unnamed	VC: 0.79		63.33%	at discharge
	LACE (validation)	0.68		44.44%	at discharge
Bradshaw et al. 2020[46]	HARRPS-tool	Score: 0.65		73.33%	at admission
Zhou et al. 2020[53]	Unnamed	0.645		62.07%	at discharge
Ehwerhemuepha et al. 2020[51]	LACE (validation)	0.7014		33.33%	at discharge
Surgical conditions related HRs					
Vo et al. 2018[45]	Unnamed	0.747	slope: 1; intercept: 0.002	68.97%	at discharge
Polites et al. 2016[44]	Unnamed	DC: 0.71; VC: 0.701	(DC): p = 0.95, O/E-Ratio= 1.03; (VC): p = 0.36, O/E-Ratio= 1.07	62.07%	at discharge
Smith et al. 2015[50]	Developed model	DC: 0.68	H&L (DC): 0.47	55.17%	at discharge
	Validated model	VC: 0.68		53.85%	at discharge
Jiang et al. 2018[56]	CCI (validation NRD)	0.63	calculation not possible	62.96%	at discharge
	VWI (validation NRD)	0.54	calculation not possible	66.67%	at discharge
	Rhee index (validation NRD)	0.58	calculation not possible	70.37%	at discharge
	CCI (validation SID)	0.63	calculation not possible	59.26%	at discharge
	VWI (validation SID)	0.54	calculation not possible	62.96%	at discharge
	Rhee index (validation SID)	0.56	calculation not possible	66.67%	at discharge

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		Performance			
Delaplain et al. 2020[48]	30-day readmission model	VC: 0.799		51.72%	at discharge
General medical conditions related HRs					
Learly et al. 2019[52]	Prediction at admission	0.65; Score: 0.65	Calibration plot	79.31%	at admission
	Prediction at discharge	0.67; Score: 0.67	Calibration plot	81.25%	at discharge
Ryan et al. 2019[54]	PASS (validation)	0.28		55.17%	at discharge

Abbreviations: CCI, Charlson Comorbidity Index; DC, derivation cohort; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; NRD, nationwide readmission database; PACR, Pediatric All-Condition Readmission; PASS, Pediatric Asthma Severity Score; SDH, social determinants of health; SID, state inpatient database; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; VWI, Van Walraven Index; VC, validation cohort

Including all 37 items, the overall median adherence per TRIPOD item across models was 62% ($P_{25} - P_{75}$ 29% - 90%, average: 58%), ranging from 0% to 100% (see table A6 in the supplementary material for a detailed description by model type). The overall adherence per TRIPOD item is illustrated in figure 2.

[…Placement of figure 2…]

Figure 2: Overall adherence per TRIPOD item across all included predictive models (n=21).

52% of the models reported the title (item 1) completely, while 62%[46, 47, 49, 52, 54-56] of the models mentioned the predictive model type in this context. 5% of the models had a completed abstract (item 2). The detailed predictors definition (item 7a) was fulfilled for more models (81%), in contrast to outcome definition (item 6a) (reported in 48%). The handling of predictors in the analysis (item 10a) showed incomplete reporting in 73% of the models. In addition, the handling (item 9, reported in 29%) and reporting of missing values (part of item 13b, reported in 14%) was not addressed in many models. Just 9% of the models displayed complete reporting of the model-building procedure (item 10b), as the majority of the models (91%) did not address the testing of interaction terms[44, 45, 47-50, 52, 53, 55]. The description (item 10d) and reporting of performance measures (item 16) was incomplete in 48% of the models. Just 52% of the models address results of calibration measures (cf. table 3). No model presented the full predictive model (item 15a) by providing an example of an intercept. An explanation for using the prediction model (item 15b, e.g., by a simplified scoring rule) was presented in 64% of the models. One model provided detailed information about a simplified scoring rule (item 15b) in supplementary material[52].

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The discriminative ability (c-statistic) of the models ranged from 0.28[54] to 0.79[48, 49]. 15 out of 21 predictive models had a c-statistic <0.7. The linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.078 (P=.74). Models with good discriminative ability (c-statistic >0.7)[44, 45, 48, 49, 51, 55] are potentially applicable at discharge and have a TRIPOD score ranging from 52%[48] to 69%[45]. The two models with the highest reporting quality (79%, 81%) are applicable for predicting 30-day unplanned HRs of children with complex chronic conditions. The c-statistic of these models was 0.65[52] and 0.67[52], respectively (see figure A1 in the supplementary material for a combined illustration of performance and TRIPOD adherence at model level).

DISCUSSION

Based on 13 studies, this systematic review identifies 21 predictive models that could potentially be used for determining individual 30-day HR risk in paediatrics. According to the models, the three most common significant risk factors in predictive models were comorbidity, (postoperative) length of stay and illness severity. Six validated predictive models have a c-statistic of >0.7. These models are potentially applicable at discharge for the prediction of all-cause or surgical condition-related 30-day unplanned HRs. The median TRIPOD adherence of the predictive models included was 63% ($P_{25} - P_{75}$ 55% - 69%), ranging from 33% to 81%. Common reasons for reduced reporting quality had to do with incomplete information (i.e., abstract, performance measures, handling of predictors, missing values, or model presentation). The two predictive models with the highest TRIPOD adherence focused on general medical condition-related unplanned HRs, but showed poor discriminative ability (i.e., a c-statistic below 0.7).

The reporting quality (median of TRIPOD adherence 63% at model level) in our review was similar to that reported in other systematic reviews which have tested TRIPOD adherence for predictive models published in high-impact journals across different domains. The median overall TRIPOD adherence for prognostic model publications was 44% ($P_{25} - P_{75}$ 35% – 53%)[12]. Furthermore, an average overall TRIPOD adherence of 74% from 2012 to 2014 and of 76% from 2016 to 2017 was found[24]. Improving the reporting quality of predictive models may facilitate their implementation and validation.

In general, reporting quality and discriminative ability can provide crucial information about the strengths and weaknesses of a predictive model for implementation in practice (see figure A1 in the supplementary material for a combined illustration). However, the results from this

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systematic review revealed considerable differences in the c-statistics (0.28[54] to 0.79[48, 49]) and in the TRIPOD scores (33%[51] to 81%[52]) at model level. When considering the available information about reporting quality and discriminative ability in relation to each other, it should be noted that the linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.078 (P=.74). Therefore, clinicians and decision-makers should independently evaluate both aspects for the selection of an appropriate predictive model.

This systematic review has certain limitations. (i) The studies included needed be to published in English or German with full-text access. (ii) Summarizing the results of the included studies quantitatively was not possible due to the heterogeneity of the predictive models (resulting from differences in sample sizes, the examined variables or variations in the periods of data collection). (iii) With regard to the inclusion criteria, we defined the outcome as 30-day HRs for predictive models that can address both planned and unplanned HRs. The resulting heterogeneity was accepted because in an earlier systematic review a wider definition of HRs had proved useful[8]. (iv) The sample size of the included studies were reported in different units(e.g., encounters, discharges), impeding the comparisons of HR rates. (v) Our assignment of the predictive models that are potentially applicable at discharge assumes that the required variables are available at the time point. If clinicians and other stakeholders decide to use a predictive model, it should be checked beforehand whether complete data collection is possible at the desired time. (vi) In addition to the identified medical risk factors (e.g., comorbidity) and several country-specific risk factors (e.g., location of residence) that result in paediatric readmissions, health-policy initiatives may also affect the readmission rates in paediatric clinical practice[57]. However, due to a lack of data, these aspects could not be captured by this review.

This systematic review did not identify predictive models for individualized risk prediction of potentially preventable unplanned HRs in paediatrics, emphasizing past discussions to expand the research field further[3].

Current external validation studies were conducted in the USA and primarily examined the applicability of existing predictive models with other outcomes or population backgrounds to paediatric 30-day HRs[49, 51, 54, 56]. Therefore, external validation studies are needed for those models that are explicitly developed to predict 30-day HRs in paediatrics.

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Non-regression-based techniques (e.g., neural networks or machine learning) are an increasing field in order to predict 30-day HRs in paediatrics, showing good discriminative ability[31-34, 51]. Future systematic reviews should summarize and critically assess existing non-regression-based HR predictive models in paediatrics, for instance by applying the TRIPOD-ML statement that is going to be published[58].

Existing studies discuss the benefit of shorter time intervals in order to identify preventable readmissions more accurately[6, 59]; one study concluded that a 30-day unplanned HR metric was more precise (c-statistic=0.799) for paediatric trauma patients than a 7-day unplanned HR metric(c-statistic=0.737)[48]. To our knowledge, there is one predictive model for 365-day[7] and one for 7-day[48] unplanned HRs in paediatrics with good discriminative ability (c-statistic>0.7). Future studies should address the evaluation of paediatric HR predictive models with different time intervals.

CONCLUSION

This systematic review revealed an increase in the development of predictive models for 30-day HRs in paediatrics in recent years. To support the implementation of the predictive models in the long term, it is essential to validate existing models in order to test their applicability in different settings. To increase accessibility for use, more attention should be placed on completeness in reporting, particularly for items that may be relevant for the implementation of paediatric 30-day HR predictive models in practice.

List of Figures

Figure 1: Flowchart for the search and study selection process (PRISMA)

Figure 2: Overall adherence per TRIPOD item across all included predictive models (n=21)

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Contributors:

IN conceptualized and designed the systematic review, participated in the literature search, study selection, quality assessment, data extraction, data analyses and drafted the initial manuscript.

NK contributed to the literature search, study selection, quality assessment, data extraction and critically reviewed the manuscript.

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SS contributed to the data analysis and critically reviewed the manuscript.

JD contributed to the study selection, data extraction, data analysis and critically reviewed the manuscript.

DM conceptualized and designed the systematic review, participated in the study selection, quality assessment, data extraction, data analyses and critically reviewed the manuscript.

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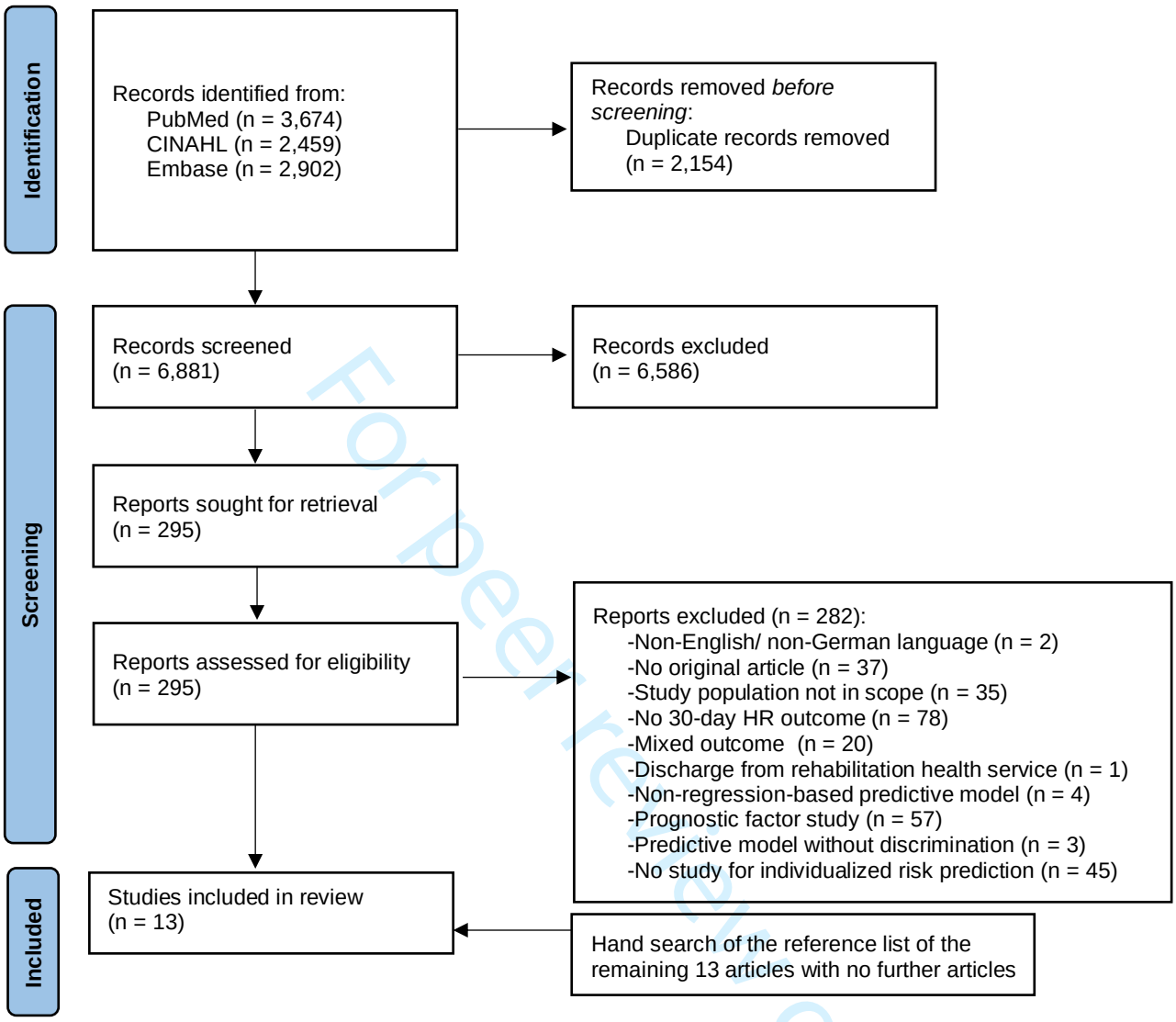
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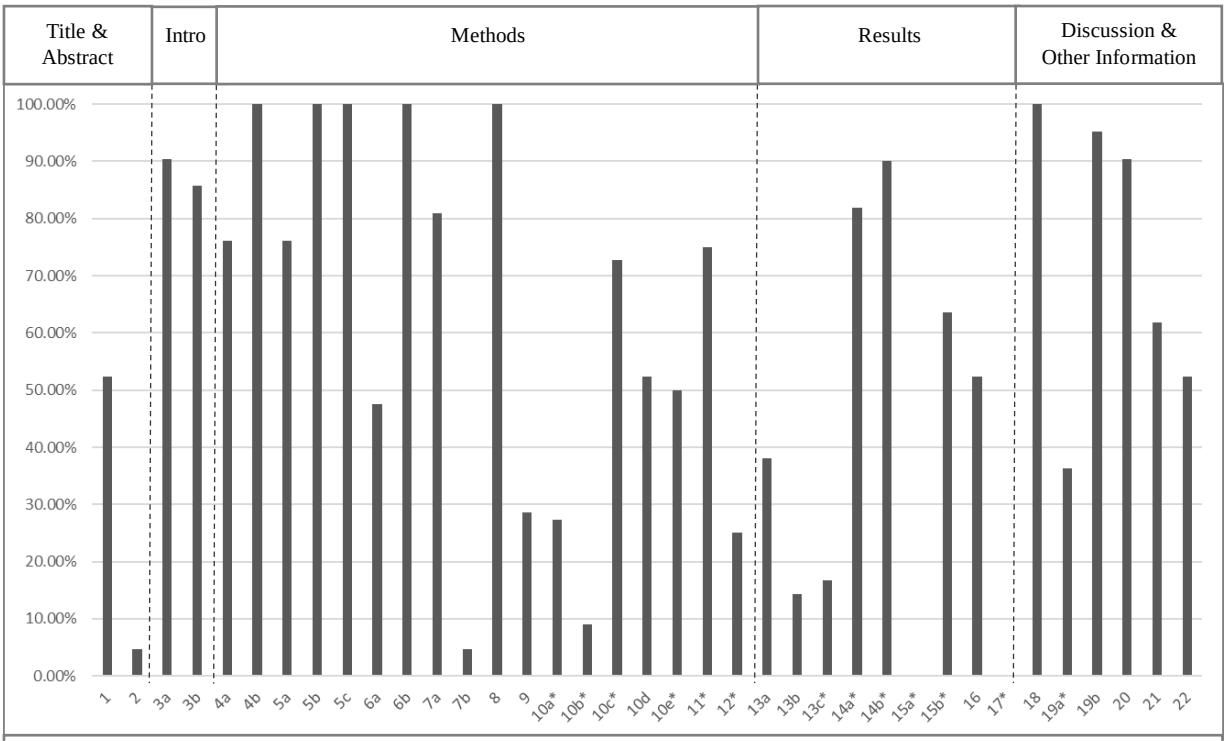
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Abbreviations: HR, hospital readmission; PRISMA, preferred reporting items for systematic reviews and meta-analyses.

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Percentages relate to the number of models for which an item was applicable (in this case, the respective item should have been reported). *Indication of derivation from the total number of models for which a TRIPOD item was applicable (N=# of models for which the TRIPOD item is applicable): 10a (N=11), 10b (N=11), 10c (N=11), 10e (N=2), 11 (N=4), 12 (N=12), 13c (N=12), 14a (N=11), 14b (N=10), 15a (N=11), 15b (N=11), 17 (N=1), 19a (N=11).

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Supplemental Material

Table A1: Search strategy for PubMed

#	PubMed (Search date: 01st January 2021)	Results
1	"Patient Readmission"[Mesh]	18,336
2	rehospitali*[Title/Abstract]	8,017
3	readmission*[Title/Abstract]	31,790
4	(hospital[Title/Abstract]) AND readmission*[Title/Abstract]	21,144
5	(unplanned[Title/Abstract]) AND readmission*[Title/Abstract]	2,360
6	(patient[Title/Abstract]) AND readmi*[Title/Abstract]	16,395
7	readmit*[Title/Abstract]	8,289
8	re-admission*[Title/Abstract]	2,205
9	(repeat*[Title/Abstract]) AND hospital*[Title/Abstract]	29,556
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	75,738
11	predictive factor*[Title/Abstract]	28,124
12	predict*[Title/Abstract]	1,646,322
13	"Predictive Value of Tests"[Mesh]	207,290
14	"ROC Curve"[Mesh]	60,417
15	model*[Title/Abstract]	3,025,019
16	c-statistic*[Title/Abstract]	6,938
17	ROC*[Title/Abstract]	55,019
18	"Sensitivity and Specificity"[Mesh]	595,955
19	Sensitivity [Title/Abstract]	827,733
20	Specificity [Title/Abstract]	478,451
21	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20	5,225,692
22	"Child"[Mesh]	1,936,577
23	"Infant"[Mesh]	1,152,685
24	"Adolescent"[Mesh]	2,058,561
25	(((((adolescen*[Title/Abstract]) OR teen*[Title/Abstract]) OR youth[Title/Abstract]) OR juvenile*[Title/Abstract]) OR young person*[Title/Abstract]) OR young people*[Title/Abstract]	464,723
26	"Pediatrics"[Mesh]	59,136
27	((((((((((child*[Title/Abstract]) OR (infant*[Title/Abstract])) OR (toddler*[Title/Abstract])) OR (bab*[Title/Abstract])) OR (newborn*[Title/Abstract])) OR (neonat*[Title/Abstract])) OR (school age*[Title/Abstract])) OR (preschool)) OR (paediatric*[Title/Abstract])) OR (pediatric*[Title/Abstract])) OR (kid*[Title/Abstract])) OR (boy*[Title/Abstract])) OR (girl*[Title/Abstract])	2,246,452
28	22 OR 23 OR 24 OR 25 OR 26 OR 27	4,365,971
29	10 AND 21 AND 28	3,674

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Table A2: Search strategy for CINAHL

	CINAHL (Search date: 01st January 2021)	Results
1	(MH "Readmission")	14,256
2	rehospitali*	2,624
3	readmission*	21,640
4	hospital AND readmission*	13,210
5	unplanned AND readmission*	1,227
6	patient AND readmi*	17,823
7	readmit*	2,588
8	re-admission*	954
9	repeat* AND hospital*	12,404
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	36,195
11	predictive factor*	10,824
12	predict*	394,485
13	(MH "Predictive Value of Tests")	52,176
14	(MH "ROC Curve")	27,863
15	model*	626,177
16	c-statistic*	2,593
17	ROC*	45,218
18	(MH "Sensitivity and Specificity")	85,260
19	sensitiv*	236,551
20	specific*	481,504
21	(MH "Predictive Validity")	5,587
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	1,337,979
23	(MH "Child")	467,217
24	(MH "Infant+")	260,190
25	(MH "Adolescence+")	535,922
26	adolescen* OR teen* OR youth OR juvenile* OR young person* OR young people*	598,199
27	(MH "Pediatrics+")	21,316
28	child* OR infant* OR toddler* OR bab* OR newborn* OR neonat* OR school age* OR preschool OR paediatric* OR pediatric* OR kid* OR boy* OR girl*	1,150,913
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	1,413,615
30	10 AND 22 AND 29	2,459

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Table A3: Search strategy for Embase

	Embase (Initial Search date: 01st January 2021*; Updated Search date: 7th June 2021)	Results from updated search
1	'hospital readmission'/exp	73,736
2	rehospitali*:ab,ti	10,922
3	readmission*:ab,ti	57,908
4	(hospital NEAR/10 readmission*):ab,ti	22,791
5	(unplanned NEAR/10 readmission*):ab,ti	3,446
6	(patient NEAR/5 readmi*):ab,ti	5,568
7	readmit*:ab,ti	1,6262
8	're admission*':ab,ti	5,802
9	(repeat* NEAR/5 hospital*):ab,ti	5,341
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	100,580
11	'predictive factor*':ab,ti	47,721
12	predict*:ab,ti	2,301,090
13	'predictive value'/exp	191,705
14	'roc curve'/exp OR 'receiver operating characteristic'/exp	149,025
15	'model'/exp	3,201,228
16	'c statistic*':ab,ti	12,892
17	roc*:ab,ti	196,628
18	'sensitivity and specificity'/exp	391,861
19	'sensitivity':ab,ti	1,109,211
20	'specificity':ab,ti	636,321
21	'predictive validity'/exp	8,691
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	6,371,499
23	'child'/exp	3,031,530
24	'infant'/exp	1,180,824
25	'adolescent'/exp	1,741,216
26	adolescen*:ab,ti OR teen*:ab,ti OR youth:ab,ti OR 'juvenile*':ab,ti OR 'young person*':ab,ti OR 'young people':ab,ti	598,758
27	'pediatrics'/exp	123,610
28	'child*':ti,ab OR 'infant*':ti,ab OR 'toddler*':ti,ab OR bab*:ti,ab OR 'newborn*':ti,ab OR neonat*:ti,ab OR 'school age*':ti,ab OR 'preschool':ti,ab OR 'pediatric*':ti,ab OR kid*:ti,ab OR 'boy*':ti,ab OR 'girl*'	3,642,407
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	5,498,512
30	#10 AND #22 AND #29 AND [1-1-1966]/sd NOT [2-1-2021]/sd	2,902
* Final result from initial search: 2,845		

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Table A4: Risk of bias assessment at study level

Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?''	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Vo et al. 2018	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - No detailed description of statistical analysis for the simplified scoring rule; Unadjusted associations for "Complications" are missing	**1**
Polites et al. 2016	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Moderate - 38.5% missing values for the ICD-9-diagnosis variable used for determining the reasons for readmission	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Brittan et al. 2018	- Moderate - Potential limitation of generalizability as just one paediatric hospital was examined; Only basic information about "age"	- Moderate - About 75% of patients were missing homecare values in the original data (the study assumption is that this patients did not received home health)	- Moderate - Assumption that 75% of patients did not receive home health; Possible omission or error of clinical documentation	- Moderate - no clear description for the determination of unplanned hospital readmissions	- Low - just PEs of the composite score are considered	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?''	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Sills et al. 2017	- Low -	- Moderate - No flow diagram	- Moderate - Area-level SDH data were not created for research and are a biased proxy for individual-level SDH measures; Handling of continuous predictors relating to ZIP-Codes (i.e. linear or non-linear) not clear described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2018	- Moderate - Potential limitation of generalizability as just one hospital was examined; No detailed information about the source of ZIP-related variables; No information about characteristics of patients in the validation cohort (Random-Split-Sample)	- Low -	- Moderate - No detailed description for the handling of continuous predictors and method for categorized predictors	- Low - The outcome was determined automatically and not validated by manual chart review	- Low -	- Moderate - No detailed definition for LACE variables; Missing information for the validation cohort	**2**
Smith et al. 2015	- Moderate - No information about characteristics of patients in the external validation set; No detailed information about the source of ZIP-related variables	- Moderate - No flow diagram; Missing information for validation cohort	- Moderate - Age categories are not clearly reported	- Low -	- Low -	- Moderate - Missing information for the validation cohort (non-random split sample)	**2**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?''	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Jiang et al. 2018	- Low -	- Moderate - Outcome events for NRD-data set are not described; No flow diagram	- Moderate - Patient distribution is presented by risk scores	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions; It is unclear, if 30-day inpatient readmissions in the NRD are also calculated from the date of the index procedure	- Low - Due to external validation of existing risk scores confounders are not relevant	- Moderate - Method of CCS is not clear; Outcome event rates and summary information of predictors are not clearly reported	**6**
Learly et al. 2019	- Moderate - Potential limitation of generalizability as just one hospital examined; No detailed information about the source of ZIP-related variables	- Low -	- Moderate - Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Low -	**2**
Bradshaw et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; Due to deidentification of patient data demographic characteristics are not captured	- Low -	- Low -	- Moderate - "The multiple categories that comprise the HARRPS-tool have different weight values associated that help prevent planned readmissions from skewing the results."(p. 53)	- Low -	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?''	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Delaplain et al. 2020	- Moderate - Focus on baseline characteristics of readmission in 7 days; No detailed characteristics for the validation cohort (random-split sample)	- Moderate - Information about the patient flow are provided at hospital level; Hospitals were excluded, if data were not collected for all variables related to the study; Missing information for validation cohort (random-split sample)	- Low - Predictors are presented by focusing on 7-day readmissions	- Moderate - Clear definition for 7-day unplanned readmissions with possible transfer to 30-day unplanned readmissions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**
Zhou et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; One-day procedures are also included (discharge within 24h)	- Low -	- Moderate - Census data might be a biased proxy for individual level measures; Method to choose the cut-points was not clearly described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2020	- Moderate - Basic information just for the training data set	- Moderate - Missing information for test data set	- High - No summary information for all predictors included in the validated model; No clear definition of predictors	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions	- Low - Due to external validation of LACE index confounders are not relevant	- High - Applied method for LACE is not clear (i.e. possible updating); Missing information for training data set	**1**
Ryan et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Low -	- Low -	- Moderate - It is unclear, how unplanned readmissions are determined	- Low - Due to external validation of PASS confounders are not relevant	- Moderate - Summary information of predictors are not clearly reported	**1**

* For the risk of bias assessment at study-level, only the predictive models that were also included in the systematic review were considered.

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Table A5: Examined and significant variables for all 21 included predictive models

Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
All-cause related readmissions					
Brittan et al. 2018	Composite Score	All-cause	30-day unplanned HRs	Age; Lenght of stay; Non-English speaking caregiver; Discharge medications; Enteral feeding (eg, gastrostomy tube feeding); Respiratory (eg, home ventilator); IV infusion (eg, infusion of IV medication); Speech therapy; Physical therapy; Occupational therapy; Skilled nursing home visits; Private duty nursing home visits; CNA nursing assistant home visits; Durable medical equipment (eg, wheel chair); Home Care ≥ 1 order; Composite score = 0; Composite score = 1; Composite score = 2; Composite score = 3	Score based on the variables: Non-English speaking caregiver, Discharge medications, Home Care ≥ 1 order: Composite score (0 vs 1) (OR=1.7; 95% CI 1.5-2); Composite score (0 vs ≥ 2) (OR=4.2; 95% CI 3.6-4.9)
Sills et al. 2017	PACR + SDH	All-cause	30-day unplanned HRs	CCI count; Infectious and parasitic disease; Neoplasms; Endocrine, nutritional, and metabolic diseases and immunity disorders; Diseases of blood and blood-forming organs; Mental disorders; Diseases of the nervous system and sense organs; Diseases of the circulatory system; Diseases of the respiratory system; Diseases of the digestive system; Diseases of the genitourinary system; Diseases of the skin and subcutaneous tissue; Diseases of the musculoskeletal system; Congenital anomalies; Certain conditions originating in the perinatal period; Symptoms, signs, and ill-defined conditions; Injury and poisoning; Factors influencing health status and contact with health services; Age group; Sex; Race; Hispanic ethnicity; Payer; Median household income; Proportion of housing units vacant; Proportion of households with children that are single-parent; Proportion of families below poverty level; Unemployment rate; Proportion of adults with less than a high school diploma or equivalent	CCI count: [2 body systems (OR=1.51; 95% CI 1.44-1.58), 3 body systems (OR=1.57; 95% CI 1.48-1.67), >3 body systems (OR=1.31; 95% CI 1.19-1.44)]; CCIs: Neoplasms (OR=0.46; 95% CI 0.44-0.48), Endocrine, nutritional, and metabolic diseases and immunity disorders (OR=0.8; 95% CI 0.77-0.83), Diseases of blood and blood-forming organs (OR=0.59; 95% CI 0.56-0.61), Mental disorders (OR=0.95; 95% CI 0.92-0.99), Diseases of the nervous system and sense organs (OR=0.78; 95% CI 0.75-0.8), Diseases of the circulatory system (OR=0.73; 95% CI 0.7-0.76), Diseases of the respiratory system (OR=1.08; 95% CI 1.05-1.12), Diseases of the digestive system (OR=0.65; 95% CI 0.63-0.68), Diseases of the genitourinary system (OR=0.7; 95% CI 0.66-0.74), Congenital anomalies (OR=0.88; 95% CI 0.85-0.91), Injury and poisoning (OR=0.63; 95% CI 0.54-0.74), Factors influencing health status and contact with health services (OR=0.51; 95% CI 0.49-0.53); Age group: [1-<5y (OR=0.84; 95% CI 0.81-0.87), 5-<8y (OR=0.73; 95% CI 0.69-0.76), 8-<12 y (OR=0.72; 95% CI 0.68-0.75), 12-<18 y (OR=0.79; 95% CI 0.76-0.83)]; male (OR=0.96; 95% CI 0.93-0.98); Race: [Asian (OR=0.91; 95% CI 0.84-0.99), Other (OR=0.92; 95% CI 0.88-0.96)], Hispanic (OR=1.06; 95% CI 1.02-1.1); Payer: [Private (OR=0.85; 95% CI 0.83-0.88)]; Median household income (per $10 000 increase) (OR=1.02; 95% CI 1.02-1.02); Vacancy rate (per 10% increase) (OR=0.96; 95% CI 0.94-0.99); Percent single-parent (per 10% increase) (OR=1.03; 95% CI 1.01-1.04)
Ehwerhemuepha et al. 2018	Unnamed	All-cause	30-day unplanned HRs	Sex; Race and/or ethnicity; Length of stay, d; Age, y; Median income by zip code, per $10 000; Percent vacant houses by zip code; Percent single parents by zip code; Low income primary medical insurance; Planned admission; ED visits within last 6 mo; Admitted through ED; Previous inpatient visits within last 6 mo; History of 30 d readmission within last 6mo; Ambulatory resource use within last 6 mo; Charlson's comorbidities; Complex chronic conditions; Was in ICU; pRI: Admitting; pRI: Discharge; pRI: Minimum; pRI: Maximum; pRI: Average; pRI: Decreasing slope across entire stay; pRI: Decreasing slope during last 24 h of hospitalization; pRI: Minimum occurred first 24 h; pRI: Maximum occurred last 24 h; Infectious and/or parasitic; Neoplasms (excluding encounters for chemotherapy); Blood and/or immune; Endocrine, nutritional, and/or metabolic; Mental and/or neurodevelopment; Nervous, eye, ear, and/or mastoid; Circulatory; Respiratory; Digestive; Skin and/or subcutaneous tissue; Musculoskeletal; Genitourinary; Perinatal period; Congenital malformations; Symptoms, signs, and/or laboratory findings not classified elsewhere; Injury and/or poison; External morbidity causes; Health status and/or services factors	Length of stay, d: [<4 (2, 3) (OR=1.23; NR% CI 1.07–1.42), < 7 (4, 5, 6) (OR=1.42; NR% CI 1.20–1.67), 7 or more (OR=1.80; NR% CI 1.51–2.14)]; Planned admission: [Yes (OR=0.65; NR% CI 0.52–0.82)]; ED and/ or department visits within last 6 mo: [1 (OR=1.27; NR% CI 1.11–1.45), 2 or more (OR=1.30; NR% CI 1.12–1.50)]; Admitted through ED: [Yes (OR=1.15; NR% CI 1.02–1.29)]; Previous inpatient visits within last 6 mo: [1 (OR=1.86; NR% CI 1.59–2.17), 2 or more (OR=2.39; NR% CI 1.94–2.94)]; History of 30 d readmission within last 6mo: [1 (OR=1.27; NR% CI 1.06–1.52), 2 or more (OR=2.75; NR% CI 2.19–3.44)]; Complex chronic conditions: [1 (OR=1.63; NR% CI 1.42–1.87), 2 or more (OR=1.66; NR% CI 1.43–1.92)]; Discharge pRI (10 pt increment) (OR=0.86; NR% CI 0.81–0.90); Maximum pRI occurred last 24 h of hospitalization (OR=0.85; NR% CI 0.75–0.95); Neoplasms (excluding encounters for chemotherapy): [Yes (OR=2.17; NR% CI 1.85–2.55)]; Blood and/or immune: [Yes (OR=1.30; NR% CI 1.14–1.48)]; Nervous, eye, ear, and/or mastoid: [Yes (OR=1.21; NR% CI 1.08–1.37)]; Circulatory: [Yes (OR=1.33; NR% CI 1.14–1.56)]; Respiratory: [Yes (OR=0.75; NR% CI 0.66–0.85)]; External morbidity causes: [Yes (OR=0.64; NR% CI 0.45–0.89)]; Health status and/or services factors: [Yes (OR=1.20; NR% CI 1.06–1.35)]
	LACE (validation)			Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Bradshaw et al. 2020	HARRPS-tool	All-cause	30-day unplanned HRs	Other Diagnosis; Anemia/Neutropenia; Appendectomy; Asthma; Bronchiolitis; Gastroenteritis; Pneumonia; Seizure; Sickle Cell Crisis; Upper Respiratory Tract Infection; Ventricular Shunt; At-risk admission diagnosis present; Chronic condition indicator; Readmitted within 30 days (history); Inpatient Admit in last 6 months; Acuity of admission: No admission acuity identified; Acuity of admission: ICN/PICU Admission; Acuity of admission: Significant psychosocial concern; Acuity of admission: Medical transport from outside facility; Acuity of admission: Other; Insurance type: Self-Pay; Insurance type: Medicaid; Insurance type: Commercial/Private; Insurance type: Medicaid & Self-Pay; Caregiver language: English; Caregiver language: Spanish; Caregiver language: Other; Medical equipment/Supplies count at home; Home nursing: Private Duty Nursing; Home nursing: Skilled Nursing; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy; Home therapy: Home Physical Therapy; Home therapy: Home Speech Therapy; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy	The final scoring system also includes risk factors, which were not significant in the multivariable analysis (i.e. At-risk admission diagnosis present; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy): Chronic condition indicator: [1 (OR=1.43; 95% CI 1.13-1.81), 2 (OR=1.75; 95% CI 1.27-2.40), 3 (OR=1.41; 95% CI 1.12-1.78)]; Previous 30-day readmission (OR=1.70; 95% CI 1.38-2.09); Inpatient admission in last 6 months (OR=2.21; 95% CI 1.76 2.54); ICN or PICU admit (OR=1.43; 95% CI 1.17-1.75); Self-Pay or Medicaid (OR=1.21; 95% CI 1.05 1.39); Medical equipment/Supplies count at home: [2 (OR=1.43; 95% CI 1.09-1.88), 3 (OR=2.68; 95% CI 1.80 4.01), 4 (OR=2.21; 95% CI 1.53 3.19)]
Zhou et al. 2020	Unnamed	All-cause	30-day unplanned HRs	Age; Gender; Admission status; Funding source as inpatients; Source of referral transport; State/Territory of residence; Care type provided; Type of health insurance; Index of Relative Social-Economic Advantage and Disadvantage (IRSAD); Interpreter service; ICU stay at index admission; Had GA at index admission; LOS at index admission; Day of index admission date (weekdays or weekend); Day of index admission date; Day of discharge from index admission; Day of discharge from index admission (weekdays or weekend); Number of Co-diagnoses	Age: [13-15y (OR=1.30; 95% CI 1.14–1.48) >=16y (OR=1.46; 95% CI 1.07–1.98)]; Private health insurance (OR=1.16; 95% CI 1.00–1.34); Aeromedical service (OR=0.47; 95% CI 0.31–0.71); IRSAD (%): 91-100 (OR=1.20; 95% CI 1.02–1.41) - based on residential postcode; GA at index admission (yes) (OR=0.67; 95% CI 0.64–0.76); LOS: [LOS 2-7 days (OR=1.42; 95% CI 1.30–1.55), LOS 8-14 days (OR=2.35; 95% CI 1.97–2.82), LOS >= 15 days (OR=2.39; 95% CI 1.88–2.98)]; Admission on Friday (OR=1.21; 95% CI 1.05–1.39); Day of discharge from index admission: [Friday (OR=1.26; 95% CI 1.10–1.44), Saturday (OR=1.34; 95% CI 1.15–1.57), Sunday (OR=1.24; 95% CI 1.05–1.47)]; Number of Co-diagnoses:[1 (OR=1.28; 95% CI 1.16–1.41), 2 (OR=1.73; 95% CI 1.53–1.95), 3 (OR=2.10; 95% CI 1.80–2.46), >=4 (OR=2.41; 95% CI 2.08–2.80)]
Ehwerhemuepha et al. 2020	LACE (validation)	All-cause	30-day HRs	Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study
Surgical conditions related readmissions					
Vo et al. 2018	Unnamed	All surgical specialities without cardiac surgery	30-day unplanned post-surgical HRs relating to noncardiac surgery	Age; BMI; ASA class; Gender; Race; Surgical specialty; Admission status; Urgency of procedure; Prematurity; Congenital Heart Disease; Bleeding complication; Wound complication; Pulmonary complication; Renal complication; Neurologic complication, Cardiac complication	ASA>=3 (OR=1.9; 95% CI 1.8-2.0); Inpatient vs outpatient (OR=3.5; 95% CI 3.3-3.7); >1 Postoperative complication (OR=3.14; 95% CI 2.92-3.34); Presence of CHD (OR=1.66; 95% CI 1.31-2.11)
Polites et al. 2016	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Age; Sex; Race; Procedure group; Procedure type; Weight percentage; Premature birth; Congenital malformation; Any comorbidity; Diabetes mellitus (Individual comorbidity); Respiratory (Individual comorbidity); GI/Hepatobiliary (Individual comorbidity); Major cardiac (Individual comorbidity); Acute renal failure (Individual comorbidity); Neurologic (Individual comorbidity); Immunosuppression (Individual comorbidity); Nutrition (Individual comorbidity); Hematologic (Individual comorbidity); Active malignancy (Individual comorbidity); Preoperative SIRS, sepsis, septic shock; Preoperative transfusion or hematocrit <32; Case status; ASA class; Operative time, minutes; Wound class; Preoperative length of stay, days; Postoperative complication; Superficial Incisional SSI; Deep SSI; Pulmonary complications; Sepsis/CL associated bloodstream infection; Postoperative length of stay, days	Procedure group: [Head and Neck (HR=2.40; 95% CI 1.48-3.89), Hepatobiliary (HR=1.69; 95% CI 1.17-2.44), Small and Large intestine (HR=1.59; 95% CI 1.33-1.90), Thoracic (HR=0.69; 95% CI 0.52-0.91)]; Comorbidity: [Preoperative acute renal failure (HR=2.47; 95% CI 1.31-4.66), Neurologic comorbidity (HR=1.3; 95% CI 1.05-1.62)]; SIRS/Sepsis/Septic Shock within 48 h prior to index admission (HR=1.2; 95% CI 1.02-1.41); Operative time, minutes: [60–140 min (HR=1.21; 95% CI 1.06-1.39), >140min (HR=1.51; 95% CI 1.26-1.81)]; Wound class: [Contaminated (HR=1.29; 95% CI 1.05-1.60), Dirty/Infected (HR=1.92; 95% CI 1.53-2.40)]; Any complication (HR=1.34; 95% CI 1.09-1.65); Postoperative length of stay, days: [2-4 days (HR=2.17; 95% CI 1.84-2.57), >4 days (HR=3.12; 95% CI 2.60-3.74)]

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Smith et al. 2015	Developed model	Cardiac surgery	30-day HRs following congenital heart surgery	[Readmission characteristics are not mentioned, as they were not a part of the modelling process] Male gender; Neonatal age group; Hospital volume, Ethnicity; Mean regional income ($USD); Payor; ICU length of stay (days); ICU LOS ≥7 days; Postoperative LOS (days); Postoperative LOS ≥14 days; ECMO support during hospitalization; RACHS-1 score; GERD diagnosis; Genetic syndrome; Vocal cord paralysis; Thrombosis; Prematurity; Renal insufficiency; Serious Infection; Mediastinitis; Sepsis; Necrotizing enterocolitis; Stroke; Tachyarrhythmia; Complete AV block; Gastrostomy tube placement; Tracheostomy; Diaphragm plication; Weekend discharge	Hispanic (OR=1.24; 95% CI 1.05–1.41); Government payor (OR=1.22; 95% CI 1.06–1.41); ICU LOS ≥7 days (OR=1.86; 95% CI 1.56–2.21); Postoperative LOS ≥14 days (OR=1.59; 95% CI 1.32–1.91); ECMO support during hospitalization (OR=1.74; 95% CI 1.16–2.60); RACHS-6 score (OR=2.60; 95% CI 1.83–3.69); Gastroesophageal reflux (OR=1.49; 95% CI 1.25–1.78); Genetic syndrome (OR=1.31; 95% CI 1.1–1.57); Thrombosis (OR=1.46; 95% CI 1.05–2.03)
	Validated model			Neonatal age group; Ethnicity; Payor; ICU LOS ≥7 days; Postoperative LOS ≥14 days; ECMO support during hospitalization; RACHS-1 score; GERD diagnosis; Genetic syndrome; Vocal cord paralysis; Thrombosis; Prematurity; Renal insufficiency; Serious Infection; Necrotizing enterocolitis; Stroke; Tachyarrhythmia; Complete AV block; Gastrostomy tube placement, Diaphragm plication; Weekend discharge	External validation study
Jiang et al. 2018	Charlson Comorbidity Index (validation NRD)	Urological surgery	30-day HRs after paediatric urological procedures	Myocardial infarction; Congestive heart failure; Peripheral vascular disease; Cerebrovascular disease; Dementia; Chronic pulmonary disease; Rheumatic disease; Peptic ulcer disease; Mild liver disease; Diabetes without chronic complication; Diabetes with chronic complication; Hemiplegia or paraplegia; Renal disease; Any malignancy, including lymphoma and leukemia, except malignant neoplasm of skin; Moderate or severe liver disease; Metastatic solid tumor; AIDS/HIV	External validation study
	Van Walraven Index (validation NRD)			Congestive heart failure; Cardiac arrhythmias; Valvular disease; Pulmonary circulation disorders; Peripheral vascular disorders; Hypertension; Paralysis; Neurodegenerative disorders; Chronic pulmonary disease; Diabetes, uncomplicated; Diabetes, complicated; Hypothyroidism; Renal failure; Liver disease; Peptic ulcer disease, no bleeding; AIDS/HIV; Lymphoma; Metastatic cancer; Solid tumour without metastasis; Rheumatoid arthritis/collagen; Coagulopathy; Obesity; Weight loss; Fluid and electrolyte disorders; Blood loss anemia; Deficiency anemia; Alcohol abuse; Drug abuse; Psychosis; Depression	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
	Rhee index (validation NRD)			Age ,24 mo (vs >= 24 mo); Female (vs male); Peri-/endo-/myocarditis, cardiomyopathy, or tamponade; Acute myocardial infarction; Coronary atherosclerosis/other ischemic heart disease; Pulmonary vascular disease (eg, PE, pulmonary HTN); Ventricular tachycardia or other cardiac dysrhythmias; Cardiac arrest or ventricular fibrillation or flutter; Congestive heart failure; Aortic, peripheral, visceral artery aneurysms/ dissection; Aortic or peripheral arterial embolism or thrombosis; Cardiac or circulatory congenital anomalies; Pneumonia; Influenza; Chronic obstructive pulmonary disease/ bronchiectasis; Asthma; Aspiration pneumonitis; Respiratory failure, insufficiency, arrest; Cystic fibrosis; Respiratory distress syndrome; Acute renal failure; Chronic renal failure; Urinary tract infections or unspecified cystitis; Thyroid disorders or other endocrine disorders; Diabetes mellitus or complications; Septicemia (except in labor); Fungal infection; HIV infection; Hepatitis; Meningitis, encephalitis, or other CNS infection; Coma, stupor, or brain damage; Acute cerebrovascular disease; Gastroduodenal ulcer, gastritis, or duodenitis; Intestinal obstruction without hernia; Peritoneal or intestinal abscess, peritonitis; Biliary tract disease; Liver disease (eg, cirrhosis, increased LFTs); Pancreatic disorders (not diabetes); Gastrointestinal hemorrhage; Leukemia; Lymphomas or reticuloendothelial neoplasms; CNS or miscellaneous intracranial or intraspinal neoplasms; Primary malignant tumor of adrenal or paraganglia; Renal tumors; Hepatic tumors; Primary malignant bone or articular cartilage tumors; Soft tissue sarcomas; Immunity disorders (except AIDS); Coagulation or hemorrhagic disorders; Disease of white blood cells (eg, lymphocytosis, lymphopenia); Chronic ulcer of skin; Spondylosis, intervertebral disc disorders; Systemic lupus erythematosus or connective tissue disorder; Crushing injury or internal injury; Shock; Drowning/submersion; Firearm; Motor vehicle traffic; Pedal cyclist, not MVT; Suffocation; Suicide or intentional self-inflicted injury; Fracture of long bone or spine; Intracranial injury; Poisoning by nonmedicinal substances; Skull or face fractures; Open wounds of head, neck, or trunk; Superficial soft tissue injury: contusion; Short gestation, low birth wt, or fetal growth retardation; Hypoxia, asphyxia, or aspiration during birth; Birth trauma; Perinatal jaundice	External validation study
	Charlson Comorbidity Index (validation SID)			Myocardial infarction; Congestive heart failure; Peripheral vascular disease; Cerebrovascular disease; Dementia; Chronic pulmonary disease; Rheumatic disease; Peptic ulcer disease; Mild liver disease; Diabetes without chronic complication; Diabetes with chronic complication; Hemiplegia or paraplegia; Renal disease; Any malignancy, including lymphoma and leukemia, except malignant neoplasm of skin; Moderate or severe liver disease; Metastatic solid tumor; AIDS/HIV	External validation study
	Van Walraven Index (validation SID)			Congestive heart failure; Cardiac arrhythmias; Valvular disease; Pulmonary circulation disorders; Peripheral vascular disorders; Hypertension; Paralysis; Neurodegenerative disorders; Chronic pulmonary disease; Diabetes, uncomplicated; Diabetes, complicated; Hypothyroidism; Renal failure; Liver disease; Peptic ulcer disease, no bleeding; AIDS/HIV; Lymphoma; Metastatic cancer; Solid tumour without metastasis; Rheumatoid arthritis/collagen; Coagulopathy; Obesity; Weight loss; Fluid and electrolyte disorders; Blood loss anemia; Deficiency anemia; Alcohol abuse; Drug abuse; Psychosis; Depression	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
	Rhee index (validation SID)			Age ,24 mo (vs >= 24 mo); Female (vs male); Peri-/endo-/myocarditis, cardiomyopathy, or tamponade; Acute myocardial infarction; Coronary atherosclerosis/other ischemic heart disease; Pulmonary vascular disease (eg, PE, pulmonary HTN); Ventricular tachycardia or other cardiac dysrhythmias; Cardiac arrest or ventricular fibrillation or flutter; Congestive heart failure; Aortic, peripheral, visceral artery aneurysms/ dissection; Aortic or peripheral arterial embolism or thrombosis; Cardiac or circulatory congenital anomalies; Pneumonia; Influenza; Chronic obstructive pulmonary disease/ bronchiectasis; Asthma; Aspiration pneumonitis; Respiratory failure, insufficiency, arrest; Cystic fibrosis; Respiratory distress syndrome; Acute renal failure; Chronic renal failure; Urinary tract infections or unspecified cystitis; Thyroid disorders or other endocrine disorders; Diabetes mellitus or complications; Septicemia (except in labor); Fungal infection; HIV infection; Hepatitis; Meningitis, encephalitis, or other CNS infection; Coma, stupor, or brain damage; Acute cerebrovascular disease; Gastroduodenal ulcer, gastritis, or duodenitis; Intestinal obstruction without hernia; Peritoneal or intestinal abscess, peritonitis; Biliary tract disease; Liver disease (eg, cirrhosis, increased LFTs); Pancreatic disorders (not diabetes); Gastrointestinal hemorrhage; Leukemia; Lymphomas or reticuloendothelial neoplasms; CNS or miscellaneous intracranial or intraspinal neoplasms; Primary malignant tumor of adrenal or paraganglia; Renal tumors; Hepatic tumors; Primary malignant bone or articular cartilage tumors; Soft tissue sarcomas; Immunity disorders (except AIDS); Coagulation or hemorrhagic disorders; Disease of white blood cells (eg, lymphocytosis, lymphopenia); Chronic ulcer of skin; Spondylosis, intervertebral disc disorders; Systemic lupus erythematosus or connective tissue disorder; Crushing injury or internal injury; Shock; Drowning/submersion; Firearm; Motor vehicle traffic; Pedal cyclist, not MVT; Suffocation; Suicide or intentional self-inflicted injury; Fracture of long bone or spine; Intracranial injury; Poisoning by nonmedicinal substances; Skull or face fractures; Open wounds of head, neck, or trunk; Superficial soft tissue injury: contusion; Short gestation, low birth wt, or fetal growth retardation; Hypoxia, asphyxia, or aspiration during birth; Birth trauma; Perinatal jaundice	External validation study
Delaplain et al. 2020	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Planned readmission; Sex; Length of stay (d); Admission source; Payer; Age; Race/ethnicity; Previous ED visits; Current/index visit is a readmission from a prior visit?; Previous visits; Readmission history; Free-standing pediatric hospital; Number of medications; Number of systems diagnoses; Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or pelvis (S30-S39); Traumatic injury (ICD-10): Ankle and foot (S90-99); Traumatic injury (ICD-10): Burns (T20-32); Traumatic injury (ICD-10): Early complications of trauma (T79); Traumatic injury (ICD-10): Elbow and forearm (S50-59); Traumatic injury (ICD-10): Foreign body entering through the natural orifice (T15-19); Traumatic injury (ICD-10): Head (S00-09); Traumatic injury (ICD-10): Hip and thigh (S70-79), Traumatic injury (ICD-10): Unspecified body region (T14); Traumatic injury (ICD-10): Knee and lower leg (S80-89); Traumatic injury (ICD-10): Multiple body regions (T07); Traumatic injury (ICD-10): Neck (S10-19); Traumatic injury (ICD-10): Unspecified effects of external causes (T66-T78); Traumatic injury (ICD-10): Poisoning (T36-50); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88); Traumatic injury (ICD-10): Thorax (S20-29); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65); Traumatic injury (ICD-10): Wrist, hand, and fingers (S60-61); Surgical procedures by system: Auditory; Surgical procedures by system: Cardiovascular; Surgical procedures by system: Digestive; Surgical procedures by system: Integumentary; Surgical procedures by system: Musculoskeletal; Surgical	LOS: [LOS 2-4 days (OR=1.298; NR% CI 1.189-1.416), LOS 4-7 days (OR=1.638; NR% CI 1.493-1.797), LOS >7 (OR=1.994; NR% CI 1.816-2.189)]; Admission source: Transfer (OR=0.768; NR% CI 0.689-0.856); Previous ED visits: [1 (OR=1.137; NR% CI 1.058-1.222), 2 (OR=1.158; NR% CI 1.037-1.294), >=3 (OR=1.305; NR% CI 1.157-1.471)]; Current/index visit is a readmission from a prior visit?: [Yes: unplanned (OR=1.404; NR% CI 1.291-1.527), Yes: planned (OR=1.646; NR% CI 1.457-1.858)]; Previous visits: [1 (OR=1.746; NR% CI 1.604-1.9), 2 (OR=2.282; NR% CI 2.02-2.578), >=3 (OR=2.614; NR% CI 2.245-3.045)]; Readmission history: [1 (OR=1.473; NR% CI 1.303-1.665), 2 (OR=1.788; NR% CI 1.504-2.125), >=3 (OR=2.578; NR% CI 2.172- 3.06)]; Number of medications (OR=1.011; NR% CI 1.009-1.014); Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or pelvis (S30-S39) (OR=0.763; NR% CI 0.655-0.89); Traumatic injury (ICD-10): Burns (T20-32) (OR=0.596; NR% CI 0.494-0.719), Traumatic injury (ICD-10): Head (S00-09) (OR=0.661; NR% CI 0.594-0.735); Traumatic injury (ICD-10): Hip and thigh (S70-79) (OR=0.709; NR% CI 0.616-0.816); Traumatic injury (ICD-10): Unspecified body region (T14) (OR=0.752; NR% CI 0.62-0.912); Traumatic injury (ICD-10): Knee and lower leg (S80-89) (OR=0.727; NR% CI 0.614-0.86); Traumatic injury (ICD-10): Neck (S10-19) (OR=1.194; NR% CI 1.008-1.45); Traumatic injury (ICD-10): Poisoning (T36-50) (OR=1.16; NR% CI 1.06-1.269); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49) (OR=0.547; NR% CI 0.461-0.649); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88) (OR=1.545; NR% CI 1.439-1.66); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65) (OR=0.538; NR% CI 0.404-0.716)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				procedures by system: Nervous; Surgical procedures by system: Respiratory; Traumatic injury: trauma due to frost bite; Traumatic injury: eye/ocular; Traumatic injury: endocrine; Surgical procedure: mediastinum/diaphragm; Surgical procedures by system: hematologic; Surgical procedures by system: lymphatic; Surgical procedures by system: urinary/reproductive	
General medical conditions related readmissions					
Learly et al. 2019	Prediction at admission	Complex chronic conditions	30-day unplanned HRs	Age; Race and/or ethnicity; Boys; Non-English primary language; Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo, Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type	Significant variables (p-value<0.05) in univariate analysis are used for the analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.15–2.45); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.32–3.10); No. CCC categories: [No. CCC categories = 2 (OR=1.73; 95% CI 1.19–2.49), No. CCC categories >=3 (OR=2.30; 95% CI 1.50–3.47)]; Medical admission (OR=1.82; 95% CI 1.30–2.63)
	Prediction at discharge			Age; Race and/or ethnicity; Boys; Non-English primary language, Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo; Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type; ICU use; Discharge disposition from the hospital; LOS in d; Weekday discharge	Significant variables (p-value<0.05) in univariate analysis are used for analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.16–2.46); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.31–3.11); No. CCC categories: [No. CCC categories = 2 (OR=1.56; 95% CI 1.06–2.26), No. CCC categories >=3 (OR=1.72; 95% CI 1.08–2.69)]; Medical admission (OR=1.75; 95% CI 1.23–2.49); Discharge disposition from the hospital: [With services (OR=1.69; 95% CI 1.17–2.44), Other facility (OR=1.15; 95% CI 0.58–2.13)]; LOS: [LOS 2-5 days (OR=1.15; 95% CI 0.78–1.72), LOS >=6 days (OR=1.45; 95% CI 0.90–2.33)] Weekday discharge was not significant (p-value≤ 0.05) in the univariate analysis, but included in the final scoring system.
Ryan et al. 2021	PASS (validation)	Asthma	30-day unplanned HRs	Respiratory rate, Oxygen requirement, Auscultation, Retractions, Dyspnea	External validation study

Abbreviations: NR, not reported

* Only significant risk factors (odds ratio/ hazard ratio>1, p-value<0.05) are considered in the analysis (cf. table 2). Significant predisposed factors (1>odds ratio/ hazard ratio>0, p-value<0.05) are listed as an additional information.

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Table A6: Adherence per TRIPOD-item at predictive model level

TRIPOD-item	Developed predictive models (n=9)			External validated predictive models (n=10)			Incremental value predictive models (n=2)			All predictive models (n=21)		
	# of PMs with applicable item	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item
1	9.00	4.00	44.44%	10.00	6.00	60.00%	2.00	1.00	50.00%	21.00	11.00	52.38%
2	9.00	1.00	11.11%	10.00	0.00	0.00%	2.00	0.00	0.00%	21.00	1.00	4.76%
3a	9.00	9.00	100.00%	10.00	8.00	80.00%	2.00	2.00	100.00%	21.00	19.00	90.48%
3b	9.00	8.00	88.89%	10.00	8.00	80.00%	2.00	2.00	100.00%	21.00	18.00	85.71%
4a	9.00	6.00	66.67%	10.00	9.00	90.00%	2.00	1.00	50.00%	21.00	16.00	76.19%
4b	9.00	9.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	21.00	21.00	100.00%
5a	9.00	7.00	77.78%	10.00	7.00	70.00%	2.00	2.00	100.00%	21.00	16.00	76.19%
5b	9.00	9.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	21.00	21.00	100.00%
5c	9.00	9.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	21.00	21.00	100.00%
6a	9.00	6.00	66.67%	10.00	2.00	20.00%	2.00	2.00	100.00%	21.00	10.00	47.62%
6b	9.00	9.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	21.00	21.00	100.00%
7a	9.00	8.00	88.89%	10.00	7.00	70.00%	2.00	2.00	100.00%	21.00	17.00	80.95%
7b	9.00	1.00	11.11%	10.00	0.00	0.00%	2.00	0.00	0.00%	21.00	1.00	4.76%
8	9.00	9.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	21.00	21.00	100.00%
9	9.00	4.00	44.44%	10.00	1.00	10.00%	2.00	1.00	50.00%	21.00	6.00	28.57%
10a	9.00	3.00	33.33%	0.00	0.00	0.00%	2.00	0.00	0.00%	11.00	3.00	27.27%
10b	9.00	1.00	11.11%	0.00	0.00	0.00%	2.00	0.00	0.00%	11.00	1.00	9.09%
10c	0.00	0.00	0.00%	10.00	7.00	70.00%	1.00	1.00	100.00%	11.00	8.00	72.73%
10d	9.00	4.00	44.44%	10.00	6.00	60.00%	2.00	1.00	50.00%	21.00	11.00	52.38%
10e	0.00	0.00	0.00%	1.00	0.00	0.00%	1.00	1.00	100.00%	2.00	1.00	50.00%
11	4.00	3.00	75.00%	0.00	0.00	0.00%	0.00	0.00	0.00%	4.00	3.00	75.00%
12	0.00	0.00	0.00%	10.00	1.00	10.00%	2.00	2.00	100.00%	12.00	3.00	25.00%
13a	9.00	6.00	66.67%	10.00	1.00	10.00%	2.00	1.00	50.00%	21.00	8.00	38.10%
13b	9.00	1.00	11.11%	10.00	0.00	0.00%	2.00	2.00	100.00%	21.00	3.00	14.29%
13c	0.00	0.00	0.00%	10.00	0.00	0.00%	2.00	2.00	100.00%	12.00	2.00	16.67%
14a	9.00	7.00	77.78%	0.00	0.00	0.00%	2.00	2.00	100.00%	11.00	9.00	81.82%
14b	8.00	7.00	87.50%	0.00	0.00	0.00%	2.00	2.00	100.00%	10.00	9.00	90.00%
15a	9.00	0.00	0.00%	0.00	0.00	0.00%	2.00	0.00	0.00%	11.00	0.00	0.00%
15b	9.00	6.00	66.67%	0.00	0.00	0.00%	2.00	1.00	50.00%	11.00	7.00	63.64%
16	9.00	4.00	44.44%	10.00	6.00	60.00%	2.00	1.00	50.00%	21.00	11.00	52.38%
17	0.00	0.00	0.00%	1.00	0.00	0.00%	0.00	0.00	0.00%	1.00	0.00	0.00%
18	9.00	9.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	21.00	21.00	100.00%
19a	0.00	0.00	0.00%	9.00	2.00	22.22%	2.00	2.00	100.00%	11.00	4.00	36.36%
19b	9.00	9.00	100.00%	10.00	9.00	90.00%	2.00	2.00	100.00%	21.00	20.00	95.24%
20	9.00	9.00	100.00%	10.00	8.00	80.00%	2.00	2.00	100.00%	21.00	19.00	90.48%
21	9.00	3.00	33.33%	10.00	8.00	80.00%	2.00	2.00	100.00%	21.00	13.00	61.90%
22	9.00	2.00	22.22%	10.00	8.00	80.00%	2.00	1.00	50.00%	21.00	11.00	52.38%

Abbreviations: PMs, predictive models; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

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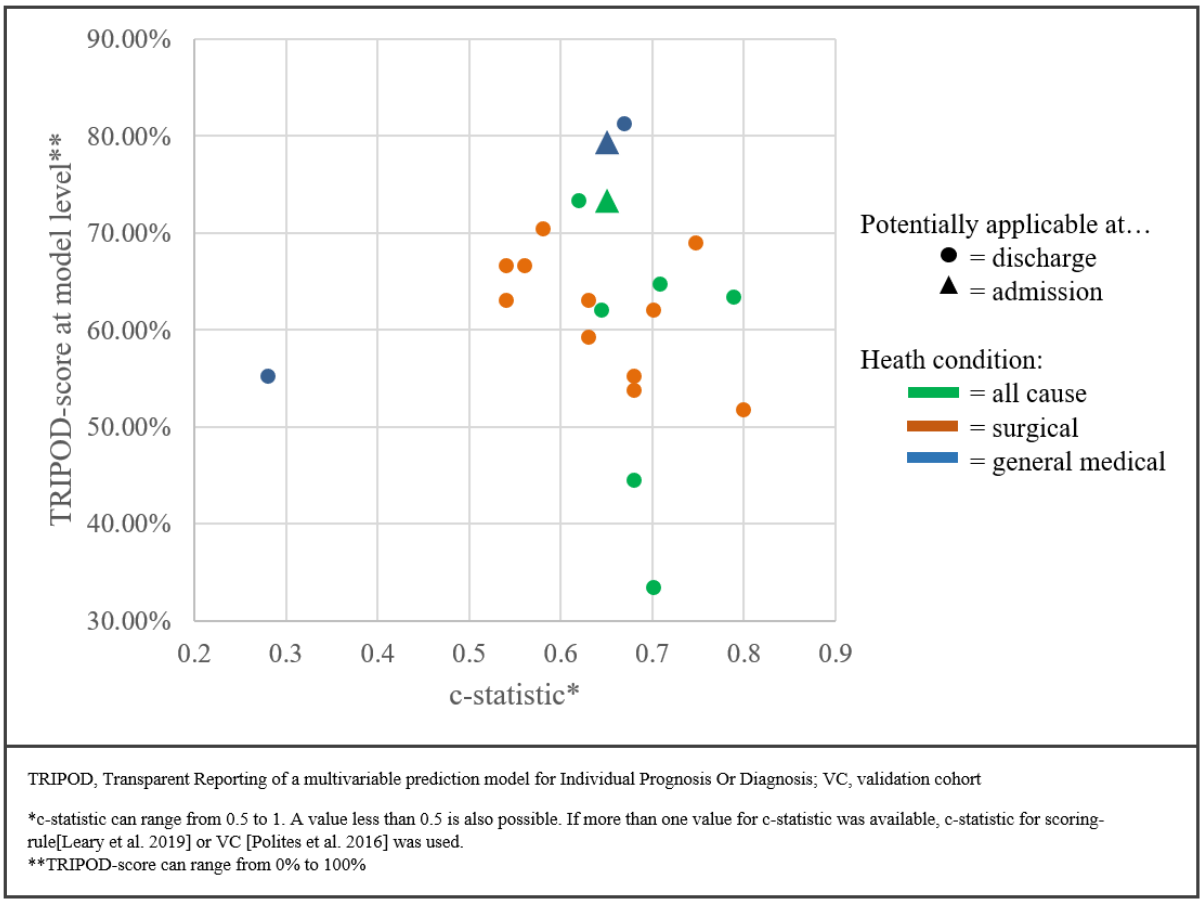


Figure A1: Discriminative ability, application and TRIPOD-adherence of 30-day HR predictive models in paediatrics (n=21)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4-5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5-6 + Supplementary File
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5 + Supplementary File
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6 + Table 1-3 + Supplementary File
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Qualitative Synthesis, Page 6-7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Qualitative Synthesis, Page 6-7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Qualitative Synthesis, Page 6-7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Qualitative Synthesis, Page 6-7, 18
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6-7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 7
Study characteristics	17	Cite each included study and present its characteristics.	Page 8-12, Table 1+3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 + Supplementary File
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1 + Supplementary File
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Qualitative Synthesis, Table 1-3, figure 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 18
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 15-17, TRIPOD Assessment
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 17-19
	23b	Discuss any limitations of the evidence included in the review.	Page 18
	23c	Discuss any limitations of the review processes used.	Page 18

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23d	Discuss implications of the results for practice, policy, and future research.	Page 17-19
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 20
Competing interests	26	Declare any competing interests of review authors.	Page 20
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 20 + Supplementary File

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Y
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Y
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Y
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Y
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Y
Synthesis of results	6	Specify the methods used to present and synthesise results.	Y
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Y
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Y
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Y
Interpretation	10	Provide a general interpretation of the results and important implications.	Y
OTHER			
Funding	11	Specify the primary source of funding for the review.	Manuscript
Registration	12	Provide the register name and registration number.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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TITLE

Applicability of predictive models for 30-day hospital readmission risk in paediatrics: a systematic review

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ABSTRACT

Objective: To summarize multivariable predictive models for 30-day hospital readmissions (HRs) in paediatrics, describe their performance and completeness in reporting, and to determine their potential for application in practice.

Design: Systematic review.

Data source: CINAHL, EMBASE and PubMed up to 7 October 2021.

Eligibility criteria: English- or German-language studies aiming to develop or validate a multivariable predictive model for 30-day paediatric HRs related to all-cause, surgical conditions or general medical conditions were included.

Data extraction and synthesis: Study characteristics, risk factors significant for predicting readmissions and information about performance measures (e.g., c-statistic) were extracted. Reporting quality was addressed by the 'Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis' (TRIPOD) adherence form. The study quality was assessed by applying six domains of potential biases. Due to expected heterogeneity among the studies, the data were qualitatively synthesized.

Results: Based on 16 studies, 29 predictive models were identified, which could potentially be used for determining individual 30-day HR risk in paediatrics. The number of study participants ranged from 328 patients to 1.4 million encounters. The two most common significant risk factors were comorbidity and (postoperative) length of stay. Eight models showed a c-statistic above 0.7 and are potentially applicable at discharge. The median TRIPOD adherence of the models was 65% ($P_{25} - P_{75}$ 55% - 69%), ranging from a minimum of 33% to a maximum of 81%. Overall, the quality of the studies was moderate in almost all domains.

Conclusions: Predictive models may be useful in identifying paediatric patients at increased risk for readmission. To support the application of predictive models, more attention should be placed on completeness in reporting, particularly for those items that may be relevant for implementation in practice.

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Strengths and limitations of this study

- Independent and standardised methodological approach for study selection, data extraction and risk of bias assessment.
- Comprehensive presentation of predictive models that provides information about applicability, performance and reporting quality at model level, differentiated by 30-day all-cause, surgical conditions and general medical conditions related paediatric hospital readmissions.
- Due to study heterogeneity, the models were only narratively synthesized.

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INTRODUCTION

Hospital readmissions (HRs) are becoming increasingly important as a quality indicator for paediatric inpatient care[1, 2]. HR is often defined as a subsequent, unplanned admission within a period of 30 days after the index hospitalization[3]. According to published evidence, all-cause 30-day unplanned HR rates for the paediatric population range from 3.4% to 18.7%[3-5]. In addition, taking 27 US states into account, it has been estimated that paediatric HRs can cost up to $2 billion annually, with approximately 40% of these occurring HRs being potentially preventable[6].

Identifying the reasons for paediatric HRs is a major challenge, as the health of children is also affected by factors aside of inpatient care[7]. Predictive models can be applied as a tool for the identification of patients with a risk of HR higher than that of the average population and for the implementation of preventive interventions to reduce the risk of HR[8]. Especially in the context of the ongoing COVID-19 pandemic, where children and adolescents are also being hospitalized with a variety of symptoms[9-11], the prevention of unplanned HRs can be beneficial, as it would allow hospital resources to be used in a more target-orientated way.

This systematic review aims to address two research gaps that have been identified. (i) Predictive models with good performance are useful in practice when clinicians and other stakeholders have all the necessary information for their application in clinical practice and critical assessment[12]. However, previous systematic reviews discussed the shortcomings in reporting the quality of prediction models[13-15] and also for paediatric clinical prediction rules[16]. (ii) A previous systematic review has already identified 36 significant risk factors for unplanned HRs in paediatric patients with different health conditions[3]. The largest number of risk factors were identified for surgical procedure related unplanned HRs. Among others, comorbidity was one of the most common risk factors across the 44 included studies[3]. The review[3] extends the findings of an earlier systematic review that focused on 29 paediatric studies targeting predictors for asthma-related unplanned HRs[17]. Both reviews[3, 17] were primarily addressed to predictor finding studies[14], while to date there is no published review of existing 30-day HR predictive models in paediatrics.

The objective of this systematic review was to determine the potential application of multivariable predictive models for individualized risk prediction of 30-day HR in the paediatric population by evaluating the models´ discriminative ability, completeness in reporting and the risk factors shown to be significant for prediction of 30-day HR.

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METHOD

The 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) statement was adhered to for conducting and reporting of this systematic review[18]. Screening of the titles and abstracts, data extraction, quality assessment and analyses (e.g., completeness in reporting) were performed by two independent reviewers, while disagreements were discussed with a third author. A protocol was pre-specified and is available from the corresponding author.

Data source and search strategy

CINAHL, EMBASE and PubMed were used for an electronic database search to identify studies published up to 7 October 2021. The key search terms include the outcome variables used for the model (i.e., readmission/rehospitalization), elements of the study design (i.e., prediction/c-statistic) and the population of interest (i.e., paediatrics/children)(see supplementary material for full search strategies–tables A1-A3).

Inclusion criteria

Studies addressing multivariable predictive models for children and adolescents (except newborns/ preterm newborns, as the index admission is the birth hospitalization) were included if they were published in English or German and available as full texts in peer-reviewed original journal articles. Studies aiming to develop a new model or to validate an existing model were included (i) if the model was potentially appropriate for the individual prediction of 30-day HR from acute healthcare service after discharge or after index procedure in paediatrics; and (ii) if the model provided at least one discrimination measure (e.g., c-statistic). Discriminative ability is a key factor in evaluating predictive models[19] and a necessary information to make well-founded conclusions about the performance of a model.

In addition, (iii) predictive model studies that developed a new model (i.e., development design) or determined the incremental or added value of a predictor for an existing model (i.e., incremental value design) had to be based on a regression modelling approach. This inclusion criterion enables us to identify significant risk factors and to apply the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) adherence form, which was originally developed for regression models[20]. This implies that predictive models using machine-learning techniques (e.g., least absolute selection and shrinkage operator[21] or random forest[22]) are excluded and coded as non-regression models. Studies that aimed to identify 30-day HR predictors are classified as prognostic factor studies

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and thus excluded from the analysis (so as not to bias them adversely in TRIPOD adherence). Prognostic factor studies, for example, are not required to present a simplified scoring rule (cf. TRIPOD item 15b[23]). Due to specific requirements of mental diseases, studies were only included (iv) if they addressed non-mental health conditions related 30-day HRs[3].

Study selection and data extraction

Potential articles were assessed for meeting the inclusion criteria. The reference lists of the included studies were examined for further potential studies. Just as in previous systematic reviews[3, 24], studies were categorized by health conditions in all tables.

Basic study characteristics were extracted in table 1. To assess the applicability of the predictive models, significant risk factors (i.e., odds ratio (OR) or hazard ratio >1 with a p-value of < 0.05) were assigned to established and revised variable categories[3] in table 2. If all variables of a predictive model are available for a patient at the time of index admission (e.g., previous health service usage before index admission), the model is applicable at admission. Applicability of predictive models at discharge is given if all variables are available at this point for a patient (e.g., length of stay, operative time).

Reporting quality and performance

Predictive models can just be used in practice when clinicians and other stakeholders have access to all information required for their application in clinical practice[12]. The newly developed 'Critical Appraisal of Models that Predict Readmission (CAMPR)' contains 15 expert recommendations for predictive model development relating to HRs. However, CAMPR should not be used as a reporting standard so far and relates to aspects that are out of the scope of this systematic review (e.g., considering different time frames for HRs)[25]. Due to the importance of high-quality information about predictive models we decided to assess the completeness of reporting by using the TRIPOD adherence form and scoring rules[12, 23, 26]. The TRIPOD adherence form consists of 22 main criteria based on the TRIPOD statement[20], resulting in 37 items that are applicable to varying degrees to the development, validation and incremental value studies [23]. We decided to apply the TRIPOD adherence form at predictive model level. Therefore, publications that report the development and validation of the same predictive model, for example, are assessed separately. According to previous research, our analysis concentrates on items that could be reported in the main text or supplements[27].

TRIPOD adherence at model level was merged with the performance results (i.e., discrimination and calibration measures) and the applicability assignment in table 3. The

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discrimination of a predictive model is often evaluated by the c-statistic or area under the receiver operating characteristic curve. The c-statistic can take a value between 0.5 and 1. A value of 0.5 indicates that the model is not superior to a random prediction of outcomes, while values between 0.7 and 0.8 indicate that the model is appropriate. A value of 0.8 or greater indicates a strong discrimination of a model.[28]

Quality assessment

Following previous systematic reviews[3, 24, 29], the refined version of the quality in prognosis studies (QUIPS) tool with its prompting items[30] was used to appraise the studies critically with regard to the included predictive models based on six domains. Each domain was rated with a 'high', 'moderate' or 'low' risk of bias.

The six domains are[30]: 'study participation', 'study attrition', 'prognostic factor measurement', 'outcome measurement', 'study confounding' and 'statistical analysis and reporting'.

Data synthesis

Because a quantitative evaluation in the form of a meta-analysis was not possible due to the high heterogeneity among the studies, the studies were qualitatively synthesized, i.e., the results for performance, completeness in reporting and significant risk factors were presented in a narrative and in a simplified quantitative form.

Patient and public involvement

Due to the study design we did not involve patients or the public.

RESULTS

Search result

From the electronic database search, 10,076 records were obtained. After duplicates had been removed, the titles and abstracts were screened for 7,694 records. Based on the predefined inclusion criteria, 7,586 records were excluded. Adding one additional recommended article[31] this results in 109 records being included in the full-text assessment. Among the 93 excluded records, twelve studies analyse 30-day HRs together with another outcome (i.e. emergency department return visits (n=5)[32-36], mortality (n=3)[37-39], other complications (n=4)[40-43]), three were predictive model studies with no discrimination metrics[44-46], and five were non-regression-based predictive model studies for 30-day HRs in paediatrics[21, 47-

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50]. Of the 70 excluded prognostic factor studies for 30-day paediatric HRs, eleven studies provided information about discriminative ability, where the c-statistic ranged from 0.61 to 0.79[31, 51-60]. Based on the full-text assessments, 16 studies were included in the systematic review, with two of them[61, 62] already presented in a previous systematic review[3] with a different focus. The results of the review process regarding the database search is provided in figure 1 (see table A7 in the supplementary material for a summary of study characteristics for selected excluded models).

[…Placement of figure 1…]

Figure 1: Flowchart for the search and study selection process (PRISMA).

Quality assessment

Overall, the quality of many studies was moderate to low for several domains. For instance, the study quality had to be reduced due to a lack of sufficient information (e.g. in the domain 'study participants' or 'study attrition'), while all studies were rated as 'low' for the domain 'study confounding'(see table A4 in the supplementary material for the results of the risk of bias assessment).

Study characteristics

All studies were based on retrospective data, with seven studies based on tertiary or paediatric hospital data[22, 63-68], and nine studies based on centralized databases[61, 62, 69-75]. Five of 16 studies additionally included census data in the analysis[63, 67-69, 73]. The period of data collection ranged from one year[65, 69, 71] to seven years[70, 72]. The majority of studies included patients up to an age of <18 or ≤18 years. Only two studies considered patients up to 21 years of age[66, 74]. The sample size was specified with different units in the individual studies (e.g., encounters, admissions) and varies between 328 patients[64] and 1.4 million encounters[70].

The 16 included studies resulted in 29 predictive models for 30-day HRs in paediatrics. Nine of 16 studies developed or validated more than one predictive model, which were in part excluded due to non-agreement with the inclusion criteria[22, 64, 67-73]. The models included were grouped into three health conditions: (i) all-cause HR (n=13)[22, 63, 65-67, 69, 70]; (ii) surgical conditions related HR (n=11)[61, 62, 71-73]; and (iii) general medical conditions related HR (n=5)[64, 68, 74, 75]. 27 of 29 predictive models explicitly aim to predict 30-day

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unplanned HRs with respect to the grouped health condition[22, 61-70, 72, 74, 75]. 30-day unplanned HR rates vary from 3.0%[64] to 41.2%[74].

Among the 29 predictive models included, 17 (43%) used a development design[22, 61-63, 65-68, 72-75], ten (47%) an external validation design[64, 67, 70, 71, 73] and two (10%) an incremental value design[68, 69]. Nine out of ten external validated models were based on existing predictive models that had been previously used in the adult population[67, 70], for different outcomes[64, 71] or both[71]. Furthermore, four of the 16 studies included did not state the primary aim to develop, validate externally or assess the incremental value of the respective 30-day HR predictive model[67, 69, 70, 72].

Of the predictive models with a development or incremental value design, two employed an apparent validation[63, 69] and 17 employed an internal validation[22, 61, 62, 65-68, 72-75]. The most commonly applied internal validation method was cross-validation (n=8)[22, 65, 66], followed by split-sample (n=6)[61, 67, 72-75] and bootstrapping (n=3)[62, 68]. In order to analyse the data, either a logistic regression[22, 62, 63, 65-69, 72-75] or a Cox proportional hazard regression[61] was used. Most models present their results by ORs with a 95% confidence interval. With a p-value of < 0.05, the results are considered as statistically significant[3]. A summary of characteristics of all included studies is provided in table 1.

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Table 1: Summary of study characteristics

Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
All-cause related HRs									
Brittan et al. 2018[66], USA	Composite Score	All-cause	30-day unplanned HRs	Retrospective/ 1 children's hospital	29,542 patients	0-21 years	2014 - 2015	4.0%	Development study/ Internal: Cross
Sills et al. 2017[69], USA	PACR + SDH	All-cause	30-day unplanned HRs	Retrospective / PHIS database, US Census's American Community Survey data, 47 hospitals	458,686 index discharges	<18 years	2014	6.1%	Incremental value study/ Apparent
Ehwerhemuepha et al. 2018[67], USA	Unnamed	All-cause	30-day unplanned HRs	Retrospective / US Census's American Community Survey data, 1 tertiary paediatric hospital	38,143 inpatient clinical encounters (DC: 19,072, VC: 19,071)	Between 28-days and 17 years	July 2013 to June 2017	10.4%	Development study/ Internal: Random-Split-Sample
	LACE (validation)				VC: 19,071 inpatient clinical encounters			NR	External validation study
Bradshaw et al. 2020[65], USA	HARRPS-tool	All-cause	30-day unplanned HRs	Retrospective / 1 paediatric hospital	5,306 patients	<18 years	May 2017-June 2018	25.3%	Development study/ Internal: Cross
Zhou et al. 2020[63], Australia	Unnamed	All-cause	30-day unplanned HRs	Retrospective/ Australian Census data, 1 tertiary paediatric hospital	73,132 patients	Age limit for admission: 15 years, special permissions by hospital executives possible	2010 - 2014	4.6%	Development study/ Apparent
Ehwerhemuepha et al. 2020[70], USA	LACE (validation)	All-cause	30-day unplanned HRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters	< 18 years	2000- 2017	12.6% (DC)	External validation study
Zhou et al. 2021[22], Australia	Model 1: GLM	All-cause	30-day unplanned HRs	Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data	940 patients	Different paediatric age groups[c]	2010-2014	4.55%[d]	Development study/ Internal: Cross
	Model 1: G-S								Development study/ Internal: Cross
	Model 2: GLM			Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records					Development study/ Internal: Cross
	Model 2: G-S								Development study/ Internal: Cross

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Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
	Model 3: GLM			Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records, written discharge documentation					Development study/ Internal: Cross
	Model 3: G-S								Development study/ Internal: Cross
Surgical conditions related HRs									
Vo et al. 2018[62], USA	Unnamed	All surgical specialties without cardiac surgery	30-day unplanned post- surgical HRs relating to noncardiac surgery	Retrospective/ ACS NSQIP-P database	182,589 patients	< 18 Years	2012-2014	4.8%	Development study/ Internal: Bootstrap
Polites et al. 2016[61], USA	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Retrospective/ ACS NSQIP-P database	54,870 patients (DC: 38,397, VC: 16473)	29 days - < 18 years	2012-2014	3.6%	Development study/ Internal: Random-Split-Sample
Smith et al. 2015[73], USA	Developed model	Cardiac surgery	30-day HRs following congenital heart surgery	Retrospective/ PHIS database, US census data, 43 not-for-profit, tertiary care paediatric hospitals	17,871 discharges (DC: 9,104, VC: 8,767)	< 18 years	2011	11.0%	Development study/ Internal: Split-Sample (Non-Random temporal)
	Validated model						2012	NR	External validation study /Split-Sample (Non-Random temporal)
Jiang et al. 2018[71], USA	CCI (validation NRD)	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ Nationwide Readmissions Database	8,006 patients	<18 years	2013	NR	External validation study
	VWI (validation NRD)								External validation study
	Rhee index (validation NRD)								External validation study
	CCI (validation SID)	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ State Inpatient Databases	6,236 patients	<18 years	2007-2010	8%[a]	External validation study
	VWI (validation SID)								External validation study
	Rhee index (validation SID)								External validation study

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Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
Delaplain et al. 2020[72], USA	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Retrospective / Cerner Health Facts database, 28 hospitals	82,532 patients (DC: 75%, VC: 25%)	< 18 years	2000-2017	8.8%	Development study/ Internal: Random-Split-Sample[b]
General medical conditions related HRs									
Learly et al. 2019[68], USA	Prediction at admission	Complex chronic conditions	30-day unplanned HRs	Retrospective / US Census Bureau data, 1 academic medical centre	2,296 index admissions	6 months to 18 years	Oct. 2010 - July 2016	8.2%	Development study/ Internal: Bootstrap
	Prediction at discharge								Incremental value study/ Internal: Bootstrap
Ryan et al. 2021[64], USA	PASS (validation)	Asthma	30-day unplanned HRs	Retrospective/ 1 university-affiliated, tertiary paediatric referral centre	328 patients	5-18 years	May 2015-Oct. 2017	3.0%	External validation study
O´Conell et al. 2021[75], USA	Unnamed	Nervous system condition	30-day unplanned HRs	Retrospective / Cerner Health Facts database, 18 hospitals	105,834 index admissions (DC: 80%. VC: 20%)	<18 years	2000-2017	12.0%	Development study/ Internal: Random-Split-Sample
Hoenk et al. 2021[74], USA	Unnamed	Oncology	30-day unplanned HRs	Retrospective / Cerner Health Facts database, 16 hospitals	10,418 patients (DC: 7,814, VC: 2,604)	<21 years	2000-2017	41.2%	Development study/ Internal: Random-Split-Sample

Abbreviations: ACS NSQIP-P, American College of Surgeons' (ACS) National Surgical Quality Improvement Program Pediatric; CCI, Charlson Comorbidity Index; DC, derivation cohort; GLM, logistic regression; G-S, stepwise logistic regression; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; NRD, nationwide readmission database; PACR, Pediatric All-Condition Readmission; PASS, Pediatric Asthma Severity Score; PHIS, Paediatric Health Information Systems; SDH, social determinants of health; SID, state inpatient database;VWI, Van Walraven Index; VC, validation cohort

[a] Own approximation resulting from Table 3 in the study based on readmission per risk score[71].
[b] Assumption for validation method: Odds ratios for 30-day unplanned HRs are displayed in a Table that is part of the DC from the 7-day unplanned HRs predictive model[72].
[c] Mean age (years): 5.2 with HR, 5.3 without HR.
[d] Based on 3330 patients from the initial data set.

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Applicability and significant risk factors in predictive models

Based on the 13 predictive models with a development or incremental value design, 24 significant risk factors associated with 30-day HRs were identified (see table 2). The most common risk factors were comorbidity (n=10), (postoperative) length of stay (n=8), illness severity (n=7) and health service usage prior to index admission (n=7). The significant risk factors were inconsistently defined across predictive models, allowing a direct comparison only to a limited extent. ORs for comorbidity ranged from 1.01[75] to 2.41[63] across predictive models. A length of stay of >=15 days (OR=2.39)[63] and a postoperative length of stay of >4 days (Hazard Ratio=3.12)[61] were considered to be a major risk factor. For health service usage priori to index admission 'three or more hospitalizations (prior six month)' (OR=3.014)[75] was one of the most pronounced risk factors, while for illness severity it was 'ICN or PICU admission' (OR=1.43)[65], respectively. For all-cause HRs, HRs related to surgical conditions and HRs related to general medical conditions 14, 15 and ten significant risk factors were found.

Most predictive models are potentially applicable at discharge (n=26), while three predictive models can be used at index admission[22, 65, 68], based on the significant and examined variables (see table A5 in the supplementary material for an overview of variables and table 3 for an application description).

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Table 2: Significant risk factors for 30-day HR predictive models with a development or incremental value design

Health condition group	**All-cause (n=5[a])**					**Surgical conditions related (n=4)**				**General medical conditions related (n=4)**			
Reference	[66]	[69]	[67]	[65]	[63]	[62]	[61]	[73]	[72]	[68][b]	[68][c]	[75]	[74]
Location of residence[d]		x			x								
Health insurance				x	x			x					
Type of index hospital						x						x	
Living environment				x									
Characteristics of primary care provider	x												
Age at admission/operation					x								
Race/Ethnicity		x						x				x	
Health service usage prior to index admission[e]			x	x					x	x	x	x	x
Comorbidity		x	x	x	x	x	x			x	x	x	x
Illness severity[f]			x	x			x	x		x	x	x	
LOS/Postop LOS			x		x		x	x	x		x	x	x
Principal diagnoses			x					x	x			x	x
Principal procedures							x					x	
Inpatient complications						x	x						
Specific medication at index admission									x				
Devices at index admission								x					
Length of operation							x						
Wound contamination before operation							x						
The ASA class						x							
Discharge on Friday or Weekend					x								
Discharge disposition											x		
Discharge with increased medication/further treatment	x												
Admission on Friday					x								
Medications at index admission												x	x

Abbreviations: ASA class, American Society of Anesthesiologists class; LOS, length of stay; PICU, paediatric intensive care unit; postop LOS, postoperative length of stay
x=risk factor (Odds ratio/Hazard ratio >1)

[a] The six predictive models of Zhou et al. 2021 [22] are not included in this analysis due to missing information about odds ratios. See table A5 in the supplementary material for a list of included variables.

[b] Admission model.

[c] Discharge model.

[d] Social determinants of health are included (e.g., median household income).

[e] Risk factor category includes, for example, the number of previous emergency department visits or hospitalizations.

[f] The risk factor category also captures the urgency of the index admission. The risk factor category includes, for example, PICU or emergency department admission.

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Completeness in reporting and discriminative ability at model level

Information about TRIPOD adherence and performance at model level are provided in table 3. The median TRIPOD adherence of the models was 65% ($P_{25} - P_{75}$ 55% - 69%, average: 63%), ranging from 33%[70] to 81%[68]. Developed predictive models had a more favourable reporting quality in comparison to external validated models (i.e., median adherence of 69% ($P_{25} - P_{75}$ 62% - 69%, average: 65%) compared with 61% ($P_{25} - P_{75}$ 54% - 66%, average: 58%), respectively). The two models with the poorest adherence in reporting were based on an external validation design, and the validation of these models was not the primary aim of the study[67, 70].

Table 3: Performance, application and TRIPOD adherence of 30-day HR predictive models in paediatrics (n=29)

Reference	Model name	Performance		TRIPOD score	Potentially applicable…
		Discrimination (c-statistic)	Calibration		
All-cause related HRs					
Brittan et al. 2018[66]	Composite Score	0.62		73.33%	at discharge
Sills et al. 2017[69]	PACR + SDH	0.708		64.71%	at discharge
Ehwerhemuepha et al. 2018[67]	Unnamed	VC: 0.79		63.33%	at discharge
	LACE (validation)	0.68		44.44%	at discharge
Bradshaw et al. 2020[65]	HARRPS-tool	Score: 0.65		73.33%	at admission
Zhou et al. 2020[63]	Unnamed	0.645		62.07%	at discharge
Ehwerhemuepha et al. 2020[70]	LACE (validation)	0.7014		33.33%	at discharge
Zhou et al. 2021[22]	Model 1: GLM	0.487		68.97%	at admission
	Model 1: G-S	0.477		68.97%	at discharge
	Model 2: GLM	0.585		68.97%	at discharge
	Model 2: G-S	0.593		68.97%	at discharge
	Model 3: GLM	0.609		68.97%	at discharge
	Model 3: G-S	0.617		68.97%	at discharge
Surgical conditions related HRs					
Vo et al. 2018[62]	Unnamed	0.747	slope: 1; intercept: 0.002	68.97%	at discharge
Polites et al. 2016[61]	Unnamed	DC: 0.71; VC: 0.701	(DC): p = 0.95, O/E-Ratio= 1.03; (VC): p = 0.36, O/E-Ratio= 1.07	62.07%	at discharge
Smith et al. 2015[73]	Developed model	DC: 0.68	H&L (DC): 0.47	55.17%	at discharge
	Validated model	VC: 0.68		53.85%	at discharge
Jiang et al. 2018[71]	CCI (validation NRD)	0.63	calculation not possible	62.96%	at discharge
	VWI (validation NRD)	0.54	calculation not possible	66.67%	at discharge

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		Performance			
	Rhee index (validation NRD)	0.58	calculation not possible	70.37%	at discharge
	CCI (validation SID)	0.63	calculation not possible	59.26%	at discharge
	VWI (validation SID)	0.54	calculation not possible	62.96%	at discharge
	Rhee index (validation SID)	0.56	calculation not possible	66.67%	at discharge
Delaplain et al. 2020[72]	30-day readmission model	VC: 0.799		51.72%	at discharge
General medical conditions related HRs					
Learly et al. 2019[68]	Prediction at admission	0.65; Score: 0.65	Calibration plot	79.31%	at admission
	Prediction at discharge	0.67; Score: 0.67	Calibration plot	81.25%	at discharge
Ryan et al. 2019[64]	PASS (validation)	0.28		55.17%	at discharge
O´Conell et al. 2021[75]	Unnamed	VC: 0.733		51.72%	at discharge
Hoenk et al. 2021[74]	Unnamed	VC: 0.714		55.17%	at discharge

Abbreviations: CCI, Charlson Comorbidity Index; DC, derivation cohort; GLM, logistic regression; G-S, stepwise logistic regression; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; NRD, nationwide readmission database; PACR, Pediatric All-Condition Readmission; PASS, Pediatric Asthma Severity Score; SDH, social determinants of health; SID, state inpatient database; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; VWI, Van Walraven Index; VC, validation cohort

Including all 37 items, the overall median adherence per TRIPOD item across models was 69% ($P_{25} - P_{75}$ 36% - 90%, average: 59%), ranging from 0% to 100% (see table A6 in the supplementary material for a detailed description by model type). The overall adherence per TRIPOD item is illustrated in figure 2.

[…Placement of figure 2…]

Figure 2: Overall adherence per TRIPOD item across all included predictive models (n=29).

38% of the models reported the title (item 1) completely, while 45%[64-69, 71] of the models mentioned the predictive model type in this context. 3% of the models had a completed abstract (item 2). The detailed predictors definition (item 7a) was fulfilled for more models (86%), in contrast to outcome definition (item 6a) (reported in 55%). The handling of predictors in the analysis (item 10a) showed incomplete reporting in 84% of the models. In addition, the handling (item 9, reported in 41%) and reporting of missing values (part of item 13b, reported in 31%) was not addressed in many models. Just 11% of the models displayed complete reporting of the model-building procedure (item 10b), as the majority of the models (89%) did not address the testing of interaction terms[22, 61-63, 66-69, 72, 73, 75]. The description (item 10d) and reporting of performance measures (item 16) was incomplete in 38% of the models. Just 38%

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of the models address results of calibration measures (cf. table 3). No model presented the full predictive model (item 15a) by providing an example of an intercept. An explanation for using the prediction model (item 15b, e.g., by a simplified scoring rule) was presented in 37% of the models. One model provided detailed information about a simplified scoring rule (item 15b) in supplementary material[68].

The discriminative ability (c-statistic) of the models ranged from 0.28[64] to 0.79[67, 72]. 21 out of 29 predictive models had a c-statistic <0.7. The linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.212 (P=.269). Models with good discriminative ability (c-statistic >0.7)[61, 62, 67, 69, 70, 72, 74, 75] are potentially applicable at discharge and have a TRIPOD score ranging from 52%[72, 75] to 69%[62]. The two models with the highest reporting quality (79%, 81%) are applicable for predicting 30-day unplanned HRs of children with complex chronic conditions. The c-statistic of these models was 0.65[68] and 0.67[68], respectively (see figure A1 in the supplementary material for a illustration of the models´ performance and TRIPOD adherence).

DISCUSSION

Based on 16 studies, this systematic review identifies 29 predictive models that could potentially be used for determining individual 30-day HR risk in paediatrics. According to the models, the four most common significant risk factors in predictive models were comorbidity, (postoperative) length of stay, illness severity and health service usage prior to index admission. Eight validated predictive models have a c-statistic of >0.7. The median TRIPOD adherence of the predictive models included was 65% ($P_{25} - P_{75}$ 55% - 69%), ranging from 33% to 81%, which is similar to that of other systematic reviews[12, 27].

Practical clinical and policy implications

In general, reporting quality and discriminative ability can provide crucial information about the strengths and weaknesses of a predictive model for implementation in practice (see figure A1 in the supplementary material for a combined illustration). However, the results from this systematic review revealed considerable differences in the c-statistics (0.28[64] to 0.79[67, 72]) and in the TRIPOD scores (33%[70] to 81%[68]) at model level. When considering the available information about reporting quality and discriminative ability in relation to each other, it should be noted that the linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.212 (P=.269). Therefore, an independent evaluation of both aspects for the selection of an appropriate predictive model is recommended.

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Clinicians and decision-makers should use predictive models with good discriminative ability (i.e. c-statistic above 0.7) and sufficient data availability. Especially predictive models that are based on census data[63, 67-69, 73] or manual data entry (e.g. written discharge documentation[22]) may be more difficult to implement than models relying on centralized databases[61, 62, 70-72, 74, 75]. The TRIPOD score at predictive model level (see table 3) can be used as a first indicator if the predictive model can be assessed and implemented with the given information.

Limitations

This systematic review has certain limitations. (i) The studies included needed be to published in English or German with full-text access. (ii) Summarizing the results of the included studies quantitatively was not possible due to the heterogeneity of the predictive models (resulting from differences in sample sizes, the examined variables or variations in the periods of data collection). (iii) With regard to the inclusion criteria, we defined the outcome as 30-day HRs for predictive models that can address both planned and unplanned HRs. The resulting heterogeneity was accepted because in an earlier systematic review a wider definition of HRs had proved useful[8]. (iv) The sample size of the included studies were reported in different units(e.g., encounters, discharges), impeding the comparisons of HR rates. (v) Our assignment of the predictive models that are potentially applicable at discharge assumes that the required variables are available at the time point. If clinicians and other stakeholders decide to use a predictive model, it should be checked beforehand whether complete data collection is possible at the desired time. (vi) In addition to the identified medical risk factors (e.g., comorbidity) and several country-specific risk factors (e.g., location of residence) that result in paediatric readmissions, health-policy initiatives may also affect the readmission rates in paediatric clinical practice[76]. However, due to a lack of data, these aspects could not be captured by this review.

Future research

This systematic review did not identify predictive models for individualized risk prediction of potentially preventable unplanned HRs in paediatrics, emphasizing past discussions to expand the research field further[3].

Current external validation studies were conducted in the USA and primarily examined the applicability of existing predictive models with other outcomes or population backgrounds to paediatric 30-day HRs[64, 67, 70, 71]. Therefore, external validation studies are needed for

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those models that are explicitly developed to predict 30-day HRs in paediatrics. Because the number of predictive models related to medical conditions-related HRs was small (n=5)[64, 68, 74, 75], with three out of five models demonstrating a c-statistic below 0.7[64, 68], there is a need for high quality models in this area. Further more, predictive models for 30-day HRs in the area of urological and cardiac surgery[71, 73] should be revised in order to achieve better discriminative ability (i.e. c-statistic>0.7).

Non-regression-based techniques (e.g., machine-learning) are an increasing field in order to predict 30-day HRs in paediatrics, most of which show good discriminative ability[21, 22, 45, 47-50, 70] (see table A7 in the supplementary material). Future systematic reviews should summarize and critically assess existing non-regression-based HR predictive models in paediatrics, for instance by applying the TRIPOD-ML statement that is going to be published[77].

Existing studies discuss the benefit of shorter time intervals in order to identify preventable readmissions more accurately[6, 78]; one study concluded that a 30-day unplanned HR metric was more precise (c-statistic=0.799) for paediatric trauma patients than a 7-day unplanned HR metric(c-statistic=0.737)[72]. To our knowledge, there is one predictive model for 365-day[7] and one for 7-day[72] unplanned HRs in paediatrics with good discriminative ability (c-statistic>0.7). Future studies should address the evaluation of paediatric HR predictive models with different time intervals.

CONCLUSION

This systematic review revealed an increase in the development of predictive models for 30-day HRs in paediatrics in recent years. To support the implementation of the predictive models in the long term, it is essential to validate existing models in order to test their applicability in different settings. To increase accessibility for use, more attention should be placed on completeness in reporting, particularly for items that may be relevant for the implementation of paediatric 30-day HR predictive models in practice.

List of Figures

Figure 1: Flowchart for the search and study selection process (PRISMA)

Figure 2: Overall adherence per TRIPOD item across all included predictive models (n=29)

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IN conceptualized and designed the systematic review, participated in the literature search, study selection, quality assessment, data extraction, data analyses and drafted the initial manuscript.

NK contributed to the literature search, study selection, quality assessment, data extraction and critically reviewed the manuscript.

SS contributed to the data analysis and critically reviewed the manuscript.

JD contributed to the study selection, data extraction, data analysis and critically reviewed the manuscript.

DM conceptualized and designed the systematic review, participated in the study selection, quality assessment, data extraction, data analyses and critically reviewed the manuscript.

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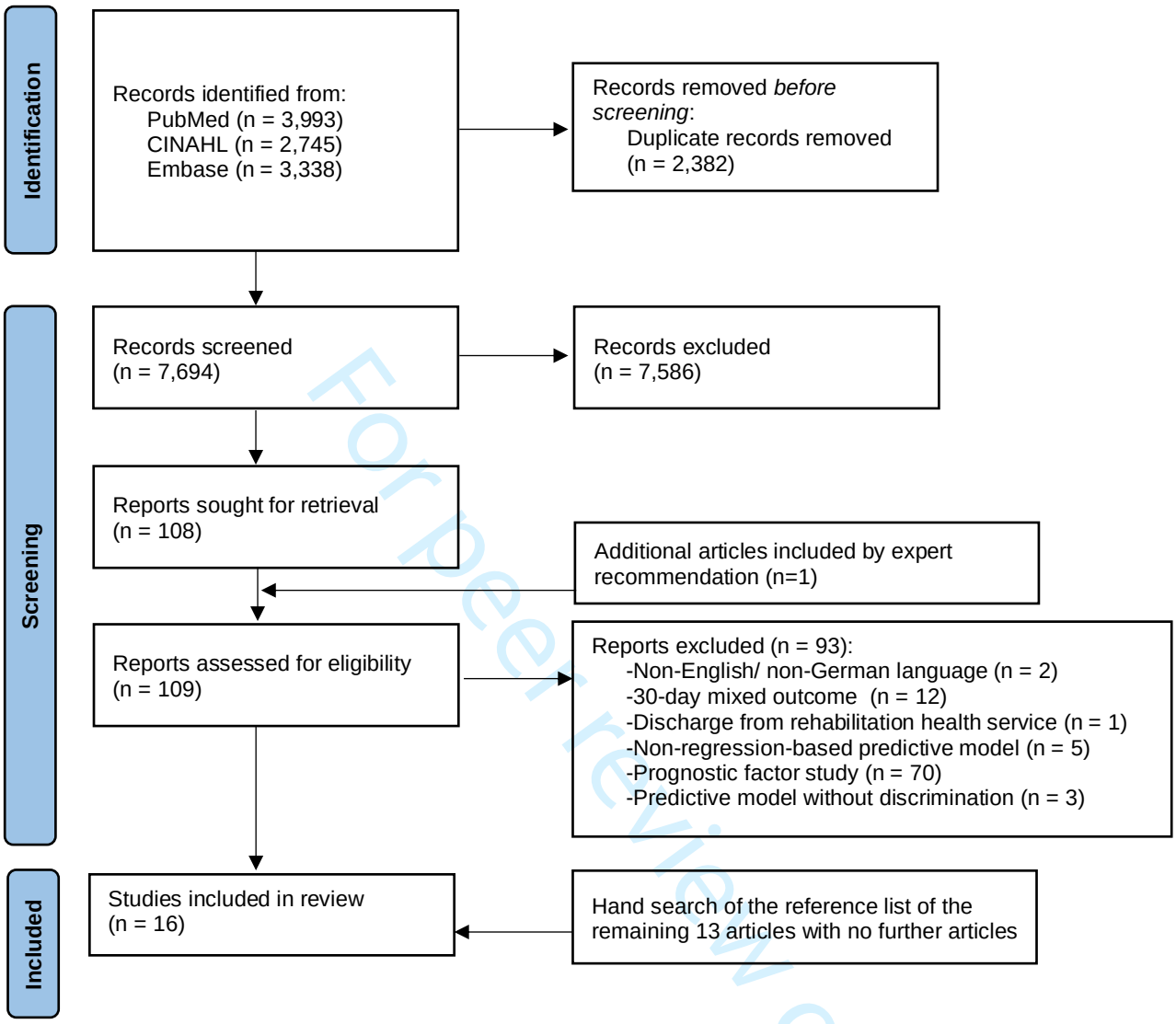
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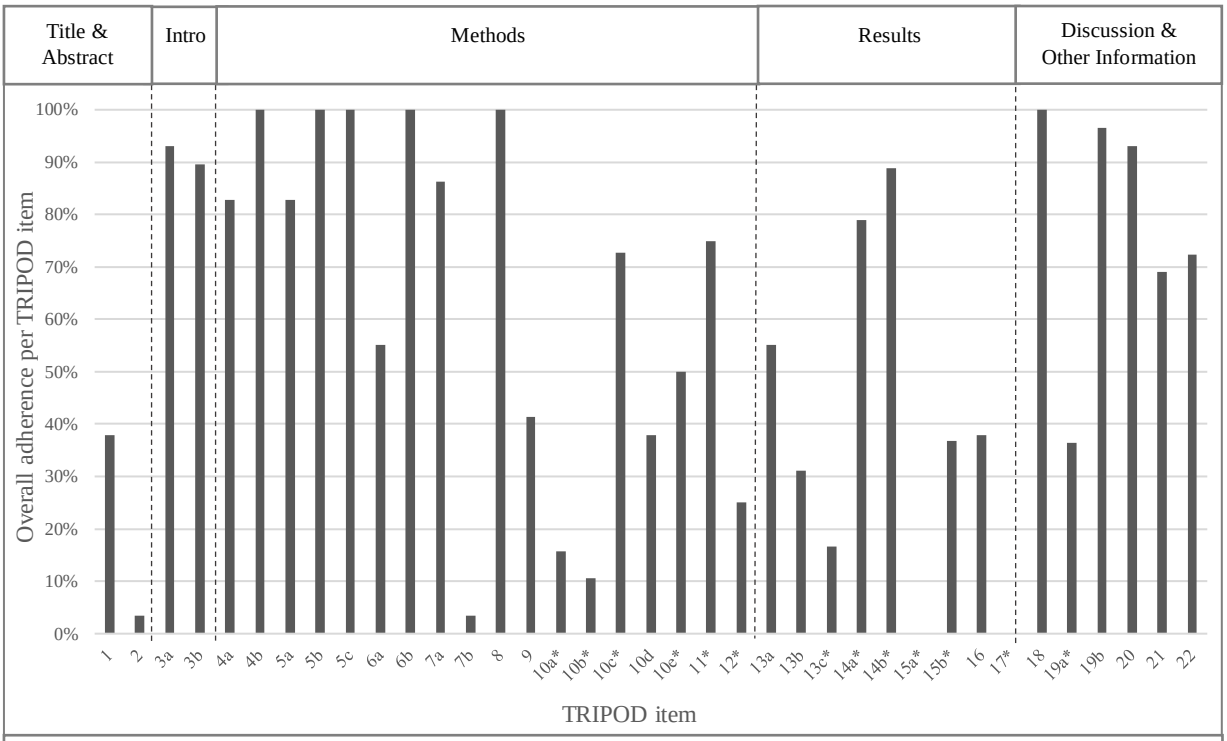
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Abbreviations: HR, hospital readmission; PRISMA, preferred reporting items for systematic reviews and meta-analyses.

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Percentages relate to the number of models for which an item was applicable (in this case, the respective item should have been reported). *Indication of derivation from the total number of models for which a TRIPOD item was applicable (N=# of models for which the TRIPOD item is applicable): 10a (N=19), 10b (N=19), 10c (N=11), 10e (N=2), 11 (N=4), 12 (N=12), 13c (N=12), 14a (N=19), 14b (N=18), 15a (N=19), 15b (N=19), 17 (N=1), 19a (N=11).

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Supplemental Material

Table A1: Search strategy for PubMed

#	PubMed (Search date: 01st January 2021; Updated Search date: 07th October 2021)	Results from initial search	Results from updated search
1	"Patient Readmission"[Mesh]	18,336	20,199
2	rehospitali*[Title/Abstract]	8,017	8,645
3	readmission*[Title/Abstract]	31,790	35,343
4	(hospital[Title/Abstract]) AND readmission*[Title/Abstract]	21,144	23,398
5	(unplanned[Title/Abstract]) AND readmission*[Title/Abstract]	2,360	2,677
6	(patient[Title/Abstract]) AND readmi*[Title/Abstract]	16,395	18,106
7	readmit*[Title/Abstract]	8,289	8,926
8	re-admission*[Title/Abstract]	2,205	2,416
9	(repeat*[Title/Abstract]) AND hospital*[Title/Abstract]	29,556	31,364
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	75,738	81,968
11	predictive factor*[Title/Abstract]	28,124	30,391
12	predict*[Title/Abstract]	1,646,322	1,765,658
13	"Predictive Value of Tests"[Mesh]	207,290	215,023
14	"ROC Curve"[Mesh]	60,417	65,077
15	model*[Title/Abstract]	3,025,019	3,240,665
16	c-statistic*[Title/Abstract]	6,938	7,765
17	ROC*[Title/Abstract]	55,019	62,608
18	"Sensitivity and Specificity"[Mesh]	595,955	619,284
19	Sensitivity [Title/Abstract]	827,733	873,715
20	Specificity [Title/Abstract]	478,451	502,627
21	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20	5,225,692	5,555,402
22	"Child"[Mesh]	1,936,577	2,010,900
23	"Infant"[Mesh]	1,152,685	1,189,047
24	"Adolescent"[Mesh]	2,058,561	2,216,512
25	(((((adolescen*[Title/Abstract]) OR teen*[Title/Abstract]) OR youth[Title/Abstract]) OR juvenile*[Title/Abstract]) OR young person*[Title/Abstract]) OR young people*[Title/Abstract]	464,723	492,107
26	"Pediatrics"[Mesh]	59,136	61,129
27	((((((((((child*[Title/Abstract]) OR (infant*[Title/Abstract])) OR (toddler*[Title/Abstract])) OR (bab*[Title/Abstract])) OR (newborn*[Title/Abstract])) OR (neonat*[Title/Abstract])) OR (school age*[Title/Abstract])) OR (preschool)) OR (paediatric*[Title/Abstract])) OR (pediatric*[Title/Abstract])) OR (kid*[Title/Abstract])) OR (boy*[Title/Abstract])) OR (girl*[Title/Abstract])	2,246,452	2,343,927
28	22 OR 23 OR 24 OR 25 OR 26 OR 27	4,365,971	4,517,928
29	10 AND 21 AND 28	3,674	3,993

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Table A2: Search strategy for CINAHL

	CINAHL (Initial Search date: 01st January 2021; Updated Search date: 07th October 2021)	Results from initial search	Results from updated Search
1	(MH "Readmission")	14,256	15,177
2	rehospitali*	2,624	2,779
3	readmission*	21,640	23,205
4	hospital AND readmission*	13,210	14,167
5	unplanned AND readmission*	1,227	1,363
6	patient AND readmi*	17,823	19,163
7	readmit*	2,588	2,791
8	re-admission*	954	1,029
9	repeat* AND hospital*	12,404	12,997
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	36,195	38,498
11	predictive factor*	10,824	11,733
12	predict*	394,485	423,169
13	(MH "Predictive Value of Tests")	52,176	54,193
14	(MH "ROC Curve")	27,863	29,859
15	model*	626,177	669,950
16	c-statistic*	2,593	2,822
17	ROC*	45,218	74,438
18	(MH "Sensitivity and Specificity")	85,260	87,853
19	sensitiv*	236,551	247,288
20	specific*	481,504	511,797
21	(MH "Predictive Validity")	5,587	5,746
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	1,337,979	1,442,565
23	(MH "Child")	467,217	486,202
24	(MH "Infant+")	260,190	268,903
25	(MH "Adolescence+")	535,922	557,353
26	adolescen* OR teen* OR youth OR juvenile* OR young person* OR young people*	598,199	624,571
27	(MH "Pediatrics+")	21,316	21,917
28	child* OR infant* OR toddler* OR bab* OR newborn* OR neonat* OR school age* OR preschool OR paediatric* OR pediatric* OR kid* OR boy* OR girl*	1,150,913	1,264,003
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	1,413,615	1,514,718
30	10 AND 22 AND 29	2,459	2745

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Table A3: Search strategy for Embase

	Embase (Initial Search date: 01st January 2021*; Updated Search dates: 7th June 2021, 07th October 2021)	Results from updated search 7th June 2021	Results from updated search 07th October 2021
1	'hospital readmission'/exp	73,736	76,850
2	rehospitali*:ab,ti	10,922	15,528
3	readmission*:ab,ti	57,908	64,311
4	(hospital NEAR/10 readmission*):ab,ti	22,791	25,140
5	(unplanned NEAR/10 readmission*):ab,ti	3,446	3,749
6	(patient NEAR/5 readmi*):ab,ti	5,568	6,357
7	readmit*:ab,ti	16,262	18,777
8	're admission*':ab,ti	5,802	5,982
9	(repeat* NEAR/5 hospital*):ab,ti	5,341	5,466
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	100,580	107,753
11	'predictive factor*':ab,ti	47,721	49,181
12	predict*:ab,ti	2,301,090	2,367,382
13	'predictive value'/exp	191,705	199,896
14	'roc curve'/exp OR 'receiver operating characteristic'/exp	149,025	157,486
15	'model'/exp	3,201,228	3,267,314
16	'c statistic*':ab,ti	12,892	13,511
17	roc*:ab,ti	196,628	207,024
18	'sensitivity and specificity'/exp	391,861	405,843
19	'sensitivity':ab,ti	1,109,211	1,135,520
20	'specificity':ab,ti	636,321	650,496
21	'predictive validity'/exp	8,691	8,839
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	6,371,499	6,524,117
23	'child'/exp	3,031,530	3,082,756
24	'infant'/exp	1,180,824	1,197,926
25	'adolescent'/exp	1,741,216	1,770,557
26	adolescen*:ab,ti OR teen*:ab,ti OR youth:ab,ti OR 'juvenile*':ab,ti OR 'young person*':ab,ti OR 'young people':ab,ti	598,758	613,037
27	'pediatrics'/exp	123,610	125,641
28	'child*':ti,ab OR 'infant*':ti,ab OR 'toddler*':ti,ab OR bab*:ti,ab OR 'newborn*':ti,ab OR neonat*:ti,ab OR 'school age*':ti,ab OR 'preschool':ti,ab OR 'pediatric*':ti,ab OR kid*:ti,ab OR 'boy*':ti,ab OR 'girl*'	3,642,407	3,715,357
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	5,498,512	5,594,597
30	#10 AND #22 AND #29	2,902**	3,338
* Final result from initial search: 2,845 ** Search term: #10 AND #22 AND #29 AND [1-1-1966]/sd NOT [2-1-2021]/sd			

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Table A4: Risk of bias assessment at study level

Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Vo et al. 2018	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - No detailed description of statistical analysis for the simplified scoring rule; Unadjusted associations for "Complications" are missing	**1**
Polites et al. 2016	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Moderate - 38.5% missing values for the ICD-9-diagnosis variable used for determining the reasons for readmission	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Brittan et al. 2018	- Moderate - Potential limitation of generalizability as just one paediatric hospital was examined; Only basic information about "age"	- Moderate - About 75% of patients were missing homecare values in the original data (the study assumption is that this patients did not received home health)	- Moderate - Assumption that 75% of patients did not receive home health; Possible omission or error of clinical documentation	- Moderate - no clear description for the determination of unplanned hospital readmissions	- Low - just PFs of the composite score are considered	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Sills et al. 2017	- Low -	- Moderate - No flow diagram	- Moderate - Area-level SDH data were not created for research and are a biased proxy for individual-level SDH measures; Handling of continuous predictors relating to ZIP-Codes (i.e. linear or non-linear) not clear described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2018	- Moderate - Potential limitation of generalizability as just one hospital was examined; No detailed information about the source of ZIP-related variables; No information about characteristics of patients in the validation cohort (Random-Split-Sample)	- Low -	- Moderate - No detailed description for the handling of continuous predictors and method for categorized predictors	- Low - The outcome was determined automatically and not validated by manual chart review	- Low -	- Moderate - No detailed definition for LACE variables; Missing information for the validation cohort	**2**
Smith et al. 2015	- Moderate - No information about characteristics of patients in the external validation set; No detailed information about the source of ZIP-related variables	- Moderate - No flow diagram; Missing information for validation cohort	- Moderate - Age categories are not clearly reported	- Low -	- Low -	- Moderate - Missing information for the validation cohort (non-random split sample)	**2**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Jiang et al. 2018	- Low -	- Moderate - Outcome events for NRD-data set are not described; No flow diagram	- Moderate - Patient distribution is presented by risk scores	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions; It is unclear, if 30-day inpatient readmissions in the NRD are also calculated from the date of the index procedure	- Low - Due to external validation of existing risk scores confounders are not relevant	- Moderate - Method of CCS is not clear; Outcome event rates and summary information of predictors are not clearly reported	**6**
Learly et al. 2019	- Moderate - Potential limitation of generalizability as just one hospital examined; No detailed information about the source of ZIP-related variables	- Low -	- Moderate - Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Low -	**2**
Bradshaw et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; Due to deidentification of patient data demographic characteristics are not captured	- Low -	- Low -	- Moderate - "The multiple categories that comprise the HARRPS-tool have different weight values associated that help prevent planned readmissions from skewing the results."(p. 53)	- Low -	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Delaplain et al. 2020	- Moderate - Focus on baseline characteristics of readmission in 7 days; No detailed characteristics for the validation cohort (random-split sample)	- Moderate - Information about the patient flow are provided at hospital level; Hospitals were excluded, if data were not collected for all variables related to the study; Missing information for validation cohort (random-split sample)	- Low - Predictors are presented by focusing on 7-day readmissions	- Moderate - Clear definition for 7-day unplanned readmissions with possible transfer to 30-day unplanned readmissions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**
Zhou et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; One-day procedures are also included (discharge within 24h)	- Low -	- Moderate - Census data might be a biased proxy for individual level measures; Method to choose the cut-points was not clearly described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2020	- Moderate - Basic information just for the training data set	- Moderate - Missing information for test data set	- High - No summary information for all predictors included in the validated model; No clear definition of predictors	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions	- Low - Due to external validation of LACE index confounders are not relevant	- High - Applied method for LACE is not clear (i.e. possible updating); Missing information for training data set	**1**
Hoenk et al. 2021	- Low -	- Moderate - Information about the patient flow are based on hospital level; Hospitals were excluded if data were not collected for all variables related to the study and if there were not at least 500 encounters for a	- Moderate - Method to choose the cut-points was not clearly presented	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
		neoplastic condition; Missing information for test data set					
Zhou et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Moderate – 470 instead of 550 patient pairs because of the burden associated with extracting data from medical records	- Low -	- Low -	- Low-	- Moderate - No detailed final model presentation (i.e. missing presentation of odds ratios in multivariable analysis)	**6**
Ryan et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Low -	- Low -	- Moderate - It is unclear, how unplanned readmissions are determined	- Low - Due to external validation of PASS confounders are not relevant	- Moderate - Summary information of predictors are not clearly reported	**1**
O´Conell et al. 2021	- Low -	- Moderate - Information about the patient flow are based on hospital level; Hospitals were excluded if data were not collected for all variables related to the study and if there were not at least 1000 encounters for a nervous system condition; No clear information on differentiation between training and test data set	- Moderate - No summary information for all predictors included in the multivariable model; Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Moderate - No clear reporting regarding the differentiation between training and test data set (i.e. summary information for predictors)	**1**

* For the risk of bias assessment at study-level, only the predictive models that were also included in the systematic review were considered.

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Table A5: Examined and significant variables for all 29 included predictive models

Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
All-cause related readmissions					
Brittan et al. 2018	Composite Score	All-cause	30-day unplanned HRs	Age; Lenght of stay; Non-English speaking caregiver; Discharge medications; Enteral feeding (eg, gastrostomy tube feeding); Respiratory (eg, home ventilator); IV infusion (eg, infusion of IV medication); Speech therapy; Physical therapy; Occupational therapy; Skilled nursing home visits; Private duty nursing home visits; CNA nursing assistant home visits; Durable medical equipment (eg, wheel chair); Home Care ≥ 1 order; Composite score = 0; Composite score = 1; Composite score = 2; Composite score = 3	Score based on the variables: Non-English speaking caregiver, Discharge medications, Home Care ≥ 1 order: Composite score (0 vs 1) (OR=1.7; 95% CI 1.5-2); Composite score (0 vs ≥ 2) (OR=4.2; 95% CI 3.6-4.9)
Sills et al. 2017	PACR + SDH	All-cause	30-day unplanned HRs	CCI count; Infectious and parasitic disease; Neoplasms; Endocrine, nutritional, and metabolic diseases and immunity disorders; Diseases of blood and blood-forming organs; Mental disorders; Diseases of the nervous system and sense organs; Diseases of the circulatory system; Diseases of the respiratory system; Diseases of the digestive system; Diseases of the genitourinary system; Diseases of the skin and subcutaneous tissue; Diseases of the musculoskeletal system; Congenital anomalies; Certain conditions originating in the perinatal period; Symptoms, signs, and ill-defined conditions; Injury and poisoning; Factors influencing health status and contact with health services; Age group; Sex; Race; Hispanic ethnicity; Payer; Median household income; Proportion of housing units vacant; Proportion of households with children that are single-parent; Proportion of families below poverty level; Unemployment rate; Proportion of adults with less than a high school diploma or equivalent	CCI count: [2 body systems (OR=1.51; 95% CI 1.44-1.58), 3 body systems (OR=1.57; 95% CI 1.48-1.67), >3 body systems (OR=1.31; 95% CI 1.19-1.44)]; CCIs: Neoplasms (OR=0.46; 95% CI 0.44-0.48), Endocrine, nutritional, and metabolic diseases and immunity disorders (OR=0.8; 95% CI 0.77-0.83), Diseases of blood and blood-forming organs (OR=0.59; 95% CI 0.56-0.61), Mental disorders (OR=0.95; 95% CI 0.92-0.99), Diseases of the nervous system and sense organs (OR=0.78; 95% CI 0.75-0.8), Diseases of the circulatory system (OR=0.73; 95% CI 0.7-0.76), Diseases of the respiratory system (OR=1.08; 95% CI 1.05-1.12), Diseases of the digestive system (OR=0.65; 95% CI 0.63-0.68), Diseases of the genitourinary system (OR=0.7; 95% CI 0.66-0.74), Congenital anomalies (OR=0.88; 95% CI 0.85-0.91), Injury and poisoning (OR=0.63; 95% CI 0.54-0.74), Factors influencing health status and contact with health services (OR=0.51; 95% CI 0.49-0.53); Age group: [1-<5y (OR=0.84; 95% CI 0.81-0.87), 5-<8y (OR=0.73; 95% CI 0.69-0.76), 8-<12 y (OR=0.72; 95% CI 0.68-0.75), 12-<18 y (OR=0.79; 95% CI 0.76-0.83)]; male (OR=0.96; 95% CI 0.93-0.98); Race: [Asian (OR=0.91; 95% CI 0.84-0.99), Other (OR=0.92; 95% CI 0.88-0.96)], Hispanic (OR=1.06; 95% CI 1.02-1.1); Payer: [Private (OR=0.85; 95% CI 0.83-0.88)]; Median household income (per $10 000 increase) (OR=1.02; 95% CI 1.02-1.02); Vacancy rate (per 10% increase) (OR=0.96; 95% CI 0.94-0.99); Percent single-parent (per 10% increase) (OR=1.03; 95% CI 1.01-1.04)
Ehwerhemuepha et al. 2018	Unnamed	All-cause	30-day unplanned HRs	Sex; Race and/or ethnicity; Length of stay, d; Age, y; Median income by zip code, per $10 000; Percent vacant houses by zip code; Percent single parents by zip code; Low income primary medical insurance; Planned admission; ED visits within last 6 mo; Admitted through ED; Previous inpatient visits within last 6 mo; History of 30 d readmission within last 6mo; Ambulatory resource use within last 6 mo; Charlson's comorbidities; Complex chronic conditions; Was in ICU; pRI: Admitting; pRI: Discharge; pRI: Minimum; pRI: Maximum; pRI: Average; pRI: Decreasing slope across entire stay; pRI: Decreasing slope during last 24 h of hospitalization; pRI: Minimum occurred first 24 h; pRI: Maximum occurred last 24 h; Infectious and/or parasitic; Neoplasms (excluding encounters for chemotherapy); Blood and/or immune; Endocrine, nutritional, and/or metabolic; Mental and/or neurodevelopment; Nervous, eye, ear, and/or mastoid; Circulatory; Respiratory; Digestive; Skin and/or subcutaneous tissue; Musculoskeletal; Genitourinary; Perinatal period; Congenital malformations; Symptoms, signs, and/or laboratory findings not classified elsewhere; Injury and/or poison; External morbidity causes; Health status and/or services factors	Length of stay, d: [<4 (2, 3) (OR=1.23; NR% CI 1.07–1.42), < 7 (4, 5, 6) (OR=1.42; NR% CI 1.20–1.67), 7 or more (OR=1.80; NR% CI 1.51–2.14)]; Planned admission: [Yes (OR=0.65; NR% CI 0.52–0.82)]; ED and/ or department visits within last 6 mo: [1 (OR=1.27; NR% CI 1.11–1.45), 2 or more (OR=1.30; NR% CI 1.12–1.50)]; Admitted through ED: [Yes (OR=1.15; NR% CI 1.02–1.29)]; Previous inpatient visits within last 6 mo: [1 (OR=1.86; NR% CI 1.59–2.17), 2 or more (OR=2.39; NR% CI 1.94–2.94)]; History of 30 d readmission within last 6mo: [1 (OR=1.27; NR% CI 1.06–1.52), 2 or more (OR=2.75; NR% CI 2.19–3.44)]; Complex chronic conditions: [1 (OR=1.63; NR% CI 1.42–1.87), 2 or more (OR=1.66; NR% CI 1.43–1.92)]; Discharge pRI (10 pt increment) (OR=0.86; NR% CI 0.81–0.90); Maximum pRI occurred last 24 h of hospitalization (OR=0.85; NR% CI 0.75–0.95); Neoplasms (excluding encounters for chemotherapy): [Yes (OR=2.17; NR% CI 1.85–2.55)]; Blood and/or immune: [Yes (OR=1.30; NR% CI 1.14–1.48)]; Nervous, eye, ear, and/or mastoid: [Yes (OR=1.21; NR% CI 1.08–1.37)]; Circulatory: [Yes (OR=1.33; NR% CI 1.14–1.56)]; Respiratory: [Yes (OR=0.75; NR% CI 0.66–0.85)]; External morbidity causes: [Yes (OR=0.64; NR% CI 0.45–0.89)]; Health status and/or services factors: [Yes (OR=1.20; NR% CI 1.06–1.35)]
	LACE (validation)			Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Bradshaw et al. 2020	HARRPS-tool	All-cause	30-day unplanned HRs	Other Diagnosis; Anemia/Neutropenia; Appendectomy; Asthma; Bronchiolitis; Gastroenteritis; Pneumonia; Seizure; Sickle Cell Crisis; Upper Respiratory Tract Infection; Ventricular Shunt; At-risk admission diagnosis present; Chronic condition indicator; Readmitted within 30 days (history); Inpatient Admit in last 6 months; Acuity of admission: No admission acuity identified; Acuity of admission: ICN/PICU Admission; Acuity of admission: Significant psychosocial concern; Acuity of admission: Medical transport from outside facility; Acuity of admission: Other; Insurance type: Self-Pay; Insurance type: Medicaid; Insurance type: Commercial/Private; Insurance type: Medicaid & Self-Pay; Caregiver language: English; Caregiver language: Spanish; Caregiver language: Other; Medical equipment/Supplies count at home; Home nursing: Private Duty Nursing; Home nursing: Skilled Nursing; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy; Home therapy: Home Physical Therapy; Home therapy: Home Speech Therapy; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy	The final scoring system also includes risk factors, which were not significant in the multivariable analysis (i.e. At-risk admission diagnosis present; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy): Chronic condition indicator: [1 (OR=1.43; 95% CI 1.13-1.81), 2 (OR=1.75; 95% CI 1.27-2.40), 3 (OR=1.41; 95% CI 1.12-1.78)]; Previous 30-day readmission (OR=1.70; 95% CI 1.38-2.09); Inpatient admission in last 6 months (OR=2.21; 95% CI 1.76 2.54); ICN or PICU admit (OR=1.43; 95% CI 1.17-1.75); Self-Pay or Medicaid (OR=1.21; 95% CI 1.05 1.39); Medical equipment/Supplies count at home: [2 (OR=1.43; 95% CI 1.09-1.88), 3 (OR=2.68; 95% CI 1.80 4.01), 4 (OR=2.21; 95% CI 1.53 3.19)]
Zhou et al. 2020	Unnamed	All-cause	30-day unplanned HRs	Age; Gender; Admission status; Funding source as inpatients; Source of referral transport; State/Territory of residence; Care type provided; Type of health insurance; Index of Relative Social-Economic Advantage and Disadvantage (IRSAD); Interpreter service; ICU stay at index admission; Had GA at index admission; LOS at index admission; Day of index admission date (weekdays or weekend); Day of index admission date; Day of discharge from index admission; Day of discharge from index admission (weekdays or weekend); Number of Co-diagnoses	Age: [13-15y (OR=1.30; 95% CI 1.14–1.48) >=16y (OR=1.46; 95% CI 1.07–1.98)]; Private health insurance (OR=1.16; 95% CI 1.00–1.34); Aeromedical service (OR=0.47; 95% CI 0.31–0.71); IRSAD (%): 91-100 (OR=1.20; 95% CI 1.02–1.41) - based on residential postcode; GA at index admission (yes) (OR=0.67; 95% CI 0.64–0.76); LOS: [LOS 2-7 days (OR=1.42; 95% CI 1.30–1.55), LOS 8-14 days (OR=2.35; 95% CI 1.97–2.82), LOS >= 15 days (OR=2.39; 95% CI 1.88–2.98)]; Admission on Friday (OR=1.21; 95% CI 1.05–1.39); Day of discharge from index admission: [Friday (OR=1.26; 95% CI 1.10–1.44), Saturday (OR=1.34; 95% CI 1.15–1.57), Sunday (OR=1.24; 95% CI 1.05–1.47)]; Number of Co-diagnoses:[1 (OR=1.28; 95% CI 1.16–1.41), 2 (OR=1.73; 95% CI 1.53–1.95), 3 (OR=2.10; 95% CI 1.80–2.46), >=4 (OR=2.41; 95% CI 2.08–2.80)]
Ehwerhemu epha et al. 2020	LACE (validation)	All-cause	30-day unplanned HRs	Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study
Zhou et al. 2021	Model 1: GLM	All-cause	30-day unplanned HRs	Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis	Variables included in the final model: Day of admission date; Day of admission (weekday/weekend and public holiday)
	Model 1: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis	Variables included in the final model: Distance from residential address to hospital; Day of discharge date; Day of admission date, Day of admission (weekday/weekend and public holiday)
	Model 2: GLM			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the	Variables included in the final model: Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				index admission: number of outpatient clinic attendances	
	Model 2: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the index admission: number of outpatient clinic attendances	Variables included in the final model: Day of discharge date; Day of admission date, Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Language spoken other than English (interpreter service required); Known allergies
	Model 3: GLM			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the index admission: number of outpatient clinic attendances; Completion of Nursing Admission and Discharge Planning Form (Admission section); Completion of Nursing Admission and Discharge Planning Form (Discharge Planning section); Operation sheet or the last entry progress note made by doctors; Clinical pathway or the last entry progress note made by nurses; Last entry progress note made by allied healthcare providers; Written evidence of discharge information given by doctors; Written evidence of discharge information given by nurses; Written evidence of discharge medications information by doctors; Written evidence of discharge medications information by nurses; Written evidence of follow-up information given by doctors; Written evidence of follow-up information given by nurses; Consistency of written discharge documentation among healthcare providers; Delay in issuing discharge summary (date of discharge summary being issued – date of discharge)	Variables included in the final model: Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Completeness of Nursing Admission and Discharge Planning Form, Discharge Planning section (incompleteness)
	Model 3: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage	Variables included in the final model: Had general anaesthetic at index admission; Source of referral transport (ambulance); Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Significant social history; Language spoken other than English (interpreter service required); Known allergies; Completeness of Nursing Admission and Discharge Planning Form, Discharge Planning section (incompleteness); Completeness of Nursing Admission and Discharge Planning Form, Admission section (incompleteness); Progress note or Clinical Pathway documentation at discharge by nurses (not recorded); Follow-up information documented by doctors (not recorded)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				of hospital services 12 months prior to the index admission: number of outpatient clinic attendances; Completion of Nursing Admission and Discharge Planning Form (Admission section); Completion of Nursing Admission and Discharge Planning Form (Discharge Planning section); Operation sheet or the last entry progress note made by doctors; Clinical pathway or the last entry progress note made by nurses; Last entry progress note made by allied healthcare providers; Written evidence of discharge information given by doctors; Written evidence of discharge information given by nurses; Written evidence of discharge medications information by doctors; Written evidence of discharge medications information by nurses; Written evidence of follow-up information given by doctors; Written evidence of follow-up information given by nurses; Consistency ofwritten discharge documentation among healthcare providers; Delay in issuing discharge summary (date of discharge summary being issued – date of discharge)	
Surgical conditions related readmissions					
Vo et al. 2018	Unnamed	All surgical specialities without cardiac surgery	30-day unplanned post-surgical HRs relating to noncardiac surgery	Age; BMI; ASA class; Gender; Race; Surgical specialty; Admission status; Urgency of procedure; Prematurity; Congenital Heart Disease; Bleeding complication; Wound complication; Pulmonary complication; Renal complication; Neurologic complication, Cardiac complication	ASA>=3 (OR=1.9; 95% CI 1.8-2.0); Inpatient vs outpatient (OR=3.5; 95% CI 3.3-3.7); >1 Postoperative complication (OR=3.14; 95% CI 2.92-3.34); Presence of CHD (OR=1.66; 95% CI 1.31-2.11)
Polites et al. 2016	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Age; Sex; Race; Procedure group; Procedure type; Weight percentage; Premature birth; Congenital malformation; Any comorbidity; Diabetes mellitus (Individual comorbidity); Respiratory (Individual comorbidity); GI/Hepatobiliary (Individual comorbidity); Major cardiac (Individual comorbidity); Acute renal failure (Individual comorbidity); Neurologic (Individual comorbidity); Immunosuppression (Individual comorbidity); Nutrition (Individual comorbidity); Hematologic (Individual comorbidity); Active malignancy (Individual comorbidity); Preoperative SIRS, sepsis, septic shock; Preoperative transfusion or hematocrit <32; Case status; ASA class; Operative time, minutes; Wound class; Preoperative length of stay, days; Postoperative complication; Superficial Incisional SSI; Deep SSI; Pulmonary complications; Sepsis/CL associated bloodstream infection; Postoperative length of stay, days	Procedure group: [Head and Neck (HR=2.40; 95% CI 1.48-3.89), Hepatobiliary (HR=1.69; 95% CI 1.17-2.44), Small and Large intestine (HR=1.59; 95% CI 1.33-1.90), Thoracic (HR=0.69; 95% CI 0.52-0.91)]; Comorbidity: [Preoperative acute renal failure (HR=2.47; 95% CI 1.31-4.66), Neurologic comorbidity (HR=1.3; 95% CI 1.05-1.62)]; SIRS/Sepsis/Septic Shock within 48 h prior to index admission (HR=1.2; 95% CI 1.02-1.41); Operative time, minutes: [60–140 min (HR=1.21; 95% CI 1.06-1.39), >140min (HR=1.51; 95% CI 1.26-1.81)]; Wound class: [Contaminated (HR=1.29; 95% CI 1.05-1.60), Dirty/Infected (HR=1.92; 95% CI 1.53-2.40)]; Any complication (HR=1.34; 95% CI 1.09-1.65); Postoperative length of stay, days: [2-4 days (HR=2.17; 95% CI 1.84-2.57), >4 days (HR=3.12; 95% CI 2.60-3.74)]
Smith et al. 2015	Developed model	Cardiac surgery	30-day HRs following congenital heart surgery	[Readmission characteristics are not mentioned, as they were not a part of the modelling process] Male gender; Neonatal age group; Hospital volume, Ethnicity; Mean regional income ($USD); Payor; ICU length of stay (days); ICU LOS ≥7 days; Postoperative LOS (days); Postoperative LOS ≥14 days; ECMO support during hospitalization; RACHS-1 score; GERD diagnosis; Genetic syndrome; Vocal cord paralysis; Thrombosis; Prematurity; Renal insufficiency; Serious Infection; Mediastinitis; Sepsis; Necrotizing enterocolitis; Stroke; Tachyarrhythmia; Complete AV block; Gastrostomy tube placement; Tracheostomy; Diaphragm plication; Weekend discharge	Hispanic (OR=1.24; 95% CI 1.05–1.41); Government payor (OR=1.22; 95% CI 1.06–1.41); ICU LOS ≥7 days (OR=1.86; 95% CI 1.56–2.21); Postoperative LOS ≥14 days (OR=1.59; 95% CI 1.32–1.91); ECMO support during hospitalization (OR=1.74; 95% CI 1.16–2.60); RACHS-6 score (OR=2.60; 95% CI 1.83–3.69); Gastroesophageal reflux (OR=1.49; 95% CI 1.25–1.78); Genetic syndrome (OR=1.31; 95% CI 1.1–1.57); Thrombosis (OR=1.46; 95% CI 1.05–2.03)
	Validated model			Neonatal age group; Ethnicity; Payor; ICU LOS ≥7 days; Postoperative LOS ≥14 days; ECMO support during hospitalization; RACHS-1 score; GERD diagnosis; Genetic syndrome; Vocal cord paralysis; Thrombosis; Prematurity; Renal insufficiency; Serious Infection; Necrotizing enterocolitis; Stroke; Tachyarrhythmia; Complete AV block; Gastrostomy tube placement, Diaphragm plication; Weekend discharge	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Jiang et al. 2018	Charlson Comorbidity Index (validation NRD)	Urological surgery	30-day HRs after paediatric urological procedures	Myocardial infarction; Congestive heart failure; Peripheral vascular disease; Cerebrovascular disease; Dementia; Chronic pulmonary disease; Rheumatic disease; Peptic ulcer disease; Mild liver disease; Diabetes without chronic complication; Diabetes with chronic complication; Hemiplegia or paraplegia; Renal disease; Any malignancy, including lymphoma and leukemia, except malignant neoplasm of skin; Moderate or severe liver disease; Metastatic solid tumor; AIDS/HIV	External validation study
	Van Walraven Index (validation NRD)			Congestive heart failure; Cardiac arrhythmias; Valvular disease; Pulmonary circulation disorders; Peripheral vascular disorders; Hypertension; Paralysis; Neurodegenerative disorders; Chronic pulmonary disease; Diabetes, uncomplicated; Diabetes, complicated; Hypothyroidism; Renal failure; Liver disease; Peptic ulcer disease, no bleeding; AIDS/HIV; Lymphoma; Metastatic cancer; Solid tumour without metastasis; Rheumatoid arthritis/collagen; Coagulopathy; Obesity; Weight loss; Fluid and electrolyte disorders; Blood loss anemia; Deficiency anemia; Alcohol abuse; Drug abuse; Psychosis; Depression	External validation study
	Rhee index (validation NRD)			Age ,24 mo (vs >= 24 mo); Female (vs male); Peri-/endo-/myocarditis, cardiomyopathy, or tamponade; Acute myocardial infarction; Coronary atherosclerosis/other ischemic heart disease; Pulmonary vascular disease (eg, PE, pulmonary HTN); Ventricular tachycardia or other cardiac dysrhythmias; Cardiac arrest or ventricular fibrillation or flutter; Congestive heart failure; Aortic, peripheral, visceral artery aneurysms/ dissection; Aortic or peripheral arterial embolism or thrombosis; Cardiac or circulatory congenital anomalies; Pneumonia; Influenza; Chronic obstructive pulmonary disease/ bronchiectasis; Asthma; Aspiration pneumonitis; Respiratory failure, insufficiency, arrest; Cystic fibrosis; Respiratory distress syndrome; Acute renal failure; Chronic renal failure; Urinary tract infections or unspecified cystitis; Thyroid disorders or other endocrine disorders; Diabetes mellitus or complications; Septicemia (except in labor); Fungal infection; HIV infection; Hepatitis; Meningitis, encephalitis, or other CNS infection; Coma, stupor, or brain damage; Acute cerebrovascular disease; Gastroduodenal ulcer, gastritis, or duodenitis; Intestinal obstruction without hernia; Peritoneal or intestinal abscess, peritonitis; Biliary tract disease; Liver disease (eg, cirrhosis, increased LFTs); Pancreatic disorders (not diabetes); Gastrointestinal hemorrhage; Leukemia; Lymphomas or reticuloendothelial neoplasms; CNS or miscellaneous intracranial or intraspinal neoplasms; Primary malignant tumor of adrenal or paraganglia; Renal tumors; Hepatic tumors; Primary malignant bone or articular cartilage tumors; Soft tissue sarcomas; Immunity disorders (except AIDS); Coagulation or hemorrhagic disorders; Disease of white blood cells (eg, lymphocytosis, lymphopenia); Chronic ulcer of skin; Spondylosis, intervertebral disc disorders; Systemic lupus erythematosus or connective tissue disorder; Crushing injury or internal injury; Shock; Drowning/submersion; Firearm; Motor vehicle traffic; Pedal cyclist, not MVT; Suffocation; Suicide or intentional self-inflicted injury; Fracture of long bone or spine; Intracranial injury; Poisoning by nonmedicinal substances; Skull or face fractures; Open wounds of head, neck, or trunk; Superficial soft tissue injury: contusion; Short gestation, low birth wt, or fetal growth retardation; Hypoxia, asphyxia, or aspiration during birth; Birth trauma; Perinatal jaundice	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
	Charlson Comorbidity Index (validation SID)			Myocardial infarction; Congestive heart failure; Peripheral vascular disease; Cerebrovascular disease; Dementia; Chronic pulmonary disease; Rheumatic disease; Peptic ulcer disease; Mild liver disease; Diabetes without chronic complication; Diabetes with chronic complication; Hemiplegia or paraplegia; Renal disease; Any malignancy, including lymphoma and leukemia, except malignant neoplasm of skin; Moderate or severe liver disease; Metastatic solid tumor; AIDS/HIV	External validation study
	Van Walraven Index (validation SID)			Congestive heart failure; Cardiac arrhythmias; Valvular disease; Pulmonary circulation disorders; Peripheral vascular disorders; Hypertension; Paralysis; Neurodegenerative disorders; Chronic pulmonary disease; Diabetes, uncomplicated; Diabetes, complicated; Hypothyroidism; Renal failure; Liver disease; Peptic ulcer disease, no bleeding; AIDS/HIV; Lymphoma; Metastatic cancer; Solid tumour without metastasis; Rheumatoid arthritis/collagen; Coagulopathy; Obesity; Weight loss; Fluid and electrolyte disorders; Blood loss anemia; Deficiency anemia; Alcohol abuse; Drug abuse; Psychosis; Depression	External validation study
	Rhee index (validation SID)			Age ,24 mo (vs >= 24 mo); Female (vs male); Peri-/endo-/myocarditis, cardiomyopathy, or tamponade; Acute myocardial infarction; Coronary atherosclerosis/other ischemic heart disease; Pulmonary vascular disease (eg, PE, pulmonary HTN); Ventricular tachycardia or other cardiac dysrhythmias; Cardiac arrest or ventricular fibrillation or flutter; Congestive heart failure; Aortic, peripheral, visceral artery aneurysms/ dissection; Aortic or peripheral arterial embolism or thrombosis; Cardiac or circulatory congenital anomalies; Pneumonia; Influenza; Chronic obstructive pulmonary disease/ bronchiectasis; Asthma; Aspiration pneumonitis; Respiratory failure, insufficiency, arrest; Cystic fibrosis; Respiratory distress syndrome; Acute renal failure; Chronic renal failure; Urinary tract infections or unspecified cystitis; Thyroid disorders or other endocrine disorders; Diabetes mellitus or complications; Septicemia (except in labor); Fungal infection; HIV infection; Hepatitis; Meningitis, encephalitis, or other CNS infection; Coma, stupor, or brain damage; Acute cerebrovascular disease; Gastroduodenal ulcer, gastritis, or duodenitis; Intestinal obstruction without hernia; Peritoneal or intestinal abscess, peritonitis; Biliary tract disease; Liver disease (eg, cirrhosis, increased LFTs); Pancreatic disorders (not diabetes); Gastrointestinal hemorrhage; Leukemia; Lymphomas or reticuloendothelial neoplasms; CNS or miscellaneous intracranial or intraspinal neoplasms; Primary malignant tumor of adrenal or paraganglia; Renal tumors; Hepatic tumors; Primary malignant bone or articular cartilage tumors; Soft tissue sarcomas; Immunity disorders (except AIDS); Coagulation or hemorrhagic disorders; Disease of white blood cells (eg, lymphocytosis, lymphopenia); Chronic ulcer of skin; Spondylosis, intervertebral disc disorders; Systemic lupus erythematosus or connective tissue disorder; Crushing injury or internal injury; Shock; Drowning/submersion; Firearm; Motor vehicle traffic; Pedal cyclist, not MVT; Suffocation; Suicide or intentional self-inflicted injury; Fracture of long bone or spine; Intracranial injury; Poisoning by nonmedicinal substances; Skull or face fractures; Open wounds of head, neck, or trunk; Superficial soft tissue injury: contusion; Short gestation, low birth wt, or fetal growth retardation; Hypoxia, asphyxia, or aspiration during birth; Birth trauma; Perinatal jaundice	External validation study
Delaplain et al. 2020	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Planned readmission; Sex; Length of stay (d); Admission source; Payer; Age; Race/ethnicity; Previous ED visits; Current/index visit is a readmission from a prior visit?; Previous visits; Readmission history; Free-standing pediatric hospital; Number of medications; Number of systems diagnoses; Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or	LOS: [LOS 2-4 days (OR=1.298; NR% CI 1.189-1.416), LOS 4-7 days (OR=1.638; NR% CI 1.493-1.797), LOS >7 (OR=1.994; NR% CI 1.816-2.189)]; Admission source: Transfer (OR=0.768; NR% CI 0.689-0.856); Previous ED visits: [1 (OR=1.137; NR% CI 1.058-1.222), 2 (OR=1.158; NR% CI 1.037-1.294), >=3 (OR=1.305; NR% CI 1.157-1.471)]; Current/index visit is a readmission from a prior visit?: [Yes: unplanned (OR=1.404; NR% CI 1.291-

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				pelvis (S30-S39); Traumatic injury (ICD-10): Ankle and foot (S90-99); Traumatic injury (ICD-10): Burns (T20-32); Traumatic injury (ICD-10): Early complications of trauma (T79); Traumatic injury (ICD-10): Elbow and forearm (S50-59); Traumatic injury (ICD-10): Foreign body entering through the natural orifice (T15-19); Traumatic injury (ICD-10): Head (S00-09); Traumatic injury (ICD-10): Hip and thigh (S70-79), Traumatic injury (ICD-10): Unspecified body region (T14); Traumatic injury (ICD-10): Knee and lower leg (S80-89); Traumatic injury (ICD-10): Multiple body regions (T07); Traumatic injury (ICD-10): Neck (S10-19); Traumatic injury (ICD-10): Unspecified effects of external causes (T66-T78); Traumatic injury (ICD-10): Poisoning (T36-50); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88); Traumatic injury (ICD-10): Thorax (S20-29); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65); Traumatic injury (ICD-10): Wrist, hand, and fingers (S60-61); Surgical procedures by system: Auditory; Surgical procedures by system: Cardiovascular; Surgical procedures by system: Digestive; Surgical procedures by system: Integumentary; Surgical procedures by system: Musculoskeletal; Surgical procedures by system: Nervous; Surgical procedures by system: Respiratory; Traumatic injury: trauma due to frost bite; Traumatic injury: eye/ocular; Traumatic injury: endocrine; Surgical procedure: mediastinum/diaphragm; Surgical procedures by system: hematologic; Surgical procedures by system: lymphatic; Surgical procedures by system: urinary/reproductive	1.527), Yes: planned (OR=1.646; NR% CI 1.457-1.858)]; Previous visits: [1 (OR=1.746; NR% CI 1.604-1.9), 2 (OR=2.282; NR% CI 2.02-2.578), >=3 (OR=2.614; NR% CI 2.245-3.045)]; Readmission history: [1 (OR=1.473; NR% CI 1.303-1.665), 2 (OR=1.788; NR% CI 1.504-2.125), >=3 (OR=2.578; NR% CI 2.172- 3.06)]; Number of medications (OR=1.011; NR% CI 1.009-1.014); Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or pelvis (S30-S39) (OR=0.763; NR% CI 0.655-0.89); Traumatic injury (ICD-10): Burns (T20-32) (OR=0.596; NR% CI 0.494-0.719), Traumatic injury (ICD-10): Head (S00-09) (OR=0.661; NR% CI 0.594-0.735); Traumatic injury (ICD-10): Hip and thigh (S70-79) (OR=0.709; NR% CI 0.616-0.816); Traumatic injury (ICD-10): Unspecified body region (T14) (OR=0.752; NR% CI 0.62-0.912); Traumatic injury (ICD-10): Knee and lower leg (S80-89) (OR=0.727; NR% CI 0.614-0.86); Traumatic injury (ICD-10): Neck (S10-19) (OR=1.194; NR% CI 1.008-1.45); Traumatic injury (ICD-10): Poisoning (T36-50) (OR=1.16; NR% CI 1.06-1.269); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49) (OR=0.547; NR% CI 0.461-0.649); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88) (OR=1.545; NR% CI 1.439-1.66); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65) (OR=0.538; NR% CI 0.404-0.716)
General medical conditions related readmissions					
Learly et al. 2019	Prediction at admission	Complex chronic conditions	30-day unplanned HRs	Age; Race and/or ethnicity; Boys; Non-English primary language; Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo, Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type	Significant variables (p-value<0.05) in univariate analysis are used for the analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.15–2.45); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.32–3.10); No. CCC categories: [No. CCC categories = 2 (OR=1.73; 95% CI 1.19–2.49), No. CCC categories >=3 (OR=2.30; 95% CI 1.50–3.47)]; Medical admission (OR=1.82; 95% CI 1.30–2.63)
	Prediction at discharge			Age; Race and/or ethnicity; Boys; Non-English primary language, Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo; Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type; ICU use; Discharge disposition from the hospital; LOS in d; Weekday discharge	Significant variables (p-value<0.05) in univariate analysis are used for analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.16–2.46); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.31–3.11); No. CCC categories: [No. CCC categories = 2 (OR=1.56; 95% CI 1.06–2.26), No. CCC categories >=3 (OR=1.72; 95% CI 1.08–2.69)]; Medical admission (OR=1.75; 95% CI 1.23–2.49); Discharge disposition from the hospital: [With services (OR=1.69; 95% CI 1.17–2.44), Other facility (OR=1.15; 95% CI 0.58–2.13)]; LOS: [LOS 2-5 days (OR=1.15; 95% CI 0.78–1.72), LOS >=6 days (OR=1.45; 95% CI 0.90–2.33)] Weekday discharge was not significant (p-value≤ 0.05) in the univariate analysis, but included in the final scoring system.
Ryan et al. 2021	PASS (validation)	Asthma	30-day unplanned HRs	Respiratory rate, Oxygen requirement, Auscultation, Retractions, Dyspnea	External validation study
O´Conell et al. 2021	Unnamed	Nervous system condition	30-day unplanned HRs	Age; Sex; race/ ethnicity; payer; Length of stay days) ; Index visit planned; Admitted through ED; Index visit is a readmission; Previouse ED visits (prior 6mo); Previous hospitalizations (prior 6mo); Previous readmissions (prior 6 mo); Number of comorbid diagnoses (by ICD 10 CM chapters); Viral Meningitis (A87); Malignant neoplasm of brain (C71); Disorders of BCAA and FA Metabolism (E71); Other AA Metabolism Disorders (E72); Other carbohydrate metabolism disorders (E74); Lipoprotein Metabolism Disorders / Lipidemias (E78); Mental Disorders Due to Physiological Condition (F06); Unspecified Intellectual Disabilities (F79); Speech & Language Development Disorders (F80); Scholastic Skill Development Disorders (F81); Motor Function Development Disorders (F82); Pervasive Developmental	Age (OR=0.993; 95% CI 0.99-0.997); Hispanic/Latino (OR=1.126; 95% CI 1.016-1.247); Self-Pay (OR=0.805; 95% CI 0.688-0.943); LOS 2-4 days (OR=1.228; 95% CI 1.167-1.293), LOS 4-6 days (OR=1.441; 95% CI 1.361-1.525), LOS 7 or more days (OR=1.585; 95% CI 1.492-1.683); Index visit planned: Yes (OR=0.898; 95% CI 0.847-0.952); Emergent Admission (OR=1.129; 95% CI 1.079-1.18); Index visit is a planned readmission (OR=1.338; 95% CI 1.269-1.411), Index visit is a unplanned readmission (OR=1.666; 95% CI 1.516-1.832); 1 previous ED visit (prior 6mo) (OR=1.106; 95% CI 1.056-1.159), 2 previous ED visits (prior 6mo) (OR=1.200; 95% CI 1.123-1.283), 3 or more previous ED visits (prior 6mo) (OR=1.297; 95% CI 1.209-1.391); 1 previous hospitalization (prior 6mo) (OR=1,695; 95% CI 1.613-1.781), 2 previous hospitalizations (prior 6mo) (OR=2.348; 95% CI 2.18-2.528), 3 or more previous hospitalizations (prior 6mo) (OR=3.014; 95% CI 2.738-3.317); 1

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				Disorders (F84); Psychological Development Disorders (F88); Bacterial Meningitis (G00); Meningitis, Other Causes (G03); Encephalitis, Myelitis, Encephalomyelitis (G04); Extrapyramidal & movement Disorders (G25); Nervous System Degenerative Diseases (G31); Epilepsy & Recurrent Seizures (G40); Migraine (G43); Headache Syndromes (G44); Sleep Disorders (G47); Polyneuropathies (G62); Primary Muscle Disorders (G71); Cerebral Palsy (G80); Hemiplegia & Hemiparesis (G81); Paraplegia & Quadriplegia (G82); Pain (G89); ANS Disorders (G90); Hydrocephalus (G91); Brain Disorders (G93); Spinal Cord Diseases (G95); CNS Disorders (G96); Postprocedural NS Disorders (G97); Visual Pathway Disorders (H47); Paralytic Strabismus (H49); Binocular Movement Disorders (H51); Nystagmus & Irregular Eye Movements (H55); Conductive & Sensorineural Hearing Loss (H90); Nontraumatic Intracerebral Hemorrhage (I61); Nontraumatic Intracranial Hemorrhage (I62); Cerebral Infarction (I63); Cerebrovascular Diseases (I67); Sequelae of Cerebrovascular Disease (I69); Neuromuscular Dysfunction of Bladder (N31); Nontraumatic Intracranial Hemorrhage Newborn (P52); Newborn Convulsions (P90); Newborn Cerebral Disturbances (P91); Newborn Muscle Tone Disorders (P94); Microcephaly (Q02); Congenital Hydrocephalus (Q03); Congenital Brain Malformations (Q04); Spina Bifida (Q05); Congenital Spinal Cord Malformations (Q06); Congenital Nervous System Malformations (Q07); Down Syndrome (Q90); Other Autosomal Trisomies (Q92); Nervous & Musculoskeletal System Symptoms (R29); Somnolence, Stupor, Coma (R40); Cognitive Function Symptoms (R41); Dizziness & Giddiness (R42); Headache (R51); Convulsions (R56); Intracranial Injury (S06); Number of medications; Free-standing pediatric hospital; Surgical procedures	previous readmission (prior 6 mo) (OR=1.179; 95% CI 1.09-1.274), 2 previous readmissions (prior 6 mo) (OR=1.496; 95% CI 1.338-1.673), 3 or more previous readmissions (prior 6 mo) (OR=2.051; 95% CI 1.831-2.298); Number of comorbid diagnoses (by ICD 10 CM chapters) (OR=1.01; 95% CI 1.006-1.014); Viral Meningitis (A87) (OR=0.446; 95% CI 0.353-0.563); Malignant neoplasm of brain (C71) (OR=1.953; 95% CI 1.788-2.133); Other AA Metabolism Disorders (E72) (OR=1.313; 95% CI 1.095-1.575); Mental Disorders Due to Physiological Condition (F06) (OR=0.768; 95% CI 0.648-0.91); Speech & Language Development Disorders (F80) (OR=0.811; 95% CI 0.725-0.907); Bacterial Meningitis (G00) (OR=0.622; 95% CI 0.487-0.796); Meningitis, Other Causes (G03) (OR=0.829; 95% CI 0.698-0.985); Sleep Disorders (G47) (OR=0.84; 95% CI 0.791-0.892); Polyneuropathies (G62) (OR=1.5; 95% CI 1.248-1.803); Hydrocephalus (G91) (OR=1.218; 95% CI 1.124-1.476); Brain Disorders (G93) (OR=1.078; 95% CI 1.019-1.14); Paralytic Strabismus (H49) (OR=1.288; 95% CI 1.124-1.476); Nystagmus & Irregular Eye Movements (H55) (OR=0.825; 95% CI 0.686-0.992); Neuromuscular Dysfunction of Bladder (N31) (OR=1.155; 95% CI 1.034-1.129); Newborn Cerebral Disturbances (P91) (OR=0.822; 95% CI 0.705-0.958); Newborn Muscle Tone Disorders (P94) (OR=0.621; 95% CI 0.567-0.68); Microcephaly (Q02) (OR=1.131; 95% CI 1.005-1.1272); Congenital Hydrocephalus (Q03) (OR=1.318; 95% CI 1.175-1.478); Down Syndrome (Q90) (OR=1.129; 95% CI 1.036-1.23); Other Autosomal Trisomies (Q92) (OR=1.214; 95% CI 1.019-1.448); Nervous & Musculoskeletal System Symptoms (R29) (OR=0.882; 95% CI 0.781-0.995); Somnolence, Stupor, Coma (R40) (OR=0.786; 95% CI 0.694-0.889); Cognitive Function Symptoms (R41) (OR=0.853; 95% CI 0.776-0.938); Convulsions (R56) (OR=0.922; 95% CI 0.874-0.972); Intracranial Injury (S06) (OR=0.476; 95% CI 0.418-0.543); Number of medications (OR=1.014; 95% CI 1.012-1.016); Free-standing pediatric hospital (Yes) (OR=1.586; 95% CI 1.103-2.28); Surgical procedures (Cardiovascular) (OR=1.204; 95% CI 1.109-1.216), Surgical procedures (Musculoskeletal) (OR=0.685; 95% CI 0.6-0.781), Surgical procedures (Nervous System) (OR=1.119; 95% CI 1.029-1.216)
Hoenk et al. 2021	Unnamed	Oncology	30-day unplanned HRs	Age; sex; race/ ethnicity; Payer; Acute lymphoid leukemia (ALL); Acute myeloid leukemia (AML); Brain cancer; Neuroblastoma; Wilms tumor; Hodgkin's lymphoma; Non-Hodgkin's lymphoma; Rhabdomyosarcoma; Bone/cartilage cancer; Other cancers; Chemotherapy; Bone marrow transplant; Number of cancer medications; Number of previous visits for chemotherapy (prior 30 d); Length of stay; Emergent admission; Is index/current visit itself a readmission; Previous ED visits (prior 6 mo); Number of previous visits without chemotherapy (prior 6 mo); Previous readmissions (prior 6 mo); Viral or bacterial infections (A00 - B99); Hematological conditions (D50-D69, D71-D79); Neutropenia (D70); Immunological conditions (D80-D89); Endocrine, nutritional, and metabolic diseases (E00-E89); Mental, behavioral, and neurodevelopmental disorders (F01-F99); Nervous system (G00-G99); Eye, adnexa, ear, mastoid (H00-H95); Circulatory (I00-I99); Respiratory (J00-J99); Digestive (K00-K95); Skin/subcutaneous tissue (L00-L99); Musculoskeletal and connective tissue (M00-M99); Genitourinary (N00-N99); Congenital/Chromosomal (Q00-Q99); Injuries and poisoning (S00-T88)	Age (OR=0.987; 95% CI 0.982-0.993); Acute lymphoid leukemia (ALL): Yes, not in remission (OR=0.788; 95% CI 0.721-0.861), Acute lymphoid leukemia (ALL): Yes, in remission (OR=0.656; 95% CI 0.590-0.729), Acute lymphoid leukemia (ALL): Yes, in relapse (OR=1.436; 95% CI 1.201-1.718); Brain cancer (OR=0.782; 95% CI 0.711-0.861); Neuroblastoma (OR=1.442; 95% CI 1.285-1.619); Rhabdomyosarcoma (OR=1.182; 95% CI 1.048-1.332); Bone/cartilage cancer (OR=1.618; 95% CI 1.455-1.799); Number of cancer medications (OR=1.46; 95% CI 1.117-1.175); LOS 2, 3 days (OR=1.292; 95% CI 1.184-1.410), LOS 4,5,6 days (OR=1.466; 95% CI 1.340-1.604), LOS 7 or more days (OR=1.344; 95% CI 1.219-1.481); Is index/current visit itself a readmission (Yes, unplanned) (OR=1.869; 95% CI 1.722-2.028), Is index/current visit itself a readmission (Yes, planned) (OR=1.397; 95% CI 1.250-1.560); Number of previous visits without chemotherapy (prior 6 mo) (OR=1.077; 95% CI 1.067-1.221); 1 previous readmission (prior 6 mo) (OR=1.277; 95% CI 1.126-1.336), 2 previous readmissions (prior 6 mo) (OR=1.281; 95% CI 1.145-1.433), 3 or more previous readmissions (prior 6 mo) (OR=1.263; 95% CI 1.087-1.468); Endocrine, nutritional, and metabolic diseases (E00-E89) (OR=1.141; 95% CI 1.067-1.221); Congenital/Chromosomal (Q00-Q99) (OR=0.745; 95% CI 0.657-0.844); Injuries and poisoning (S00-T88) (OR=0.838; 95% CI 0.760-0.925)

Abbreviations: NR, not reported

* Only significant risk factors (odds ratio/ hazard ratio>1, p-value<0.05) are considered in the analysis (cf. table 2). Significant predisposed factors (1>odds ratio/ hazard ratio>0, p-value<0.05) are listed as an additional information.

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Table A6: Adherence per TRIPOD-item at predictive model level

TRIPOD-item	Developed predictive models (n=17)			External validated predictive models (n=10)			Incremental value predictive models (n=2)			All predictive models (n=29)		
	# of PMs with applicable item	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item
1	17.00	4.00	23.53%	10.00	6.00	60.00%	2.00	1.00	50.00%	29.00	11.00	37.93%
2	17.00	1.00	5.88%	10.00	0.00	0.00%	2.00	0.00	0.00%	29.00	1.00	3.45%
3a	17.00	17.00	100.00%	10.00	8.00	80.00%	2.00	2.00	100.00%	29.00	27.00	93.10%
3b	17.00	16.00	94.12%	10.00	8.00	80.00%	2.00	2.00	100.00%	29.00	26.00	89.66%
4a	17.00	14.00	82.35%	10.00	9.00	90.00%	2.00	1.00	50.00%	29.00	24.00	82.76%
4b	17.00	17.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	29.00	29.00	100.00%
5a	17.00	15.00	88.24%	10.00	7.00	70.00%	2.00	2.00	100.00%	29.00	24.00	82.76%
5b	17.00	17.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	29.00	29.00	100.00%
5c	17.00	17.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	29.00	29.00	100.00%
6a	17.00	12.00	70.59%	10.00	2.00	20.00%	2.00	2.00	100.00%	29.00	16.00	55.17%
6b	17.00	17.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	29.00	29.00	100.00%
7a	17.00	16.00	94.12%	10.00	7.00	70.00%	2.00	2.00	100.00%	29.00	25.00	86.21%
7b	17.00	1.00	5.88%	10.00	0.00	0.00%	2.00	0.00	0.00%	29.00	1.00	3.45%
8	17.00	17.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	29.00	29.00	100.00%
9	17.00	10.00	58.82%	10.00	1.00	10.00%	2.00	1.00	50.00%	29.00	12.00	41.38%
10a	17.00	3.00	17.65%	0.00	0.00	0.00%	2.00	0.00	0.00%	19.00	3.00	15.79%
10b	17.00	2.00	11.76%	0.00	0.00	0.00%	2.00	0.00	0.00%	19.00	2.00	10.53%
10c	0.00	0.00	0.00%	10.00	7.00	70.00%	1.00	1.00	100.00%	11.00	8.00	72.73%
10d	17.00	4.00	23.53%	10.00	6.00	60.00%	2.00	1.00	50.00%	29.00	11.00	37.93%
10e	0.00	0.00	0.00%	1.00	0.00	0.00%	1.00	1.00	100.00%	2.00	1.00	50.00%
11	4.00	3.00	75.00%	0.00	0.00	0.00%	0.00	0.00	0.00%	4.00	3.00	75.00%
12	0.00	0.00	0.00%	10.00	1.00	10.00%	2.00	2.00	100.00%	12.00	3.00	25.00%
13a	17.00	14.00	82.35%	10.00	1.00	10.00%	2.00	1.00	50.00%	29.00	16.00	55.17%
13b	17.00	7.00	41.18%	10.00	0.00	0.00%	2.00	2.00	100.00%	29.00	9.00	31.03%
13c	0.00	0.00	0.00%	10.00	0.00	0.00%	2.00	2.00	100.00%	12.00	2.00	16.67%
14a	17.00	13.00	76.47%	0.00	0.00	0.00%	2.00	2.00	100.00%	19.00	15.00	78.95%
14b	16.00	14.00	87.50%	0.00	0.00	0.00%	2.00	2.00	100.00%	18.00	16.00	88.89%
15a	17.00	0.00	0.00%	0.00	0.00	0.00%	2.00	0.00	0.00%	19.00	0.00	0.00%
15b	17.00	6.00	35.29%	0.00	0.00	0.00%	2.00	1.00	50.00%	19.00	7.00	36.84%
16	17.00	4.00	23.53%	10.00	6.00	60.00%	2.00	1.00	50.00%	29.00	11.00	37.93%
17	0.00	0.00	0.00%	1.00	0.00	0.00%	0.00	0.00	0.00%	1.00	0.00	0.00%
18	17.00	17.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	29.00	29.00	100.00%
19a	0.00	0.00	0.00%	9.00	2.00	22.22%	2.00	2.00	100.00%	11.00	4.00	36.36%
19b	17.00	17.00	100.00%	10.00	9.00	90.00%	2.00	2.00	100.00%	29.00	28.00	96.55%
20	17.00	17.00	100.00%	10.00	8.00	80.00%	2.00	2.00	100.00%	29.00	27.00	93.10%
21	17.00	10.00	58.82%	10.00	8.00	80.00%	2.00	2.00	100.00%	29.00	20.00	68.97%
22	17.00	12.00	70.59%	10.00	8.00	80.00%	2.00	1.00	50.00%	29.00	21.00	72.41%

PM, predictive model; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

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Table A7: Summary of study characteristics for excluded predictive models relating to machine learning studies (n=8), predictive model studies without discrimination (n=3) and prognostic factor studies with discrimination (n=11).

Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
All-cause related HRs								
Desai et al. 2014, USA	PM without D	All-cause	30 day unplanned HRs	Prospective/ Survey data (PedsQL Infant Scales instrument or PedsQL 4.0 Generic Core Scales instrument)	19,139 eligible patients/ NR	1 month - 18 years	1. Oct. 2011 - 31. Dec. 2013	-
Santos et al. 2020, Brazil	PM without D	All-cause	30 day HRs	Prospective/ 1 public hospital, StrongKids data (nutritional screening)	641 patients/ 2%	1 month - 17 years	2014 - 2018	-
Jovanovic et al. 2016, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	66,994 patients/ 17%	general paediatric population	2009-2011	0.783 (Lasso), 0.779 (Tree-Lasso)
Stiglic et al. 2015, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	61,111 discharge records/ 18%	≤ 10 years	2009-2011	0.750-0.771
Stiglic et al. 2014, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	66,994 discharge records/ 17%	≤ 10 years	2009-2011	0.787 (distributed model), 0.789 (elastic net model)
Taylor et al. 2020, USA	ML-PM	All-cause	30 day unplanned HRs	Retrospective/ PHIS database	1,111,323 children/ 4.4%	< 18 years	2016-2017	0.811
Zhou et al. 2021, Australia	ML-PM	All-cause	30 day unplanned HRs	Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records, written discharge documentation	940 patients/ 4.55%	different paediatric age groups	2010 - 2014	Model 1: 0.519 (random forest), 0.5 (elastic net), 0.509 (gradient bossted tree); Model 2: 0.603 (random forest), 0.616 (elastic net), 0.624 (gradient bossted tree); Model 3: 0.642 (random forest), 0.635 (elastic net), 0.654 (gradient bossted tree)
Ehwerhemuepha et al. 2020, USA	ML-PM	All-cause	30 day (unplanned) HRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters/ 12.6% (DC)	< 18 years	2000-2017	M1 (single-center): 0.8226, M2 (single-center): 0.8756, M3 (multi-center): 0.8451

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Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
Wolff et al. 2019, Chile	ML-PM	All-cause	30 day HRs	Retrospective/ administrative data, 1 paediatric hospital	56,558 admissions/ 3.7%	general paediatric population	July 2011 - Oct. 2017	0.47 - 0.65
Surgical conditions related HRs								
Chotai et al. 2017, USA	PF with D	Neurosurgery	30 day unplanned HRs	Retrospective/ administrative data, 1 paediatric hospital	536 children/ 11.9%	< 18 years	Jan. 2012 - March 2015	0.72
Davidson et al. 2021, USA	PF with D	Ureteroscopy	30 day unplanned HRs	Retrospective/ NSQIP-P database	2,510 patients/ 6.5%	≤ 18 years	2015 - 2018	0.73
Garcia et al. 2018, USA	PF with D	Kasai procedure	30 day unplanned HRs	Retrospective/ NSQIP-P database	190 children/ 15.3%	< 1 year	2012 - 2015	0.703
Lee et al. 2020, USA	PF with D	Fusion surgery	30 day HRs	Retrospective/ Nationwide Readmissions Database	30,677 patients/ 2.9%	10 - 18 years	2012 - 2015	0.712
Minhas et al. 2016, USA	PF with D	Spinal Surgeries (Scoliosis)	30 day unplanned HRs	Retrospective/ NSQIP-P database	3,482 children/ 3.4%	≤ 18 years	2012 - 2013	0.760-0.769
Roddy & Diab 2017, USA	PF with D	Spine fusion surgery	30 day HRs	Retrospective/ State Inpatient Database	13,287 patients/ 4.7%	<21 years	2006-2010 (New York, Utah, Nebraska, Florida, North Carolina), 2006-2011 (California)	0.79
Sherrod et al. 2016, USA	PF with D	Neurosurgery	30 day unplanned HRs	Retrospective/ NSQIP-P database	9,799 cases/ 11.2%	< 18 years	2012-2013	0.759
Tahiri et al. 2015, USA	PF with D	Plastic Surgery	30 day HRs	Retrospective/ NSQIP-P database	5,376 patients/ 2.40%	≤ 18 years	2012	0.784
Wheeler et al. 2018, USA	PF with D	Burn diagnosis	30 day unscheduled HRs	Retrospective/ Nationwide Readmissions Database	11,940 patients/ 2.7%	1-17 years	2013-2014	0.72
Vedantam et al. 2018, USA	PF with D	Epilepsy surgery	30 day unplanned HRs	Retrospective/ NSQIP-P database	280 surgeries/ 7.1%	≤ 18 years	2015	0.71

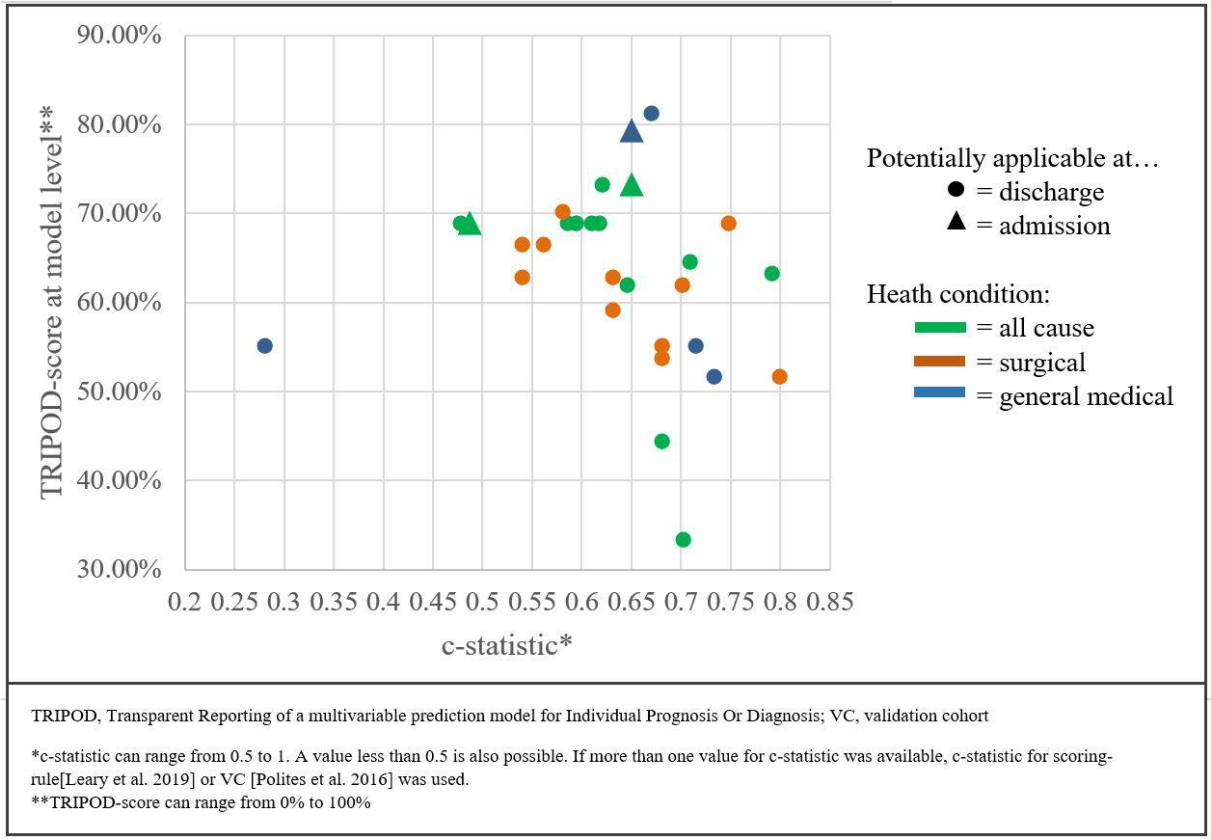
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Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
Janjua et al. 2019, USA	ML-PM without D	Spinal cord tumor surgery	30 day HRs	Retrospective/ Nationwide Readmissions Database	397 patients/ 10.8%	≤ 20 years	2010-2015	-
General medical conditions related HRs								
Sanchez-Luna et al. 2016, Spain	PF with D	Acute bronchiolitis due to respiratory syncytial virus	30 day HRs	Retrospective/ Spanish National Health Service records	63,948 discharges/ 7.5%	< 1 year	2004 - 2012	0.611 (all cases), 0.481 (due to RSV bronchiolitis), 0.612 (due to other causes)

Abbreviations: D, discrimination; DC, derivation cohort; HR, hospital readmission; ML, machine learning; NR, not reported; NSQIP-P, National Surgical Quality Improvement Program Pediatric; PF, prognostic factor study; PHIS, Paediatric Health Information Systems; PM, predictive model

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TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; VC, validation cohort

*c-statistic can range from 0.5 to 1. A value less than 0.5 is also possible. If more than one value for c-statistic was available, c-statistic for scoring-rule[Leary et al. 2019] or VC [Polites et al. 2016] was used.
**TRIPOD-score can range from 0% to 100%

Figure A1: Discriminative ability, application and TRIPOD-adherence of 30-day HR predictive models in paediatrics (n=29)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5-6 + Supplementary File
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5 + Supplementary File
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6 + Table 1-3 + Supplementary File
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Qualitative Synthesis, Page 6-7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Qualitative Synthesis, Page 6-7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Qualitative Synthesis, Page 6-7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Qualitative Synthesis, Page 6-7, 18
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6-7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7-8
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 7-8 + Supplementary File
Study characteristics	17	Cite each included study and present its characteristics.	Page 8-12, Table 1+3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 + Supplementary File
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1 + Supplementary File
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Qualitative Synthesis, Table 1-3, figure 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 18
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 15-17, TRIPOD Assessment
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 17-19

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23b	Discuss any limitations of the evidence included in the review.	Page 18
	23c	Discuss any limitations of the review processes used.	Page 18
	23d	Discuss implications of the results for practice, policy, and future research.	Page 17-19
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 20
Competing interests	26	Declare any competing interests of review authors.	Page 20
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 20 + Supplementary File

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Y
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Y
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Y
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Y
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Y
Synthesis of results	6	Specify the methods used to present and synthesise results.	Y
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Y
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Y
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Y
Interpretation	10	Provide a general interpretation of the results and important implications.	Y
OTHER			
Funding	11	Specify the primary source of funding for the review.	Manuscript
Registration	12	Provide the register name and registration number.	NA

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From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Applicability of predictive models for 30-day unplanned and planned hospital readmission risk in paediatrics: a systematic review

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TITLE

Applicability of predictive models for 30-day unplanned and planned hospital readmission risk in paediatrics: a systematic review

AUTHORS

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Keywords: Hospital readmission, paediatric, prediction, systematic review

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ABSTRACT

Objective: To summarize multivariable predictive models for 30-day hospital readmissions (HRs) in paediatrics, describe their performance and completeness in reporting, and to determine their potential for application in practice.

Design: Systematic review.

Data source: CINAHL, EMBASE and PubMed up to 7 October 2021.

Eligibility criteria: English- or German-language studies aiming to develop or validate a multivariable predictive model for 30-day paediatric HRs related to all-cause, surgical conditions or general medical conditions were included.

Data extraction and synthesis: Study characteristics, risk factors significant for predicting readmissions and information about performance measures (e.g., c-statistic) were extracted. Reporting quality was addressed by the 'Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis' (TRIPOD) adherence form. The study quality was assessed by applying six domains of potential biases. Due to expected heterogeneity among the studies, the data were qualitatively synthesized.

Results: Based on 30 studies, 45 predictive models were identified, which could potentially be used for determining individual 30-day HR risk in paediatrics. 28 of 30 studies explicitly aim to predict 30-day unplanned HRs. The number of study participants ranged from 280 surgeries to 1.4 million encounters. The two most common significant risk factors were comorbidity and (postoperative) length of stay. 23 models showed a c-statistic above 0.7 and are primary applicable at discharge. The median TRIPOD adherence of the models was 59% ($P_{25} - P_{75}$ 55% - 69%), ranging from a minimum of 33% to a maximum of 81%. Overall, the quality of many studies was moderate to low in all six domains.

Conclusions: Predictive models may be useful in identifying paediatric patients at increased risk for readmission. To support the application of predictive models, more attention should be placed on completeness in reporting, particularly for those items that may be relevant for implementation in practice.

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Strengths and limitations of this study

- Independent and standardised methodological approach for study selection, data extraction and risk of bias assessment.
- Comprehensive presentation of predictive models that provides information about applicability, performance and reporting quality at model level, differentiated by 30-day all-cause, surgical conditions and general medical conditions related paediatric hospital readmissions.
- Due to study heterogeneity, the models were only narratively synthesized.

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INTRODUCTION

Hospital readmissions (HRs) are becoming increasingly important as a quality indicator for paediatric inpatient care[1, 2]. HR is often defined as a subsequent, unplanned admission within a period of 30 days after the index hospitalization[3]. According to published evidence, all-cause 30-day unplanned HR rates for the paediatric population range from 3.4% to 18.7%[3-5]. In addition, taking 27 US states into account, it has been estimated that paediatric HRs can cost up to $2 billion annually, with approximately 40% of these occurring HRs being potentially preventable[6].

Identifying the reasons for paediatric HRs is a major challenge, as the health of children is also affected by factors aside of inpatient care[7]. Predictive models can be applied as a tool for the identification of patients with a risk of HR higher than that of the average population and for the implementation of preventive interventions to reduce the risk of HR[8]. Especially in the context of the ongoing COVID-19 pandemic, where children and adolescents are also being hospitalized with a variety of symptoms[9-11], the prevention of unplanned HRs can be beneficial, as it would allow hospital resources to be used in a more target-orientated way.

This systematic review aims to address two research gaps that have been identified. (i) Predictive models with good performance are useful in practice when clinicians and other stakeholders have all the necessary information for their application in clinical practice and critical assessment[12]. However, previous systematic reviews discussed the shortcomings in reporting the quality of prediction models[13-15] and also for paediatric clinical prediction rules[16]. (ii) A previous systematic review has already identified 36 significant risk factors for unplanned HRs in paediatric patients with different health conditions[3]. The largest number of risk factors were identified for surgical procedure related unplanned HRs. Among others, comorbidity was one of the most common risk factors across the 44 included studies[3]. The review[3] extends the findings of an earlier systematic review that focused on 29 paediatric studies targeting predictors for asthma-related unplanned HRs[17]. Both reviews[3, 17] were primarily addressed to predictor finding studies[14], while to date there is no published review of existing 30-day HR predictive models in paediatrics.

The objective of this systematic review was to determine the potential application of multivariable predictive models for individualized risk prediction of 30-day unplanned and planned HR in the paediatric population by evaluating the models´ discriminative ability,

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completeness in reporting and the risk factors shown to be significant for prediction of 30-day HR.

METHOD

The 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) statement was adhered to for conducting and reporting of this systematic review[18]. Screening of the titles and abstracts, data extraction, quality assessment and analyses (e.g., completeness in reporting) were performed by two independent reviewers, while disagreements were discussed with a third author. A protocol was pre-specified and is available from the corresponding author.

Data source and search strategy

CINAHL, EMBASE and PubMed were used for an electronic database search to identify studies published up to 7 October 2021. The key search terms include the outcome variables used for the model (i.e., readmission/rehospitalization), elements of the study design (i.e., prediction/c-statistic) and the population of interest (i.e., paediatrics/children)(see supplementary material for full search strategies–tables A1-A3). The reference lists of the included studies and of comparable systematic reviews[3, 17] were examined for further potential studies.

Inclusion criteria

Studies addressing multivariable predictive models for children and adolescents (except newborns/ preterm newborns, as the index admission is the birth hospitalization) were included if they were published in English or German and available as full texts in peer-reviewed original journal articles. Studies aiming to develop a new model or to validate an existing model were included (i) if the model was potentially appropriate for the individual prediction of 30-day unplanned and/ or planned HR from acute healthcare service after discharge or after index procedure in paediatrics; and (ii) if the model provided at least one discrimination measure (e.g., c-statistic). Discriminative ability is a key factor in evaluating predictive models[19] and a necessary information to make well-founded conclusions about the performance of a model.

In addition, (iii) predictive model studies that developed a new model (i.e., development design) or determined the incremental or added value of a predictor for an existing model (i.e., incremental value design) had to be based on a regression modelling approach. This

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inclusion criterion enables us to identify significant risk factors and to apply the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) adherence form, which was originally developed for regression models[20]. This implies that predictive models using machine-learning techniques (e.g., least absolute selection and shrinkage operator[21] or random forest[22]) are excluded and coded as non-regression models. Due to specific requirements of mental diseases, studies were only included (iv) if they addressed non-mental health conditions related 30-day HRs[3].

Data extraction

Just as in previous systematic reviews[3, 23], studies were categorized by health conditions in all tables. Basic study characteristics were extracted according to criteria in table 1 and 2. To assess the applicability of the predictive models, significant risk factors (i.e., odds ratio (OR) or hazard ratio >1 with a p-value of < 0.05) were assigned to established and revised variable categories[3] in table 3. If all variables of a predictive model are available for a patient at the time of index admission (e.g., previous health service usage before index admission), the model is applicable at admission. Applicability of predictive models at discharge is given if all variables are available at this point for a patient (e.g., length of stay, operative time).

Reporting quality and performance

Predictive models can just be used in practice when clinicians and other stakeholders have access to all information required for their application in clinical practice[12]. The newly developed 'Critical Appraisal of Models that Predict Readmission (CAMPR)' contains 15 expert recommendations for predictive model development relating to HRs. However, CAMPR should not be used as a reporting standard so far and relates to aspects that are out of the scope of this systematic review (e.g., considering different time frames for HRs)[24]. Due to the importance of high-quality information about predictive models we decided to assess the completeness of reporting by using the TRIPOD adherence form and scoring rules[12, 25, 26]. The TRIPOD adherence form consists of 22 main criteria based on the TRIPOD statement[20], resulting in 37 items that are applicable to varying degrees to the development, validation and incremental value studies [25]. We decided to apply the TRIPOD adherence form at predictive model level. Therefore, publications that report the development and validation of the same predictive model, for example, are assessed separately. According to previous research, our analysis concentrates on items that could be reported in the main text or supplements[27].

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TRIPOD adherence at model level was merged with the performance results (i.e., discrimination and calibration measures) and the applicability assignment in table 4. The discrimination of a predictive model is often evaluated by the c-statistic or area under the receiver operating characteristic curve. The c-statistic can take a value between 0.5 and 1. A value of 0.5 indicates that the model is not superior to a random prediction of outcomes, while values between 0.7 and 0.8 indicate that the model is appropriate. A value of 0.8 or greater indicates a strong discrimination of a model.[28]

Quality assessment

Following previous systematic reviews[3, 23, 29], the refined version of the quality in prognosis studies (QUIPS) tool with its prompting items[30] was used to appraise the studies critically with regard to the included predictive models based on six domains. Each domain was rated with a 'high', 'moderate' or 'low' risk of bias.

The six domains are[30]: 'study participation', 'study attrition', 'prognostic factor measurement', 'outcome measurement', 'study confounding' and 'statistical analysis and reporting'.

Data synthesis

Because a quantitative evaluation in the form of a meta-analysis was not possible due to the high heterogeneity among the studies, the studies were qualitatively synthesized, i.e., the results for performance, completeness in reporting and significant risk factors were presented in a narrative and in a simplified quantitative form.

Patient and public involvement

Due to the study design we did not involve patients or the public.

RESULTS

Search result

From the electronic database search, 10,076 records were obtained. After duplicates had been removed, the titles and abstracts were screened for 7,694 records. Based on the predefined inclusion criteria, 7,586 records were excluded. Adding one additional recommended article[31] this results in 109 records being included in the full-text assessment. Among the 82 excluded records, twelve studies analyse 30-day HRs together with another outcome (i.e. emergency department return visits (n=5)[32-36], mortality (n=3)[37-39], other complications

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(n=4)[40-43]), three were predictive model studies with no discrimination metrics[44-46], five were non-regression-based predictive model studies for 30-day HRs in paediatrics[21, 47-50], and 59 were prognostic factor studies for 30-day paediatric HRs. Based on the full-text assessments (n=27) and the hand search of reference lists (n=3, [51-53]), 30 studies were included in the systematic review, with six of them[53-58] already presented in a previous systematic review[3] with a different focus. The results of the review process regarding the database search is provided in figure A1 in the supplementary material (see table A4 in the supplementary material for a summary of study characteristics for selected excluded models).

Quality assessment

Overall, the quality of many studies was moderate to low for several domains. For instance, the study quality had to be reduced due to a lack of sufficient information (e.g. in the domain 'study participants' or 'study attrition'), while all studies were rated as 'low' for the domain 'study confounding'(see table A5 in the supplementary material for the results of the risk of bias assessment).

Study characteristics

All studies were based on retrospective data, with nine studies based on tertiary or paediatric hospital data[22, 53, 59-65], and 21 studies based on centralized databases[31, 51, 52, 54-58, 66-78]. Five of 30 studies additionally included census data in the analysis[59, 63, 64, 66, 70]. The period of data collection ranged from one year[31, 51, 52, 58, 61, 66, 68] to seven years[67, 69]. The majority of studies included patients up to an age of <18 or ≤18 years. Only five studies considered patients up to 21 years of age[57, 62, 71] or younger than one year[74, 76]. The sample size was specified with different units in the individual studies (e.g., encounters, admissions) and varies between 280 surgeries[31] and 1.4 million encounters[67].

The 30 included studies resulted in 45 predictive models for 30-day HRs in paediatrics. Twelve of 30 studies developed or validated more than one predictive model for HRs[22, 56, 57, 63-70, 75], which were in part excluded due to non-agreement with the inclusion criteria. The models included were grouped into three health conditions: (i) all-cause HR (n=13)[22, 59, 61-63, 66, 67]; (ii) surgical conditions related HR (n=25)[31, 51, 52, 54-58, 65, 68-70, 73-75, 77, 78]; and (iii) general medical conditions related HR (n=7)[53, 60, 64, 71, 72, 76]. 28 of 30 studies explicitly aim to predict 30-day unplanned HRs with respect to the grouped health condition[22, 31, 51-67, 69, 71-78]. 30-day unplanned HR rates vary from 1.5%[51] to 41.2%[71].

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Among the 45 predictive models included, 33 (73%) used a development design[22, 31, 51-59, 61-65, 69-78], ten (22%) an external validation design[60, 63, 67, 68, 70] and two (~5%) an incremental value design[64, 66]. Nine out of ten external validated models were based on existing predictive models that had been previously used in the adult population[63, 67], for different outcomes[60, 68] or both[68]. Furthermore, five of the 30 studies included did not state the primary aim to develop, validate externally or assess the incremental value of the respective 30-day HR predictive model[63, 65-67, 69].

Of the predictive models with a development or incremental value design, 18 employed an apparent validation[31, 51-53, 56-59, 65, 66, 73-78] and 17 employed an internal validation[22, 54, 55, 61-64, 69-72]. The most commonly applied internal validation method was cross-validation (n=8)[22, 61, 62], followed by split-sample (n=6)[54, 63, 69-72] and bootstrapping (n=3)[55, 64]. In order to analyse the data, either a logistic regression[22, 31, 51-53, 55-59, 61-66, 69-78] or a Cox proportional hazard regression[54] was used. Most models present their results by ORs with a 95% confidence interval. With a p-value of < 0.05, the results are considered as statistically significant[3]. A summary of characteristics of all included studies is provided in table 1 and 2.

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Table 1: Summary of study characteristics for all-cause 30-day hospital readmissions predictive models

Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
All-cause related HRs									
Brittan et al. 2018[62], USA	Composite Score	All-cause	30-day unplanned HRs	Retrospective/ 1 children's hospital	29,542 patients	0-21 years	2014 - 2015	4.0%	Development study/ Internal: Cross
Sills et al. 2017[66], USA	PACR + SDH	All-cause	30-day unplanned HRs	Retrospective / PHIS database, US Census's American Community Survey data, 47 hospitals	458,686 index discharges	<18 years	2014	6.1%	Incremental value study/ Apparent
Ehwerhemuepha et al. 2018[63], USA	Unnamed	All-cause	30-day unplanned HRs	Retrospective / US Census's American Community Survey data, 1 tertiary paediatric hospital	38,143 inpatient clinical encounters (DC: 19,072, VC: 19,071)	Between 28-days and 17 years	July 2013 to June 2017	10.4%	Development study/ Internal: Random-Split-Sample
	LACE (validation)				VC: 19,071 inpatient clinical encounters			NR	External validation study
Bradshaw et al. 2020[61], USA	HARRPS-tool	All-cause	30-day unplanned HRs	Retrospective / 1 paediatric hospital	5,306 patients	<18 years	May 2017- June 2018	25.3%	Development study/ Internal: Cross
Zhou et al. 2020[59], Australia	Unnamed	All-cause	30-day unplanned HRs	Retrospective/ Australian Census data, 1 tertiary paediatric hospital	73,132 patients	Age limit for admission: 15 years, special permissions by hospital executives possible	2010 - 2014	4.6%	Development study/ Apparent
Ehwerhemuepha et al. 2020[67], USA	LACE (validation)	All-cause	30-day unplanned HRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters	< 18 years	2000- 2017	12.6% (DC)	External validation study
Zhou et al. 2021[22], Australia	Model 1: GLM	All-cause	30-day unplanned HRs	Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data	940 patients	Different paediatric age groups[a]	2010-2014	4.55%[b]	Development study/ Internal: Cross
	Model 1: G-S								Development study/ Internal: Cross
	Model 2: GLM			Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records					Development study/ Internal: Cross
	Model 2: G-S								Development study/ Internal: Cross
	Model 3: GLM			Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records, written discharge documentation					Development study/ Internal: Cross
	Model 3: G-S								Development study/ Internal: Cross

Abbreviations: DC, derivation cohort; GLM, logistic regression; G-S, stepwise logistic regression; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; PACR, Pediatric All-Condition Readmission; PASS, PHIS, Paediatric Health Information Systems; SDH, social determinants of health; VC, validation cohort

[a] Mean age (years): 5.2 with HR, 5.3 without HR.

[b] Based on 3330 patients from the initial data set.

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Table 2: Summary of study characteristics for surgical and general medical conditions related 30-day hospital readmissions predictive models

Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
Surgical conditions related HRs									
Vo et al. 2018[55], USA	Unnamed	All surgical specialties without cardiac surgery	30-day unplanned post-surgical HRs relating to noncardiac surgery	Retrospective/ ACS NSQIP-P database	182,589 patients	< 18 Years	2012-2014	4.8%	Development study/ Internal: Bootstrap
Polites et al. 2016[54], USA	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Retrospective/ ACS NSQIP-P database	54,870 patients (DC: 38,397, VC: 16473)	29 days - < 18 years	2012-2014	3.6%	Development study/ Internal: Random-Split-Sample
Smith et al. 2015[70], USA	Developed model	Cardiac surgery	30-day HRs following congenital heart surgery	Retrospective/ PHIS database, US census data, 43 not-for-profit, tertiary care paediatric hospitals	17,871 discharges (DC: 9,104, VC: 8,767)	< 18 years	2011	11.0%	Development study/ Internal: Split-Sample (Non-Random temporal)
	Validated model						2012	NR	External validation study /Split-Sample (Non-Random temporal)
Jiang et al. 2018[68], USA	CCI (validation NRD)	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ Nationwide Readmissions Database	8,006 patients	<18 years	2013	NR	External validation study
	VWI (validation NRD)								External validation study
	Rhee index (validation NRD)								External validation study
	CCI (validation SID)	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ State Inpatient Databases	6,236 patients	<18 years	2007-2010	8%[a]	External validation study
	VWI (validation SID)								External validation study
	Rhee index (validation SID)								External validation study
Delaplain et al. 2020[69], USA	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Retrospective / Cerner Health Facts database, 28 hospitals	82,532 patients (DC: 75%, VC: 25%)	< 18 years	2000-2017	8.8%	Development study/ Internal: Random-Split-Sample[b]
Chotai et al. 2017[65], USA	Unnamed	Neurosurgery	30-day unplanned HRs following index surgery for neurosurgical diagnoses	Retrospective/ 1 paediatric hospital	536 children	< 18 years	Jan. 2012 - March 2015	11.9%	Development study/ Apparent
Davidson et al. 2021[73], USA	Unnamed	Ureteroscopy	30-day unplanned HRs after ureteroscopy	Retrospective/ NSQIP-P database	2,510 patients	≤ 18 years	2015 - 2018	6.5%	Development study/ Apparent
Garcia et al. 2018[74], USA	Unnamed	Kasai procedure	30-day unplanned HRs related to kasai procedure	Retrospective/ NSQIP-P database	190 children	< 1 year	2012 - 2015	15.3%	Development study/ Apparent
Lee et al. 2020[75], USA	Unnamed	Adolescent idiopathic scoliosis surgery	30-day unplanned HRs after adolescent idiopathic scoliosis surgery	Retrospective/ Nationwide Readmissions Database	30,677 patients	10 - 18 years	2012 - 2015	2.9%	Development study/ Apparent
Minhas et al. 2016[56], USA	Idiopathic scoliosis	Spinal Surgeries (Scoliosis)	30-day unplanned HRs	Retrospective/ NSQIP-P database	3,482 children	≤ 18 years	2012 - 2013	3.4%	Development study/ Apparent

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Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
	Progressive infantile scoliosis Scoliosis due to other conditions								
Roddy & Diab 2017[57], USA	Unnamed	Spine fusion	30-day unplanned HRs	Retrospective/ State Inpatient Database	13,287 patients	<21 years	2006-2010 (New York, Utah, Nebraska, Florida, North Carolina), 2006-2011 (California)	4.7%	Development study/ Apparent
Sherrod et al. 2016[77], USA	Unnamed	Neurosurgery	30-day unplanned HRs after neurosurgery	Retrospective/ NSQIP-P database	9,799 cases	< 18 years	2012-2013	11.2%	Development study/ Apparent
Tahiri et al. 2015[58], USA	Unnamed	Plastic Surgery	30-day unplanned HRs following paediatric plastic surgery procedures	Retrospective/ NSQIP database	5,376 patients	≤ 18 years	2012	2.4%	Development study/ Apparent
Wheeler et al. 2018[78], USA	Unnamed	Burn diagnosis	30-day unplanned HRs	Retrospective/ Nationwide Readmissions Database	11,940 patients	1-17 years	Jan. - Nov. 2013, Jan. - Nov. 2014	2.7%	Development study/ Apparent
Vedantam et al. 2018[31], USA	Unnamed	Epilepsy surgery	30-day unplanned HRs after epilepsy surgery	Retrospective/ NSQIP-P database	280 surgeries	≤ 18 years	2015	7.1%	Development study/ Apparent
Basques et al. 2015[51], USA	Unnamed	Posterior spinal fusion	30-day unplanned HRs after posterior spinal fusion	Retrospective/ NSQIP-P database	733 patients	11-18 years	2012	1.5%	Development study/ Apparent
Martin et al. 2015[52], USA	Unnamed	Spinal deformity surgery	30-day unplanned HRs after spinal deformity surgery	Retrospective/ NSQIP-P database	1,890 patients	<18 years	2012	3.96%	Development study/ Apparent
General medical conditions related HRs									
Learly et al. 2019[64], USA	Prediction at admission Prediction at discharge	Complex chronic conditions	30-day unplanned HRs	Retrospective / US Census Bureau data, 1 academic medical centre	2,296 index admissions	6 months to 18 years	Oct. 2010 - July 2016	8.2%	Development study/ Internal: Bootstrap Incremental value study/ Internal: Bootstrap
Ryan et al. 2021[60], USA	PASS (validation)	Asthma	30-day unplanned HRs	Retrospective/ 1 university-affiliated, tertiary paediatric referral centre	328 patients	5-18 years	May 2015-Oct. 2017	3.0%	External validation study
O´Conell et al. 2021[72], USA	Unnamed	Nervous system condition	30-day unplanned HRs	Retrospective / Cerner Health Facts database, 18 hospitals	105,834 index admissions (DC: 80%. VC: 20%)	<18 years	2000-2017	12.0%	Development study/ Internal: Random-Split-Sample
Hoenk et al. 2021[71], USA	Unnamed	Oncology	30-day unplanned HRs	Retrospective / Cerner Health Facts database, 16 hospitals	10,418 patients (DC: 7,814, VC: 2,604)	<21 years	2000-2017	41.2%	Development study/ Internal: Random-Split-Sample
Sanchez-Luna et al. 2016[76], Spain	Unnamed	Acute bronchiolitis due to respiratory syncytial virus	30-day unplanned HRs	Retrospective/ Spanish National Health Service records	63,948 discharges	< 1 year	2004 - 2012	7.5%	Development study/ Apparent
Sacks et al. 2017[53], USA	Unnamed	Cardiac conditions	30-day unplanned HRs	Retrospective/ 1 academic children´s hospital	1,993 hospitalizations	0-12.9 years	2012-2014	20.5%	Development study/ Apparent

Abbreviations: ACS NSQIP-P, American College of Surgeons' (ACS) National Surgical Quality Improvement Program Pediatric; CCI, Charlson Comorbidity Index; DC, derivation cohort; HR, hospital readmission; NR, not reported; NRD, nationwide readmission database; PASS, Pediatric Asthma Severity Score; PHIS, Paediatric Health Information Systems; SID, state inpatient database;VWI, Van Walraven Index; VC, validation cohort

[a] Own approximation resulting from Table 3 in the study based on readmission per risk score[68].

[b] Assumption for validation method: Odds ratios for 30-day unplanned HRs are displayed in a Table that is part of the DC from the 7-day unplanned HRs predictive model[69].

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Applicability and significant risk factors in predictive models

Based on the 29 predictive models with a development or incremental value design, 26 significant risk factors associated with 30-day HRs were identified (see table 3). The most common risk factors were comorbidity (n=18), (postoperative) length of stay (n=11) and illness severity (n=10). The significant risk factors were inconsistently defined across predictive models, allowing a direct comparison only to a limited extent. ORs for comorbidity ranged from 1.01[72] to 10.08[56] across predictive models. A length of stay of >=15 days (OR=2.39)[59] and a postoperative length of stay of >4 days (Hazard Ratio=3.12)[54] were considered to be a major risk factor. For illness severity 'ICU stay'(OR=3.302)[65] was one of the most pronounced risk factors. The risk factor with the highest OR value was 'any inpatient complication'(OR=180.44)[51]. For all-cause HRs, HRs related to surgical conditions and HRs related to general medical conditions 14, 20 and twelve significant risk factors were found, respectively.

Most predictive models are potentially applicable at discharge (n=41), while four predictive models can be used at index admission[22, 61, 64, 76], based on the significant and examined variables (see table A6 in the supplementary material for an overview of variables and table 4 for an application description).

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Table 3: Significant risk factors for 30-day HR predictive models with a development or incremental value design

Health condition group	All-cause (n=5[a])					Surgical conditions related (n=18)																		General medical conditions related (n=6)					
Reference	[62]	[66]	[63]	[61]	[59]	[55]	[54]	[70]	[69]	[65]	[73]	[74]	[75]	[56][b]	[56][c]	[56][d]	[57]	[77]	[58]	[78]	[31]	[51]	[52]	[64][e]	[64][f]	[72]	[71]	[76]	[53]
Location of residence[g]		x			x															x									
Health insurance				x	x			x									x												
Type of index hospital						x					x						x		x							x			
Living environment				x																									
Characteristics of primary care provider	x																												
Age at admission/operation					x																							x	x
Sex											x						x												
Race/Ethnicity		x						x		x								x								x			
Health service usage prior to index admission[h]			x	x					x									x						x	x	x	x		
Prematurity												x																x	
Comorbidity		x	x	x	x	x	x						x	x	x		x	x					x	x	x	x	x	x	x	x
Illness severity[i]			x	x			x	x		x								x		x				x	x	x				
LOS/Postop LOS			x		x		x	x	x				x				x			x					x	x	x			
Principal diagnoses			x					x	x								x									x	x			
Principal procedures							x							x		x	x	x		x	x		x			x				
Inpatient complications						x	x					x	x				x	x	x			x								
(Specific) medication at index admission									x																	x	x			
Devices at index admission								x																						
Length of operation							x											x	x										
Wound contamination before operation							x												x										
The ASA class						x										x			x				x						
Discharge on Friday or Weekend					x																								
Discharge disposition																	x	x							x				
Discharge with increased medication/further treatment	x																												
Admission on Friday					x																								
Surgical location											x																		

Abbreviations: ASA class, American Society of Anesthesiologists class; LOS, length of stay; PICU, paediatric intensive care unit; postop LOS, postoperative length of stay
x=risk factor (Odds ratio/Hazard ratio >1)
[a] The six predictive models of Zhou et al. 2021 [22] are not included in this analysis due to missing information about odds ratios. See table A6 in the supplementary material for a list of included variables.
[b] Model for idiopathic scoliosis.
[c] Model for progressive infantile scoliosis.
[d] Model for scoliosis due to other conditions.
[e] Admission model.
[f] Discharge model.
[g] Social determinants of health are included (e.g., median household income).
[h] Risk factor category includes, for example, the number of previous emergency department visits or hospitalizations.
[i] The risk factor category also captures the urgency of the index admission. The risk factor category includes, for example, PICU or emergency department admission.

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Completeness in reporting and discriminative ability at model level

Information about TRIPOD adherence and performance at model level are provided in table 4. The median TRIPOD adherence of the models was 59% ($P_{25} - P_{75}$ 55% - 69%, average: 60%), ranging from 33%[67] to 81%[64]. Externally validated predictive models and development models have a comparable reporting quality with a median adherence of 61% ($P_{25} - P_{75}$ 54% - 66%, average: 58%) for externally validated models and with 59% ($P_{25} - P_{75}$ 55% - 69%, average: 60%) for development models, respectively. Two models with poor adherence in reporting were based on an external validation design, and the validation of these models was not the primary aim of the study[63, 67].

Table 4: Performance, application and TRIPOD adherence of 30-day HR predictive models in paediatrics (n=45)

Reference	Model name	Performance		TRIPOD score	Potentially applicable…
		Discrimination (c-statistic)	Calibration		
All-cause related HRs					
Brittan et al. 2018[62]	Composite Score	0.62		73.33%	at discharge
Sills et al. 2017[66]	PACR + SDH	0.708		64.71%	at discharge
Ehwerhemuepha et al. 2018[63]	Unnamed	VC: 0.79		63.33%	at discharge
	LACE (validation)	0.68		44.44%	at discharge
Bradshaw et al. 2020[61]	HARRPS-tool	Score: 0.65		73.33%	at admission
Zhou et al. 2020[59]	Unnamed	0.645		62.07%	at discharge
Ehwerhemuepha et al. 2020[67]	LACE (validation)	0.7014		33.33%	at discharge
Zhou et al. 2021[22]	Model 1: GLM	0.487		68.97%	at admission
	Model 1: G-S	0.477		68.97%	at discharge
	Model 2: GLM	0.585		68.97%	at discharge
	Model 2: G-S	0.593		68.97%	at discharge
	Model 3: GLM	0.609		68.97%	at discharge
	Model 3: G-S	0.617		68.97%	at discharge
Surgical conditions related HRs					
Vo et al. 2018[55]	Unnamed	0.747	slope: 1; intercept: 0.002	68.97%	at discharge
Polites et al. 2016[54]	Unnamed	DC: 0.71; VC: 0.701	(DC): p = 0.95, O/E-Ratio= 1.03; (VC): p = 0.36, O/E-Ratio= 1.07	62.07%	at discharge
Smith et al. 2015[70]	Developed model	DC: 0.68	H&L (DC): 0.47	55.17%	at discharge
	Validated model	VC: 0.68		53.85%	at discharge
Jiang et al. 2018[68]	CCI (validation NRD)	0.63	calculation not possible	62.96%	at discharge
	VWI (validation NRD)	0.54	calculation not possible	66.67%	at discharge

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		Performance			
	Rhee index (validation NRD)	0.58	calculation not possible	70.37%	at discharge
	CCI (validation SID)	0.63	calculation not possible	59.26%	at discharge
	VWI (validation SID)	0.54	calculation not possible	62.96%	at discharge
	Rhee index (validation SID)	0.56	calculation not possible	66.67%	at discharge
Delaplain et al. 2020[69]	30-day readmission model	VC: 0.799		51.72%	at discharge
Chotai et al. 2017[65]	Unnamed	0.72		42.86%	at discharge
Davidson et al. 2021[73]	Unnamed	0.73	H&L χ^2: 7.5 (p=0.4474)	58.62%	at discharge
Garcia et al. 2018[74]	Unnamed	0.703		51.72%	at discharge
Lee et al. 2020[75]	Unnamed	0.712	H&L: 0.0974	58.62%	at discharge
Minhas et al. 2016[56]	Idiopathic scoliosis	0.760-0.769		55.17%	at discharge[a]
	Progressive infantile scoliosis			55.17%	at discharge[a]
	Scoliosis due to other conditions			55.17%	at discharge[a]
Roddy & Diab 2017[57]	Unnamed	0.75	H&L (p-value): 0.46	55.17%	at discharge
Sherrod et al. 2016[77]	Unnamed	0.759		55.17%	at discharge
Tahiri et al. 2015[58]	Unnamed	0.784		55.17%	at discharge
Wheeler et al. 2018[78]	Unnamed	0.72		55.17%	at discharge
Vedantam et al. 2018[31]	Unnamed	0.71	H&L (p-value): 0.94	41.38%	at discharge
Basques et al. 2015[51]	Unnamed	0.87	H&L: value not reported[b]	68.97%	at discharge
Martin et al. 2015[52]	Unnamed	0.77		62.07%	at discharge
General medical conditions related HRs					
Learly et al. 2019[64]	Prediction at admission	0.65; Score: 0.65	Calibration plot	79.31%	at admission
	Prediction at discharge	0.67; Score: 0.67	Calibration plot	81.25%	at discharge
Ryan et al. 2019[60]	PASS (validation)	0.28		55.17%	at discharge
O´Conell et al. 2021[72]	Unnamed	VC: 0.733		51.72%	at discharge
Hoenk et al. 2021[71]	Unnamed	VC: 0.714		55.17%	at discharge
Sanchez-Luna et al. 2016[76]	Unnamed	0.611		56.67%	at admission
Sacks et al. 2017[53]	Unnamed	0.75		58.62%	at discharge

Abbreviations: CCI, Charlson Comorbidity Index; DC, derivation cohort; GLM, logistic regression; G-S, stepwise logistic regression; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; NRD, nationwide readmission database; PACR, Pediatric All-Condition Readmission; PASS, Pediatric Asthma Severity Score; SDH, social determinants of health; SID, state inpatient database; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; VWI, Van Walraven Index; VC, validation cohort

[a]Assumption for applicability based on variables included in the univariable analysis.

[b]H&L shows 'no evidence of a lack of fit'([51] p. 290).

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Including all 37 items, the overall median adherence per TRIPOD item across models was 60% ($P_{25} - P_{75}$ 24% - 91%, average: 56%), ranging from 0% to 100% (see table A7 in the supplementary material for a detailed description by model type). The overall adherence per TRIPOD item is illustrated in figure 1.

[…Placement of figure 1…]

Figure 1: Overall adherence per TRIPOD item across all included predictive models (n=45).

24% of the models reported the title (item 1) completely, while 29%[60-64, 66, 68] of the models mentioned the predictive model type in this context. 2% of the models had a completed abstract (item 2). The detailed predictors definition (item 7a) was fulfilled for more models (91%), in contrast to outcome definition (item 6a) (reported in 62%). The handling of predictors in the analysis (item 10a) showed incomplete reporting in 83% of the models. In addition, the handling (item 9, reported in 29%) and reporting of missing values (part of item 13b, reported in 27%) was not addressed in many models. Just 9% of the models displayed complete reporting of the model-building procedure (item 10b), as the majority of the models (91%) did not address the testing of interaction terms[22, 31, 51-59, 62-66, 69, 70, 72-75, 77, 78]. The description (item 10d) and reporting of performance measures (item 16) was incomplete in 58% and 76% of the models. Just 36% of the models address results of calibration measures (cf. table 4). No model presented the full predictive model (item 15a) by providing an example of an intercept. An explanation for using the prediction model (item 15b, e.g., by a simplified scoring rule) was presented in 20% of the models. One model provided detailed information about a simplified scoring rule (item 15b) in supplementary material[64].

The discriminative ability (c-statistic) of the models ranged from 0.28[60] to 0.87[51]. 22 out of 45 predictive models had a c-statistic <0.7. The linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.287 (P=.06). Models with good discriminative ability (c-statistic >0.7)[31, 51-58, 63, 65-67, 69, 71-75, 77, 78] are primary applicable at discharge and have a TRIPOD score ranging from 41%[31] to 69%[55]. The two models with the highest reporting quality (79%, 81%) are applicable for predicting 30-day unplanned HRs of children with complex chronic conditions. The c-statistic of these models was 0.65[64] and 0.67[64], respectively (see figure A2 in the supplementary material for a illustration of the models´ performance and TRIPOD adherence).

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DISCUSSION

Based on 30 studies, this systematic review identifies 45 predictive models that could potentially be used for determining individual 30-day HR risk in paediatrics. According to the models, the three most common significant risk factors in predictive models were comorbidity, (postoperative) length of stay and illness severity. 23 validated predictive models have a c-statistic of >0.7. The median TRIPOD adherence of the predictive models included was 59% ($P_{25} - P_{75}$ 55% - 69%), ranging from 33% to 81%, which is similar to that of other systematic reviews[12, 27].

Practical clinical and policy implications

In general, reporting quality and discriminative ability can provide crucial information about the strengths and weaknesses of a predictive model for implementation in practice (see figure A2 in the supplementary material for a combined illustration). However, the results from this systematic review revealed considerable differences in the c-statistics (0.28[60] to 0.87[51]) and in the TRIPOD scores (33%[67] to 81%[64]) at model level. When considering the available information about reporting quality and discriminative ability in relation to each other, it should be noted that the linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.287 (P=.06). Therefore, an independent evaluation of both aspects for the selection of an appropriate predictive model is recommended.

Clinicians and decision-makers should use predictive models with good discriminative ability (i.e. c-statistic above 0.7) and sufficient data availability. Especially predictive models that are based on census data[59, 63, 64, 66, 70] or manual data entry (e.g. written discharge documentation[22]) may be more difficult to implement than models relying on centralized databases[31, 51, 52, 54-58, 67-69, 71-78]. The TRIPOD score at predictive model level (see table 4) can be used as a first indicator if the predictive model can be assessed and implemented with the given information.

Limitations

This systematic review has certain limitations. (i) The studies included needed be to published in English or German with full-text access. (ii) Summarizing the results of the included studies quantitatively was not possible due to the heterogeneity of the predictive models (resulting from differences in sample sizes, the examined variables or variations in the periods of data collection). (iii) With regard to the inclusion criteria, we defined the outcome as 30-

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day HRs for predictive models that can address both planned and unplanned HRs. The resulting heterogeneity was accepted because in an earlier systematic review a wider definition of HRs had proved useful[8]. (iv) The sample size of the included studies were reported in different units(e.g., encounters, discharges), impeding the comparisons of HR rates. (v) Our assignment of the predictive models that are potentially applicable at discharge assumes that the required variables are available at the time point. If clinicians and other stakeholders decide to use a predictive model, it should be checked beforehand whether complete data collection is possible at the desired time. (vi) In addition to the identified medical risk factors (e.g., comorbidity) and several country-specific risk factors (e.g., location of residence) that result in paediatric readmissions, health-policy initiatives may also affect the readmission rates in paediatric clinical practice[79]. However, due to a lack of data, these aspects could not be captured by this review.

Future research

This systematic review did not identify predictive models for individualized risk prediction of potentially preventable unplanned HRs in paediatrics, emphasizing past discussions to expand the research field further[3].

Current external validation studies were conducted in the USA and primarily examined the applicability of existing predictive models with other outcomes or population backgrounds to paediatric 30-day HRs[60, 63, 67, 68]. Therefore, external validation studies are needed for those models that are explicitly developed to predict 30-day HRs in paediatrics. Because the number of predictive models related to medical conditions-related HRs was small (n=7)[53, 60, 64, 71, 72, 76], with four out of seven models demonstrating a c-statistic below 0.7[60, 64, 76], there is a need for high quality models in this area. Further more, predictive models for 30-day HRs in the area of urological and cardiac surgery[68, 70] should be revised in order to achieve better discriminative ability (i.e. c-statistic>0.7).

Non-regression-based techniques (e.g., machine-learning) are an increasing field in order to predict 30-day HRs in paediatrics, most of which show good discriminative ability[21, 22, 45, 47-50, 67] (see table A4 in the supplementary material). Future systematic reviews should summarize and critically assess existing non-regression-based HR predictive models in paediatrics, for instance by applying the TRIPOD-ML statement that is going to be published[80].

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Existing studies discuss the benefit of shorter time intervals in order to identify preventable readmissions more accurately[6, 81]; one study concluded that a 30-day unplanned HR metric was more precise (c-statistic=0.799) for paediatric trauma patients than a 7-day unplanned HR metric(c-statistic=0.737)[69]. To our knowledge, there is one predictive model for 365-day[7], three for 90-day[57, 65, 75] and one for 7-day[69] unplanned HRs in paediatrics with good discriminative ability (c-statistic>0.7). Future studies should address the evaluation of paediatric HR predictive models with different time intervals.

CONCLUSION

This systematic review revealed an increase in the development of predictive models for 30-day HRs in paediatrics in recent years. To support the implementation of the predictive models in the long term, it is essential to validate existing models in order to test their applicability in different settings. To increase accessibility for use, more attention should be given on completeness in reporting, particularly for items that may be relevant for the implementation of paediatric 30-day HR predictive models in practice.

List of Figures

Figure 1: Overall adherence per TRIPOD item across all included predictive models (n=45)

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Contributors:

IN conceptualized and designed the systematic review, participated in the literature search, study selection, quality assessment, data extraction, data analyses and drafted the initial manuscript.

NK contributed to the literature search, study selection, quality assessment, data extraction and critically reviewed the manuscript.

SS contributed to the data analysis and critically reviewed the manuscript.

JD contributed to the study selection, data extraction, data analysis and critically reviewed the manuscript.

DM conceptualized and designed the systematic review, participated in the study selection, quality assessment, data extraction, data analyses and critically reviewed the manuscript.

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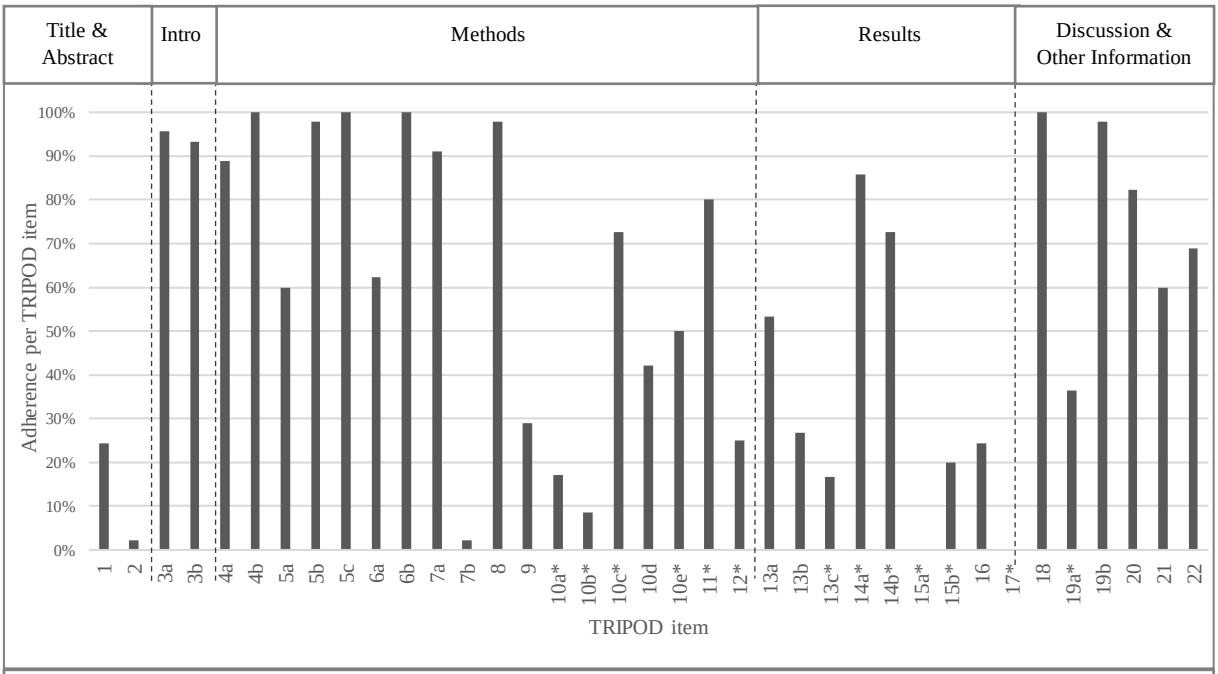
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Percentages relate to the number of models for which an item was applicable (in this case, the respective item should have been reported). *Indication of derivation from the total number of models for which a TRIPOD item was applicable (N=# of models for which the TRIPOD item is applicable): 10a (N=35), 10b (N=35), 10c (N=11), 10e (N=2), 11 (N=5), 12 (N=12), 13c (N=12), 14a (N=35), 14b (N=33), 15a (N=35), 15b (N=35), 17 (N=1), 19a (N=11).

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Supplemental Material

Table A1: Search strategy for PubMed

#	PubMed (Search date: 01st January 2021; Updated Search date: 07th October 2021)	Results from initial search	Results from updated search
1	"Patient Readmission"[Mesh]	18,336	20,199
2	rehospitali*[Title/Abstract]	8,017	8,645
3	readmission*[Title/Abstract]	31,790	35,343
4	(hospital[Title/Abstract]) AND readmission*[Title/Abstract]	21,144	23,398
5	(unplanned[Title/Abstract]) AND readmission*[Title/Abstract]	2,360	2,677
6	(patient[Title/Abstract]) AND readmi*[Title/Abstract]	16,395	18,106
7	readmit*[Title/Abstract]	8,289	8,926
8	re-admission*[Title/Abstract]	2,205	2,416
9	(repeat*[Title/Abstract]) AND hospital*[Title/Abstract]	29,556	31,364
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	75,738	81,968
11	predictive factor*[Title/Abstract]	28,124	30,391
12	predict*[Title/Abstract]	1,646,322	1,765,658
13	"Predictive Value of Tests"[Mesh]	207,290	215,023
14	"ROC Curve"[Mesh]	60,417	65,077
15	model*[Title/Abstract]	3,025,019	3,240,665
16	c-statistic*[Title/Abstract]	6,938	7,765
17	ROC*[Title/Abstract]	55,019	62,608
18	"Sensitivity and Specificity"[Mesh]	595,955	619,284
19	Sensitivity [Title/Abstract]	827,733	873,715
20	Specificity [Title/Abstract]	478,451	502,627
21	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20	5,225,692	5,555,402
22	"Child"[Mesh]	1,936,577	2,010,900
23	"Infant"[Mesh]	1,152,685	1,189,047
24	"Adolescent"[Mesh]	2,058,561	2,216,512
25	(((((adolescen*[Title/Abstract]) OR teen*[Title/Abstract]) OR youth[Title/Abstract]) OR juvenile*[Title/Abstract]) OR young person*[Title/Abstract]) OR young people*[Title/Abstract]	464,723	492,107
26	"Pediatrics"[Mesh]	59,136	61,129
27	((((((((((child*[Title/Abstract]) OR (infant*[Title/Abstract])) OR (toddler*[Title/Abstract])) OR (bab*[Title/Abstract])) OR (newborn*[Title/Abstract])) OR (neonat*[Title/Abstract])) OR (school age*[Title/Abstract])) OR (preschool)) OR (paediatric*[Title/Abstract])) OR (pediatric*[Title/Abstract])) OR (kid*[Title/Abstract])) OR (boy*[Title/Abstract])) OR (girl*[Title/Abstract])	2,246,452	2,343,927
28	22 OR 23 OR 24 OR 25 OR 26 OR 27	4,365,971	4,517,928
29	10 AND 21 AND 28	3,674	3,993

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Table A2: Search strategy for CINAHL

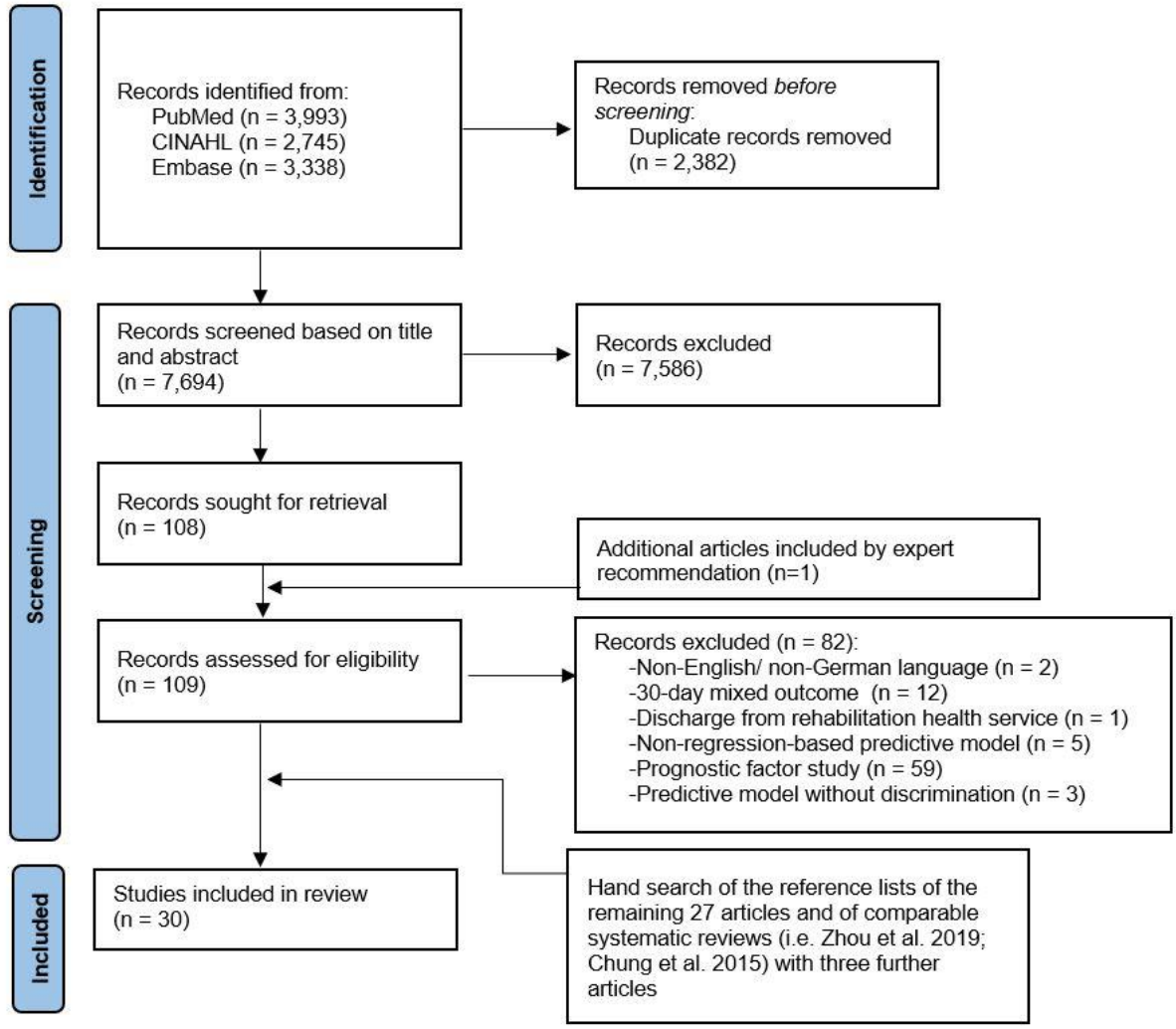
	CINAHL (Initial Search date: 01st January 2021; Updated Search date: 07th October 2021)	Results from initial search	Results from updated Search
1	(MH "Readmission")	14,256	15,177
2	rehospitali*	2,624	2,779
3	readmission*	21,640	23,205
4	hospital AND readmission*	13,210	14,167
5	unplanned AND readmission*	1,227	1,363
6	patient AND readmi*	17,823	19,163
7	readmit*	2,588	2,791
8	re-admission*	954	1,029
9	repeat* AND hospital*	12,404	12,997
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	36,195	38,498
11	predictive factor*	10,824	11,733
12	predict*	394,485	423,169
13	(MH "Predictive Value of Tests")	52,176	54,193
14	(MH "ROC Curve")	27,863	29,859
15	model*	626,177	669,950
16	c-statistic*	2,593	2,822
17	ROC*	45,218	74,438
18	(MH "Sensitivity and Specificity")	85,260	87,853
19	sensitiv*	236,551	247,288
20	specific*	481,504	511,797
21	(MH "Predictive Validity")	5,587	5,746
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	1,337,979	1,442,565
23	(MH "Child")	467,217	486,202
24	(MH "Infant+")	260,190	268,903
25	(MH "Adolescence+")	535,922	557,353
26	adolescen* OR teen* OR youth OR juvenile* OR young person* OR young people*	598,199	624,571
27	(MH "Pediatrics+")	21,316	21,917
28	child* OR infant* OR toddler* OR bab* OR newborn* OR neonat* OR school age* OR preschool OR paediatric* OR pediatric* OR kid* OR boy* OR girl*	1,150,913	1,264,003
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	1,413,615	1,514,718
30	10 AND 22 AND 29	2,459	2745

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Table A3: Search strategy for Embase

	Embase (Initial Search date: 01st January 2021*; Updated Search dates: 7th June 2021, 07th October 2021)	Results from updated search 7th June 2021	Results from updated search 07th October 2021
1	'hospital readmission'/exp	73,736	76,850
2	rehospitali*:ab,ti	10,922	15,528
3	readmission*:ab,ti	57,908	64,311
4	(hospital NEAR/10 readmission*):ab,ti	22,791	25,140
5	(unplanned NEAR/10 readmission*):ab,ti	3,446	3,749
6	(patient NEAR/5 readmi*):ab,ti	5,568	6,357
7	readmit*:ab,ti	16,262	18,777
8	're admission*':ab,ti	5,802	5,982
9	(repeat* NEAR/5 hospital*):ab,ti	5,341	5,466
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	100,580	107,753
11	'predictive factor*':ab,ti	47,721	49,181
12	predict*:ab,ti	2,301,090	2,367,382
13	'predictive value'/exp	191,705	199,896
14	'roc curve'/exp OR 'receiver operating characteristic'/exp	149,025	157,486
15	'model'/exp	3,201,228	3,267,314
16	'c statistic*':ab,ti	12,892	13,511
17	roc*:ab,ti	196,628	207,024
18	'sensitivity and specificity'/exp	391,861	405,843
19	'sensitivity':ab,ti	1,109,211	1,135,520
20	'specificity':ab,ti	636,321	650,496
21	'predictive validity'/exp	8,691	8,839
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	6,371,499	6,524,117
23	'child'/exp	3,031,530	3,082,756
24	'infant'/exp	1,180,824	1,197,926
25	'adolescent'/exp	1,741,216	1,770,557
26	adolescen*:ab,ti OR teen*:ab,ti OR youth:ab,ti OR 'juvenile*':ab,ti OR 'young person*':ab,ti OR 'young people':ab,ti	598,758	613,037
27	'pediatrics'/exp	123,610	125,641
28	'child*':ti,ab OR 'infant*':ti,ab OR 'toddler*':ti,ab OR bab*:ti,ab OR 'newborn*':ti,ab OR neonat*:ti,ab OR 'school age*':ti,ab OR 'preschool':ti,ab OR 'pediatric*':ti,ab OR kid*:ti,ab OR 'boy*':ti,ab OR 'girl*'	3,642,407	3,715,357
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	5,498,512	5,594,597
30	#10 AND #22 AND #29	2,902**	3,338
* Final result from initial search: 2,845 ** Search term: #10 AND #22 AND #29 AND [1-1-1966]/sd NOT [2-1-2021]/sd			

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Abbreviations: PRISMA, preferred reporting items for systematic reviews and meta-analyses.

Figure A1: Flowchart for the search and study selection process (PRISMA)

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Table A4: Summary of study characteristics for excluded predictive models relating to machine learning studies (n=8) and predictive model studies without discrimination (n=3).

Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
All-cause related HRs								
Desai et al. 2014, USA	PM without D	All-cause	30 day unplanned HRs	Prospective/ Survey data (PedsQL Infant Scales instrument or PedsQL 4.0 Generic Core Scales instrument)	19,139 eligible patients/ NR	1 month - 18 years	1. Oct. 2011 - 31. Dec. 2013	-
Santos et al. 2020, Brazil	PM without D	All-cause	30 day HRs	Prospective/ 1 public hospital, StrongKids data (nutritional screening)	641 patients/ 2%	1 month - 17 years	2014 - 2018	-
Jovanovic et al. 2016, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	66,994 patients/ 17%	general paediatric population	2009-2011	0.783 (Lasso), 0.779 (Tree-Lasso)
Stiglic et al. 2015, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	61,111 discharge records/ 18%	≤ 10 years	2009-2011	0.750-0.771
Stiglic et al. 2014, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	66,994 discharge records/ 17%	≤ 10 years	2009-2011	0.787 (distributed model), 0.789 (elastic net model)
Taylor et al. 2020, USA	ML-PM	All-cause	30 day unplanned HRs	Retrospective/ PHIS database	1,111,323 children/ 4.4%	< 18 years	2016-2017	0.811
Zhou et al. 2021, Australia	ML-PM	All-cause	30 day unplanned HRs	Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records, written discharge documentation	940 patients/ 4.55%	different paediatric age groups	2010 - 2014	Model 1: 0.519 (random forest), 0.5 (elastic net), 0.509 (gradient bossted tree); Model 2: 0.603 (random forest), 0.616 (elastic net), 0.624 (gradient bossted tree); Model 3: 0.642 (random forest), 0.635 (elastic net), 0.654 (gradient bossted tree)
Ehwerhemuepha et al. 2020, USA	ML-PM	All-cause	30 day (unplanned) HRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters/ 12.6% (DC)	< 18 years	2000-2017	M1 (single-center): 0.8226, M2 (single-center): 0.8756, M3 (multi-center): 0.8451

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Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
Wolff et al. 2019, Chile	ML-PM	All-cause	30 day HRs	Retrospective/ administrative data, 1 paediatric hospital	56,558 admissions/ 3.7%	general paediatric population	July 2011 - Oct. 2017	0.47 - 0.65
Surgical conditions related HRs								
Janjua et al. 2019, USA	ML-PM without D	Spinal cord tumor surgery	30 day HRs	Retrospective/ Nationwide Readmissions Database	397 patients/ 10.8%	≤ 20 years	2010-2015	-

Abbreviations: D, discrimination; DC, derivation cohort; HR, hospital readmission; ML, machine learning; NR, not reported; PHIS, Paediatric Health Information Systems; PM, predictive model

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Table A5: Risk of bias assessment at study level

Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Vo et al. 2018	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - No detailed description of statistical analysis for the simplified scoring rule; Unadjusted associations for "Complications" are missing	**1**
Polites et al. 2016	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Moderate - 38.5% missing values for the ICD-9-diagnosis variable used for determining the reasons for readmission	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Brittan et al. 2018	- Moderate - Potential limitation of generalizability as just one paediatric hospital was examined; Only basic information about "age"	- Moderate - About 75% of patients were missing homecare values in the original data (the study assumption is that this patients did not received home health)	- Moderate - Assumption that 75% of patients did not receive home health; Possible omission or error of clinical documentation	- Moderate - no clear description for the determination of unplanned hospital readmissions	- Low - just PEs of the composite score are considered	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Sills et al. 2017	- Low -	- Moderate - No flow diagram	- Moderate - Area-level SDH data were not created for research and are a biased proxy for individual-level SDH measures; Handling of continuous predictors relating to ZIP-Codes (i.e. linear or non-linear) not clear described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2018	- Moderate - Potential limitation of generalizability as just one hospital was examined; No detailed information about the source of ZIP-related variables; No information about characteristics of patients in the validation cohort (Random-Split-Sample)	- Low -	- Moderate - No detailed description for the handling of continuous predictors and method for categorized predictors	- Low - The outcome was determined automatically and not validated by manual chart review	- Low -	- Moderate - No detailed definition for LACE variables; Missing information for the validation cohort	**2**
Smith et al. 2015	- Moderate - No information about characteristics of patients in the external validation set; No detailed information about the source of ZIP-related variables	- Moderate - No flow diagram; Missing information for validation cohort	- Moderate - Age categories are not clearly reported	- Low -	- Low -	- Moderate - Missing information for the validation cohort (non-random split sample)	**2**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Jiang et al. 2018	- Low -	- Moderate - Outcome events for NRD-data set are not described; No flow diagram	- Moderate - Patient distribution is presented by risk scores	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions; It is unclear, if 30-day inpatient readmissions in the NRD are also calculated from the date of the index procedure	- Low - Due to external validation of existing risk scores confounders are not relevant	- Moderate - Method of CCS is not clear; Outcome event rates and summary information of predictors are not clearly reported	**6**
Learly et al. 2019	- Moderate - Potential limitation of generalizability as just one hospital examined; No detailed information about the source of ZIP-related variables	- Low -	- Moderate - Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Low -	**2**
Bradshaw et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; Due to deidentification of patient data demographic characteristics are not captured	- Low -	- Low -	- Moderate - "The multiple categories that comprise the HARRPS-tool have different weight values associated that help prevent planned readmissions from skewing the results."(p. 53)	- Low -	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Delaplain et al. 2020	- Moderate - Focus on baseline characteristics of readmission in 7 days; No detailed characteristics for the validation cohort (random-split sample)	- Moderate - Information about the patient flow are provided at hospital level; Hospitals were excluded, if data were not collected for all variables related to the study; Missing information for validation cohort (random-split sample)	- Low - Predictors are presented by focusing on 7-day readmissions	- Moderate - Clear definition for 7-day unplanned readmissions with possible transfer to 30-day unplanned readmissions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**
Zhou et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; One-day procedures are also included (discharge within 24h)	- Low -	- Moderate - Census data might be a biased proxy for individual level measures; Method to choose the cut-points was not clearly described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2020	- Moderate - Basic information just for the training data set	- Moderate - Missing information for test data set	- High - No summary information for all predictors included in the validated model; No clear definition of predictors	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions	- Low - Due to external validation of LACE index confounders are not relevant	- High - Applied method for LACE is not clear (i.e. possible updating); Missing information for training data set	**1**
Hoenk et al. 2021	- Low -	- Moderate - Information about the patient flow are based on hospital level; Hospitals were excluded if data were not collected for all variables related to the study and if there were not at least 500 encounters for a	- Moderate - Method to choose the cut-points was not clearly presented	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
		neoplastic condition; Missing information for test data set					
Zhou et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Moderate – 470 instead of 550 patient pairs because of the burden associated with extracting data from medical records	- Low -	- Low -	- Low-	- Moderate - No detailed final model presentation (i.e. missing presentation of odds ratios in multivariable analysis)	**6**
Ryan et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Low -	- Low -	- Moderate - It is unclear, how unplanned readmissions are determined	- Low - Due to external validation of PASS confounders are not relevant	- Moderate - Summary information of predictors are not clearly reported	**1**
O´Conell et al. 2021	- Low -	- Moderate - Information about the patient flow are based on hospital level; Hospitals were excluded if data were not collected for all variables related to the study and if there were not at least 1000 encounters for a nervous system condition; No clear information on differentiation between training and test data set	- Moderate - No summary information for all predictors included in the multivariable model; Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Moderate - No clear reporting regarding the differentiation between training and test data set (i.e. summary information for predictors)	**1**
Chotai et al. 2017	- Moderate - Potential limitation of generalizability as just one hospital examined; Focus on baseline characteristics of readmission in 90 days	- Low -	- Low -	- Moderate - Detailed description for the determination of unplanned readmission is missing	- Low -	- Moderate - Apparent validation with small sample size	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Davidson et al. 2021	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Garcia et al. 2018	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - Apparent validation with small sample size	**1**
Lee et al. 2020	- Moderate – Possible selection bias due to exclusion of patients which were admitted within the last quarter of the year	- Moderate - No detailed information about the patient flow	- Low -	- Low -	- Low -	- Low -	**1**
Minhas et al. 2016	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - No complete reporting of the model development (e.g. uni- and multivariable analysis)	**3**
Roddy & Diab 2017	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Moderate - No detailed information about the patient flow	- Low -	- Moderate - Detailed description for the determination of unplanned readmission is missing	- Low -	- Low -	**1**
Sherrod et al. 2016	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Tahiri et al 2015	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Wheeler et al. 2018	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Moderate – Patients which were admitted in December were excluded.	- Low -	- Low -	- Low -	- Moderate – Incomplete reporting of statistical analysis (e.g. p-values)	**1**
Vedantam et al. 2018	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - Apparent validation with small sample size	**1**
Basques et al. 2015	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate – Incomplete reporting of statistical analysis (e.g. univariable analysis, control variables)	**1**
Martin et al. 2015	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Sanchez-Luna et al. 2016	- Low -	- Moderate - No information about the "gender" characteristics of the study sample	- Low -	- Moderate - Detailed description for the determination of "non-staged" readmission is missing	- Low -	- Moderate – No detailed presentation of results	**1**
Sacks et al. 2015	- Moderate - Potential limitation of generalizability as just one hospital examined	- Moderate – study just included patients, which are living close to the hospital	- Low -	- Moderate - Detailed description for the determination of unplanned readmission is missing	- Low -	- Low -	**1**

* For the risk of bias assessment at study-level, only the predictive models that were also included in the systematic review were considered.

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Table A6: Examined and significant variables for all 45 included predictive models

Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
All-cause related readmissions					
Brittan et al. 2018	Composite Score	All-cause	30-day unplanned HRs	Age; Lenght of stay; Non-English speaking caregiver; Discharge medications; Enteral feeding (eg, gastrostomy tube feeding); Respiratory (eg, home ventilator); IV infusion (eg, infusion of IV medication); Speech therapy; Physical therapy; Occupational therapy; Skilled nursing home visits; Private duty nursing home visits; CNA nursing assistant home visits; Durable medical equipment (eg, wheel chair); Home Care ≥ 1 order; Composite score = 0; Composite score = 1; Composite score = 2; Composite score = 3	Score based on the variables: Non-English speaking caregiver, Discharge medications, Home Care ≥ 1 order: Composite score (0 vs 1) (OR=1.7; 95% CI 1.5-2); Composite score (0 vs ≥ 2) (OR=4.2; 95% CI 3.6-4.9)
Sills et al. 2017	PACR + SDH	All-cause	30-day unplanned HRs	CCI count; Infectious and parasitic disease; Neoplasms; Endocrine, nutritional, and metabolic diseases and immunity disorders; Diseases of blood and blood-forming organs; Mental disorders; Diseases of the nervous system and sense organs; Diseases of the circulatory system; Diseases of the respiratory system; Diseases of the digestive system; Diseases of the genitourinary system; Diseases of the skin and subcutaneous tissue; Diseases of the musculoskeletal system; Congenital anomalies; Certain conditions originating in the perinatal period; Symptoms, signs, and ill-defined conditions; Injury and poisoning; Factors influencing health status and contact with health services; Age group; Sex; Race; Hispanic ethnicity; Payer; Median household income; Proportion of housing units vacant; Proportion of households with children that are single-parent; Proportion of families below poverty level; Unemployment rate; Proportion of adults with less than a high school diploma or equivalent	CCI count: [2 body systems (OR=1.51; 95% CI 1.44-1.58), 3 body systems (OR=1.57; 95% CI 1.48-1.67), >3 body systems (OR=1.31; 95% CI 1.19-1.44)]; CCIs: Neoplasms (OR=0.46; 95% CI 0.44-0.48), Endocrine, nutritional, and metabolic diseases and immunity disorders (OR=0.8; 95% CI 0.77-0.83), Diseases of blood and blood-forming organs (OR=0.59; 95% CI 0.56-0.61), Mental disorders (OR=0.95; 95% CI 0.92-0.99), Diseases of the nervous system and sense organs (OR=0.78; 95% CI 0.75-0.8), Diseases of the circulatory system (OR=0.73; 95% CI 0.7-0.76), Diseases of the respiratory system (OR=1.08; 95% CI 1.05-1.12), Diseases of the digestive system (OR=0.65; 95% CI 0.63-0.68), Diseases of the genitourinary system (OR=0.7; 95% CI 0.66-0.74), Congenital anomalies (OR=0.88; 95% CI 0.85-0.91), Injury and poisoning (OR=0.63; 95% CI 0.54-0.74), Factors influencing health status and contact with health services (OR=0.51; 95% CI 0.49-0.53); Age group: [1-<5y (OR=0.84; 95% CI 0.81-0.87), 5-<8y (OR=0.73; 95% CI 0.69-0.76), 8-<12 y (OR=0.72; 95% CI 0.68-0.75), 12-<18 y (OR=0.79; 95% CI 0.76-0.83)]; male (OR=0.96; 95% CI 0.93-0.98); Race: [Asian (OR=0.91; 95% CI 0.84-0.99), Other (OR=0.92; 95% CI 0.88-0.96)], Hispanic (OR=1.06; 95% CI 1.02-1.1); Payer: [Private (OR=0.85; 95% CI 0.83-0.88)]; Median household income (per $10 000 increase) (OR=1.02; 95% CI 1.02-1.02); Vacancy rate (per 10% increase) (OR=0.96; 95% CI 0.94-0.99); Percent single-parent (per 10% increase) (OR=1.03; 95% CI 1.01-1.04)
Ehwerhemuepha et al. 2018	Unnamed	All-cause	30-day unplanned HRs	Sex; Race and/or ethnicity; Length of stay, d; Age, y; Median income by zip code, per $10 000; Percent vacant houses by zip code; Percent single parents by zip code; Low income primary medical insurance; Planned admission; ED visits within last 6 mo; Admitted through ED; Previous inpatient visits within last 6 mo; History of 30 d readmission within last 6mo; Ambulatory resource use within last 6 mo; Charlson's comorbidities; Complex chronic conditions; Was in ICU; pRI: Admitting; pRI: Discharge; pRI: Minimum; pRI: Maximum; pRI: Average; pRI: Decreasing slope across entire stay; pRI: Decreasing slope during last 24 h of hospitalization; pRI: Minimum occurred first 24 h; pRI: Maximum occurred last 24 h; Infectious and/or parasitic; Neoplasms (excluding encounters for chemotherapy); Blood and/or immune; Endocrine, nutritional, and/or metabolic; Mental and/or neurodevelopment; Nervous, eye, ear, and/or mastoid; Circulatory; Respiratory; Digestive; Skin and/or subcutaneous tissue; Musculoskeletal; Genitourinary; Perinatal period; Congenital malformations; Symptoms, signs, and/or laboratory findings not classified elsewhere; Injury and/or poison; External morbidity causes; Health status and/or services factors	Length of stay, d: [<4 (2, 3) (OR=1.23; NR% CI 1.07–1.42), < 7 (4, 5, 6) (OR=1.42; NR% CI 1.20–1.67), 7 or more (OR=1.80; NR% CI 1.51–2.14)]; Planned admission: [Yes (OR=0.65; NR% CI 0.52–0.82)]; ED and/ or department visits within last 6 mo: [1 (OR=1.27; NR% CI 1.11–1.45), 2 or more (OR=1.30; NR% CI 1.12–1.50)]; Admitted through ED: [Yes (OR=1.15; NR% CI 1.02–1.29)]; Previous inpatient visits within last 6 mo: [1 (OR=1.86; NR% CI 1.59–2.17), 2 or more (OR=2.39; NR% CI 1.94–2.94)]; History of 30 d readmission within last 6mo: [1 (OR=1.27; NR% CI 1.06–1.52), 2 or more (OR=2.75; NR% CI 2.19–3.44)]; Complex chronic conditions: [1 (OR=1.63; NR% CI 1.42–1.87), 2 or more (OR=1.66; NR% CI 1.43–1.92)]; Discharge pRI (10 pt increment) (OR=0.86; NR% CI 0.81–0.90); Maximum pRI occurred last 24 h of hospitalization (OR=0.85; NR% CI 0.75–0.95); Neoplasms (excluding encounters for chemotherapy): [Yes (OR=2.17; NR% CI 1.85–2.55)]; Blood and/or immune: [Yes (OR=1.30; NR% CI 1.14–1.48)]; Nervous, eye, ear, and/or mastoid: [Yes (OR=1.21; NR% CI 1.08–1.37)]; Circulatory: [Yes (OR=1.33; NR% CI 1.14–1.56)]; Respiratory: [Yes (OR=0.75; NR% CI 0.66–0.85)]; External morbidity causes: [Yes (OR=0.64; NR% CI 0.45–0.89)]; Health status and/or services factors: [Yes (OR=1.20; NR% CI 1.06–1.35)]
	LACE (validation)			Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Bradshaw et al. 2020	HARRPS-tool	All-cause	30-day unplanned HRs	Other Diagnosis; Anemia/Neutropenia; Appendectomy; Asthma; Bronchiolitis; Gastroenteritis; Pneumonia; Seizure; Sickle Cell Crisis; Upper Respiratory Tract Infection; Ventricular Shunt; At-risk admission diagnosis present; Chronic condition indicator; Readmitted within 30 days (history); Inpatient Admit in last 6 months; Acuity of admission: No admission acuity identified; Acuity of admission: ICN/PICU Admission; Acuity of admission: Significant psychosocial concern; Acuity of admission: Medical transport from outside facility; Acuity of admission: Other; Insurance type: Self-Pay; Insurance type: Medicaid; Insurance type: Commercial/Private; Insurance type: Medicaid & Self-Pay; Caregiver language: English; Caregiver language: Spanish; Caregiver language: Other; Medical equipment/Supplies count at home; Home nursing: Private Duty Nursing; Home nursing: Skilled Nursing; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy; Home therapy: Home Physical Therapy; Home therapy: Home Speech Therapy; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy	The final scoring system also includes risk factors, which were not significant in the multivariable analysis (i.e. At-risk admission diagnosis present; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy): Chronic condition indicator: [1 (OR=1.43; 95% CI 1.13-1.81), 2 (OR=1.75; 95% CI 1.27-2.40), 3 (OR=1.41; 95% CI 1.12-1.78)]; Previous 30-day readmission (OR=1.70; 95% CI 1.38-2.09); Inpatient admission in last 6 months (OR=2.21; 95% CI 1.76 2.54); ICN or PICU admit (OR=1.43; 95% CI 1.17-1.75); Self-Pay or Medicaid (OR=1.21; 95% CI 1.05 1.39); Medical equipment/Supplies count at home: [2 (OR=1.43; 95% CI 1.09-1.88), 3 (OR=2.68; 95% CI 1.80 4.01), 4 (OR=2.21; 95% CI 1.53 3.19)]
Zhou et al. 2020	Unnamed	All-cause	30-day unplanned HRs	Age; Gender; Admission status; Funding source as inpatients; Source of referral transport; State/Territory of residence; Care type provided; Type of health insurance; Index of Relative Social-Economic Advantage and Disadvantage (IRSAD); Interpreter service; ICU stay at index admission; Had GA at index admission; LOS at index admission; Day of index admission date (weekdays or weekend); Day of index admission date; Day of discharge from index admission; Day of discharge from index admission (weekdays or weekend); Number of Co-diagnoses	Age: [13-15y (OR=1.30; 95% CI 1.14–1.48) >=16y (OR=1.46; 95% CI 1.07–1.98)]; Private health insurance (OR=1.16; 95% CI 1.00–1.34); Aeromedical service (OR=0.47; 95% CI 0.31–0.71); IRSAD (%): 91-100 (OR=1.20; 95% CI 1.02–1.41) - based on residential postcode; GA at index admission (yes) (OR=0.67; 95% CI 0.64–0.76); LOS: [LOS 2-7 days (OR=1.42; 95% CI 1.30–1.55), LOS 8-14 days (OR=2.35; 95% CI 1.97–2.82), LOS >= 15 days (OR=2.39; 95% CI 1.88–2.98)]; Admission on Friday (OR=1.21; 95% CI 1.05–1.39); Day of discharge from index admission: [Friday (OR=1.26; 95% CI 1.10–1.44), Saturday (OR=1.34; 95% CI 1.15–1.57), Sunday (OR=1.24; 95% CI 1.05–1.47)]; Number of Co-diagnoses:[1 (OR=1.28; 95% CI 1.16–1.41), 2 (OR=1.73; 95% CI 1.53–1.95), 3 (OR=2.10; 95% CI 1.80–2.46), >=4 (OR=2.41; 95% CI 2.08–2.80)]
Ehwerhemu epha et al. 2020	LACE (validation)	All-cause	30-day unplanned HRs	Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study
Zhou et al. 2021	Model 1: GLM	All-cause	30-day unplanned HRs	Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis	Variables included in the final model: Day of admission date; Day of admission (weekday/weekend and public holiday)
	Model 1: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis	Variables included in the final model: Distance from residential address to hospital; Day of discharge date; Day of admission date, Day of admission (weekday/weekend and public holiday)
	Model 2: GLM			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the	Variables included in the final model: Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				index admission: number of outpatient clinic attendances	
	Model 2: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the index admission: number of outpatient clinic attendances	Variables included in the final model: Day of discharge date; Day of admission date, Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Language spoken other than English (interpreter service required); Known allergies
	Model 3: GLM			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the index admission: number of outpatient clinic attendances; Completion of Nursing Admission and Discharge Planning Form (Admission section); Completion of Nursing Admission and Discharge Planning Form (Discharge Planning section); Operation sheet or the last entry progress note made by doctors; Clinical pathway or the last entry progress note made by nurses; Last entry progress note made by allied healthcare providers; Written evidence of discharge information given by doctors; Written evidence of discharge information given by nurses; Written evidence of discharge medications information by doctors; Written evidence of discharge medications information by nurses; Written evidence of follow-up information given by doctors; Written evidence of follow-up information given by nurses; Consistency of written discharge documentation among healthcare providers; Delay in issuing discharge summary (date of discharge summary being issued – date of discharge)	Variables included in the final model: Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Completeness of Nursing Admission and Discharge Planning Form, Discharge Planning section (incompleteness)
	Model 3: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage	Variables included in the final model: Had general anaesthetic at index admission; Source of referral transport (ambulance); Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Significant social history; Language spoken other than English (interpreter service required); Known allergies; Completeness of Nursing Admission and Discharge Planning Form, Discharge Planning section (incompleteness); Completeness of Nursing Admission and Discharge Planning Form, Admission section (incompleteness); Progress note or Clinical Pathway documentation at discharge by nurses (not recorded); Follow-up information documented by doctors (not recorded)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				of hospital services 12 months prior to the index admission: number of outpatient clinic attendances; Completion of Nursing Admission and Discharge Planning Form (Admission section); Completion of Nursing Admission and Discharge Planning Form (Discharge Planning section); Operation sheet or the last entry progress note made by doctors; Clinical pathway or the last entry progress note made by nurses; Last entry progress note made by allied healthcare providers; Written evidence of discharge information given by doctors; Written evidence of discharge information given by nurses; Written evidence of discharge medications information by doctors; Written evidence of discharge medications information by nurses; Written evidence of follow-up information given by doctors; Written evidence of follow-up information given by nurses; Consistency ofwritten discharge documentation among healthcare providers; Delay in issuing discharge summary (date of discharge summary being issued – date of discharge)	
Surgical conditions related readmissions					
Vo et al. 2018	Unnamed	All surgical specialities without cardiac surgery	30-day unplanned post-surgical HRs relating to noncardiac surgery	Age; BMI; ASA class; Gender; Race; Surgical specialty; Admission status; Urgency of procedure; Prematurity; Congenital Heart Disease; Bleeding complication; Wound complication; Pulmonary complication; Renal complication; Neurologic complication, Cardiac complication	ASA>=3 (OR=1.9; 95% CI 1.8-2.0); Inpatient vs outpatient (OR=3.5; 95% CI 3.3-3.7); >1 Postoperative complication (OR=3.14; 95% CI 2.92-3.34); Presence of CHD (OR=1.66; 95% CI 1.31-2.11)
Polites et al. 2016	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Age; Sex; Race; Procedure group; Procedure type; Weight percentage; Premature birth; Congenital malformation; Any comorbidity; Diabetes mellitus (Individual comorbidity); Respiratory (Individual comorbidity); GI/Hepatobiliary (Individual comorbidity); Major cardiac (Individual comorbidity); Acute renal failure (Individual comorbidity); Neurologic (Individual comorbidity); Immunosuppression (Individual comorbidity); Nutrition (Individual comorbidity); Hematologic (Individual comorbidity); Active malignancy (Individual comorbidity); Preoperative SIRS, sepsis, septic shock; Preoperative transfusion or hematocrit <32; Case status; ASA class; Operative time, minutes; Wound class; Preoperative length of stay, days; Postoperative complication; Superficial Incisional SSI; Deep SSI; Pulmonary complications; Sepsis/CL associated bloodstream infection; Postoperative length of stay, days	Procedure group: [Head and Neck (HR=2.40; 95% CI 1.48-3.89), Hepatobiliary (HR=1.69; 95% CI 1.17-2.44), Small and Large intestine (HR=1.59; 95% CI 1.33-1.90), Thoracic (HR=0.69; 95% CI 0.52-0.91)]; Comorbidity: [Preoperative acute renal failure (HR=2.47; 95% CI 1.31-4.66), Neurologic comorbidity (HR=1.3; 95% CI 1.05-1.62)]; SIRS/Sepsis/Septic Shock within 48 h prior to index admission (HR=1.2; 95% CI 1.02-1.41); Operative time, minutes: [60–140 min (HR=1.21; 95% CI 1.06-1.39), >140min (HR=1.51; 95% CI 1.26-1.81)]; Wound class: [Contaminated (HR=1.29; 95% CI 1.05-1.60), Dirty/Infected (HR=1.92; 95% CI 1.53-2.40)]; Any complication (HR=1.34; 95% CI 1.09-1.65); Postoperative length of stay, days: [2-4 days (HR=2.17; 95% CI 1.84-2.57), >4 days (HR=3.12; 95% CI 2.60-3.74)]
Smith et al. 2015	Developed model	Cardiac surgery	30-day HRs following congenital heart surgery	[Readmission characteristics are not mentioned, as they were not a part of the modelling process] Male gender; Neonatal age group; Hospital volume, Ethnicity; Mean regional income ($USD); Payor; ICU length of stay (days); ICU LOS ≥7 days; Postoperative LOS (days); Postoperative LOS ≥14 days; ECMO support during hospitalization; RACHS-1 score; GERD diagnosis; Genetic syndrome; Vocal cord paralysis; Thrombosis; Prematurity; Renal insufficiency; Serious Infection; Mediastinitis; Sepsis; Necrotizing enterocolitis; Stroke; Tachyarrhythmia; Complete AV block; Gastrostomy tube placement; Tracheostomy; Diaphragm plication; Weekend discharge	Hispanic (OR=1.24; 95% CI 1.05–1.41); Government payor (OR=1.22; 95% CI 1.06–1.41); ICU LOS ≥7 days (OR=1.86; 95% CI 1.56–2.21); Postoperative LOS ≥14 days (OR=1.59; 95% CI 1.32–1.91); ECMO support during hospitalization (OR=1.74; 95% CI 1.16–2.60); RACHS-6 score (OR=2.60; 95% CI 1.83–3.69); Gastroesophageal reflux (OR=1.49; 95% CI 1.25–1.78); Genetic syndrome (OR=1.31; 95% CI 1.1–1.57); Thrombosis (OR=1.46; 95% CI 1.05–2.03)
	Validated model			Neonatal age group; Ethnicity; Payor; ICU LOS ≥7 days; Postoperative LOS ≥14 days; ECMO support during hospitalization; RACHS-1 score; GERD diagnosis; Genetic syndrome; Vocal cord paralysis; Thrombosis; Prematurity; Renal insufficiency; Serious Infection; Necrotizing enterocolitis; Stroke; Tachyarrhythmia; Complete AV block; Gastrostomy tube placement, Diaphragm plication; Weekend discharge	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Jiang et al. 2018	Charlson Comorbidity Index (validation NRD)	Urological surgery	30-day HRs after paediatric urological procedures	Myocardial infarction; Congestive heart failure; Peripheral vascular disease; Cerebrovascular disease; Dementia; Chronic pulmonary disease; Rheumatic disease; Peptic ulcer disease; Mild liver disease; Diabetes without chronic complication; Diabetes with chronic complication; Hemiplegia or paraplegia; Renal disease; Any malignancy, including lymphoma and leukemia, except malignant neoplasm of skin; Moderate or severe liver disease; Metastatic solid tumor; AIDS/HIV	External validation study
	Van Walraven Index (validation NRD)			Congestive heart failure; Cardiac arrhythmias; Valvular disease; Pulmonary circulation disorders; Peripheral vascular disorders; Hypertension; Paralysis; Neurodegenerative disorders; Chronic pulmonary disease; Diabetes, uncomplicated; Diabetes, complicated; Hypothyroidism; Renal failure; Liver disease; Peptic ulcer disease, no bleeding; AIDS/HIV; Lymphoma; Metastatic cancer; Solid tumour without metastasis; Rheumatoid arthritis/collagen; Coagulopathy; Obesity; Weight loss; Fluid and electrolyte disorders; Blood loss anemia; Deficiency anemia; Alcohol abuse; Drug abuse; Psychosis; Depression	External validation study
	Rhee index (validation NRD)			Age ,24 mo (vs >= 24 mo); Female (vs male); Peri-/endo-/myocarditis, cardiomyopathy, or tamponade; Acute myocardial infarction; Coronary atherosclerosis/other ischemic heart disease; Pulmonary vascular disease (eg, PE, pulmonary HTN); Ventricular tachycardia or other cardiac dysrhythmias; Cardiac arrest or ventricular fibrillation or flutter; Congestive heart failure; Aortic, peripheral, visceral artery aneurysms/ dissection; Aortic or peripheral arterial embolism or thrombosis; Cardiac or circulatory congenital anomalies; Pneumonia; Influenza; Chronic obstructive pulmonary disease/ bronchiectasis; Asthma; Aspiration pneumonitis; Respiratory failure, insufficiency, arrest; Cystic fibrosis; Respiratory distress syndrome; Acute renal failure; Chronic renal failure; Urinary tract infections or unspecified cystitis; Thyroid disorders or other endocrine disorders; Diabetes mellitus or complications; Septicemia (except in labor); Fungal infection; HIV infection; Hepatitis; Meningitis, encephalitis, or other CNS infection; Coma, stupor, or brain damage; Acute cerebrovascular disease; Gastroduodenal ulcer, gastritis, or duodenitis; Intestinal obstruction without hernia; Peritoneal or intestinal abscess, peritonitis; Biliary tract disease; Liver disease (eg, cirrhosis, increased LFTs); Pancreatic disorders (not diabetes); Gastrointestinal hemorrhage; Leukemia; Lymphomas or reticuloendothelial neoplasms; CNS or miscellaneous intracranial or intraspinal neoplasms; Primary malignant tumor of adrenal or paraganglia; Renal tumors; Hepatic tumors; Primary malignant bone or articular cartilage tumors; Soft tissue sarcomas; Immunity disorders (except AIDS); Coagulation or hemorrhagic disorders; Disease of white blood cells (eg, lymphocytosis, lymphopenia); Chronic ulcer of skin; Spondylosis, intervertebral disc disorders; Systemic lupus erythematosus or connective tissue disorder; Crushing injury or internal injury; Shock; Drowning/submersion; Firearm; Motor vehicle traffic; Pedal cyclist, not MVT; Suffocation; Suicide or intentional self-inflicted injury; Fracture of long bone or spine; Intracranial injury; Poisoning by nonmedicinal substances; Skull or face fractures; Open wounds of head, neck, or trunk; Superficial soft tissue injury: contusion; Short gestation, low birth wt, or fetal growth retardation; Hypoxia, asphyxia, or aspiration during birth; Birth trauma; Perinatal jaundice	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
	Charlson Comorbidity Index (validation SID)			Myocardial infarction; Congestive heart failure; Peripheral vascular disease; Cerebrovascular disease; Dementia; Chronic pulmonary disease; Rheumatic disease; Peptic ulcer disease; Mild liver disease; Diabetes without chronic complication; Diabetes with chronic complication; Hemiplegia or paraplegia; Renal disease; Any malignancy, including lymphoma and leukemia, except malignant neoplasm of skin; Moderate or severe liver disease; Metastatic solid tumor; AIDS/HIV	External validation study
	Van Walraven Index (validation SID)			Congestive heart failure; Cardiac arrhythmias; Valvular disease; Pulmonary circulation disorders; Peripheral vascular disorders; Hypertension; Paralysis; Neurodegenerative disorders; Chronic pulmonary disease; Diabetes, uncomplicated; Diabetes, complicated; Hypothyroidism; Renal failure; Liver disease; Peptic ulcer disease, no bleeding; AIDS/HIV; Lymphoma; Metastatic cancer; Solid tumour without metastasis; Rheumatoid arthritis/collagen; Coagulopathy; Obesity; Weight loss; Fluid and electrolyte disorders; Blood loss anemia; Deficiency anemia; Alcohol abuse; Drug abuse; Psychosis; Depression	External validation study
	Rhee index (validation SID)			Age ,24 mo (vs >= 24 mo); Female (vs male); Peri-/endo-/myocarditis, cardiomyopathy, or tamponade; Acute myocardial infarction; Coronary atherosclerosis/other ischemic heart disease; Pulmonary vascular disease (eg, PE, pulmonary HTN); Ventricular tachycardia or other cardiac dysrhythmias; Cardiac arrest or ventricular fibrillation or flutter; Congestive heart failure; Aortic, peripheral, visceral artery aneurysms/ dissection; Aortic or peripheral arterial embolism or thrombosis; Cardiac or circulatory congenital anomalies; Pneumonia; Influenza; Chronic obstructive pulmonary disease/ bronchiectasis; Asthma; Aspiration pneumonitis; Respiratory failure, insufficiency, arrest; Cystic fibrosis; Respiratory distress syndrome; Acute renal failure; Chronic renal failure; Urinary tract infections or unspecified cystitis; Thyroid disorders or other endocrine disorders; Diabetes mellitus or complications; Septicemia (except in labor); Fungal infection; HIV infection; Hepatitis; Meningitis, encephalitis, or other CNS infection; Coma, stupor, or brain damage; Acute cerebrovascular disease; Gastroduodenal ulcer, gastritis, or duodenitis; Intestinal obstruction without hernia; Peritoneal or intestinal abscess, peritonitis; Biliary tract disease; Liver disease (eg, cirrhosis, increased LFTs); Pancreatic disorders (not diabetes); Gastrointestinal hemorrhage; Leukemia; Lymphomas or reticuloendothelial neoplasms; CNS or miscellaneous intracranial or intraspinal neoplasms; Primary malignant tumor of adrenal or paraganglia; Renal tumors; Hepatic tumors; Primary malignant bone or articular cartilage tumors; Soft tissue sarcomas; Immunity disorders (except AIDS); Coagulation or hemorrhagic disorders; Disease of white blood cells (eg, lymphocytosis, lymphopenia); Chronic ulcer of skin; Spondylosis, intervertebral disc disorders; Systemic lupus erythematosus or connective tissue disorder; Crushing injury or internal injury; Shock; Drowning/submersion; Firearm; Motor vehicle traffic; Pedal cyclist, not MVT; Suffocation; Suicide or intentional self-inflicted injury; Fracture of long bone or spine; Intracranial injury; Poisoning by nonmedicinal substances; Skull or face fractures; Open wounds of head, neck, or trunk; Superficial soft tissue injury: contusion; Short gestation, low birth wt, or fetal growth retardation; Hypoxia, asphyxia, or aspiration during birth; Birth trauma; Perinatal jaundice	External validation study
Delaplain et al. 2020	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Planned readmission; Sex; Length of stay (d); Admission source; Payer; Age; Race/ethnicity; Previous ED visits; Current/index visit is a readmission from a prior visit?; Previous visits; Readmission history; Free-standing pediatric hospital; Number of medications; Number of systems diagnoses; Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or	LOS: [LOS 2-4 days (OR=1.298; NR% CI 1.189-1.416), LOS 4-7 days (OR=1.638; NR% CI 1.493-1.797), LOS >7 (OR=1.994; NR% CI 1.816-2.189)]; Admission source: Transfer (OR=0.768; NR% CI 0.689-0.856); Previous ED visits: [1 (OR=1.137; NR% CI 1.058-1.222), 2 (OR=1.158; NR% CI 1.037-1.294), >=3 (OR=1.305; NR% CI 1.157-1.471)]; Current/index visit is a readmission from a prior visit?: [Yes: unplanned (OR=1.404; NR% CI 1.291-

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				pelvis (S30-S39); Traumatic injury (ICD-10): Ankle and foot (S90-99); Traumatic injury (ICD-10): Burns (T20-32); Traumatic injury (ICD-10): Early complications of trauma (T79); Traumatic injury (ICD-10): Elbow and forearm (S50-59); Traumatic injury (ICD-10): Foreign body entering through the natural orifice (T15-19); Traumatic injury (ICD-10): Head (S00-09); Traumatic injury (ICD-10): Hip and thigh (S70-79), Traumatic injury (ICD-10): Unspecified body region (T14); Traumatic injury (ICD-10): Knee and lower leg (S80-89); Traumatic injury (ICD-10): Multiple body regions (T07); Traumatic injury (ICD-10): Neck (S10-19); Traumatic injury (ICD-10): Unspecified effects of external causes (T66-T78); Traumatic injury (ICD-10): Poisoning (T36-50); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88); Traumatic injury (ICD-10): Thorax (S20-29); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65); Traumatic injury (ICD-10): Wrist, hand, and fingers (S60-61); Surgical procedures by system: Auditory; Surgical procedures by system: Cardiovascular; Surgical procedures by system: Digestive; Surgical procedures by system: Integumentary; Surgical procedures by system: Musculoskeletal; Surgical procedures by system: Nervous; Surgical procedures by system: Respiratory; Traumatic injury: trauma due to frost bite; Traumatic injury: eye/ocular; Traumatic injury: endocrine; Surgical procedure: mediastinum/diaphragm; Surgical procedures by system: hematologic; Surgical procedures by system: lymphatic; Surgical procedures by system: urinary/reproductive	1.527), Yes: planned (OR=1.646; NR% CI 1.457-1.858)]; Previous visits: [1 (OR=1.746; NR% CI 1.604-1.9), 2 (OR=2.282; NR% CI 2.02-2.578), >=3 (OR=2.614; NR% CI 2.245-3.045)]; Readmission history: [1 (OR=1.473; NR% CI 1.303-1.665), 2 (OR=1.788; NR% CI 1.504-2.125), >=3 (OR=2.578; NR% CI 2.172- 3.06)]; Number of medications (OR=1.011; NR% CI 1.009-1.014); Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or pelvis (S30-S39) (OR=0.763; NR% CI 0.655-0.89); Traumatic injury (ICD-10): Burns (T20-32) (OR=0.596; NR% CI 0.494-0.719), Traumatic injury (ICD-10): Head (S00-09) (OR=0.661; NR% CI 0.594-0.735); Traumatic injury (ICD-10): Hip and thigh (S70-79) (OR=0.709; NR% CI 0.616-0.816); Traumatic injury (ICD-10): Unspecified body region (T14) (OR=0.752; NR% CI 0.62-0.912); Traumatic injury (ICD-10): Knee and lower leg (S80-89) (OR=0.727; NR% CI 0.614-0.86); Traumatic injury (ICD-10): Neck (S10-19) (OR=1.194; NR% CI 1.008-1.45); Traumatic injury (ICD-10): Poisoning (T36-50) (OR=1.16; NR% CI 1.06-1.269); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49) (OR=0.547; NR% CI 0.461-0.649); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88) (OR=1.545; NR% CI 1.439-1.66); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65) (OR=0.538; NR% CI 0.404-0.716)
Chotai et al. 2017	Unnamed	Neurosurgery	30 day unplanned HRs following index surgery for neurosurgical diagnoses	Age; Sex; Race; Preterm birth at <37 wks; Brain tumor/cyst resection or biopsy; Craniectomy or craniotomy for epilepsy, vascular, trauma, CM-I, & other brain lesions; Shunt surgery or ETV-CPC; Spine surgery; Preop or intraop EVD; Median LOS in days (range); No. w/ ICU stay; No. w/ postop complications	Race: other vs white (OR=5.916; 95% CI 1.304-26.84); ICU stay (OR=3.302; 95% CI 1.325-8.231)
Davidson et al. 2021	Unnamed	Ureteroscopy	30-day unplanned HRs after ureteroscopy	Age; Sex; Weight; Stone location; Ureteral stent placement during procedure; Anaesthesia time; Operative time; Unplanned reoperation; Organ space SSI; Pneumonia; Unplanned reintubation; Progressive renal insufficiency; Urinary tract infection; Sepsis; Septic shock; Occurrence of transfusion ; Preterm birth; Previous cardiac surgery; Case type; Patient status at time of surgery; ASA class	Female (RR=2.03; 95% CI 1.34-3.07); Renal stone location (RR=1.77; 95% CI 1.10-2.83), Both ureteric and renal stone location (RR=1.29; 95% CI 0.74-2.25); Inpatient at time of surgery (RR=1.61; 95% CI 1.03-2.51)
Garcia et al. 2018	Unnamed	Kasai procedure	30-day unplanned HRs related to kasai procedure	Age; Female (vs male) ; White race; Hispanic; Prematurity; Preoperative comorbidities: Major cardiac risk factors; Respiratory comorbidity; Non-hepatobiliary GI disease; Renal disease; Neurologic comorbidity; Preoperative steroid use; Preoperative nutritional support; Systemic inflammatory response syndrome; Preoperative blood transfusion; Anemia; Hypoalbuminemia ; Total bilirubin > 8; Operative time, minutes; ASA class ≥ III; Perioperative blood transfusion; 30-day post-operative complication; Reoperation	Prematurity (OR=3.88; 95% CI 1.08-13.95); 30-day post-operative complication (OR=4.09; 95% CI 1.41-11.87)
Lee at al-2020	Unnamed	Adolescent idiopathic scoliosis surgery	30-day unplanned HRs after adolescent idiopathic scoliosis surgery	Age; Sex; Income; Primary payer; Anemia; Coagulopathy; Chronic pulmonary disease; Depression; Diabetes; Hypothyroidism; Hypertension; Liver disease; Fluid & electrolyte disorders; Pulmonary vascular disorders; Renal failure; Valvular disease; Smoker; Obesity; Weight loss; Chronic steroid use; Chronic use of antiplatelets, antithrombotics, anticoagulants; Blood transfusion; Autograft; BMP use; Osteotomy; Fusion levels; Hospital teaching status; Hospital ownership; Disposition; Index complication: Cardiac; Index complication: Pulmonary; Index complication: Neurological; Index complication: Urinary tract infection; Index complication: Paralytic ileus; Index complication: SIADH; Index complication: SMA syndrome; Index complication: Thromboembolic complications; Index complication: Wound-related complications; Index complication: Intraop hemorrhage or hematoma; Index complication: Mechanical implant-related complications; Index complication: Dural tear; LOS >5 days	Anemia (OR=2.0; 95% CI 1.6-2.5); Hypothyroidism (OR=3.0; 95% CI 2.0-4.5); Fluid & electrolyte disorders (OR=1.8; 95% CI 1.5-2.3); Obesity (OR=2.9; 95% CI 2.2-4.0); Chronic use of anticoagulants (OR=7; 95% CI 3.0-16.4); Index complication: SIADH (OR=4.7; 95% CI 2.6-8.4); Index complication: Dural tear (OR=2.7; 95% CI 1.6-4.7); LOS >5 days (OR=1.8; 95% CI 1.6-2.2)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Minhas et al. 2016	Idiopathic scoliosis Progressive infantile scoliosis Scoliosis due to other conditions	Spinal Surgeries (Scoliosis)	30-day unplanned HRs	Age; Sex; Underweight; Obesity; Diabetes; Preterm birth; Ventilator requirement; Asthma; Cystic fibrosis; CLD; Oxygen requirement; Tracheostomy; Structural pulmonary abnormality; Esophageal/GI disease; Hepatobiliary/pancreatic disease; Cardiac risk factors; History of cerebrovascular event; Childhood malignancy; CNS tumor; Impaired cognition; History of seizure; Cerebral palsy; Structural CNS abnormality; Neuromuscular Disorder; History of intraventricular hemorrhage; Immunity disorder; Chronic steroid use; Bone marrow disorder; History of organ transplant; Open wound; Weight loss; Nutritional support requirement; Bleeding disorder; Hematological disorder; Chemotherapy; Preoperative sepsis; Preoperative inotrope requirement; Prior operation within last 30 d; Preoperative transfusion requirement; ASA>=3; Posterior fusion up to 6 levels; Posterior fusion 7-12 levels; Posterior fusion 13 or more levels; Pelvic fixation; Progressive infantile scoliosis; Idiopathic scoliosis; Scoliosis due to other condition; LOS>5 d	Idiopathic scoliosis: Obesity (OR=3.09; 95% CI 1.83-5.21); Posterior fusion 13 or more levels (OR=1.86; 95% CI 1.07-3.23) Progressive infantile scoliosis: Impaired cognition (OR=10.08; 95% CI 2.78-14.23) Scoliosis due to other conditions: ASA>=3 (OR=5.92; 95% CI 1.02-10.74); Pelvic fixation (OR=2.80; 95% CI 1.14-6.89)
Roddy & Diab 2017	Unnamed	Spine fusion	30-day unplanned HRs	Male versus female; Race; Age; Diagnosis (vs. idiopathic); Insurance type; Approach (vs. posterior); LOS; Hospital volume per year; Discharge disposition; Number of comorbidities; Teaching hospital; Children's hospital; Infection on index admission; Mechanical complication on index admission; Discharge on the weekend; 8+ levels fused (vs. 3–7); Wound dehiscence ; Pulmonary complication; VTE; Hematoma; Ileus; Non-mechanical complication of internal prosthesis	Male versus female (OR=1.28; 95% CI 1.07–1.54); Age 12–13y (OR=0.74; 95% CI 0.57–0.97); Neuromuscular (OR=2.99; 95% CI 2.39–3.75), Congenital (OR=1.66; 95% CI 1.16-2.38), Scheuermann kyphosis (OR=2.00; 95% CI 1.27-3.17), Other diagnosis (OR=1.68; 95% CI 1.32-2.15); Medicais (OR=1.50; 95% CI 1.24-1.82); Anterior approach (OR=1.55; 95% CI 1.01-2.36); LOS <= 3 days (OR=1.89; 95% CI 1.37-2.59), LOS 6-124 days (OR=1.66; 95% CI 1.35-2.02); hospital volume 41-60 fusions (OR=0.72; 95% CI 0.54-0.97), hospital volume >80 fusions (OR=0.66; 95% CI 0.50-0.88); Short-term care hospital (OR=12.41; 95% CI 8.51-18.10), Home health care (OR=1.44; 95% CI 1.09-1.91), other discharge dispositions (OR=3.79; 95% CI 1.14-12.61); Number of comorbidities >=1 (OR=1.21; 95% CI 1.01-1.50); Teaching hospital (OR=1.59; 95% CI 1.10-2.18); Infection on index admission (OR=2.12; 95% CI 1.22-3.69); Mechanical complication on index admission (OR=3.79; 95% CI 1.71-8.39)
Sherrod et al. 2016	Unnamed	Neurosurgery	30-day unplanned HRs after neurosurgery	Shunt/ventricular catheter revision, removal, irrigation; MMC repair; Shunt/ventricular catheter placement; Craniotomy for neoplasm; Other procedures (e.g. primarily baclofen pump placement) ; Craniotomy for Chiari malformation; Spine; Craniotomy for craniosynostosis; Skin lesion; Age; Neonate; LOS; Sex; Race; Patient status; Prior operation w/in 30 days ; Concurrent procedure; Transfer status ; Discharge destination; Any comorbidity; Any non-CNS comorbidity; Obesity; Pulmonary comorbidity; Ventilator dependent; Pneumonia; Asthma; Cystic fibrosis; Bronchopulmonary dysplasia; Oxygen support; Structural pulmonary abnormality; GI comorbidity; Esophageal, gastric, intestinal disease; Biliary, liver, pancreatic disease; Renal comorbidity; Renal failure; Dialysis; CNS comorbidity; Coma >24 hrs; History of CVA or TBI; CNS tumor; Developmental delay; Cerebral palsy; Neuromuscular disorder; Seizure disorder; Structural CNS abnormality; Cardiac comorbidity; Steroid use; Chemotherapy w/in 30 days before surgery; Radiotherapy w/in 90 days before surgery; Open wound (w/ or w/o infection); Tracheostomy at time of surgery; Immune disease or immunosuppressant use; Nutritional support (IV or NG tube); Bleeding disorder; Hematological disorder; Current or previous malignancy; History of prematurity; Intraventricular hemorrhage; Congenital malformation, any system; SIRS/sepsis w/in 48 hrs before surgery; Hypoalbuminemia; Hyponatremia; Hypernatremia; Elevated WBC count; Thrombocytopenia; Elevated AST; Elevated BUN; Abnormal PT; Abnormal PTT; Anemia ; Length of operation; Triage; ASA Class; Periop blood transfusion ; Wound classification; Any complication; Any infection; Superficial SSI; Deep SSI; Organ/space SSI; (postop) Sepsis; (postop) UTI; (postop) Pneumonia; CLABI; Wound disruption; Unplanned intubation; Renal insufficiency; Acute renal failure; Coma >24 hrs; CVA/intracranial hemorrhage; (postop) Seizure; Peripheral nerve injury; Cardiac arrest; Graft or prosthesis failure; DVT; PE	Shunt/ventricular catheter revision, removal, or irrigation procedure (OR=2.283; 95% CI 1.679–3.103); MMC procedure (OR=1.979; 95% CI 1.066–3.675); Shunt/ventricular catheter placement procedure (OR=2.128; 95% CI 1.542–2.937); Craniotomy for craniosynostosis (OR=0.291; 95% CI 0.151–0.560); Spine procedure (OR=0.703; 95% CI 0.503–0.984); Native American race (OR=2.363; 95% CI 1.149–4.861); Prior operation w/in 30 days of index procedure (OR=1.378; 95% CI 1.001–1.897); Transfer from ER (OR=1.273; 95% CI 1.046–1.549); Home discharge (OR=1.885; 95% CI 1.208–2.942); Presence of any comorbidity (OR=1.943; 95% CI 1.086–3.478); Oxygen supplementation (OR=1.645; 95% CI 1.128–2.399); Preexisting seizure disorder (OR=1.250; 95% CI 1.034–1.510); Steroid use >10 days (OR=1.411; 95% CI 1.087–1.831); Nutritional support (IV or NG tube) (OR=1.403; 95% CI 1.088–1.809); Operation time (per hr increase) (OR=1.059; 95% CI 1.006–1.114); Superficial incisional SSI (OR=12.151; 95% CI 7.783–18.973); Deep incisional SSI (OR=25.547; 95% CI 10.229–63.373); Organ/space SSI (OR=19.156; 95% CI 11.618–31.585); Postop sepsis (OR=2.616; 95% CI 1.321–5.181); Postop UTI (OR=4.262; 95% CI 2.598–6.992); Postop pneumonia (OR=4.294; 95% CI 2.045–9.017); Wound disruption (OR=17.582; 95% CI 10.750–28.756); Postop seizure (OR=2.532; 95% CI 1.398–4.587); Graft/prosthesis failure (OR=11.074; 95% CI 2.882–42.548)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Tahiri et al. 2015	Unnamed	Plastic Surgery	30-day unplanned HRs following pediatric plastic surgery procedures	Sex; Race; Inpatient vs outpatient; Type of procedure; Discharge destination; Respiratory history; GI history; Cardiac history; CNS history; Nutritional history; Hematologic history; One comorbidity; Multiple comorbidities (>2); Congenital malformation; Operation within 30 days; Anesthesia type; Triage; RVUs; Wound class; ASA class; Operative time; LOS; Any complications; Surgical complications; Medical complications	Inpatient procedure (OR=1.569; 95% CI 1.028–2.395); RVUs: 19.66–87.09 (OR=0.149; 95% CI 0.057–0.387); Wound contamination (OR=2.328; 95% CI 1.347–4.024); ASA class IV (OR=7.700; 95% CI 1.479–40.079); Operative time: 93–174 min (OR=2.511; 95% CI 1.494–4.219), Operative time: >175 min (OR=3.887; 95% CI 2.220–6.808); Surgical complications (OR=6.936 ; 95% CI 3.702–12.994); Medical complications (OR=11.922; 95% CI 4.706–30.208)
Wheeler et al. 2018	Unnamed	Burn diagnosis	30-day unplanned HRs	Age; Gender; Median household income by zip code; Primary expected payer; Patient location/urban-rural; No. of chronic conditions; TBSA burned ; Burn degree; Burn to eye and adnexa; Burn of face head and neck; Burn of trunk; Burn of upper limb except wrist and hand; Burn of wrist(s) and hand(s); Burn of lower limb(s); Burn of internal organs; Burn unspecified site(s); Minor loss of function; Moderate loss of function; Major/extreme loss of function; Burn mechanism; Annual no. of admitted burn patients ; Teaching status of hospital; Major operating room procedure (index); LOS; Disposition of patient	No detailed reporting of p-values that were accepted to be significant. Predictors that were assigned as significant are included in the analysis: Patient residence: Medium metropolitan county (OR=1.03; 95% CI 1.14–3.29), Patient residence: Small metropolitan county (OR=2.04; 95% CI 1.06–3.92); TBSA burned (%) ≥ 10 (OR=1.81; 95% CI 1.18–2.79); Third burn degree (OR=2.68; 95% CI 1.69–4.24); Major operating room procedure (OR=1.76; 95% CI 1.14–2.70); LOS 2-3 days (OR=1.72; 95% CI 1.03–2.88)
Vedantam et al. 2018	Unnamed	Epilepsy surgery	30-day unplanned HRs after epilepsy surgery	Age; Gender; Race; ASA classification; Operative time; LOS; Weight ; Neurologic and neuromuscular; Other complex chronic conditions; Discharge destination; Procedure	Hemispherectomy (OR=4.11; 95% CI 1.48–11.42)
Basques et al. 2015	Unnamed	Posterior spinal fusion	30-day unplanned HRs after posterior spinal fusion	Age; Sex; BMI for age; History of asthma; Number of levels fused; Osteotomy performed; Operative time; Any inpatient complication; Serious adverse event; Return to the operating room; Wound dehiscence; Deep surgical site infection; Nervous injury; Minor adverse event; Superficial surgical site infection; Urinary tract infection; Pneumonia; Length of stay more than 6 days	Any inpatient complication (OR=180.44; 95% CI 35.47–917.97)
Martin et al. 2015	Unnamed	Spinal deformity surgery	30-day unplanned HRs after spinal deformity surgery	Age; Sex; Race; BMI ; Diabetes; Ventilator dependence; Asthma; Cystic fi brosis; Chronic lung disease; Oxygen support; Tracheostomy; Airway abnormalities; GI disease; Hepatobiliary disease; Cardiac risk factors; Previous cardiac surgery; Acute renal failure; Dialysis; History of stroke; Tumor involving CNS; Developmental delay; Seizure disorder; Cerebral palsy; CNS abnormality; Neuromuscular disorder; Immune disorder; Steroid use w/in 30 d; Bone marrow transplant; Solid organ transplant; Recent weight loss; Nutritional support; Bleeding disorder; Hematologic disorder; Chemo w/in 30 d; Rad therapy w/in 90 d; Prior operation w/in 30 d; Congenital malformation; Hx childhood malignancy; Requires inotropic support; Sodium; BUN; WBC; Hematocrit; INR; Creatinine; Albumin; ASA class; Received blood transfusion; Blood transfused; Operative time; Total case RVUs; LOS; Surgeon specialty; Isolated primary posterior arthrodesis; Revision posterior arthrodesis; Isolated primary anterior arthrodesis; Revision anterior arthrodesis; Combined anterior and posterior arthrodesis; Instrumentation extending to pelvis; Insertion of intervertebral device; Osteotomy; Bone grafting; Diagnosis	Structural pulmonary abnormalities (OR=2.53; 95% CI 1.22–5.23); ASA class (3 or 4 vs . 1 or 2) (OR=2.18; 95% CI 1.07–4.47); Isolated primary anterior spinal fusion (OR=7.65; 95% CI 1.32–44.3)
General medical conditions related readmissions					
Learly et al. 2019	Prediction at admission	Complex chronic conditions	30-day unplanned HRs	Age; Race and/or ethnicity; Boys; Non-English primary language; Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo, Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type	Significant variables (p-value<0.05) in univariate analysis are used for the analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.15–2.45); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.32–3.10); No. CCC categories: [No. CCC categories = 2 (OR=1.73; 95% CI 1.19–2.49), No. CCC categories >=3 (OR=2.30; 95% CI 1.50–3.47)]; Medical admission (OR=1.82; 95% CI 1.30–2.63)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
	Prediction at discharge			Age; Race and/or ethnicity; Boys; Non-English primary language, Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo; Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type; ICU use; Discharge disposition from the hospital; LOS in d; Weekday discharge	Significant variables (p-value<0.05) in univariate analysis are used for analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.16–2.46); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.31–3.11); No. CCC categories: [No. CCC categories = 2 (OR=1.56; 95% CI 1.06–2.26), No. CCC categories >=3 (OR=1.72; 95% CI 1.08–2.69)]; Medical admission (OR=1.75; 95% CI 1.23–2.49); Discharge disposition from the hospital: [With services (OR=1.69; 95% CI 1.17–2.44), Other facility (OR=1.15; 95% CI 0.58–2.13)]; LOS: [LOS 2-5 days (OR=1.15; 95% CI 0.78–1.72), LOS >=6 days (OR=1.45; 95% CI 0.90–2.33)] Weekday discharge was not significant (p-value≤ 0.05) in the univariate analysis, but included in the final scoring system.
Ryan et al. 2021	PASS (validation)	Asthma	30-day unplanned HRs	Respiratory rate, Oxygen requirement, Auscultation, Retractions, Dyspnea	External validation study
O´Conell et al. 2021	Unnamed	Nervous system condition	30-day unplanned HRs	Age; Sex; race/ ethnicity; payer; Length of stay days) ; Index visit planned; Admitted through ED; Index visit is a readmission; Previouse ED visits (prior 6mo); Previous hospitalizations (prior 6mo); Previous readmissions (prior 6 mo); Number of comorbid diagnoses (by ICD 10 CM chapters); Viral Meningitis (A87); Malignant neoplasm of brain (C71); Disorders of BCAA and FA Metabolism (E71); Other AA Metabolism Disorders (E72); Other carbohydrate metabolism disorders (E74); Lipoprotein Metabolism Disorders / Lipidemias (E78); Mental Disorders Due to Physiological Condition (F06); Unspecified Intellectual Disabilities (F79); Speech & Language Development Disorders (F80); Scholastic Skill Development Disorders (F81); Motor Function Development Disorders (F82); Pervasive Developmental Disorders (F84); Psychological Development Disorders (F88); Bacterial Meningitis (G00); Meningitis, Other Causes (G03); Encephalitis, Myelitis, Encephalomyelitis (G04); Extrapyramidal & movement Disorders (G25); Nervous System Degenerative Diseases (G31); Epilepsy & Recurrent Seizures (G40); Migraine (G43); Headache Syndromes (G44); Sleep Disorders (G47); Polyneuropathies (G62); Primary Muscle Disorders (G71); Cerebral Palsy (G80); Hemiplegia & Hemiparesis (G81); Paraplegia & Quadriplegia (G82); Pain (G89); ANS Disorders (G90); Hydrocephalus (G91); Brain Disorders (G93); Spinal Cord Diseases (G95); CNS Disorders (G96); Postprocedural NS Disorders (G97); Visual Pathway Disorders (H47); Paralytic Strabismus (H49); Binocular Movement Disorders (H51); Nystagmus & Irregular Eye Movements (H55); Conductive & Sensorineural Hearing Loss (H90); Nontraumatic Intracerebral Hemorrhage (I61); Nontraumatic Intracranial Hemorrhage (I62); Cerebral Infarction (I63); Cerebrovascular Diseases (I67); Sequelae of Cerebrovascular Disease (I69); Neuromuscular Dysfunction of Bladder (N31); Nontraumatic Intracranial Hemorrhage Newborn (P52); Newborn Convulsions (P90); Newborn Cerebral Disturbances (P91); Newborn Muscle Tone Disorders (P94); Microcephaly (Q02); Congenital Hydrocephalus (Q03); Congenital Brain Malformations (Q04); Spina Bifida (Q05); Congenital Spinal Cord Malformations (Q06); Congenital Nervous System Malformations (Q07); Down Syndrome (Q90); Other Autosomal Trisomies (Q92); Nervous & Musculoskeletal System Symptoms (R29); Somnolence, Stupor, Coma (R40); Cognitive Function Symptoms (R41); Dizziness & Giddiness (R42); Headache (R51); Convulsions (R56); Intracranial Injury (S06); Number of medications; Free-standing pediatric hospital; Surgical procedures	Age (OR=0.993; 95% CI 0.99-0.997); Hispanic/Latino (OR=1.126; 95% CI 1.016-1.247); Self-Pay (OR=0.805; 95% CI 0.688-0.943); LOS 2-4 days (OR=1.228; 95% CI 1.167-1.293), LOS 4-6 days (OR=1.441; 95% CI 1.361-1.525), LOS 7 or more days (OR=1.585; 95% CI 1.492-1.683); Index visit planned: Yes (OR=0.898; 95% CI 0.847-0.952); Emergent Admission (OR=1.129; 95% CI 1.079-1.18); Index visit is a planned readmission (OR=1.338; 95% CI 1.269-1.411), Index visit is a unplanned readmission (OR=1.666; 95% CI 1.516-1.832); 1 previous ED visit (prior 6mo) (OR=1.106; 95% CI 1.056-1.159), 2 previous ED visits (prior 6mo) (OR=1.200; 95% CI 1.123-1.283), 3 or more previous ED visits (prior 6mo) (OR=1.297; 95% CI 1.209-1.391); 1 previous hospitalization (prior 6mo) (OR=1,695; 95% CI 1.613-1.781), 2 previous hospitalizations (prior 6mo) (OR=2.348; 95% CI 2.18-2.528), 3 or more previous hospitalizations (prior 6mo) (OR=3.014; 95% CI 2.738-3.317); 1 previous readmission (prior 6 mo) (OR=1.179; 95% CI 1.09-1.274), 2 previous readmissions (prior 6 mo) (OR=1.496; 95% CI 1.338-1.673), 3 or more previous readmissions (prior 6 mo) (OR=2.051; 95% CI 1.831-2.298); Number of comorbid diagnoses (by ICD 10 CM chapters) (OR=1.01; 95% CI 1.006-1.014); Viral Meningitis (A87) (OR=0.446; 95% CI 0.353-0.563); Malignant neoplasm of brain (C71) (OR=1.953; 95% CI 1.788-2.133); Other AA Metabolism Disorders (E72) (OR=1.313; 95% CI 1.095-1.575); Mental Disorders Due to Physiological Condition (F06) (OR=0.768; 95% CI 0.648-0.91); Speech & Language Development Disorders (F80) (OR=0.811; 95% CI 0.725-0.907); Bacterial Meningitis (G00) (OR=0.622; 95% CI 0.487-0.796); Meningitis, Other Causes (G03) (OR=0.829; 95% CI 0.698-0.985); Sleep Disorders (G47) (OR=0.84; 95% CI 0.791-0.892); Polyneuropathies (G62) (OR=1.5; 95% CI 1.248-1.803); Hydrocephalus (G91) (OR=1.218; 95% CI 1.124-1.476); Brain Disorders (G93) (OR=1.078; 95% CI 1.019-1.14); Paralytic Strabismus (H49) (OR=1.288; 95% CI 1.124-1.476); Nystagmus & Irregular Eye Movements (H55) (OR=0.825; 95% CI 0.686-0.992); Neuromuscular Dysfunction of Bladder (N31) (OR=1.155; 95% CI 1.034-1.129); Newborn Cerebral Disturbances (P91) (OR=0.822; 95% CI 0.705-0.958); Newborn Muscle Tone Disorders (P94) (OR=0.621; 95% CI 0.567-0.68); Microcephaly (Q02) (OR=1.131; 95% CI 1.005-1.1272); Congenital Hydrocephalus (Q03) (OR=1.318; 95% CI 1.175-1.478); Down Syndrome (Q90) (OR=1.129; 95% CI 1.036-1.23); Other Autosomal Trisomies (Q92) (OR=1.214; 95% CI 1.019-1.448); Nervous & Musculoskeletal System Symptoms (R29) (OR=0.882; 95% CI 0.781-0.995); Somnolence, Stupor, Coma (R40) (OR=0.786; 95% CI 0.694-0.889); Cognitive Function Symptoms (R41) (OR=0.853; 95% CI 0.776-0.938); Convulsions (R56) (OR=0.922; 95% CI 0.874-0.972); Intracranial Injury (S06) (OR=0.476; 95% CI 0.418-0.543); Number of medications (OR=1.014; 95% CI 1.012-1.016); Free-standing pediatric hospital (Yes) (OR=1.586; 95% CI 1.103-2.28); Surgical procedures (Cardiovascular) (OR=1.204; 95% CI 1.109-1.216), Surgical procedures (Musculoskeletal) (OR=0.685; 95% CI 0.6-0.781), Surgical procedures (Nervous System) (OR=1.119; 95% CI 1.029-1.216)
Hoenk et al. 2021	Unnamed	Oncology	30-day unplanned HRs	Age; sex; race/ ethnicity; Payer; Acute lymphoid leukemia (ALL); Acute myeloid leukemia (AML); Brain cancer; Neuroblastoma; Wilms tumor; Hodgkin's lymphoma; Non-Hodgkin's lymphoma; Rhabdomyosarcoma; Bone/cartilage cancer; Other cancers; Chemotherapy; Bone marrow transplant; Number of cancer medications; Number of previous visits for chemotherapy (prior 30 d); Length of stay; Emergent	Age (OR=0.987; 95% CI 0.982-0.993); Acute lymphoid leukemia (ALL): Yes, not in remission (OR=0.788; 95% CI 0.721-0.861), Acute lymphoid leukemia (ALL): Yes, in remission (OR=0.656; 95% CI 0.590-0.729), Acute lymphoid leukemia (ALL): Yes, in relapse (OR=1.436; 95% CI 1.201-1.718); Brain cancer (OR=0.782; 95% CI 0.711-0.861); Neuroblastoma (OR=1.442; 95% CI 1.285-1.619); Rhabdomyosarcoma (OR=1.182; 95% CI 1.048-1.332); Bone/cartilage cancer (OR=1.618; 95% CI

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				admission; Is index/current visit itself a readmission; Previous ED visits (prior 6 mo); Number of previous visits without chemotherapy (prior 6 mo); Previous readmissions (prior 6 mo); Viral or bacterial infections (A00 - B99); Hematological conditions (D50-D69, D71-D79); Neutropenia (D70); Immunological conditions (D80-D89); Endocrine, nutritional, and metabolic diseases (E00-E89); Mental, behavioral, and neurodevelopmental disorders (F01-F99); Nervous system (G00-G99); Eye, adnexa, ear, mastoid (H00-H95); Circulatory (I00-I99); Respiratory (J00-J99); Digestive (K00-K95); Skin/subcutaneous tissue (L00-L99); Musculoskeletal and connective tissue (M00-M99); Genitourinary (N00-N99); Congenital/Chromosomal (Q00-Q99); Injuries and poisoning (S00-T88)	1.455-1.799); Number of cancer medications (OR=1.46; 95% CI 1.117-1.175); LOS 2, 3 days (OR=1.292; 95% CI 1.184-1.410), LOS 4,5,6 days (OR=1.466; 95% CI 1.340-1.604), LOS 7 or more days (OR=1.344; 95% CI 1.219-1.481); Is index/current visit itself a readmission (Yes, unplanned) (OR=1.869; 95% CI 1.722-2.028), Is index/current visit itself a readmission (Yes, planned) (OR=1.397; 95% CI 1.250-1.560); Number of previous visits without chemotherapy (prior 6 mo) (OR=1.077; 95% CI 1.067-1.221); 1 previous readmission (prior 6 mo) (OR=1.277; 95% CI 1.126-1.336), 2 previous readmissions (prior 6 mo) (OR=1.281; 95% CI 1.145-1.433), 3 or more previous readmissions (prior 6 mo) (OR=1.263; 95% CI 1.087-1.468); Endocrine, nutritional, and metabolic diseases (E00-E89) (OR=1.141; 95% CI 1.067-1.221); Congenital/Chromosomal (Q00-Q99) (OR=0.745; 95% CI 0.657-0.844); Injuries and poisoning (S00-T88) (OR=0.838; 95% CI 0.760-0.925)
Sanchez-Luna et al. 2016	Unnamed	Acute bronchiolitis due to respiratory syncytial virus	30-day unplanned HRs	Prematurity; Gestational age; Congenital heart disease; Chronic lung disease; Down's syndrome; Velo-cardio-facial syndrome; Neuromuscular disorders; Inmunodeficiency; Number of risk factors; Heart transplant	Significant odds ratios before multilevel modelling: Prematurity (OR=3.66; 95% CI 3.15-4,27); Gestational age: <=28w (OR=12.47; 95% CI 1.76-88.51), Gestational age: 29-32w (OR=4.99; 95% CI 0.97-25.71), Gestational age: 33-36w (OR=8.48; 95% CI 4.57-15.71); Congenital heart disease (OR=3.05; 95% CI 2.51-3.71); Chronic lung disease (OR=6.76; 95% CI 3.87-11.83); Down's syndrome (OR=2.65; 95% CI 1.83-3.84); Velo-cardio-facial syndrome (OR=12.42; 95% CI 1.75-88.21); Neuromuscular disorders (OR=5.42; 95% CI 3.72-7.89); Inmunodeficiency (OR=4.35; 95% CI 1.84-10.3); Number of risk factors = 1 (OR=3.64; 95% CI 3.12-4.25), Number of risk factors >=2 (OR=3.86; 95% CI 2.28-6.54)
Sacks et al. 2017	Unnamed	Cardiac conditions	30-day unplanned HRs	Age; LOS; Diagnosis count; Procedure count; First weight; Last weight; Medication count; Catheterization; Electrophysiology study; Surgery (noncardiac); ACE/ARB antihypertensive; Antacid; Antiarrhythmic; Antibiotic; Anticoagulant; Beta-blocker; Calcium channel blocker; Diuretic; Lipid management; Milrinone; Neuroactive; Pulmonary antihypertensive; Steroid; Gender; Race; Ethnicity ; Language; Insurance status; Distance from center (miles); Season of discharge; Day of discharge	Age: 1mo-1year (OR=4.11; 95% CI 2.83–5.98); Diagnosis count (OR=1.10; 95% CI 1.07–1.13); Antibiotic (OR=0.60; 95% CI 0.40–0.90)

Abbreviations: NR, not reported

* Only significant risk factors (odds ratio/ hazard ratio>1, p-value<0.05) are considered in the analysis (cf. table 2). Significant predisposed factors (1>odds ratio/ hazard ratio>0, p-value<0.05) are listed as an additional information.

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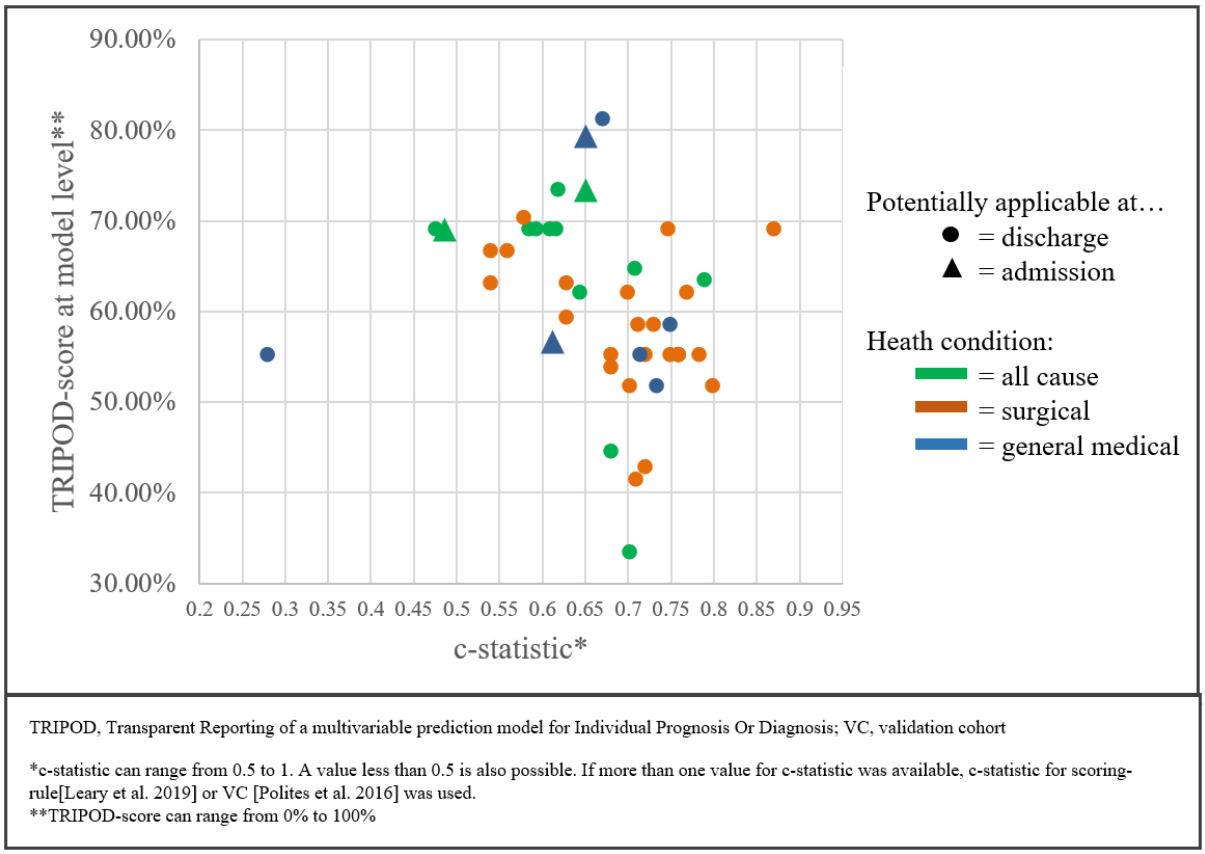
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Table A7: Adherence per TRIPOD-item at predictive model level

TRIPOD-item	Developed predictive models (n=33)			External validated predictive models (n=10)			Incremental value predictive models (n=2)			All predictive models (n=45)		
	# of PMs with applicable item	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item
1	33.00	4.00	12.12%	10.00	6.00	60.00%	2.00	1.00	50.00%	45.00	11.00	24.44%
2	33.00	1.00	3.03%	10.00	0.00	0.00%	2.00	0.00	0.00%	45.00	1.00	2.22%
3a	33.00	33.00	100.00%	10.00	8.00	80.00%	2.00	2.00	100.00%	45.00	43.00	95.56%
3b	33.00	32.00	96.97%	10.00	8.00	80.00%	2.00	2.00	100.00%	45.00	42.00	93.33%
4a	33.00	30.00	90.91%	10.00	9.00	90.00%	2.00	1.00	50.00%	45.00	40.00	88.89%
4b	33.00	33.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	45.00	45.00	100.00%
5a	33.00	18.00	54.55%	10.00	7.00	70.00%	2.00	2.00	100.00%	45.00	27.00	60.00%
5b	33.00	32.00	96.97%	10.00	10.00	100.00%	2.00	2.00	100.00%	45.00	44.00	97.78%
5c	33.00	33.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	45.00	45.00	100.00%
6a	33.00	24.00	72.73%	10.00	2.00	20.00%	2.00	2.00	100.00%	45.00	28.00	62.22%
6b	33.00	33.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	45.00	45.00	100.00%
7a	33.00	32.00	96.97%	10.00	7.00	70.00%	2.00	2.00	100.00%	45.00	41.00	91.11%
7b	33.00	1.00	3.03%	10.00	0.00	0.00%	2.00	0.00	0.00%	45.00	1.00	2.22%
8	33.00	32.00	96.97%	10.00	10.00	100.00%	2.00	2.00	100.00%	45.00	44.00	97.78%
9	33.00	11.00	33.33%	10.00	1.00	10.00%	2.00	1.00	50.00%	45.00	13.00	28.89%
10a	33.00	6.00	18.18%	0.00	0.00	0.00%	2.00	0.00	0.00%	35.00	6.00	17.14%
10b	33.00	3.00	9.09%	0.00	0.00	0.00%	2.00	0.00	0.00%	35.00	3.00	8.57%
10c	0.00	0.00	0.00%	10.00	7.00	70.00%	1.00	1.00	100.00%	11.00	8.00	72.73%
10d	33.00	12.00	36.36%	10.00	6.00	60.00%	2.00	1.00	50.00%	45.00	19.00	42.22%
10e	0.00	0.00	0.00%	1.00	0.00	0.00%	1.00	1.00	100.00%	2.00	1.00	50.00%
11	5.00	4.00	80.00%	0.00	0.00	0.00%	0.00	0.00	0.00%	5.00	4.00	80.00%
12	0.00	0.00	0.00%	10.00	1.00	10.00%	2.00	2.00	100.00%	12.00	3.00	25.00%
13a	33.00	22.00	66.67%	10.00	1.00	10.00%	2.00	1.00	50.00%	45.00	24.00	53.33%
13b	33.00	10.00	30.30%	10.00	0.00	0.00%	2.00	2.00	100.00%	45.00	12.00	26.67%
13c	0.00	0.00	0.00%	10.00	0.00	0.00%	2.00	2.00	100.00%	12.00	2.00	16.67%
14a	33.00	28.00	84.85%	0.00	0.00	0.00%	2.00	2.00	100.00%	35.00	30.00	85.71%
14b	31.00	22.00	70.97%	0.00	0.00	0.00%	2.00	2.00	100.00%	33.00	24.00	72.73%
15a	33.00	0.00	0.00%	0.00	0.00	0.00%	2.00	0.00	0.00%	35.00	0.00	0.00%
15b	33.00	6.00	18.18%	0.00	0.00	0.00%	2.00	1.00	50.00%	35.00	7.00	20.00%
16	33.00	4.00	12.12%	10.00	6.00	60.00%	2.00	1.00	50.00%	45.00	11.00	24.44%
17	0.00	0.00	0.00%	1.00	0.00	0.00%	0.00	0.00	0.00%	1.00	0.00	0.00%
18	33.00	33.00	100.00%	10.00	10.00	100.00%	2.00	2.00	100.00%	45.00	45.00	100.00%
19a	0.00	0.00	0.00%	9.00	2.00	22.22%	2.00	2.00	100.00%	11.00	4.00	36.36%
19b	33.00	33.00	100.00%	10.00	9.00	90.00%	2.00	2.00	100.00%	45.00	44.00	97.78%
20	33.00	27.00	81.82%	10.00	8.00	80.00%	2.00	2.00	100.00%	45.00	37.00	82.22%
21	33.00	17.00	51.52%	10.00	8.00	80.00%	2.00	2.00	100.00%	45.00	27.00	60.00%
22	33.00	22.00	66.67%	10.00	8.00	80.00%	2.00	1.00	50.00%	45.00	31.00	68.89%

PM, predictive model; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

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TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; VC, validation cohort

*c-statistic can range from 0.5 to 1. A value less than 0.5 is also possible. If more than one value for c-statistic was available, c-statistic for scoring-rule[Leary et al. 2019] or VC [Polites et al. 2016] was used.
**TRIPOD-score can range from 0% to 100%

Figure A2: Discriminative ability, application and TRIPOD-adherence of 30-day HR predictive models in paediatrics (n=45)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4-5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5 + Supplementary File
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5 + Supplementary File
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5-6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6-7 + Table 1-4 + Supplementary File
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5+7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Qualitative Synthesis, Page 6-7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Qualitative Synthesis, Page 6-7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Qualitative Synthesis, Page 6-7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Qualitative Synthesis, Page 6-7, 18-19
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6-7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7-8 + Figure A1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 7-8 + Supplementary File
Study characteristics	17	Cite each included study and present its characteristics.	Page 8-12, Table 1+2+3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 + Supplementary File
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1-4 + Supplementary File
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Qualitative Synthesis, Table 1-4, figure 1
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 18-19
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 15-17, TRIPOD Assessment
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 18-20

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23b	Discuss any limitations of the evidence included in the review.	Page 18-19
	23c	Discuss any limitations of the review processes used.	Page 18-19
	23d	Discuss implications of the results for practice, policy, and future research.	Page 18-20
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 21
Competing interests	26	Declare any competing interests of review authors.	Page 21
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 21 + Supplementary File

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Y
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Y
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Y
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Y
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Y
Synthesis of results	6	Specify the methods used to present and synthesise results.	Y
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Y
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Y
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Y
Interpretation	10	Provide a general interpretation of the results and important implications.	Y
OTHER			
Funding	11	Specify the primary source of funding for the review.	Manuscript
Registration	12	Provide the register name and registration number.	NA

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TITLE

Applicability of predictive models for 30-day unplanned hospital readmission risk in paediatrics: a systematic review

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Keywords: Hospital readmission, paediatric, prediction, systematic review

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ABSTRACT

Objective: To summarize multivariable predictive models for 30-day unplanned hospital readmissions (UHRs) in paediatrics, describe their performance and completeness in reporting, and to determine their potential for application in practice.

Design: Systematic review.

Data source: CINAHL, EMBASE and PubMed up to 7 October 2021.

Eligibility criteria: English- or German-language studies aiming to develop or validate a multivariable predictive model for 30-day paediatric UHRs related to all-cause, surgical conditions or general medical conditions were included.

Data extraction and synthesis: Study characteristics, risk factors significant for predicting readmissions and information about performance measures (e.g., c-statistic) were extracted. Reporting quality was addressed by the 'Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis' (TRIPOD) adherence form. The study quality was assessed by applying six domains of potential biases. Due to expected heterogeneity among the studies, the data were qualitatively synthesized.

Results: Based on 28 studies, 37 predictive models were identified, which could potentially be used for determining individual 30-day UHR risk in paediatrics. The number of study participants ranged from 190 children to 1.4 million encounters. The two most common significant risk factors were comorbidity and (postoperative) length of stay. 23 models showed a c-statistic above 0.7 and are primary applicable at discharge. The median TRIPOD adherence of the models was 59% ($P_{25} - P_{75}$ 55% - 69%), ranging from a minimum of 33% to a maximum of 81%. Overall, the quality of many studies was moderate to low in all six domains.

Conclusions: Predictive models may be useful in identifying paediatric patients at increased risk for readmission. To support the application of predictive models, more attention should be placed on completeness in reporting, particularly for those items that may be relevant for implementation in practice.

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Strengths and limitations of this study

- Independent and standardised methodological approach for study selection, data extraction and risk of bias assessment.
- Comprehensive presentation of predictive models that provides information about applicability, performance and reporting quality at model level, differentiated by 30-day all-cause, surgical conditions and general medical conditions related paediatric unplanned hospital readmissions.
- Due to study heterogeneity, the models were only narratively synthesized.

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INTRODUCTION

Hospital readmissions (HRs) are becoming increasingly important as a quality indicator for paediatric inpatient care[1, 2]. HR is often defined as a subsequent, unplanned admission within a period of 30 days after the index hospitalization[3]. For paediatric populations, rates of all-cause 30-day unplanned hospital readmission (UHR) ranged from 3.4% to 18.7%[3-5]. In addition, taking 27 US states into account, it has been estimated that paediatric HRs can cost up to $2 billion annually, with approximately 40% of these occurring HRs being potentially preventable[6].

Identifying the reasons for paediatric HRs is a major challenge, as the health of children is also affected by factors aside of inpatient care[7]. Predictive models can be applied as a tool for the identification of patients with a risk of HR higher than that of the average population and for the implementation of preventive interventions to reduce the risk of HR[8]. Especially in the context of the ongoing COVID-19 pandemic, where children and adolescents are also being hospitalized with a variety of symptoms[9-11], the prevention of UHRs can be beneficial, as it would allow hospital resources to be used in a more target-orientated way.

This systematic review aims to address two research gaps that have been identified. (i) Predictive models with good performance are useful in practice when clinicians and other stakeholders have all the necessary information for their application in clinical practice and critical assessment[12]. However, previous systematic reviews discussed the shortcomings in reporting the quality of prediction models[13-15] and also for paediatric clinical prediction rules[16]. (ii) A previous systematic review has already identified 36 significant risk factors for UHRs in paediatric patients with different health conditions[3]. The largest number of risk factors were identified for surgical procedure related UHRs. Among others, comorbidity was one of the most common risk factors across the 44 included studies[3]. The review[3] extends the findings of an earlier systematic review that focused on 29 paediatric studies targeting predictors for asthma-related UHRs[17]. Both reviews[3, 17] were primarily addressed to predictor finding studies[14], while to date there is no published review of existing 30-day UHR predictive models in paediatrics.

The objective of this systematic review was to determine the potential application of multivariable predictive models for individualized risk prediction of 30-day UHR in the paediatric population by evaluating the models´ discriminative ability, completeness in reporting and the risk factors shown to be significant for prediction of 30-day UHR.

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METHOD

The 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis) statement was adhered to for conducting and reporting of this systematic review[18]. Screening of the titles and abstracts, data extraction, quality assessment and analyses (e.g., completeness in reporting) were performed by two independent reviewers, while disagreements were discussed with a third author. A protocol for this non-registered systematic review was pre-specified and is available from the corresponding author. Based on expert recommendation, the analysis was subsequently focused on 30-day UHRs instead of 30-day HRs (i.e. planned and unplanned HRs), deviating from the pre-specified protocol.

Data source and search strategy

CINAHL, EMBASE and PubMed were used for an electronic database search to identify studies published up to 7 October 2021. The key search terms include the outcome variables used for the model (i.e., readmission/rehospitalization), elements of the study design (i.e., prediction/c-statistic) and the population of interest (i.e., paediatrics/children)(see supplementary material for full search strategies–tables A1-A3). The reference lists of the included studies and of comparable systematic reviews[3, 17] were examined for further potential studies.

Inclusion criteria

Studies addressing multivariable predictive models for children and adolescents (except newborns/ preterm newborns, as the index admission is the birth hospitalization) were included if they were published in English or German and available as full texts in peer-reviewed original journal articles. Studies aiming to develop a new model or to validate an existing model were included (i) if the model was potentially appropriate for the individual prediction of 30-day UHR from acute healthcare service after discharge or after index procedure in paediatrics; and (ii) if the model provided at least one discrimination measure (e.g., c-statistic). Discriminative ability is a key factor in evaluating predictive models[19] and a necessary information to make well-founded conclusions about the performance of a model.

In addition, (iii) predictive model studies that developed a new model (i.e., development design) or determined the incremental or added value of a predictor for an existing model (i.e., incremental value design) had to be based on a regression modelling approach. This inclusion criterion enables us to identify significant risk factors and to apply the Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

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(TRIPOD) adherence form, which was originally developed for regression models[20]. This implies that predictive models using machine-learning techniques (e.g., least absolute selection and shrinkage operator[21] or random forest[22]) are excluded and coded as non-regression models. Studies that aimed to identify 30-day UHR predictors and did not provide a discrimination measure are classified as prognostic factor studies and thus excluded from the analysis (so as not to bias them adversely in TRIPOD adherence). Prognostic factor studies, for example, are not required to present a simplified scoring rule (cf. TRIPOD item 15b[23]). Due to specific requirements of mental diseases, studies were only included (iv) if they addressed non-mental health conditions related 30-day UHRs[3].

Data extraction

Just as in previous systematic reviews[3, 24], studies were categorized by health conditions in all tables. Basic study characteristics were extracted according to criteria in table 1 and 2. To assess the applicability of the predictive models, significant risk factors (i.e., odds ratio (OR) or hazard ratio >1 with a p-value of < 0.05) were assigned to established and revised variable categories[3] in table 3. If all variables of a predictive model are available for a patient at the time of index admission (e.g., previous health service usage before index admission), the model is applicable at admission. Applicability of predictive models at discharge is given if all variables are available at this point for a patient (e.g., length of stay, operative time).

Reporting quality and performance

Predictive models can just be used in practice when clinicians and other stakeholders have access to all information required for their application in clinical practice[12]. The newly developed 'Critical Appraisal of Models that Predict Readmission (CAMPR)' contains 15 expert recommendations for predictive model development relating to HRs. However, CAMPR should not be used as a reporting standard so far and relates to aspects that are out of the scope of this systematic review (e.g., considering different time frames for UHRs)[25]. Due to the importance of high-quality information about predictive models we decided to assess the completeness of reporting by using the TRIPOD adherence form and scoring rules[12, 23, 26]. The TRIPOD adherence form consists of 22 main criteria based on the TRIPOD statement[20], resulting in 37 items that are applicable to varying degrees to the development, validation and incremental value studies[23]. We decided to apply the TRIPOD adherence form at predictive model level. Therefore, publications that report the development and validation of the same predictive model, for example, are assessed separately. According

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to previous research, our analysis concentrates on items that could be reported in the main text or supplements[27].

TRIPOD adherence at model level was merged with the performance results (i.e., discrimination and calibration measures) and the applicability assignment in table 4. The discrimination of a predictive model is often evaluated by the c-statistic or area under the receiver operating characteristic curve. The c-statistic can take a value between 0.5 and 1. A value of 0.5 indicates that the model is not superior to a random prediction of outcomes, while values between 0.7 and 0.8 indicate that the model is appropriate. A value of 0.8 or greater indicates a strong discrimination of a model.[28]

Quality assessment

Following previous systematic reviews[3, 24, 29], the refined version of the quality in prognosis studies (QUIPS) tool with its prompting items[30] was used to appraise the studies critically with regard to the included predictive models based on six domains. Each domain was rated with a 'high', 'moderate' or 'low' risk of bias.

The six domains are[30]: 'study participation', 'study attrition', 'prognostic factor measurement', 'outcome measurement', 'study confounding' and 'statistical analysis and reporting'.

Data synthesis

Because a quantitative evaluation in the form of a meta-analysis was not possible due to the high heterogeneity among the studies, the studies were qualitatively synthesized, i.e., the results for performance, completeness in reporting and significant risk factors were presented in a narrative and in a simplified quantitative form.

Patient and public involvement

Due to the study design we did not involve patients or the public.

RESULTS

Search result

From the electronic database search, 10,076 records were obtained. After duplicates had been removed, the titles and abstracts were screened for 7,694 records. Based on the predefined inclusion criteria, 7,586 records were excluded. Adding one additional recommended article[31] this results in 109 records being included in the full-text assessment. Among the 84

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excluded records, two were predictive model studies for 30-day HRs (i.e. unplanned and planned HRs) with discrimination metrics[32, 33], twelve studies analyse 30-day UHRs or 30-day HRs combined with another outcome (i.e. emergency department return visits (n=5)[34-38], mortality (n=3)[39-41], other complications (n=4)[42-45]), three were predictive model studies for 30-day UHRs or 30-day HRs with no discrimination metrics[46-48], five were non-regression-based predictive model studies for 30-day UHRs or 30-day HRs in paediatrics[21, 49-52], and 59 were prognostic factor studies for 30-day UHRs or 30-day HRs. Based on the full-text assessments (n=25) and the hand search of reference lists (n=3, [53-55]), 28 studies were included in the systematic review, with six of them[55-60] already presented in a previous systematic review[3] with a different focus. The results of the review process regarding the database search is provided in figure A1 in the supplementary material (see table A4 in the supplementary material for a summary of study characteristics for selected excluded models).

Quality assessment

Overall, the quality of many studies was moderate to low for several domains. For instance, the study quality had to be reduced due to a lack of sufficient information (e.g. in the domain 'study participants' or 'study attrition'), while all studies were rated as 'low' for the domain 'study confounding'(see table A5 in the supplementary material for the results of the risk of bias assessment).

Study characteristics

All studies were based on retrospective data, with nine studies based on tertiary or paediatric hospital data[22, 55, 61-67], and 19 studies based on centralized databases[31, 53, 54, 56-60, 68-78]. Four of 28 studies additionally included census data in the analysis[61, 65, 66, 68]. The period of data collection ranged from one year[31, 53, 54, 60, 63, 68] to seven years[69, 70]. The majority of studies included patients up to an age of <18 or ≤18 years. Only five studies considered patients up to 21 years of age[59, 64, 71] or younger than one year[74, 76]. The sample size was specified with different units in the individual studies (e.g., encounters, admissions) and varies between 190 children[74] and 1.4 million encounters[69].

The 28 included studies resulted in 37 predictive models for 30-day UHRs in paediatrics. Ten of 28 studies developed or validated more than one predictive model for UHRs[22, 58, 59, 65-70, 75], which were in part excluded due to non-agreement with the inclusion criteria. The models included were grouped into three health conditions: (i) all-cause UHR (n=13)[22, 61,

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63-65, 68, 69]; (ii) surgical conditions related UHR (n=17)[31, 53, 54, 56-60, 67, 70, 73-75, 77, 78]; and (iii) general medical conditions related UHR (n=7)[55, 62, 66, 71, 72, 76]. The 30-day UHR rates vary from 1.5%[53] to 41.2%[71].

Among the 37 predictive models included, 32 (87%) used a development design[22, 31, 53-61, 63-67, 70-78], three (8%) an external validation design[62, 65, 69] and two (5%) an incremental value design[66, 68]. All external validated models were based on existing predictive models that had been previously used in the adult population[65, 69] or for different outcomes[62]. Furthermore, five of the 28 studies included did not state the primary aim to develop, validate externally or assess the incremental value of the respective 30-day UHR predictive model[65, 67-70].

Of the predictive models with a development or incremental value design, 18 employed an apparent validation[31, 53-55, 58-61, 67, 68, 73-78] and 16 employed an internal validation[22, 56, 57, 63-66, 70-72]. The most commonly applied internal validation method was cross-validation (n=8)[22, 63, 64], followed by split-sample (n=5)[56, 65, 70-72] and bootstrapping (n=3)[57, 66]. In order to analyse the data, either a logistic regression[22, 31, 53-55, 57-61, 63-68, 70-78] or a Cox proportional hazard regression[56] was used. Most models present their results by ORs with a 95% confidence interval. With a p-value of < 0.05, the results are considered as statistically significant[3]. A summary of characteristics of all included studies is provided in table 1 and 2.

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Table 1: Summary of study characteristics for all-cause 30-day UHR predictive models

Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
All-cause related UHRs									
Brittan et al. 2018[64], USA	Composite Score	All-cause	30-day UHRs	Retrospective/ 1 children's hospital	29,542 patients	0-21 years	2014 - 2015	4.0%	Development study/ Internal: Cross
Sills et al. 2017[68], USA	PACR + SDH	All-cause	30-day UHRs	Retrospective / PHIS database, US Census's American Community Survey data, 47 hospitals	458,686 index discharges	<18 years	2014	6.1%	Incremental value study/ Apparent
Ehwerhemuepha et al. 2018[65], USA	Unnamed	All-cause	30-day UHRs	Retrospective / US Census's American Community Survey data, 1 tertiary paediatric hospital	38,143 inpatient clinical encounters (DC: 19,072, VC: 19,071)	Between 28-days and 17 years	July 2013 to June 2017	10.4%	Development study/ Internal: Random-Split-Sample
	LACE (validation)				VC: 19,071 inpatient clinical encounters			NR	External validation study
Bradshaw et al. 2020[63], USA	HARRPS-tool	All-cause	30-day UHRs	Retrospective / 1 paediatric hospital	5,306 patients	<18 years	May 2017- June 2018	25.3%	Development study/ Internal: Cross
Zhou et al. 2020[61], Australia	Unnamed	All-cause	30-day UHRs	Retrospective/ Australian Census data, 1 tertiary paediatric hospital	73,132 patients	Age limit for admission: 15 years, special permissions by hospital executives possible	2010 - 2014	4.6%	Development study/ Apparent
Ehwerhemuepha et al. 2020[69], USA	LACE (validation)	All-cause	30-day UHRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters	< 18 years	2000- 2017	12.6% (DC)	External validation study
Zhou et al. 2021[22], Australia	Model 1: GLM	All-cause	30-day UHRs	Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data	940 patients	Different paediatric age groups[a]	2010-2014	4.55%[b]	Development study/ Internal: Cross
	Model 1: G-S								Development study/ Internal: Cross
	Model 2: GLM			Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records					Development study/ Internal: Cross
	Model 2: G-S								Development study/ Internal: Cross
	Model 3: GLM			Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records, written discharge documentation					Development study/ Internal: Cross
	Model 3: G-S								Development study/ Internal: Cross

Abbreviations: DC, derivation cohort; GLM, logistic regression; G-S, stepwise logistic regression; HARRPS, High Acuity Readmission Risk Pediatric Screen; HR, hospital readmission; NR, not reported; PACR, Pediatric All-Condition Readmission; PASS, PHIS, Paediatric Health Information Systems; SDH, social determinants of health; UHR, unplanned hospital readmission; VC, validation cohort

[a] Mean age (years): 5.2 with HR, 5.3 without HR.

[b] Based on 3330 patients from the initial data set.

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Table 2: Summary of study characteristics for surgical and general medical conditions related 30-day UHR predictive models

Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
Surgical conditions related UHRs									
Vo et al. 2018[57], USA	Unnamed	All surgical specialties without cardiac surgery	30-day unplanned post-surgical HRs relating to noncardiac surgery	Retrospective/ ACS NSQIP-P database	182,589 patients	< 18 Years	2012-2014	4.8%	Development study/ Internal: Bootstrap
Polites et al. 2017[56], USA	Unnamed	General and thoracic surgery	30-day UHRs related to the index surgical procedure	Retrospective/ ACS NSQIP-P database	54,870 patients (DC: 38,397, VC: 16473)	29 days - < 18 years	2012-2014	3.6%	Development study/ Internal: Random-Split-Sample
Delaplain et al. 2020[70], USA	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Retrospective / Cerner Health Facts database, 28 hospitals	82,532 patients (DC: 75%, VC: 25%)	< 18 years	2000-2017	8.8%	Development study/ Internal: Random-Split-Sample[a]
Chotai et al. 2017[67], USA	Unnamed	Neurosurgery	30-day UHRs following index surgery for neurosurgical diagnoses	Retrospective/ 1 paediatric hospital	536 children	< 18 years	Jan. 2012 - March 2015	11.9%	Development study/ Apparent
Davidson et al. 2021[73], USA	Unnamed	Ureteroscopy	30-day UHRs after ureteroscopy	Retrospective/ NSQIP-P database	2,510 patients	≤ 18 years	2015 - 2018	6.5%	Development study/ Apparent
Garcia et al. 2018[74], USA	Unnamed	Kasai procedure	30-day UHRs related to kasai procedure	Retrospective/ NSQIP-P database	190 children	< 1 year	2012 - 2015	15.3%	Development study/ Apparent
Lee et al. 2021[75], USA	Unnamed	Adolescent idiopathic scoliosis surgery	30-day UHRs after adolescent idiopathic scoliosis surgery	Retrospective/ Nationwide Readmissions Database	30,677 patients	10 - 18 years	2012 - 2015	2.9%	Development study/ Apparent
Minhas et al. 2016[58], USA	Idiopathic scoliosis Progressive infantile scoliosis Scoliosis due to other conditions	Spinal Surgeries (Scoliosis)	30-day UHRs	Retrospective/ NSQIP-P database	3,482 children	≤ 18 years	2012 - 2013	3.4%	Development study/ Apparent
Roddy & Diab 2017[59], USA	Unnamed	Spine fusion	30-day UHRs	Retrospective/ State Inpatient Database	13,287 patients	<21 years	2006-2010 (New York, Utah, Nebraska, Florida, North Carolina), 2006-2011 (California)	4.7%	Development study/ Apparent
Sherrod et al. 2016[77], USA	Unnamed	Neurosurgery	30-day UHRs after neurosurgery	Retrospective/ NSQIP-P database	9,799 cases	< 18 years	2012-2013	11.2%	Development study/ Apparent
Tahiri et al. 2015[60], USA	Unnamed	Plastic Surgery	30-day UHRs following paediatric plastic surgery procedures	Retrospective/ NSQIP database	5,376 patients	≤ 18 years	2012	2.4%	Development study/ Apparent
Wheeler et al. 2018[78], USA	Unnamed	Burn diagnosis	30-day UHRs	Retrospective/ Nationwide Readmissions Database	11,940 patients	1-17 years	Jan. - Nov. 2013, Jan. - Nov. 2014	2.7%	Development study/ Apparent
Vedantam et al. 2018[31], USA	Unnamed	Epilepsy surgery	30-day UHRs after epilepsy surgery	Retrospective/ NSQIP-P database	280 surgeries	≤ 18 years	2015	7.1%	Development study/ Apparent

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Reference	Model name	Medical condition	Model outcome	Study design/ data source	Sample size	Age group	Period of data collection	Readmission rate	Model type/ validation method
Basques et al. 2015[53], USA	Unnamed	Posterior spinal fusion	30-day UHRs after posterior spinal fusion	Retrospective/ NSQIP-P database	733 patients	11-18 years	2012	1.5%	Development study/ Apparent
Martin et al. 2015[54], USA	Unnamed	Spinal deformity surgery	30-day UHRs after spinal deformity surgery	Retrospective/ NSQIP-P database	1,890 patients	<18 years	2012	3.96%	Development study/ Apparent
General medical conditions related UHRs									
Learly et al. 2019[66], USA	Prediction at admission	Complex chronic conditions	30-day UHRs	Retrospective / US Census Bureau data, 1 academic medical centre	2,296 index admissions	6 months to 18 years	Oct. 2010 - July 2016	8.2%	Development study/ Internal: Bootstrap
	Prediction at discharge								Incremental value study/ Internal: Bootstrap
Ryan et al. 2021[62], USA	PASS (validation)	Asthma	30-day UHRs	Retrospective/ 1 university-affiliated, tertiary paediatric referral centre	328 patients	5-18 years	May 2015-Oct. 2017	3.0%	External validation study
O´Connell et al. 2021[72], USA	Unnamed	Nervous system condition	30-day UHRs	Retrospective / Cerner Health Facts database, 18 hospitals	105,834 index admissions (DC: 80%. VC: 20%)	<18 years	2000-2017	12.0%	Development study/ Internal: Random-Split-Sample
Hoenk et al. 2021[71], USA	Unnamed	Oncology	30-day UHRs	Retrospective / Cerner Health Facts database, 16 hospitals	10,418 patients (DC: 7,814, VC: 2,604)	<21 years	2000-2017	41.2%	Development study/ Internal: Random-Split-Sample
Sanchez-Luna et al. 2016[76], Spain	Unnamed	Acute bronchiolitis due to respiratory syncytial virus	30-day UHRs	Retrospective/ Spanish National Health Service records	63,948 discharges	< 1 year	2004 - 2012	7.5%	Development study/ Apparent
Sacks et al. 2017[55], USA	Unnamed	Cardiac conditions	30-day UHRs	Retrospective/ 1 academic children´s hospital	1,993 hospitalizations	0-12.9 years	2012-2014	20.5%	Development study/ Apparent

Abbreviations: ACS NSQIP-P, American College of Surgeons' (ACS) National Surgical Quality Improvement Program Pediatric; DC, derivation cohort; HR, hospital readmission; NR, not reported; PASS, Pediatric Asthma Severity Score; PHIS, Paediatric Health Information Systems; UHR, unplanned hospital readmission; VC, validation cohort

[a] Assumption for validation method: Odds ratios for 30-day unplanned HRs are displayed in a Table that is part of the DC from the 7-day unplanned HRs predictive model[70].

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Applicability and significant risk factors in predictive models

Based on the 28 predictive models with a development or incremental value design, 25 significant risk factors associated with 30-day UHRs were identified (see table 3). The most common risk factors were comorbidity (n=18), (postoperative) length of stay (n=10), illness severity (n=9) and principal procedures (n=9). The significant risk factors were inconsistently defined across predictive models, allowing a direct comparison only to a limited extent. ORs for comorbidity ranged from 1.01[72] to 10.08[58] across predictive models. A length of stay of >=15 days (OR=2.39)[61] and a postoperative length of stay of >4 days (Hazard Ratio=3.12)[56] were considered to be a major risk factor. For illness severity 'ICU stay'(OR=3.302)[67] and for principal procedures 'isolated primary anterior spinal fusion'(OR=7.65)[54] were one of the most pronounced risk factors, respectively. The risk factor with the highest OR value was 'any inpatient complication'(OR=180.44)[53]. For all-cause UHRs, UHRs related to surgical conditions and UHRs related to general medical conditions 14, 19 and twelve significant risk factors were found, respectively.

Most predictive models are potentially applicable at discharge (n=33), while four predictive models can be used at index admission[22, 63, 66, 76], based on the significant and examined variables (see table A6 in the supplementary material for an overview of variables and table 4 for an application description).

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Table 3: Significant risk factors for 30-day UHR predictive models with a development or incremental value design

Health condition group	All-cause (n=5[a])					Surgical conditions related (n=17)																	General medical conditions related (n=6)					
Reference	[64]	[68]	[65]	[63]	[61]	[57]	[56]	[70]	[67]	[73]	[74]	[75]	[58][b]	[58][c]	[58][d]	[59]	[77]	[60]	[78]	[31]	[53]	[54]	[66][e]	[66][f]	[72]	[71]	[76]	[55]
Location of residence[g]		x			x														x									
Health insurance				x	x											x												
Type of index hospital						x				x						x		x							x			
Living environment				x																								
Characteristics of primary care provider	x																											
Age at admission/operation					x																						x	x
Sex										x						x												
Race/Ethnicity		x							x								x								x			
Health service usage prior to index admission[h]			x	x				x									x						x	x	x	x		
Prematurity											x																x	
Comorbidity		x	x	x	x	x	x					x	x	x		x	x					x	x	x	x	x	x	x	x
Illness severity[i]			x	x			x		x								x		x				x	x	x				
LOS/Postop LOS			x		x		x	x				x				x			x					x	x	x			
Principal diagnoses			x					x								x									x	x			
Principal procedures							x						x		x	x	x		x	x		x			x				
Inpatient complications						x	x				x	x				x	x	x			x								
(Specific) medication at index admission								x																	x	x			
Length of operation							x										x	x											
Wound contamination before operation							x											x											
The ASA class						x									x			x				x							
Discharge on Friday or Weekend					x																								
Discharge disposition																x	x							x					
Discharge with increased medication/further treatment	x																												
Admission on Friday					x																								
Surgical location										x																			

Abbreviations: ASA class, American Society of Anesthesiologists class; LOS, length of stay; PICU, paediatric intensive care unit; postop LOS, postoperative length of stay

x=risk factor (Odds ratio/Hazard ratio >1)

[a] The six predictive models of Zhou et al. 2021 [22] are not included in this analysis due to missing information about odds ratios. See table A6 in the supplementary material for a list of included variables.

[b] Model for idiopathic scoliosis.

[c] Model for progressive infantile scoliosis.

[d] Model for scoliosis due to other conditions.

[e] Admission model.

[f] Discharge model.

[g] Social determinants of health are included (e.g., median household income).

[h] Risk factor category includes, for example, the number of previous emergency department visits or hospitalizations.

[i] The risk factor category also captures the urgency of the index admission. The risk factor category includes, for example, PICU or emergency department admission.

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Completeness in reporting and discriminative ability at model level

Information about TRIPOD adherence and performance at model level are provided in table 4. The median TRIPOD adherence of the models was 59% ($P_{25} - P_{75}$ 55% - 69%, average: 60%), ranging from 33%[69] to 81%[66]. Developed predictive models had a more favourable reporting quality in comparison to external validated models (i.e. 59% ($P_{25} - P_{75}$ 55% - 69%, average: 61%) compared with 44% ($P_{25} - P_{75}$ 39% - 50%, average: 44%), respectively). Two models with poor adherence in reporting were based on an external validation design, and the validation of these models was not the primary aim of the study[65, 69].

Table 4: Performance, application and TRIPOD adherence of 30-day UHR predictive models in paediatrics (n=37)

Reference	Model name	Performance		TRIPOD score	Potentially applicable…
		Discrimination (c-statistic)	Calibration		
All-cause related UHRs					
Brittan et al. 2018[64]	Composite Score	0.62		73.33%	at discharge
Sills et al. 2017[68]	PACR + SDH	0.708		64.71%	at discharge
Ehwerhemuepha et al. 2018[65]	Unnamed	VC: 0.79		63.33%	at discharge
	LACE (validation)	0.68		44.44%	at discharge
Bradshaw et al. 2020[63]	HARRPS-tool	Score: 0.65		73.33%	at admission
Zhou et al. 2020[61]	Unnamed	0.645		62.07%	at discharge
Ehwerhemuepha et al. 2020[69]	LACE (validation)	0.7014		33.33%	at discharge
Zhou et al. 2021[22]	Model 1: GLM	0.487		68.97%	at admission
	Model 1: G-S	0.477		68.97%	at discharge
	Model 2: GLM	0.585		68.97%	at discharge
	Model 2: G-S	0.593		68.97%	at discharge
	Model 3: GLM	0.609		68.97%	at discharge
	Model 3: G-S	0.617		68.97%	at discharge
Surgical conditions related UHRs					
Vo et al. 2018[57]	Unnamed	0.747	slope: 1; intercept: 0.002	68.97%	at discharge
Polites et al. 2017[56]	Unnamed	DC: 0.71; VC: 0.701	(DC): p = 0.95, O/E-Ratio= 1.03; (VC): p = 0.36, O/E-Ratio= 1.07	62.07%	at discharge
Delaplain et al. 2020[70]	30-day readmission model	VC: 0.799		51.72%	at discharge
Chotai et al. 2017[67]	Unnamed	0.72		42.86%	at discharge
Davidson et al. 2021[73]	Unnamed	0.73	H&L χ^2: 7.5 (p=0.4474)	58.62%	at discharge
Garcia et al. 2018[74]	Unnamed	0.703		51.72%	at discharge

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		Performance			
Lee et al. 2021[75]	Unnamed	0.712	H&L: 0.0974	58.62%	at discharge
Minhas et al. 2016[58]	Idiopathic scoliosis	0.760-0.769		55.17%	at discharge[a]
	Progressive infantile scoliosis			55.17%	at discharge[a]
	Scoliosis due to other conditions			55.17%	at discharge[a]
Roddy & Diab 2017[59]	Unnamed	0.75	H&L (p-value): 0.46	55.17%	at discharge
Sherrod et al. 2016[77]	Unnamed	0.759		55.17%	at discharge
Tahiri et al. 2015[60]	Unnamed	0.784		55.17%	at discharge
Wheeler et al. 2018[78]	Unnamed	0.72		55.17%	at discharge
Vedantam et al. 2018[31]	Unnamed	0.71	H&L (p-value): 0.94	41.38%	at discharge
Basques et al. 2015[53]	Unnamed	0.87	H&L: value not reported[b]	68.97%	at discharge
Martin et al. 2015[54]	Unnamed	0.77		62.07%	at discharge
General medical conditions related UHRs					
Learly et al. 2019[66]	Prediction at admission	0.65; Score: 0.65	Calibration plot	79.31%	at admission
	Prediction at discharge	0.67; Score: 0.67	Calibration plot	81.25%	at discharge
Ryan et al. 2021[62]	PASS (validation)	0.28		55.17%	at discharge
O´Connell et al. 2021[72]	Unnamed	VC: 0.733		51.72%	at discharge
Hoenk et al. 2021[71]	Unnamed	VC: 0.714		55.17%	at discharge
Sanchez-Luna et al. 2016[76]	Unnamed	0.611		56.67%	at admission
Sacks et al. 2017[55]	Unnamed	0.75		58.62%	at discharge

Abbreviations: DC, derivation cohort; GLM, logistic regression; G-S, stepwise logistic regression; HARRPS, High Acuity Readmission Risk Pediatric Screen; NR, not reported; PACR, Pediatric All-Condition Readmission; PASS, Pediatric Asthma Severity Score; SDH, social determinants of health; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; UHR, unplanned hospital readmission; VC, validation cohort

[a]Assumption for applicability based on variables included in the univariable analysis.

[b]H&L shows 'no evidence of a lack of fit'([53] p. 290).

Including all 37 items, the overall median adherence per TRIPOD item across models was 65% ($P_{25} - P_{75}$ 32% - 92%, average: 57%), ranging from 0% to 100% (see table A7 in the supplementary material for a detailed description by model type). The overall adherence per TRIPOD item is illustrated in figure 1.

[…Placement of figure 1…]

Figure 1: Overall adherence per TRIPOD item across all included predictive models (n=37).

14% of the models reported the title (item 1) completely, while 19%[62-66, 68] of the models mentioned the predictive model type in this context. 3% of the models had a completed abstract (item 2). The detailed predictors definition (item 7a) was fulfilled for more models

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(95%), in contrast to outcome definition (item 6a) (reported in 70%). The handling of predictors in the analysis (item 10a) showed incomplete reporting in 82% of the models. In addition, the handling (item 9, reported in 35%) and reporting of missing values (part of item 13b, reported in 32%) was not addressed in many models. Just 9% of the models displayed complete reporting of the model-building procedure (item 10b), as the majority of the models (91%) did not address the testing of interaction terms[22, 31, 53-61, 64-68, 70, 72-75, 77, 78]. The description (item 10d) and reporting of performance measures (item 16) was incomplete in 68% and 89% of the models. Just 24% of the models address results of calibration measures (cf. table 4). No model presented the full predictive model (item 15a) by providing an example of an intercept. An explanation for using the prediction model (item 15b, e.g., by a simplified scoring rule) was presented in 21% of the models. One model provided detailed information about a simplified scoring rule (item 15b) in supplementary material[66].

The discriminative ability (c-statistic) of the models ranged from 0.28[62] to 0.87[53]. 14 out of 37 predictive models had a c-statistic <0.7. The linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.241 (P=.15). Models with good discriminative ability (c-statistic >0.7)[31, 53-60, 65, 67-75, 77, 78] are primary applicable at discharge and have a TRIPOD score ranging from 41%[31] to 69%[57]. The two models with the highest reporting quality (79%, 81%) are applicable for predicting 30-day UHRs of children with complex chronic conditions. The c-statistic of these models was 0.65[66] and 0.67[66], respectively (see figure A2 in the supplementary material for a illustration of the models´ performance and TRIPOD adherence).

DISCUSSION

Based on 28 studies, this systematic review identifies 37 predictive models that could potentially be used for determining individual 30-day UHR risk in paediatrics. According to the models, the four most common significant risk factors in predictive models were comorbidity, (postoperative) length of stay, illness severity and principal procedures. 23 validated predictive models have a c-statistic of >0.7. The median TRIPOD adherence of the predictive models included was 59% ($P_{25} - P_{75}$ 55% - 69%), ranging from 33% to 81%, which is similar to that of other systematic reviews[12, 27].

Practical clinical and policy implications

In general, reporting quality and discriminative ability can provide crucial information about the strengths and weaknesses of a predictive model for implementation in practice (see figure

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A2 in the supplementary material for a combined illustration). However, the results from this systematic review revealed considerable differences in the c-statistics (0.28[62] to 0.87[53]) and in the TRIPOD scores (33%[69] to 81%[66]) at model level. When considering the available information about reporting quality and discriminative ability in relation to each other, it should be noted that the linear correlation between c-statistic and TRIPOD score at model level was not statistically significant, r=-0.241 (P=.15). Therefore, an independent evaluation of both aspects for the selection of an appropriate predictive model is recommended.

Clinicians and decision-makers should use predictive models with good discriminative ability (i.e. c-statistic above 0.7) and sufficient data availability. Especially predictive models that are based on census data[61, 65, 66, 68] or manual data entry (e.g. written discharge documentation[22]) may be more difficult to implement than models relying on centralized databases[31, 53, 54, 56-60, 69-78]. The TRIPOD score at predictive model level (see table 4) can be used as a first indicator if the predictive model can be assessed and implemented with the given information.

Similar to a previous systematic review[3], comorbidity and (postoperative) length of stay were identified as consistently cited risk factors across the included studies. In addition, illness severity was one main risk factor among all three health condition groups. For surgical conditions related UHR, the principal procedure has shown to be crucial as a risk factor. The practical application of risk factors should be made with caution because risk factors are often inconsistently defined across studies. Therefore, knowledge about study-related predictor definitions is required before application.

Limitations

This systematic review has certain limitations. (i) The studies included needed be to published in English or German with full-text access. (ii) Summarizing the results of the included studies quantitatively was not possible due to the heterogeneity of the predictive models (resulting from differences in sample sizes, the examined variables or variations in the periods of data collection). (iii) The sample size of the included studies were reported in different units (e.g., encounters, discharges), impeding the comparisons of UHR rates. (iv) Our assignment of the predictive models that are potentially applicable at discharge assumes that the required variables are available at the time point. If clinicians and other stakeholders decide to use a predictive model, it should be checked beforehand whether complete data

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collection is possible at the desired time. (v) In addition to the identified medical risk factors (e.g., comorbidity) and several country-specific risk factors (e.g., location of residence) that result in paediatric readmissions, health-policy initiatives may also affect the readmission rates in paediatric clinical practice[79]. However, due to a lack of data, these aspects could not be captured by this review.

Future research

This systematic review did not identify predictive models for individualized risk prediction of potentially preventable UHRs in paediatrics, emphasizing past discussions to expand the research field further[3].

Current external validation studies were conducted in the USA and examined the applicability of existing predictive models with other outcomes or population backgrounds to paediatric 30-day UHRs[62, 65, 69]. Therefore, external validation studies are needed for those models that are explicitly developed to predict 30-day UHRs in paediatrics. Because the number of predictive models related to medical conditions-related UHRs was small (n=7)[55, 62, 66, 71, 72, 76], with four out of seven models demonstrating a c-statistic below 0.7[62, 66, 76], there is a need for high quality models in this area.

Non-regression-based techniques (e.g., machine-learning) are an increasing field in order to predict 30-day HRs in paediatrics, most of which show good discriminative ability[21, 22, 47, 49-52, 69] (see table A4 in the supplementary material). Future systematic reviews should summarize and critically assess existing non-regression-based HR predictive models in paediatrics, for instance by applying the TRIPOD-ML statement that is going to be published[80].

Existing studies discuss the benefit of shorter time intervals in order to identify preventable readmissions more accurately[6, 81]; one study concluded that a 30-day UHR metric was more precise (c-statistic=0.799) for paediatric trauma patients than a 7-day UHR metric(c-statistic=0.737)[70]. To our knowledge, there is one predictive model for 365-day[7], three for 90-day[59, 67, 75] and one for 7-day[70] UHRs in paediatrics with good discriminative ability (c-statistic>0.7). Future studies should address the evaluation of paediatric UHR predictive models with different time intervals.

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CONCLUSION

This systematic review revealed an increase in the development of predictive models for 30-day UHRs in paediatrics in recent years. To support the implementation of the predictive models in the long term, it is essential to validate existing models in order to test their applicability in different settings. To increase accessibility for use, more attention should be given on completeness in reporting, particularly for items that may be relevant for the implementation of paediatric 30-day UHR predictive models in practice (i.e., those relating to outcome and predictor definitions, handling of missing values, full predictive model presentation, and an explanation for its use).

List of Figures

Figure 1: Overall adherence per TRIPOD item across all included predictive models (n=37)

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Contributors:

IN conceptualized and designed the systematic review, participated in the literature search, study selection, quality assessment, data extraction, data analyses and drafted the initial manuscript.

NK contributed to the literature search, study selection, quality assessment, data extraction and critically reviewed the manuscript.

SS contributed to the data analysis and critically reviewed the manuscript.

JD contributed to the study selection, data extraction, data analysis and critically reviewed the manuscript.

DM conceptualized and designed the systematic review, participated in the study selection, quality assessment, data extraction, data analyses and critically reviewed the manuscript.

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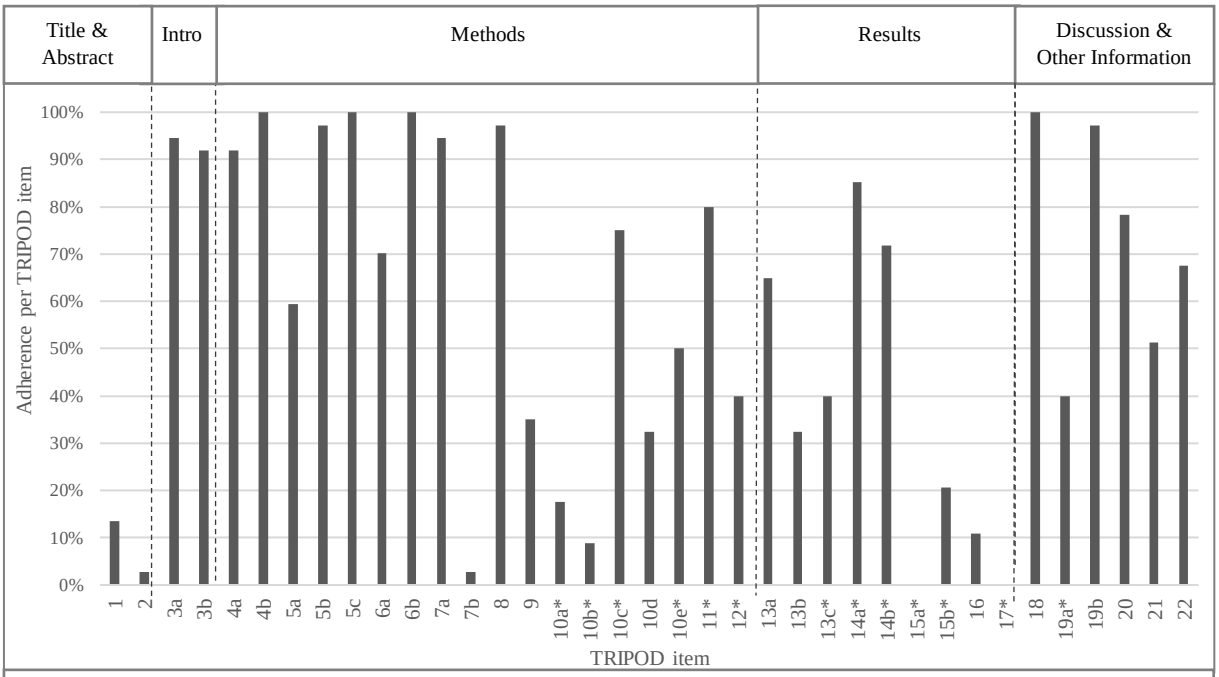
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Percentages relate to the number of models for which an item was applicable (in this case, the respective item should have been reported). *Indication of derivation from the total number of models for which a TRIPOD item was applicable (N=# of models for which the TRIPOD item is applicable): 10a (N=34), 10b (N=34), 10c (N=4), 10e (N=2), 11 (N=5), 12 (N=5), 13c (N=5), 14a (N=34), 14b (N=32), 15a (N=34), 15b (N=34), 17 (N=1), 19a (N=5).

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Supplemental Material

Table A1: Search strategy for PubMed

#	PubMed (Search date: 01st January 2021; Updated Search date: 07th October 2021)	Results from initial search	Results from updated search
1	"Patient Readmission"[Mesh]	18,336	20,199
2	rehospitali*[Title/Abstract]	8,017	8,645
3	readmission*[Title/Abstract]	31,790	35,343
4	(hospital[Title/Abstract]) AND readmission*[Title/Abstract]	21,144	23,398
5	(unplanned[Title/Abstract]) AND readmission*[Title/Abstract]	2,360	2,677
6	(patient[Title/Abstract]) AND readmi*[Title/Abstract]	16,395	18,106
7	readmit*[Title/Abstract]	8,289	8,926
8	re-admission*[Title/Abstract]	2,205	2,416
9	(repeat*[Title/Abstract]) AND hospital*[Title/Abstract]	29,556	31,364
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	75,738	81,968
11	predictive factor*[Title/Abstract]	28,124	30,391
12	predict*[Title/Abstract]	1,646,322	1,765,658
13	"Predictive Value of Tests"[Mesh]	207,290	215,023
14	"ROC Curve"[Mesh]	60,417	65,077
15	model*[Title/Abstract]	3,025,019	3,240,665
16	c-statistic*[Title/Abstract]	6,938	7,765
17	ROC*[Title/Abstract]	55,019	62,608
18	"Sensitivity and Specificity"[Mesh]	595,955	619,284
19	Sensitivity [Title/Abstract]	827,733	873,715
20	Specificity [Title/Abstract]	478,451	502,627
21	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20	5,225,692	5,555,402
22	"Child"[Mesh]	1,936,577	2,010,900
23	"Infant"[Mesh]	1,152,685	1,189,047
24	"Adolescent"[Mesh]	2,058,561	2,216,512
25	(((((adolescen*[Title/Abstract]) OR teen*[Title/Abstract]) OR youth[Title/Abstract]) OR juvenile*[Title/Abstract]) OR young person*[Title/Abstract]) OR young people*[Title/Abstract]	464,723	492,107
26	"Pediatrics"[Mesh]	59,136	61,129
27	((((((((((child*[Title/Abstract]) OR (infant*[Title/Abstract])) OR (toddler*[Title/Abstract])) OR (bab*[Title/Abstract])) OR (newborn*[Title/Abstract])) OR (neonat*[Title/Abstract])) OR (school age*[Title/Abstract])) OR (preschool)) OR (paediatric*[Title/Abstract])) OR (pediatric*[Title/Abstract])) OR (kid*[Title/Abstract])) OR (boy*[Title/Abstract])) OR (girl*[Title/Abstract])	2,246,452	2,343,927
28	22 OR 23 OR 24 OR 25 OR 26 OR 27	4,365,971	4,517,928
29	10 AND 21 AND 28	3,674	3,993

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Table A2: Search strategy for CINAHL

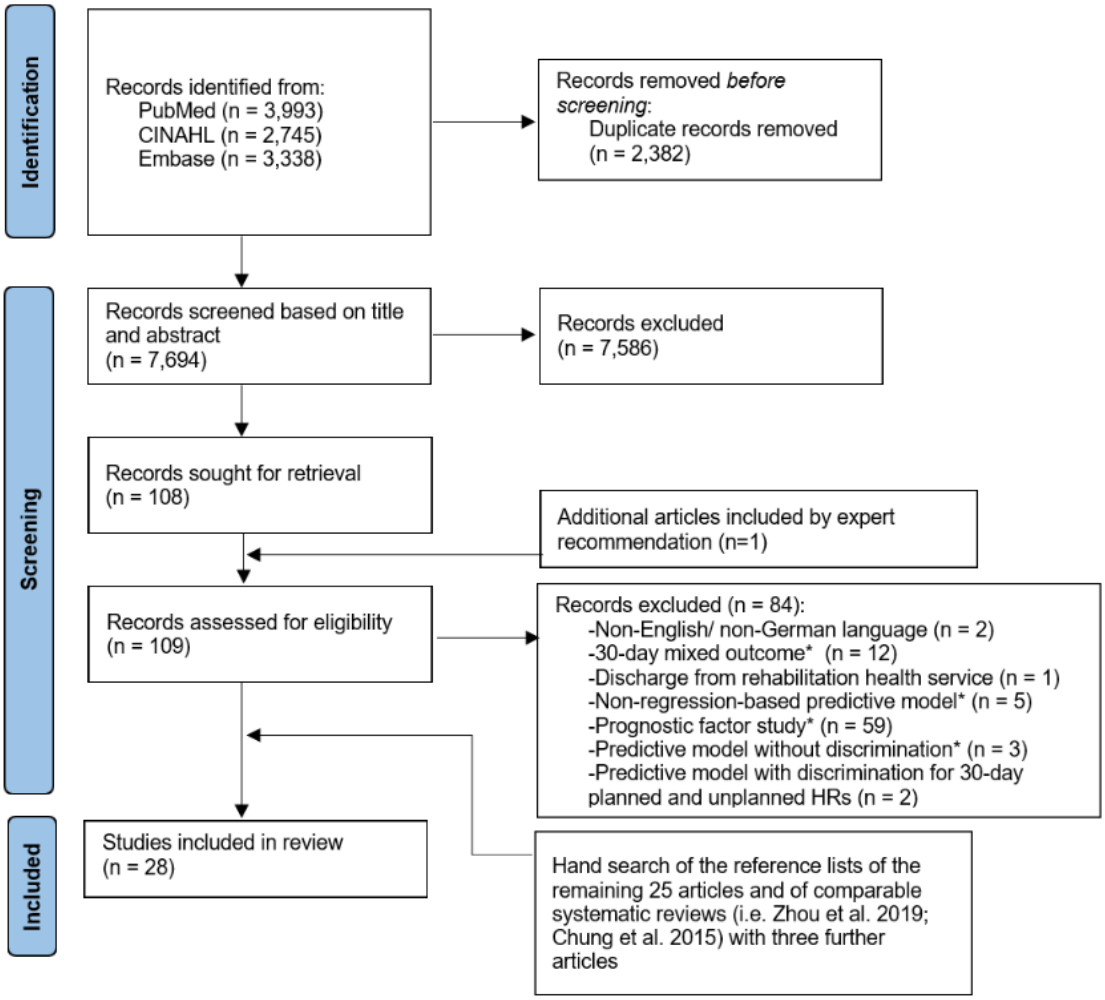
	CINAHL (Initial Search date: 01st January 2021; Updated Search date: 07th October 2021)	Results from initial search	Results from updated Search
1	(MH "Readmission")	14,256	15,177
2	rehospitali*	2,624	2,779
3	readmission*	21,640	23,205
4	hospital AND readmission*	13,210	14,167
5	unplanned AND readmission*	1,227	1,363
6	patient AND readmi*	17,823	19,163
7	readmit*	2,588	2,791
8	re-admission*	954	1,029
9	repeat* AND hospital*	12,404	12,997
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	36,195	38,498
11	predictive factor*	10,824	11,733
12	predict*	394,485	423,169
13	(MH "Predictive Value of Tests")	52,176	54,193
14	(MH "ROC Curve")	27,863	29,859
15	model*	626,177	669,950
16	c-statistic*	2,593	2,822
17	ROC*	45,218	74,438
18	(MH "Sensitivity and Specificity")	85,260	87,853
19	sensitiv*	236,551	247,288
20	specific*	481,504	511,797
21	(MH "Predictive Validity")	5,587	5,746
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	1,337,979	1,442,565
23	(MH "Child")	467,217	486,202
24	(MH "Infant+")	260,190	268,903
25	(MH "Adolescence+")	535,922	557,353
26	adolescen* OR teen* OR youth OR juvenile* OR young person* OR young people*	598,199	624,571
27	(MH "Pediatrics+")	21,316	21,917
28	child* OR infant* OR toddler* OR bab* OR newborn* OR neonat* OR school age* OR preschool OR paediatric* OR pediatric* OR kid* OR boy* OR girl*	1,150,913	1,264,003
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	1,413,615	1,514,718
30	10 AND 22 AND 29	2,459	2745

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Table A3: Search strategy for Embase

	Embase (Initial Search date: 01st January 2021*; Updated Search dates: 7th June 2021, 07th October 2021)	Results from updated search 7th June 2021	Results from updated search 07th October 2021
1	'hospital readmission'/exp	73,736	76,850
2	rehospitali*:ab,ti	10,922	15,528
3	readmission*:ab,ti	57,908	64,311
4	(hospital NEAR/10 readmission*):ab,ti	22,791	25,140
5	(unplanned NEAR/10 readmission*):ab,ti	3,446	3,749
6	(patient NEAR/5 readmi*):ab,ti	5,568	6,357
7	readmit*:ab,ti	16,262	18,777
8	're admission*':ab,ti	5,802	5,982
9	(repeat* NEAR/5 hospital*):ab,ti	5,341	5,466
10	1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8 OR 9	100,580	107,753
11	'predictive factor*':ab,ti	47,721	49,181
12	predict*:ab,ti	2,301,090	2,367,382
13	'predictive value'/exp	191,705	199,896
14	'roc curve'/exp OR 'receiver operating characteristic'/exp	149,025	157,486
15	'model'/exp	3,201,228	3,267,314
16	'c statistic*':ab,ti	12,892	13,511
17	roc*:ab,ti	196,628	207,024
18	'sensitivity and specificity'/exp	391,861	405,843
19	'sensitivity':ab,ti	1,109,211	1,135,520
20	'specificity':ab,ti	636,321	650,496
21	'predictive validity'/exp	8,691	8,839
22	11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21	6,371,499	6,524,117
23	'child'/exp	3,031,530	3,082,756
24	'infant'/exp	1,180,824	1,197,926
25	'adolescent'/exp	1,741,216	1,770,557
26	adolescen*:ab,ti OR teen*:ab,ti OR youth:ab,ti OR 'juvenile*':ab,ti OR 'young person*':ab,ti OR 'young people':ab,ti	598,758	613,037
27	'pediatrics'/exp	123,610	125,641
28	'child*':ti,ab OR 'infant*':ti,ab OR 'toddler*':ti,ab OR bab*:ti,ab OR 'newborn*':ti,ab OR neonat*:ti,ab OR 'school age*':ti,ab OR 'preschool':ti,ab OR 'pediatric*':ti,ab OR kid*:ti,ab OR 'boy*':ti,ab OR 'girl*'	3,642,407	3,715,357
29	23 OR 24 OR 25 OR 26 OR 27 OR 28	5,498,512	5,594,597
30	#10 AND #22 AND #29	2,902**	3,338
* Final result from initial search: 2,845 ** Search term: #10 AND #22 AND #29 AND [1-1-1966]/sd NOT [2-1-2021]/sd			

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Abbreviations: HRs, hospital readmissions; PRISMA, preferred reporting items for systematic reviews and meta-analyses.

* The study outcome definition for hospital readmissions was (i) 30-day unplanned hospital readmission or (ii) 30-day planned and unplanned hospital readmission.

Figure A1: Flowchart for the search and study selection process (PRISMA)

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Table A4: Summary of study characteristics for excluded predictive models relating to machine learning studies (n=8), predictive model studies without discrimination (n=3) and predictive model studies with discrimination for 30-day planned and unplanned HRs (n=2).

Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
All-cause related HRs								
Desai et al. 2014, USA	PM without D	All-cause	30 day unplanned HRs	Prospective/ Survey data (PedsQL Infant Scales instrument or PedsQL 4.0 Generic Core Scales instrument)	19,139 eligible patients/ NR	1 month - 18 years	1. Oct. 2011 - 31. Dec. 2013	-
Santos et al. 2020, Brazil	PM without D	All-cause	30 day HRs	Prospective/ 1 public hospital, StrongKids data (nutritional screening)	641 patients/ 2%	1 month - 17 years	2014 - 2018	-
Jovanovic et al. 2016, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	66,994 patients/ 17%	general paediatric population	2009-2011	0.783 (Lasso), 0.779 (Tree-Lasso)
Stiglic et al. 2015, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	61,111 discharge records/ 18%	≤ 10 years	2009-2011	0.750-0.771
Stiglic et al. 2014, USA	ML-PM	All-cause	30 day HRs	Retrospective/ State Inpatient Databases, Healthcare Cost and Utilization Project	66,994 discharge records/ 17%	≤ 10 years	2009-2011	0.787 (distributed model), 0.789 (elastic net model)
Taylor et al. 2020, USA	ML-PM	All-cause	30 day unplanned HRs	Retrospective/ PHIS database	1,111,323 children/ 4.4%	< 18 years	2016-2017	0.811
Zhou et al. 2021, Australia*	ML-PM	All-cause	30 day unplanned HRs	Retrospective matched case-control / 1 tertiary paediatric facility, administrative inpatient data, medical records, written discharge documentation	940 patients/ 4.55%	different paediatric age groups	2010 - 2014	Model 1: 0.519 (random forest), 0.5 (elastic net), 0.509 (gradient bossted tree); Model 2: 0.603 (random forest), 0.616 (elastic net), 0.624 (gradient bossted tree); Model 3: 0.642 (random forest), 0.635 (elastic net), 0.654 (gradient bossted tree)
Ehwerhemuepha et al. 2020, USA*	ML-PM	All-cause	30 day (unplanned) HRs	Retrospective/ Cerner Health Facts Database, 48 hospitals	1.4 million encounters/ 12.6% (DC)	< 18 years	2000-2017	M1 (single-center): 0.8226, M2 (single-center): 0.8756, M3 (multi-center): 0.8451

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Reference	Model type	Medical condition	Model Outcome	Study design/ data source	Sample size/ readmission rate	Age group	Period of data collection	Discrimination (c-statistic)
Wolff et al. 2019, Chile	ML-PM	All-cause	30 day HRs	Retrospective/ administrative data, 1 paediatric hospital	56,558 admissions/ 3.7%	general paediatric population	July 2011 - Oct. 2017	0.47 - 0.65
Surgical conditions related HRs								
Janjua et al. 2019, USA	ML-PM without D	Spinal cord tumor surgery	30 day HRs	Retrospective/ Nationwide Readmissions Database	397 patients/ 10.8%	≤ 20 years	2010-2015	-
Jiang et al. 2018, USA	PM with D	Urological surgery	30-day HRs after paediatric urological procedures	Retrospective/ Nationwide Readmissions Database/ State Inpatient Databases	Nationwide Readmissions Database: 8,006 patients/ NR; State Inpatient Databases: 6,236 patients/ NR	<18 years	Nationwide Readmissions Database: 2013; State Inpatient Databases: 2007-2010	Nationwide Readmissions Database: 0.63 (CCI), 0.54 (VWI), 0.58 (Rhee index); State Inpatient Databases: 0.63 (CCI), 0.54 (VWI), 0.56 (Rhee index)
Smith et al. 2015, USA	PM with D	Cardiac surgery	30-day HRs following congenital heart surgery	Retrospective/ PHIS database, US census data, 43 not-for-profit, tertiary care paediatric hospitals	17,871 discharges (DC: 9,104, VC: 8,767)/ 11.0% (DC), NR (VC)	< 18 years	2011 (DC), 2012 (VC)	0.68 (DC), 0.68 (VC)

Abbreviations: CCI, Charlson Comorbidity Index; D, discrimination; DC, derivation cohort; HR, hospital readmission; ML, machine learning; NR, not reported; PHIS, Paediatric Health Information Systems; PM, predictive model; VC, validation cohort; VWI, Van Walraven Index

* This study contains one or more predictive models that were included in the systematic review.

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Table A5: Risk of bias assessment at study level

Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Vo et al. 2018	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - No detailed description of statistical analysis for the simplified scoring rule; Unadjusted associations for "Complications" are missing	**1**
Polites et al. 2017	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data (i.e. facility type), but all hospitals are from the USA	- Moderate - 38.5% missing values for the ICD-9-diagnosis variable used for determining the reasons for readmission	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Brittan et al. 2018	- Moderate - Potential limitation of generalizability as just one paediatric hospital was examined; Only basic information about "age"	- Moderate - About 75% of patients were missing homecare values in the original data (the study assumption is that this patients did not received home health)	- Moderate - Assumption that 75% of patients did not receive home health; Possible omission or error of clinical documentation	- Moderate - no clear description for the determination of unplanned hospital readmissions	- Low - just PEs of the composite score are considered	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Sills et al. 2017	- Low -	- Moderate - No flow diagram	- Moderate - Area-level SDH data were not created for research and are a biased proxy for individual-level SDH measures; Handling of continuous predictors relating to ZIP-Codes (i.e. linear or non-linear) not clear described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2018	- Moderate - Potential limitation of generalizability as just one hospital was examined; No detailed information about the source of ZIP-related variables; No information about characteristics of patients in the validation cohort (Random-Split-Sample)	- Low -	- Moderate - No detailed description for the handling of continuous predictors and method for categorized predictors	- Low - The outcome was determined automatically and not validated by manual chart review	- Low -	- Moderate - No detailed definition for LACE variables; Missing information for the validation cohort	**2**
Learly et al. 2019	- Moderate - Potential limitation of generalizability as just one hospital examined; No detailed information about the source of ZIP-related variables	- Low -	- Moderate - Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Low -	**2**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Bradshaw et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; Due to deidentification of patient data demographic characteristics are not captured	- Low -	- Low -	- Moderate - "The multiple categories that comprise the HARRPS-tool have different weight values associated that help prevent planned readmissions from skewing the results."(p. 53)	- Low -	- Low -	**1**
Delaplain et al. 2020	- Moderate - Focus on baseline characteristics of readmission in 7 days; No detailed characteristics for the validation cohort (random-split sample)	- Moderate - Information about the patient flow are provided at hospital level; Hospitals were excluded, if data were not collected for all variables related to the study; Missing information for validation cohort (random-split sample)	- Low - Predictors are presented by focusing on 7-day readmissions	- Moderate - Clear definition for 7-day unplanned readmissions with possible transfer to 30-day unplanned readmissions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**
Zhou et al. 2020	- Moderate - Potential limitation of generalizability as just one hospital was examined; One-day procedures are also included (discharge within 24h)	- Low -	- Moderate - Census data might be a biased proxy for individual level measures; Method to choose the cut-points was not clearly described	- Low -	- Low -	- Low -	**1**
Ehwerhemuepha et al. 2020	- Moderate - Basic information just for the training data set	- Moderate - Missing information for test data set	- High - No summary information for all predictors included in the validated model; No clear definition of predictors	- Moderate – It is unclear, if the definition captures readmissions to the	- Low - Due to external validation of LACE index confounders are not relevant	- High - Applied method for LACE is not clear (i.e. possible updating); Missing information for training data set	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
				same or also to other institutions			
Hoenk et al. 2021	- Low -	- Moderate - Information about the patient flow are based on hospital level; Hospitals were excluded if data were not collected for all variables related to the study and if there were not at least 500 encounters for a neoplastic condition; Missing information for test data set	- Moderate - Method to choose the cut-points was not clearly presented	- Moderate – It is unclear, if the definition captures readmissions to the same or also to other institutions	- Low -	- Moderate - Missing information for validation cohort (random-split sample)	**1**
Zhou et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Moderate – 470 instead of 550 patient pairs because of the burden associated with extracting data from medical records	- Low -	- Low -	- Low-	- Moderate - No detailed final model presentation (i.e. missing presentation of odds ratios in multivariable analysis)	**6**
Ryan et al. 2021	- Moderate - Potential limitation of generalizability as just one hospital was examined	- Low -	- Low -	- Moderate - It is unclear, how unplanned readmissions are determined	- Low - Due to external validation of PASS confounders are not relevant	- Moderate - Summary information of predictors are not clearly reported	**1**
O´Connell et al. 2021	- Low -	- Moderate - Information about the patient flow are based on hospital level; Hospitals were excluded if data were not collected for all variables related to the study and if there were	- Moderate - No summary information for all predictors included in the multivariable model; Method to choose the cut-points was not clearly presented	- Low -	- Low -	- Moderate - No clear reporting regarding the differentiation between training and test data set (i.e. summary information for predictors)	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
		not at least 1000 encounters for a nervous system condition; No clear information on differentiation between training and test data set					
Chotai et al. 2017	- Moderate - Potential limitation of generalizability as just one hospital examined; Focus on baseline characteristics of readmission in 90 days	- Low -	- Low -	- Moderate - Detailed description for the determination of unplanned readmission is missing	- Low -	- Moderate - Apparent validation with small sample size	**1**
Davidson et al. 2021	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Garcia et al. 2018	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - Apparent validation with small sample size	**1**
Lee et al. 2021	- Moderate – Possible selection bias due to exclusion of patients which were admitted within the last quarter of the year	- Moderate - No detailed information about the patient flow	- Low -	- Low -	- Low -	- Low -	**1**
Minhas et al. 2016	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - No complete reporting of the model development (e.g. uni- and multivariable analysis)	**3**
Roddy & Diab 2017	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Moderate - No detailed information about the patient flow	- Low -	- Moderate - Detailed description for the determination of unplanned readmission is missing	- Low -	- Low -	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Sherrod et al. 2016	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Tahiri et al 2015	- Low - surgery was conducted in inpatient or outpatient setting; Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Wheeler et al. 2018	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Moderate – Patients which were admitted in December were excluded.	- Low -	- Low -	- Low -	- Moderate – Incomplete reporting of statistical analysis (e.g. p-values)	**1**
Vedantam et al. 2018	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate - Apparent validation with small sample size	**1**
Basques et al. 2015	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Moderate – Incomplete reporting of statistical analysis (e.g. univariable analysis, control variables)	**1**
Martin et al. 2015	- Low - Deidentification of hospital-related data, but all hospitals are from the USA	- Low -	- Low -	- Moderate - varied lengths at which patients were at risk for readmission	- Low -	- Low -	**1**
Sanchez-Luna et al. 2016	- Low -	- Moderate - No information about the "gender" characteristics of the study sample	- Low -	- Moderate - Detailed description for the determination of "non-staged" readmission is missing	- Low -	- Moderate – No detailed presentation of results	**1**

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Reference	1. Domain: Study participation	2. Domain: Study attrition	3. Domain: Prognostic factor measurement	4. Domain: Outcome measurement	5. Domain: Study confounding	6. Domain: Statistical analysis and reporting	# predictive models considered*
	'Did the study population adequately represent the population of interest?'	'Did the study data available adequately represent the study sample?'	'Were the prognostic factors measured in a similar way for all participants?'	'Was outcome of interest measured in a similar way for all participants?'	'Were important potential confounding factors appropriately accounted for?'	'Was the statistical analysis appropriate?' and 'Were all primary outcomes reported?'	
Sacks et al. 2017	- Moderate - Potential limitation of generalizability as just one hospital examined	- Moderate – study just included patients, which are living close to the hospital	- Low -	- Moderate - Detailed description for the determination of unplanned readmission is missing	- Low -	- Low -	**1**

* For the risk of bias assessment at study-level, only the predictive models that were also included in the systematic review were considered.

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Table A6: Examined and significant variables for all 37 included predictive models

Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
All-cause related readmissions					
Brittan et al. 2018	Composite Score	All-cause	30-day unplanned HRs	Age; Lenght of stay; Non-English speaking caregiver; Discharge medications; Enteral feeding (eg, gastrostomy tube feeding); Respiratory (eg, home ventilator); IV infusion (eg, infusion of IV medication); Speech therapy; Physical therapy; Occupational therapy; Skilled nursing home visits; Private duty nursing home visits; CNA nursing assistant home visits; Durable medical equipment (eg, wheel chair); Home Care ≥ 1 order; Composite score = 0; Composite score = 1; Composite score = 2; Composite score = 3	Score based on the variables: Non-English speaking caregiver, Discharge medications, Home Care ≥ 1 order: Composite score (0 vs 1) (OR=1.7; 95% CI 1.5-2); Composite score (0 vs ≥ 2) (OR=4.2; 95% CI 3.6-4.9)
Sills et al. 2017	PACR + SDH	All-cause	30-day unplanned HRs	CCI count; Infectious and parasitic disease; Neoplasms; Endocrine, nutritional, and metabolic diseases and immunity disorders; Diseases of blood and blood-forming organs; Mental disorders; Diseases of the nervous system and sense organs; Diseases of the circulatory system; Diseases of the respiratory system; Diseases of the digestive system; Diseases of the genitourinary system; Diseases of the skin and subcutaneous tissue; Diseases of the musculoskeletal system; Congenital anomalies; Certain conditions originating in the perinatal period; Symptoms, signs, and ill-defined conditions; Injury and poisoning; Factors influencing health status and contact with health services; Age group; Sex; Race; Hispanic ethnicity; Payer; Median household income; Proportion of housing units vacant; Proportion of households with children that are single-parent; Proportion of families below poverty level; Unemployment rate; Proportion of adults with less than a high school diploma or equivalent	CCI count: [2 body systems (OR=1.51; 95% CI 1.44-1.58), 3 body systems (OR=1.57; 95% CI 1.48-1.67), >3 body systems (OR=1.31; 95% CI 1.19-1.44)]; CCIs: Neoplasms (OR=0.46; 95% CI 0.44-0.48), Endocrine, nutritional, and metabolic diseases and immunity disorders (OR=0.8; 95% CI 0.77-0.83), Diseases of blood and blood-forming organs (OR=0.59; 95% CI 0.56-0.61), Mental disorders (OR=0.95; 95% CI 0.92-0.99), Diseases of the nervous system and sense organs (OR=0.78; 95% CI 0.75-0.8), Diseases of the circulatory system (OR=0.73; 95% CI 0.7-0.76), Diseases of the respiratory system (OR=1.08; 95% CI 1.05-1.12), Diseases of the digestive system (OR=0.65; 95% CI 0.63-0.68), Diseases of the genitourinary system (OR=0.7; 95% CI 0.66-0.74), Congenital anomalies (OR=0.88; 95% CI 0.85-0.91), Injury and poisoning (OR=0.63; 95% CI 0.54-0.74), Factors influencing health status and contact with health services (OR=0.51; 95% CI 0.49-0.53); Age group: [1-<5y (OR=0.84; 95% CI 0.81-0.87), 5-<8y (OR=0.73; 95% CI 0.69-0.76), 8-<12 y (OR=0.72; 95% CI 0.68-0.75), 12-<18 y (OR=0.79; 95% CI 0.76-0.83)]; male (OR=0.96; 95% CI 0.93-0.98); Race: [Asian (OR=0.91; 95% CI 0.84-0.99), Other (OR=0.92; 95% CI 0.88-0.96)], Hispanic (OR=1.06; 95% CI 1.02-1.1); Payer: [Private (OR=0.85; 95% CI 0.83-0.88)]; Median household income (per $10 000 increase) (OR=1.02; 95% CI 1.02-1.02); Vacancy rate (per 10% increase) (OR=0.96; 95% CI 0.94-0.99); Percent single-parent (per 10% increase) (OR=1.03; 95% CI 1.01-1.04)
Ehwerhemuepha et al. 2018	Unnamed	All-cause	30-day unplanned HRs	Sex; Race and/or ethnicity; Length of stay, d; Age, y; Median income by zip code, per $10 000; Percent vacant houses by zip code; Percent single parents by zip code; Low income primary medical insurance; Planned admission; ED visits within last 6 mo; Admitted through ED; Previous inpatient visits within last 6 mo; History of 30 d readmission within last 6mo; Ambulatory resource use within last 6 mo; Charlson's comorbidities; Complex chronic conditions; Was in ICU; pRI: Admitting; pRI: Discharge; pRI: Minimum; pRI: Maximum; pRI: Average; pRI: Decreasing slope across entire stay; pRI: Decreasing slope during last 24 h of hospitalization; pRI: Minimum occurred first 24 h; pRI: Maximum occurred last 24 h; Infectious and/or parasitic; Neoplasms (excluding encounters for chemotherapy); Blood and/or immune; Endocrine, nutritional, and/or metabolic; Mental and/or neurodevelopment; Nervous, eye, ear, and/or mastoid; Circulatory; Respiratory; Digestive; Skin and/or subcutaneous tissue; Musculoskeletal; Genitourinary; Perinatal period; Congenital malformations; Symptoms, signs, and/or laboratory findings not classified elsewhere; Injury and/or poison; External morbidity causes; Health status and/or services factors	Length of stay, d: [<4 (2, 3) (OR=1.23; NR% CI 1.07–1.42), < 7 (4, 5, 6) (OR=1.42; NR% CI 1.20–1.67), 7 or more (OR=1.80; NR% CI 1.51–2.14)]; Planned admission: [Yes (OR=0.65; NR% CI 0.52–0.82)]; ED and/ or department visits within last 6 mo: [1 (OR=1.27; NR% CI 1.11–1.45), 2 or more (OR=1.30; NR% CI 1.12–1.50)]; Admitted through ED: [Yes (OR=1.15; NR% CI 1.02–1.29)]; Previous inpatient visits within last 6 mo: [1 (OR=1.86; NR% CI 1.59–2.17), 2 or more (OR=2.39; NR% CI 1.94–2.94)]; History of 30 d readmission within last 6mo: [1 (OR=1.27; NR% CI 1.06–1.52), 2 or more (OR=2.75; NR% CI 2.19–3.44)]; Complex chronic conditions: [1 (OR=1.63; NR% CI 1.42–1.87), 2 or more (OR=1.66; NR% CI 1.43–1.92)]; Discharge pRI (10 pt increment) (OR=0.86; NR% CI 0.81–0.90); Maximum pRI occurred last 24 h of hospitalization (OR=0.85; NR% CI 0.75–0.95); Neoplasms (excluding encounters for chemotherapy): [Yes (OR=2.17; NR% CI 1.85–2.55)]; Blood and/or immune: [Yes (OR=1.30; NR% CI 1.14–1.48)]; Nervous, eye, ear, and/or mastoid: [Yes (OR=1.21; NR% CI 1.08–1.37)]; Circulatory: [Yes (OR=1.33; NR% CI 1.14–1.56)]; Respiratory: [Yes (OR=0.75; NR% CI 0.66–0.85)]; External morbidity causes: [Yes (OR=0.64; NR% CI 0.45–0.89)]; Health status and/or services factors: [Yes (OR=1.20; NR% CI 1.06–1.35)]
	LACE (validation)			Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Bradshaw et al. 2020	HARRPS-tool	All-cause	30-day unplanned HRs	Other Diagnosis; Anemia/Neutropenia; Appendectomy; Asthma; Bronchiolitis; Gastroenteritis; Pneumonia; Seizure; Sickle Cell Crisis; Upper Respiratory Tract Infection; Ventricular Shunt; At-risk admission diagnosis present; Chronic condition indicator; Readmitted within 30 days (history); Inpatient Admit in last 6 months; Acuity of admission: No admission acuity identified; Acuity of admission: ICN/PICU Admission; Acuity of admission: Significant psychosocial concern; Acuity of admission: Medical transport from outside facility; Acuity of admission: Other; Insurance type: Self-Pay; Insurance type: Medicaid; Insurance type: Commercial/Private; Insurance type: Medicaid & Self-Pay; Caregiver language: English; Caregiver language: Spanish; Caregiver language: Other; Medical equipment/Supplies count at home; Home nursing: Private Duty Nursing; Home nursing: Skilled Nursing; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy; Home therapy: Home Physical Therapy; Home therapy: Home Speech Therapy; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy	The final scoring system also includes risk factors, which were not significant in the multivariable analysis (i.e. At-risk admission diagnosis present; Home nursing: Private Duty Nursing and/or Skilled Nursing; Home therapy: Home Occupational Therapy, Home Physical Therapy and/or Home Speech Therapy): Chronic condition indicator: [1 (OR=1.43; 95% CI 1.13-1.81), 2 (OR=1.75; 95% CI 1.27-2.40), 3 (OR=1.41; 95% CI 1.12-1.78)]; Previous 30-day readmission (OR=1.70; 95% CI 1.38-2.09); Inpatient admission in last 6 months (OR=2.21; 95% CI 1.76 2.54); ICN or PICU admit (OR=1.43; 95% CI 1.17-1.75); Self-Pay or Medicaid (OR=1.21; 95% CI 1.05 1.39); Medical equipment/Supplies count at home: [2 (OR=1.43; 95% CI 1.09-1.88), 3 (OR=2.68; 95% CI 1.80 4.01), 4 (OR=2.21; 95% CI 1.53 3.19)]
Zhou et al. 2020	Unnamed	All-cause	30-day unplanned HRs	Age; Gender; Admission status; Funding source as inpatients; Source of referral transport; State/Territory of residence; Care type provided; Type of health insurance; Index of Relative Social-Economic Advantage and Disadvantage (IRSAD); Interpreter service; ICU stay at index admission; Had GA at index admission; LOS at index admission; Day of index admission date (weekdays or weekend); Day of index admission date; Day of discharge from index admission; Day of discharge from index admission (weekdays or weekend); Number of Co-diagnoses	Age: [13-15y (OR=1.30; 95% CI 1.14–1.48) >=16y (OR=1.46; 95% CI 1.07–1.98)]; Private health insurance (OR=1.16; 95% CI 1.00–1.34); Aeromedical service (OR=0.47; 95% CI 0.31–0.71); IRSAD (%): 91-100 (OR=1.20; 95% CI 1.02–1.41) - based on residential postcode; GA at index admission (yes) (OR=0.67; 95% CI 0.64–0.76); LOS: [LOS 2-7 days (OR=1.42; 95% CI 1.30–1.55), LOS 8-14 days (OR=2.35; 95% CI 1.97–2.82), LOS >= 15 days (OR=2.39; 95% CI 1.88–2.98)]; Admission on Friday (OR=1.21; 95% CI 1.05–1.39); Day of discharge from index admission: [Friday (OR=1.26; 95% CI 1.10–1.44), Saturday (OR=1.34; 95% CI 1.15–1.57), Sunday (OR=1.24; 95% CI 1.05–1.47)]; Number of Co-diagnoses:[1 (OR=1.28; 95% CI 1.16–1.41), 2 (OR=1.73; 95% CI 1.53–1.95), 3 (OR=2.10; 95% CI 1.80–2.46), >=4 (OR=2.41; 95% CI 2.08–2.80)]
Ehwerhemu epha et al. 2020	LACE (validation)	All-cause	30-day unplanned HRs	Length of stay, d; Admitted through ED; Charlson's comorbidities; ED visits within last 6 mo	External validation study
Zhou et al. 2021	Model 1: GLM	All-cause	30-day unplanned HRs	Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis	Variables included in the final model: Day of admission date; Day of admission (weekday/weekend and public holiday)
	Model 1: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis	Variables included in the final model: Distance from residential address to hospital; Day of discharge date; Day of admission date, Day of admission (weekday/weekend and public holiday)
	Model 2: GLM			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the	Variables included in the final model: Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				index admission: number of outpatient clinic attendances	
	Model 2: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the index admission: number of outpatient clinic attendances	Variables included in the final model: Day of discharge date; Day of admission date, Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Language spoken other than English (interpreter service required); Known allergies
	Model 3: GLM			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage of hospital services 12 months prior to the index admission: number of outpatient clinic attendances; Completion of Nursing Admission and Discharge Planning Form (Admission section); Completion of Nursing Admission and Discharge Planning Form (Discharge Planning section); Operation sheet or the last entry progress note made by doctors; Clinical pathway or the last entry progress note made by nurses; Last entry progress note made by allied healthcare providers; Written evidence of discharge information given by doctors; Written evidence of discharge information given by nurses; Written evidence of discharge medications information by doctors; Written evidence of discharge medications information by nurses; Written evidence of follow-up information given by doctors; Written evidence of follow-up information given by nurses; Consistency of written discharge documentation among healthcare providers; Delay in issuing discharge summary (date of discharge summary being issued – date of discharge)	Variables included in the final model: Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Completeness of Nursing Admission and Discharge Planning Form, Discharge Planning section (incompleteness)
	Model 3: G-S			Age; Sex; Admission status; Length of hospital say (LOS); Funding source as an inpatient; Health insurance status; Source of referral transport ; State/Territory of residence; Care type; Socioeconomic indexes for areas (SEIFA); Distance to hospital; Had general anaesthetic; Had intensive care unit (ICU) stay; Day of admission date; Day of discharge date; Number of co-diagnosis; Significant social history (legal custody or patient was under the care of Department for Child Protection); Language other than English; Significant laboratory result; Significant imaging result; Significant vital signs; Added new medication at discharge upon existing regular medication regime; Number of past medical history recorded in the patient progress notes; Known allergies; Usage of hospital services 12 months prior to the index admission: number of emergency department (ED) presentations; Usage of hospital services 12 months prior to the index admission: number of hospitalisations; Usage	Variables included in the final model: Had general anaesthetic at index admission; Source of referral transport (ambulance); Day of discharge date; Day of admission date; Day of admission (weekday/weekend and public holiday); No. admissions in the previous 12 months; No. emergency department presentations in the previous 12 months; No. past medical histories recorded in the progress notes; Significant social history; Language spoken other than English (interpreter service required); Known allergies; Completeness of Nursing Admission and Discharge Planning Form, Discharge Planning section (incompleteness); Completeness of Nursing Admission and Discharge Planning Form, Admission section (incompleteness); Progress note or Clinical Pathway documentation at discharge by nurses (not recorded); Follow-up information documented by doctors (not recorded)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				of hospital services 12 months prior to the index admission: number of outpatient clinic attendances; Completion of Nursing Admission and Discharge Planning Form (Admission section); Completion of Nursing Admission and Discharge Planning Form (Discharge Planning section); Operation sheet or the last entry progress note made by doctors; Clinical pathway or the last entry progress note made by nurses; Last entry progress note made by allied healthcare providers; Written evidence of discharge information given by doctors; Written evidence of discharge information given by nurses; Written evidence of discharge medications information by doctors; Written evidence of discharge medications information by nurses; Written evidence of follow-up information given by doctors; Written evidence of follow-up information given by nurses; Consistency ofwritten discharge documentation among healthcare providers; Delay in issuing discharge summary (date of discharge summary being issued – date of discharge)	
Surgical conditions related readmissions					
Vo et al. 2018	Unnamed	All surgical specialities without cardiac surgery	30-day unplanned post-surgical HRs relating to noncardiac surgery	Age; BMI; ASA class; Gender; Race; Surgical specialty; Admission status; Urgency of procedure; Prematurity; Congenital Heart Disease; Bleeding complication; Wound complication; Pulmonary complication; Renal complication; Neurologic complication, Cardiac complication	ASA>=3 (OR=1.9; 95% CI 1.8-2.0); Inpatient vs outpatient (OR=3.5; 95% CI 3.3-3.7); >1 Postoperative complication (OR=3.14; 95% CI 2.92-3.34); Presence of CHD (OR=1.66; 95% CI 1.31-2.11)
Polites et al. 2017	Unnamed	General and thoracic surgery	30-day unplanned HRs related to the index surgical procedure	Age; Sex; Race; Procedure group; Procedure type; Weight percentage; Premature birth; Congenital malformation; Any comorbidity; Diabetes mellitus (Individual comorbidity); Respiratory (Individual comorbidity); GI/Hepatobiliary (Individual comorbidity); Major cardiac (Individual comorbidity); Acute renal failure (Individual comorbidity); Neurologic (Individual comorbidity); Immunosuppression (Individual comorbidity); Nutrition (Individual comorbidity); Hematologic (Individual comorbidity); Active malignancy (Individual comorbidity); Preoperative SIRS, sepsis, septic shock; Preoperative transfusion or hematocrit <32; Case status; ASA class; Operative time, minutes; Wound class; Preoperative length of stay, days; Postoperative complication; Superficial Incisional SSI; Deep SSI; Pulmonary complications; Sepsis/CL associated bloodstream infection; Postoperative length of stay, days	Procedure group: [Head and Neck (HR=2.40; 95% CI 1.48-3.89), Hepatobiliary (HR=1.69; 95% CI 1.17-2.44), Small and Large intestine (HR=1.59; 95% CI 1.33-1.90), Thoracic (HR=0.69; 95% CI 0.52-0.91)]; Comorbidity: [Preoperative acute renal failure (HR=2.47; 95% CI 1.31-4.66), Neurologic comorbidity (HR=1.3; 95% CI 1.05-1.62)]; SIRS/Sepsis/Septic Shock within 48 h prior to index admission (HR=1.2; 95% CI 1.02-1.41); Operative time, minutes: [60–140 min (HR=1.21; 95% CI 1.06-1.39), >140min (HR=1.51; 95% CI 1.26-1.81)]; Wound class: [Contaminated (HR=1.29; 95% CI 1.05-1.60), Dirty/Infected (HR=1.92; 95% CI 1.53-2.40)]; Any complication (HR=1.34; 95% CI 1.09-1.65); Postoperative length of stay, days: [2-4 days (HR=2.17; 95% CI 1.84-2.57), >4 days (HR=3.12; 95% CI 2.60-3.74)]
Delaplain et al. 2020	30-day readmission model	Trauma-related conditions	30-day unplanned trauma HRs	Planned readmission; Sex; Length of stay (d); Admission source; Payer; Age; Race/ethnicity; Previous ED visits; Current/index visit is a readmission from a prior visit?; Previous visits; Readmission history; Free-standing pediatric hospital; Number of medications; Number of systems diagnoses; Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or pelvis (S30-S39); Traumatic injury (ICD-10): Ankle and foot (S90-99); Traumatic injury (ICD-10): Burns (T20-32); Traumatic injury (ICD-10): Early complications of trauma (T79); Traumatic injury (ICD-10): Elbow and forearm (S50-59); Traumatic injury (ICD-10): Foreign body entering through the natural orifice (T15-19); Traumatic injury (ICD-10): Head (S00-09); Traumatic injury (ICD-10): Hip and thigh (S70-79), Traumatic injury (ICD-10): Unspecified body region (T14); Traumatic injury (ICD-10): Knee and lower leg (S80-89); Traumatic injury (ICD-10): Multiple body regions (T07); Traumatic injury (ICD-10): Neck (S10-19); Traumatic injury (ICD-10): Unspecified effects of external causes (T66-T78); Traumatic injury (ICD-10): Poisoning (T36-50); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88); Traumatic injury (ICD-10): Thorax (S20-29); Traumatic injury (ICD-10): Toxic effects of nonmedical substances (T51-65); Traumatic injury (ICD-10): Wrist, hand, and fingers (S60-61); Surgical procedures by system: Auditory; Surgical procedures by	LOS: [LOS 2-4 days (OR=1.298; NR% CI 1.189-1.416), LOS 4-7 days (OR=1.638; NR% CI 1.493-1.797), LOS >7 (OR=1.994; NR% CI 1.816-2.189)]; Admission source: Transfer (OR=0.768; NR% CI 0.689-0.856); Previous ED visits: [1 (OR=1.137; NR% CI 1.058-1.222), 2 (OR=1.158; NR% CI 1.037-1.294), >=3 (OR=1.305; NR% CI 1.157-1.471)]; Current/index visit is a readmission from a prior visit?: [Yes: unplanned (OR=1.404; NR% CI 1.291-1.527), Yes: planned (OR=1.646; NR% CI 1.457-1.858)]; Previous visits: [1 (OR=1.746; NR% CI 1.604-1.9), 2 (OR=2.282; NR% CI 2.02-2.578), >=3 (OR=2.614; NR% CI 2.245-3.045)]; Readmission history: [1 (OR=1.473; NR% CI 1.303-1.665), 2 (OR=1.788; NR% CI 1.504-2.125), >=3 (OR=2.578; NR% CI 2.172- 3.06)]; Number of medications (OR=1.011; NR% CI 1.009-1.014); Traumatic injury (ICD-10): Abdomen, back, lumbar spine, genitalia, or pelvis (S30-S39) (OR=0.763; NR% CI 0.655-0.89); Traumatic injury (ICD-10): Burns (T20-32) (OR=0.596; NR% CI 0.494-0.719), Traumatic injury (ICD-10): Head (S00-09) (OR=0.661; NR% CI 0.594-0.735); Traumatic injury (ICD-10): Hip and thigh (S70-79) (OR=0.709; NR% CI 0.616-0.816); Traumatic injury (ICD-10): Unspecified body region (T14) (OR=0.752; NR% CI 0.62-0.912); Traumatic injury (ICD-10): Knee and lower leg (S80-89) (OR=0.727; NR% CI 0.614-0.86); Traumatic injury (ICD-10): Neck (S10-19) (OR=1.194; NR% CI 1.008-1.45); Traumatic injury (ICD-10): Poisoning (T36-50) (OR=1.16; NR% CI 1.06-1.269); Traumatic injury (ICD-10): Shoulder and upper arm (S40-49) (OR=0.547; NR% CI 0.461-0.649); Traumatic injury (ICD-10): Complications of surgical/medical care (T80-88) (OR=1.545; NR% CI 1.439-1.66); Traumatic injury (ICD-10): Toxic effects of

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				system: Cardiovascular; Surgical procedures by system: Digestive; Surgical procedures by system: Integumentary; Surgical procedures by system: Musculoskeletal; Surgical procedures by system: Nervous; Surgical procedures by system: Respiratory; Traumatic injury: trauma due to frost bite; Traumatic injury: eye/ocular; Traumatic injury: endocrine; Surgical procedure: mediastinum/diaphragm; Surgical procedures by system: hematologic; Surgical procedures by system: lymphatic; Surgical procedures by system: urinary/reproductive	nonmedical substances (T51-65) (OR=0.538; NR% CI 0.404-0.716)
Chotai et al. 2017	Unnamed	Neurosurgery	30 day unplanned HRs following index surgery for neurosurgical diagnoses	Age; Sex; Race; Preterm birth at <37 wks; Brain tumor/cyst resection or biopsy; Craniectomy or craniotomy for epilepsy, vascular, trauma, CM-I, & other brain lesions; Shunt surgery or ETV-CPC; Spine surgery; Preop or intraop EVD; Median LOS in days (range); No. w/ ICU stay; No. w/ postop complications	Race: other vs white (OR=5.916; 95% CI 1.304-26.84); ICU stay (OR=3.302; 95% CI 1.325-8.231)
Davidson et al. 2021	Unnamed	Ureteroscopy	30-day unplanned HRs after ureteroscopy	Age; Sex; Weight; Stone location; Ureteral stent placement during procedure; Anaesthesia time; Operative time; Unplanned reoperation; Organ space SSI; Pneumonia; Unplanned reintubation; Progressive renal insufficiency; Urinary tract infection; Sepsis; Septic shock; Occurrence of transfusion ; Preterm birth; Previous cardiac surgery; Case type; Patient status at time of surgery; ASA class	Female (RR=2.03; 95% CI 1.34-3.07); Renal stone location (RR=1.77; 95% CI 1.10-2.83), Both ureteric and renal stone location (RR=1.29; 95% CI 0.74-2.25); Inpatient at time of surgery (RR=1.61; 95% CI 1.03-2.51)
Garcia et al. 2018	Unnamed	Kasai procedure	30-day unplanned HRs related to kasai procedure	Age; Female (vs male) ; White race; Hispanic; Prematurity; Preoperative comorbidities: Major cardiac risk factors; Respiratory comorbidity; Non-hepatobiliary GI disease; Renal disease; Neurologic comorbidity; Preoperative steroid use; Preoperative nutritional support; Systemic inflammatory response syndrome; Preoperative blood transfusion; Anemia; Hypoalbuminemia ; Total bilirubin > 8; Operative time, minutes; ASA class ≥ III; Perioperative blood transfusion; 30-day post-operative complication; Reoperation	Prematurity (OR=3.88; 95% CI 1.08-13.95); 30-day post-operative complication (OR=4.09; 95% CI 1.41-11.87)
Lee at al-2021	Unnamed	Adolescent idiopathic scoliosis surgery	30-day unplanned HRs after adolescent idiopathic scoliosis surgery	Age; Sex; Income; Primary payer; Anemia; Coagulopathy; Chronic pulmonary disease; Depression; Diabetes; Hypothyroidism; Hypertension; Liver disease; Fluid & electrolyte disorders; Pulmonary vascular disorders; Renal failure; Valvular disease; Smoker; Obesity; Weight loss; Chronic steroid use; Chronic use of antiplatelets, antithrombotics, anticoagulants; Blood transfusion; Autograft; BMP use; Osteotomy; Fusion levels; Hospital teaching status; Hospital ownership; Disposition; Index complication: Cardiac; Index complication: Pulmonary; Index complication: Neurological; Index complication: Urinary tract infection; Index complication: Paralytic ileus; Index complication: SIADH; Index complication: SMA syndrome; Index complication: Thromboembolic complications; Index complication: Wound-related complications; Index complication: Intraop hemorrhage or hematoma; Index complication: Mechanical implant-related complications; Index complication: Dural tear; LOS >5 days	Anemia (OR=2.0; 95% CI 1.6-2.5); Hypothyroidism (OR=3.0; 95% CI 2.0-4.5); Fluid & electrolyte disorders (OR=1.8; 95% CI 1.5-2.3); Obesity (OR=2.9; 95% CI 2.2-4.0); Chronic use of anticoagulants (OR=7; 95% CI 3.0-16.4); Index complication: SIADH (OR=4.7; 95% CI 2.6-8.4); Index complication: Dural tear (OR=2.7; 95% CI 1.6-4.7); LOS >5 days (OR=1.8; 95% CI 1.6-2.2)
Minhas et al. 2016	Idiopathic scoliosis Progressive infantile scoliosis Scoliosis due to other conditions	Spinal Surgeries (Scoliosis)	30-day unplanned HRs	Age; Sex; Underweight; Obesity; Diabetes; Preterm birth; Ventilator requirement; Asthma; Cystic fibrosis; CLD; Oxygen requirement; Tracheostomy; Structural pulmonary abnormality; Esophageal/GI disease; Hepatobiliary/pancreatic disease; Cardiac risk factors; History of cerebrovascular event; Childhood malignancy; CNS tumor; Impaired cognition; History of seizure; Cerebral palsy; Structural CNS abnormality; Neuromuscular Disorder; History of intraventricular hemorrhage; Immunity disorder; Chronic steroid use; Bone marrow disorder; History of organ transplant; Open wound; Weight loss; Nutritional support requirement; Bleeding disorder; Hematological disorder; Chemotherapy; Preoperative sepsis; Preoperative inotrope requirement; Prior operation within last 30 d; Preoperative transfusion requirement; ASA>=3; Posterior fusion up to 6 levels; Posterior fusion 7-12 levels; Posterior fusion 13 or more levels; Pelvic fixation; Progressive infantile scoliosis; Idiopathic scoliosis; Scoliosis due to other condition; LOS>5 d	Idiopathic scoliosis: Obesity (OR=3.09; 95% CI 1.83-5.21); Posterior fusion 13 or more levels (OR=1.86; 95% CI 1.07-3.23) Progressive infantile scoliosis: Impaired cognition (OR=10.08; 95% CI 2.78-14.23) Scoliosis due to other conditions: ASA>=3 (OR=5.92; 95% CI 1.02-10.74); Pelvic fixation (OR=2.80; 95% CI 1.14-6.89)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Roddy & Diab 2017	Unnamed	Spine fusion	30-day unplanned HRs	Male versus female; Race; Age; Diagnosis (vs. idiopathic); Insurance type; Approach (vs. posterior); LOS; Hospital volume per year; Discharge disposition; Number of comorbidities; Teaching hospital; Children's hospital; Infection on index admission; Mechanical complication on index admission; Discharge on the weekend; 8+ levels fused (vs. 3–7); Wound dehiscence ; Pulmonary complication; VTE; Hematoma; Ileus; Non-mechanical complication of internal prosthesis	Male versus female (OR=1.28; 95% CI 1.07–1.54); Age 12–13y (OR=0.74; 95% CI 0.57–0.97); Neuromuscular (OR=2.99; 95% CI 2.39–3.75), Congenital (OR=1.66; 95% CI 1.16-2.38), Scheuermann kyphosis (OR=2.00; 95% CI 1.27-3.17), Other diagnosis (OR=1.68; 95% CI 1.32-2.15); Medicais (OR=1.50; 95% CI 1.24-1.82); Anterior approach (OR=1.55; 95% CI 1.01-2.36); LOS <= 3 days (OR=1.89; 95% CI 1.37-2.59), LOS 6-124 days (OR=1.66; 95% CI 1.35-2.02); hospital volume 41-60 fusions (OR=0.72; 95% CI 0.54-0.97), hospital volume >80 fusions (OR=0.66; 95% CI 0.50-0.88); Short-term care hospital (OR=12.41; 95% CI 8.51-18.10), Home health care (OR=1.44; 95% CI 1.09-1.91), other discharge dispositions (OR=3.79; 95% CI 1.14-12.61); Number of comorbidities >=1 (OR=1.21; 95% CI 1.01-1.50); Teaching hospital (OR=1.59; 95% CI 1.10-2.18); Infection on index admission (OR=2.12; 95% CI 1.22-3.69); Mechanical complication on index admission (OR=3.79; 95% CI 1.71-8.39)
Sherrod et al. 2016	Unnamed	Neurosurgery	30-day unplanned HRs after neurosurgery	Shunt/ventricular catheter revision, removal, irrigation; MMC repair; Shunt/ventricular catheter placement; Craniotomy for neoplasm; Other procedures (e.g. primarily baclofen pump placement) ; Craniotomy for Chiari malformation; Spine; Craniotomy for craniosynostosis; Skin lesion; Age; Neonate; LOS; Sex; Race; Patient status; Prior operation w/in 30 days ; Concurrent procedure; Transfer status ; Discharge destination; Any comorbidity; Any non-CNS comorbidity; Obesity; Pulmonary comorbidity; Ventilator dependent; Pneumonia; Asthma; Cystic fibrosis; Bronchopulmonary dysplasia; Oxygen support; Structural pulmonary abnormality; GI comorbidity; Esophageal, gastric, intestinal disease; Biliary, liver, pancreatic disease; Renal comorbidity; Renal failure; Dialysis; CNS comorbidity; Coma >24 hrs; History of CVA or TBI; CNS tumor; Developmental delay; Cerebral palsy; Neuromuscular disorder; Seizure disorder; Structural CNS abnormality; Cardiac comorbidity; Steroid use; Chemotherapy w/in 30 days before surgery; Radiotherapy w/in 90 days before surgery; Open wound (w/ or w/o infection); Tracheostomy at time of surgery; Immune disease or immunosuppressant use; Nutritional support (IV or NG tube); Bleeding disorder; Hematological disorder; Current or previous malignancy; History of prematurity; Intraventricular hemorrhage; Congenital malformation, any system; SIRS/sepsis w/in 48 hrs before surgery; Hypoalbuminemia; Hyponatremia; Hypernatremia; Elevated WBC count; Thrombocytopenia; Elevated AST; Elevated BUN; Abnormal PT; Abnormal PTT; Anemia ; Length of operation; Triage; ASA Class; Periop blood transfusion ; Wound classification; Any complication; Any infection; Superficial SSI; Deep SSI; Organ/space SSI; (postop) Sepsis; (postop) UTI; (postop) Pneumonia; CLABI; Wound disruption; Unplanned intubation; Renal insufficiency; Acute renal failure; Coma >24 hrs; CVA/intracranial hemorrhage; (postop) Seizure; Peripheral nerve injury; Cardiac arrest; Graft or prosthesis failure; DVT; PE	Shunt/ventricular catheter revision, removal, or irrigation procedure (OR=2.283; 95% CI 1.679–3.103); MMC procedure (OR=1.979; 95% CI 1.066–3.675); Shunt/ventricular catheter placement procedure (OR=2.128; 95% CI 1.542–2.937); Craniotomy for craniosynostosis (OR=0.291; 95% CI 0.151–0.560); Spine procedure (OR=0.703; 95% CI 0.503–0.984); Native American race (OR=2.363; 95% CI 1.149–4.861); Prior operation w/in 30 days of index procedure (OR=1.378; 95% CI 1.001–1.897); Transfer from ER (OR=1.273; 95% CI 1.046–1.549); Home discharge (OR=1.885; 95% CI 1.208–2.942); Presence of any comorbidity (OR=1.943; 95% CI 1.086–3.478); Oxygen supplementation (OR=1.645; 95% CI 1.128–2.399); Preexisting seizure disorder (OR=1.250; 95% CI 1.034–1.510); Steroid use >10 days (OR=1.411; 95% CI 1.087–1.831); Nutritional support (IV or NG tube) (OR=1.403; 95% CI 1.088–1.809); Operation time (per hr increase) (OR=1.059; 95% CI 1.006–1.114); Superficial incisional SSI (OR=12.151; 95% CI 7.783–18.973); Deep incisional SSI (OR=25.547; 95% CI 10.229–63.373); Organ/space SSI (OR=19.156; 95% CI 11.618–31.585); Postop sepsis (OR=2.616; 95% CI 1.321–5.181); Postop UTI (OR=4.262; 95% CI 2.598–6.992); Postop pneumonia (OR=4.294; 95% CI 2.045–9.017); Wound disruption (OR=17.582; 95% CI 10.750–28.756); Postop seizure (OR=2.532; 95% CI 1.398–4.587); Graft/prosthesis failure (OR=11.074; 95% CI 2.882–42.548)
Tahiri et al. 2015	Unnamed	Plastic Surgery	30-day unplanned HRs following pediatric plastic surgery procedures	Sex; Race; Inpatient vs outpatient; Type of procedure; Discharge destination; Respiratory history; GI history; Cardiac history; CNS history; Nutritional history; Hematologic history; One comorbidity; Multiple comorbidities (>2); Congenital malformation; Operation within 30 days; Anesthesia type; Triage; RVUs; Wound class; ASA class; Operative time; LOS; Any complications; Surgical complications; Medical complications	Inpatient procedure (OR=1.569; 95% CI 1.028–2.395); RVUs: 19.66–87.09 (OR=0.149; 95% CI 0.057–0.387); Wound contamination (OR=2.328; 95% CI 1.347–4.024); ASA class IV (OR=7.700; 95% CI 1.479–40.079); Operative time: 93–174 min (OR=2.511; 95% CI 1.494–4.219), Operative time: >175 min (OR=3.887; 95% CI 2.220–6.808); Surgical complications (OR=6.936 ; 95% CI 3.702–12.994); Medical complications (OR=11.922; 95% CI 4.706–30.208)
Wheeler et al. 2018	Unnamed	Burn diagnosis	30-day unplanned HRs	Age; Gender; Median household income by zip code; Primary expected payer; Patient location/urban-rural; No. of chronic conditions; TBSA burned ; Burn degree; Burn to eye and adnexa; Burn of face head and neck; Burn of trunk; Burn of upper limb except wrist and hand; Burn of wrist(s) and hand(s); Burn of lower limb(s); Burn of internal organs; Burn unspecified site(s); Minor loss of function; Moderate loss of function; Major/extreme loss of function; Burn mechanism; Annual no. of admitted burn patients ; Teaching status of hospital; Major operating room procedure (index); LOS; Disposition of patient	No detailed reporting of p-values that were accepted to be significant. Predictors that were assigned as significant are included in the analysis: Patient residence: Medium metropolitan county (OR=1.03; 95% CI 1.14–3.29), Patient residence: Small metropolitan county (OR=2.04; 95% CI 1.06–3.92); TBSA burned (%) ≥ 10 (OR=1.81; 95% CI 1.18–2.79); Third burn degree (OR=2.68; 95% CI 1.69–4.24); Major operating room procedure (OR=1.76; 95% CI 1.14–2.70); LOS 2-3 days (OR=1.72; 95% CI 1.03–2.88)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Vedantam et al. 2018	Unnamed	Epilepsy surgery	30-day unplanned HRs after epilepsy surgery	Age; Gender; Race; ASA classification; Operative time; LOS; Weight ; Neurologic and neuromuscular; Other complex chronic conditions; Discharge destination; Procedure	Hemispherectomy (OR=4.11; 95% CI 1.48–11.42)
Basques et al. 2015	Unnamed	Posterior spinal fusion	30-day unplanned HRs after posterior spinal fusion	Age; Sex; BMI for age; History of asthma; Number of levels fused; Osteotomy performed; Operative time; Any inpatient complication; Serious adverse event; Return to the operating room; Wound dehiscence; Deep surgical site infection; Nervous injury; Minor adverse event; Superficial surgical site infection; Urinary tract infection; Pneumonia; Length of stay more than 6 days	Any inpatient complication (OR=180.44; 95% CI 35.47–917.97)
Martin et al. 2015	Unnamed	Spinal deformity surgery	30-day unplanned HRs after spinal deformity surgery	Age; Sex; Race; BMI ; Diabetes; Ventilator dependence; Asthma; Cystic fi brosis; Chronic lung disease; Oxygen support; Tracheostomy; Airway abnormalities; GI disease; Hepatobiliary disease; Cardiac risk factors; Previous cardiac surgery; Acute renal failure; Dialysis; History of stroke; Tumor involving CNS; Developmental delay; Seizure disorder; Cerebral palsy; CNS abnormality; Neuromuscular disorder; Immune disorder; Steroid use w/in 30 d; Bone marrow transplant; Solid organ transplant; Recent weight loss; Nutritional support; Bleeding disorder; Hematologic disorder; Chemo w/in 30 d; Rad therapy w/in 90 d; Prior operation w/in 30 d; Congenital malformation; Hx childhood malignancy; Requires inotropic support; Sodium; BUN; WBC; Hematocrit; INR; Creatinine; Albumin; ASA class; Received blood transfusion; Blood transfused; Operative time; Total case RVUs; LOS; Surgeon specialty; Isolated primary posterior arthrodesis; Revision posterior arthrodesis; Isolated primary anterior arthrodesis; Revision anterior arthrodesis; Combined anterior and posterior arthrodesis; Instrumentation extending to pelvis; Insertion of intervertebral device; Osteotomy; Bone grafting; Diagnosis	Structural pulmonary abnormalities (OR=2.53; 95% CI 1.22–5.23); ASA class (3 or 4 vs . 1 or 2) (OR=2.18; 95% CI 1.07–4.47); Isolated primary anterior spinal fusion (OR=7.65; 95% CI 1.32–44.3)
General medical conditions related readmissions					
Learly et al. 2019	Prediction at admission	Complex chronic conditions	30-day unplanned HRs	Age; Race and/or ethnicity; Boys; Non-English primary language; Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo, Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type	Significant variables (p-value<0.05) in univariate analysis are used for the analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.15–2.45); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.32–3.10); No. CCC categories: [No. CCC categories = 2 (OR=1.73; 95% CI 1.19–2.49), No. CCC categories >=3 (OR=2.30; 95% CI 1.50–3.47)]; Medical admission (OR=1.82; 95% CI 1.30–2.63)
	Prediction at discharge			Age; Race and/or ethnicity; Boys; Non-English primary language, Insurance type (Private); Insurance type (Public); Insurance type (uninsured); Neighborhood per capita income, $; Any admissions in previous 6mo; Any ED visits in previous 6 mo; No. home medications at admission; CCC category (Neurologic); CCC category (Cardiovascular); CCC category (Gastrointestinal); CCC category (Other); No. CCC categories; Technology assistance; Admission type; ICU use; Discharge disposition from the hospital; LOS in d; Weekday discharge	Significant variables (p-value<0.05) in univariate analysis are used for analysis of the final model: Any admissions in previous 6 mo (OR=1.70; 95% CI 1.16–2.46); Any ED visits in previous 6 mo (OR=2.04; 95% CI 1.31–3.11); No. CCC categories: [No. CCC categories = 2 (OR=1.56; 95% CI 1.06–2.26), No. CCC categories >=3 (OR=1.72; 95% CI 1.08–2.69)]; Medical admission (OR=1.75; 95% CI 1.23–2.49); Discharge disposition from the hospital: [With services (OR=1.69; 95% CI 1.17–2.44), Other facility (OR=1.15; 95% CI 0.58–2.13)]; LOS: [LOS 2-5 days (OR=1.15; 95% CI 0.78–1.72), LOS >=6 days (OR=1.45; 95% CI 0.90–2.33)] Weekday discharge was not significant (p-value≤ 0.05) in the univariate analysis, but included in the final scoring system.
Ryan et al. 2021	PASS (validation)	Asthma	30-day unplanned HRs	Respiratory rate, Oxygen requirement, Auscultation, Retractions, Dyspnea	External validation study
O´Connell et al. 2021	Unnamed	Nervous system condition	30-day unplanned HRs	Age; Sex; race/ ethnicity; payer; Length of stay days) ; Index visit planned; Admitted through ED; Index visit is a readmission; Previouse ED visits (prior 6mo); Previous hospitalizations (prior 6mo); Previous readmissions (prior 6 mo); Number of comorbid diagnoses (by ICD 10 CM chapters); Viral Meningitis (A87); Malignant neoplasm of brain (C71); Disorders of BCAA and FA Metabolism (E71); Other AA Metabolism Disorders (E72); Other carbohydrate metabolism disorders (E74); Lipoprotein Metabolism Disorders / Lipidemias (E78); Mental Disorders Due to Physiological Condition (F06); Unspecified Intellectual Disabilities (F79); Speech &	Age (OR=0.993; 95% CI 0.99-0.997); Hispanic/Latino (OR=1.126; 95% CI 1.016-1.247); Self-Pay (OR=0.805; 95% CI 0.688-0.943); LOS 2-4 days (OR=1.228; 95% CI 1.167-1.293), LOS 4-6 days (OR=1.441; 95% CI 1.361-1.525), LOS 7 or more days (OR=1.585; 95% CI 1.492-1.683); Index visit planned: Yes (OR=0.898; 95% CI 0.847-0.952); Emergent Admission (OR=1.129; 95% CI 1.079-1.18); Index visit is a planned readmission (OR=1.338; 95% CI 1.269-1.411), Index visit is a unplanned readmission (OR=1.666; 95% CI 1.516-1.832); 1 previous ED visit (prior 6mo) (OR=1.106; 95% CI 1.056-1.159), 2 previous ED visits (prior 6mo) (OR=1.200; 95% CI 1.123-1.283), 3 or more previous ED visits (prior 6mo) (OR=1.297; 95% CI 1.209-1.391); 1 previous hospitalization (prior 6mo)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
				Language Development Disorders (F80); Scholastic Skill Development Disorders (F81); Motor Function Development Disorders (F82); Pervasive Developmental Disorders (F84); Psychological Development Disorders (F88); Bacterial Meningitis (G00); Meningitis, Other Causes (G03); Encephalitis, Myelitis, Encephalomyelitis (G04); Extrapyramidal & movement Disorders (G25); Nervous System Degenerative Diseases (G31); Epilepsy & Recurrent Seizures (G40); Migraine (G43); Headache Syndromes (G44); Sleep Disorders (G47); Polyneuropathies (G62); Primary Muscle Disorders (G71); Cerebral Palsy (G80); Hemiplegia & Hemiparesis (G81); Paraplegia & Quadriplegia (G82); Pain (G89); ANS Disorders (G90); Hydrocephalus (G91); Brain Disorders (G93); Spinal Cord Diseases (G95); CNS Disorders (G96); Postprocedural NS Disorders (G97); Visual Pathway Disorders (H47); Paralytic Strabismus (H49); Binocular Movement Disorders (H51); Nystagmus & Irregular Eye Movements (H55); Conductive & Sensorineural Hearing Loss (H90); Nontraumatic Intracerebral Hemorrhage (I61); Nontraumatic Intracranial Hemorrhage (I62); Cerebral Infarction (I63); Cerebrovascular Diseases (I67); Sequelae of Cerebrovascular Disease (I69); Neuromuscular Dysfunction of Bladder (N31); Nontraumatic Intracranial Hemorrhage Newborn (P52); Newborn Convulsions (P90); Newborn Cerebral Disturbances (P91); Newborn Muscle Tone Disorders (P94); Microcephaly (Q02); Congenital Hydrocephalus (Q03); Congenital Brain Malformations (Q04); Spina Bifida (Q05); Congenital Spinal Cord Malformations (Q06); Congenital Nervous System Malformations (Q07); Down Syndrome (Q90); Other Autosomal Trisomies (Q92); Nervous & Musculoskeletal System Symptoms (R29); Somnolence, Stupor, Coma (R40); Cognitive Function Symptoms (R41); Dizziness & Giddiness (R42); Headache (R51); Convulsions (R56); Intracranial Injury (S06); Number of medications; Free-standing pediatric hospital; Surgical procedures	(OR=1,695; 95% CI 1.613-1.781), 2 previous hospitalizations (prior 6mo) (OR=2.348; 95% CI 2.18-2.528), 3 or more previous hospitalizations (prior 6mo) (OR=3.014; 95% CI 2.738-3.317); 1 previous readmission (prior 6 mo) (OR=1.179; 95% CI 1.09-1.274), 2 previous readmissions (prior 6 mo) (OR=1.496; 95% CI 1.338-1.673), 3 or more previous readmissions (prior 6 mo) (OR=2.051; 95% CI 1.831-2.298); Number of comorbid diagnoses (by ICD 10 CM chapters) (OR=1.01; 95% CI 1.006-1.014); Viral Meningitis (A87) (OR=0.446; 95% CI 0.353-0.563); Malignant neoplasm of brain (C71) (OR=1.953; 95% CI 1.788-2.133); Other AA Metabolism Disorders (E72) (OR=1.313; 95% CI 1.095-1.575); Mental Disorders Due to Physiological Condition (F06) (OR=0.768; 95% CI 0.648-0.91); Speech & Language Development Disorders (F80) (OR=0.811; 95% CI 0.725-0.907); Bacterial Meningitis (G00) (OR=0.622; 95% CI 0.487-0.796); Meningitis, Other Causes (G03) (OR=0.829; 95% CI 0.698-0.985); Sleep Disorders (G47) (OR=0.84; 95% CI 0.791-0.892); Polyneuropathies (G62) (OR=1.5; 95% CI 1.248-1.803); Hydrocephalus (G91) (OR=1.218; 95% CI 1.124-1.476); Brain Disorders (G93) (OR=1.078; 95% CI 1.019-1.14); Paralytic Strabismus (H49) (OR=1.288; 95% CI 1.124-1.476); Nystagmus & Irregular Eye Movements (H55) (OR=0.825; 95% CI 0.686-0.992); Neuromuscular Dysfunction of Bladder (N31) (OR=1.155; 95% CI 1.034-1.129); Newborn Cerebral Disturbances (P91) (OR=0.822; 95% CI 0.705-0.958); Newborn Muscle Tone Disorders (P94) (OR=0.621; 95% CI 0.567-0.68); Microcephaly (Q02) (OR=1.131; 95% CI 1.005-1.1272); Congenital Hydrocephalus (Q03) (OR=1.318; 95% CI 1.175-1.478); Down Syndrome (Q90) (OR=1.129; 95% CI 1.036-1.23); Other Autosomal Trisomies (Q92) (OR=1.214; 95% CI 1.019-1.448); Nervous & Musculoskeletal System Symptoms (R29) (OR=0.882; 95% CI 0.781-0.995); Somnolence, Stupor, Coma (R40) (OR=0.786; 95% CI 0.694-0.889); Cognitive Function Symptoms (R41) (OR=0.853; 95% CI 0.776-0.938); Convulsions (R56) (OR=0.922; 95% CI 0.874-0.972); Intracranial Injury (S06) (OR=0.476; 95% CI 0.418-0.543); Number of medications (OR=1.014; 95% CI 1.012-1.016); Free-standing pediatric hospital (Yes) (OR=1.586; 95% CI 1.103-2.28); Surgical procedures (Cardiovascular) (OR=1.204; 95% CI 1.109-1.216), Surgical procedures (Musculoskeletal) (OR=0.685; 95% CI 0.6-0.781), Surgical procedures (Nervous System) (OR=1.119; 95% CI 1.029-1.216)
Hoenk et al. 2021	Unnamed	Oncology	30-day unplanned HRs	Age; sex; race/ ethnicity; Payer; Acute lymphoid leukemia (ALL); Acute myeloid leukemia (AML); Brain cancer; Neuroblastoma; Wilms tumor; Hodgkin's lymphoma; Non-Hodgkin's lymphoma; Rhabdomyosarcoma; Bone/cartilage cancer; Other cancers; Chemotherapy; Bone marrow transplant; Number of cancer medications; Number of previous visits for chemotherapy (prior 30 d); Length of stay; Emergent admission; Is index/current visit itself a readmission; Previous ED visits (prior 6 mo); Number of previous visits without chemotherapy (prior 6 mo); Previous readmissions (prior 6 mo); Viral or bacterial infections (A00 - B99); Hematological conditions (D50-D69, D71-D79); Neutropenia (D70); Immunological conditions (D80-D89); Endocrine, nutritional, and metabolic diseases (E00-E89); Mental, behavioral, and neurodevelopmental disorders (F01-F99); Nervous system (G00-G99); Eye, adnexa, ear, mastoid (H00-H95); Circulatory (I00-I99); Respiratory (J00-J99); Digestive (K00-K95); Skin/subcutaneous tissue (L00-L99); Musculoskeletal and connective tissue (M00-M99); Genitourinary (N00-N99); Congenital/Chromosomal (Q00-Q99); Injuries and poisoning (S00-T88)	Age (OR=0.987; 95% CI 0.982-0.993); Acute lymphoid leukemia (ALL): Yes, not in remission (OR=0.788; 95% CI 0.721-0.861), Acute lymphoid leukemia (ALL): Yes, in remission (OR=0.656; 95% CI 0.590-0.729), Acute lymphoid leukemia (ALL): Yes, in relapse (OR=1.436; 95% CI 1.201-1.718); Brain cancer (OR=0.782; 95% CI 0.711-0.861); Neuroblastoma (OR=1.442; 95% CI 1.285-1.619); Rhabdomyosarcoma (OR=1.182; 95% CI 1.048-1.332); Bone/cartilage cancer (OR=1.618; 95% CI 1.455-1.799); Number of cancer medications (OR=1.46; 95% CI 1.117-1.175); LOS 2, 3 days (OR=1.292; 95% CI 1.184-1.410), LOS 4,5,6 days (OR=1.466; 95% CI 1.340-1.604), LOS 7 or more days (OR=1.344; 95% CI 1.219-1.481); Is index/current visit itself a readmission (Yes, unplanned) (OR=1.869; 95% CI 1.722-2.028), Is index/current visit itself a readmission (Yes, planned) (OR=1.397; 95% CI 1.250-1.560); Number of previous visits without chemotherapy (prior 6 mo) (OR=1.077; 95% CI 1.067-1.221); 1 previous readmission (prior 6 mo) (OR=1.277; 95% CI 1.126-1.336), 2 previous readmissions (prior 6 mo) (OR=1.281; 95% CI 1.145-1.433), 3 or more previous readmissions (prior 6 mo) (OR=1.263; 95% CI 1.087-1.468); Endocrine, nutritional, and metabolic diseases (E00-E89) (OR=1.141; 95% CI 1.067-1.221); Congenital/Chromosomal (Q00-Q99) (OR=0.745; 95% CI 0.657-0.844); Injuries and poisoning (S00-T88) (OR=0.838; 95% CI 0.760-0.925)
Sanchez-Luna et al. 2016	Unnamed	Acute bronchiolitis due to respiratory syncytial virus	30-day unplanned HRs	Prematurity; Gestational age; Congenital heart disease; Chronic lung disease; Down's syndrome; Velo-cardio-facial syndrome; Neuromuscular disorders; Inmunodeficiency; Number of risk factors; Heart transplant	Significant odds ratios before multilevel modelling: Prematurity (OR=3.66; 95% CI 3.15-4,27); Gestational age: <=28w (OR=12.47; 95% CI 1.76-88.51), Gestational age: 29-32w (OR=4.99; 95% CI 0.97-25.71), Gestational age: 33-36w (OR=8.48; 95% CI 4.57-15.71); Congenital heart disease (OR=3.05; 95% CI 2.51-3.71); Chronic lung disease (OR=6.76; 95% CI 3.87-11.83); Down's syndrome (OR=2.65; 95% CI 1.83-3.84); Velo-cardio-facial syndrome (OR=12.42; 95% CI 1.75-88.21); Neuromuscular disorders (OR=5.42; 95% CI 3.72-7.89); Inmunodeficiency (OR=4.35; 95% CI 1.84-10.3); Number of risk factors = 1 (OR=3.64; 95% CI 3.12-4.25), Number of risk factors >=2 (OR=3.86; 95% CI 2.28-6.54)

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Reference	Model name	Medical condition	Model outcome	Examined variables	Significant variables*
Sacks et al. 2017	Unnamed	Cardiac conditions	30-day unplanned HRs	Age; LOS; Diagnosis count; Procedure count; First weight; Last weight; Medication count; Catheterization; Electrophysiology study; Surgery (noncardiac); ACE/ARB antihypertensive; Antacid; Antiarrhythmic; Antibiotic; Anticoagulant; Beta-blocker; Calcium channel blocker; Diuretic; Lipid management; Milrinone; Neuroactive; Pulmonary antihypertensive; Steroid; Gender; Race; Ethnicity ; Language; Insurance status; Distance from center (miles); Season of discharge; Day of discharge	Age: 1mo-1year (OR=4.11; 95% CI 2.83–5.98); Diagnosis count (OR=1.10; 95% CI 1.07–1.13); Antibiotic (OR=0.60; 95% CI 0.40–0.90)

Abbreviations: NR, not reported
* Only significant risk factors (odds ratio/ hazard ratio>1, p-value<0.05) are considered in the analysis (cf. table 2). Significant predisposed factors (1>odds ratio/ hazard ratio>0, p-value<0.05) are listed as an additional information.

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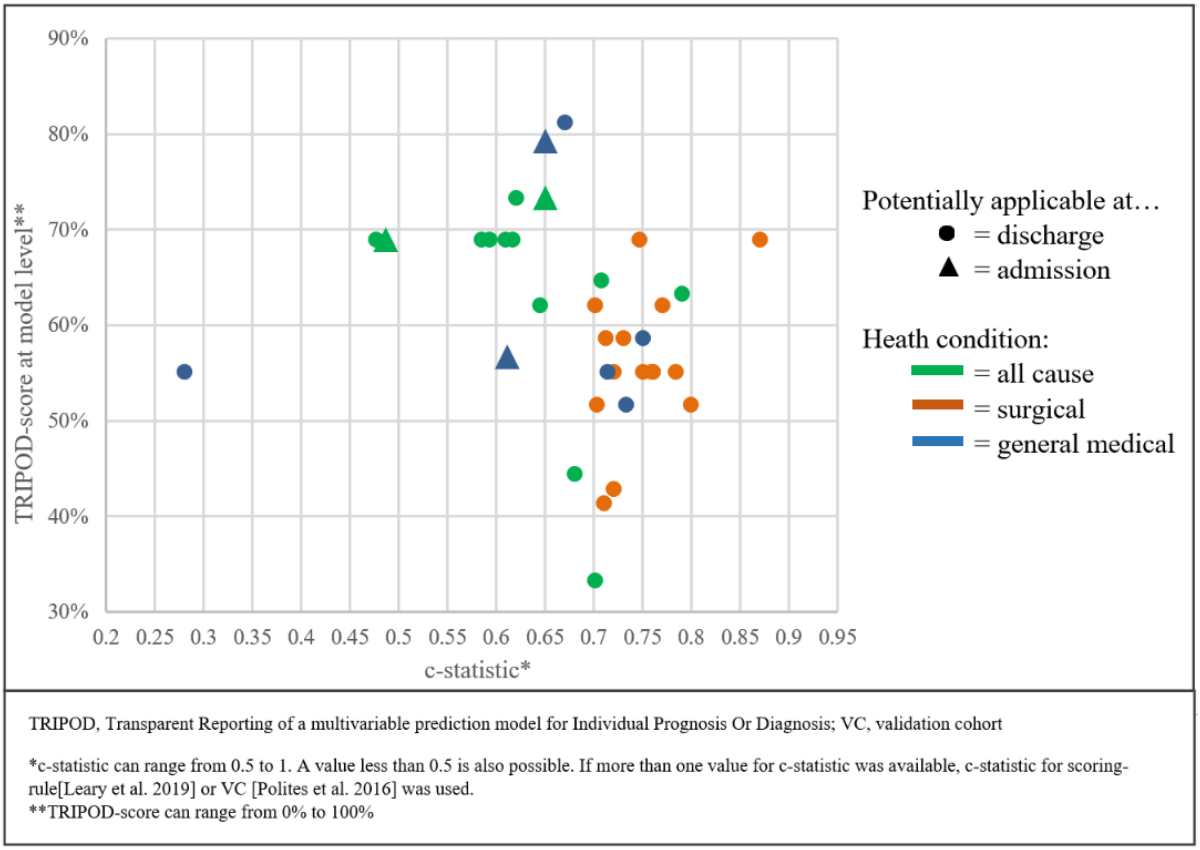
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Table A7: Adherence per TRIPOD-item at predictive model level

TRIPOD-item	Developed predictive models (n=32)			External validated predictive models (n=3)			Incremental value predictive models (n=2)			All predictive models (n=37)		
	# of PMs with applicable item	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item	# of PMs with applicable items	# of PMs adhered to item	Adherence per item
1	32.00	4.00	12.50%	3.00	0.00	0.00%	2.00	1.00	50.00%	37.00	5.00	13.51%
2	32.00	1.00	3.13%	3.00	0.00	0.00%	2.00	0.00	0.00%	37.00	1.00	2.70%
3a	32.00	32.00	100.00%	3.00	1.00	33.33%	2.00	2.00	100.00%	37.00	35.00	94.59%
3b	32.00	31.00	96.88%	3.00	1.00	33.33%	2.00	2.00	100.00%	37.00	34.00	91.89%
4a	32.00	30.00	93.75%	3.00	3.00	100.00%	2.00	1.00	50.00%	37.00	34.00	91.89%
4b	32.00	32.00	100.00%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	37.00	100.00%
5a	32.00	17.00	53.13%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	22.00	59.46%
5b	32.00	31.00	96.88%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	36.00	97.30%
5c	32.00	32.00	100.00%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	37.00	100.00%
6a	32.00	23.00	71.88%	3.00	1.00	33.33%	2.00	2.00	100.00%	37.00	26.00	70.27%
6b	32.00	32.00	100.00%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	37.00	100.00%
7a	32.00	32.00	100.00%	3.00	1.00	33.33%	2.00	2.00	100.00%	37.00	35.00	94.59%
7b	32.00	1.00	3.13%	3.00	0.00	0.00%	2.00	0.00	0.00%	37.00	1.00	2.70%
8	32.00	31.00	96.88%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	36.00	97.30%
9	32.00	11.00	34.38%	3.00	1.00	33.33%	2.00	1.00	50.00%	37.00	13.00	35.14%
10a	32.00	6.00	18.75%	0.00	0.00	0.00%	2.00	0.00	0.00%	34.00	6.00	17.65%
10b	32.00	3.00	9.38%	0.00	0.00	0.00%	2.00	0.00	0.00%	34.00	3.00	8.82%
10c	0.00	0.00	0.00%	3.00	2.00	66.67%	1.00	1.00	100.00%	4.00	3.00	75.00%
10d	32.00	11.00	34.38%	3.00	0.00	0.00%	2.00	1.00	50.00%	37.00	12.00	32.43%
10e	0.00	0.00	0.00%	1.00	0.00	0.00%	1.00	1.00	100.00%	2.00	1.00	50.00%
11	5.00	4.00	80.00%	0.00	0.00	0.00%	0.00	0.00	0.00%	5.00	4.00	80.00%
12	0.00	0.00	0.00%	3.00	0.00	0.00%	2.00	2.00	100.00%	5.00	2.00	40.00%
13a	32.00	22.00	68.75%	3.00	1.00	33.33%	2.00	1.00	50.00%	37.00	24.00	64.86%
13b	32.00	10.00	31.25%	3.00	0.00	0.00%	2.00	2.00	100.00%	37.00	12.00	32.43%
13c	0.00	0.00	0.00%	3.00	0.00	0.00%	2.00	2.00	100.00%	5.00	2.00	40.00%
14a	32.00	27.00	84.38%	0.00	0.00	0.00%	2.00	2.00	100.00%	34.00	29.00	85.29%
14b	30.00	21.00	70.00%	0.00	0.00	0.00%	2.00	2.00	100.00%	32.00	23.00	71.88%
15a	32.00	0.00	0.00%	0.00	0.00	0.00%	2.00	0.00	0.00%	34.00	0.00	0.00%
15b	32.00	6.00	18.75%	0.00	0.00	0.00%	2.00	1.00	50.00%	34.00	7.00	20.59%
16	32.00	3.00	9.38%	3.00	0.00	0.00%	2.00	1.00	50.00%	37.00	4.00	10.81%
17	0.00	0.00	0.00%	1.00	0.00	0.00%	0.00	0.00	0.00%	1.00	0.00	0.00%
18	32.00	32.00	100.00%	3.00	3.00	100.00%	2.00	2.00	100.00%	37.00	37.00	100.00%
19a	0.00	0.00	0.00%	3.00	0.00	0.00%	2.00	2.00	100.00%	5.00	2.00	40.00%
19b	32.00	32.00	100.00%	3.00	2.00	66.67%	2.00	2.00	100.00%	37.00	36.00	97.30%
20	32.00	26.00	81.25%	3.00	1.00	33.33%	2.00	2.00	100.00%	37.00	29.00	78.38%
21	32.00	16.00	50.00%	3.00	1.00	33.33%	2.00	2.00	100.00%	37.00	19.00	51.35%
22	32.00	22.00	68.75%	3.00	2.00	66.67%	2.00	1.00	50.00%	37.00	25.00	67.57%

PM, predictive model; TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis

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TRIPOD, Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis; VC, validation cohort

*c-statistic can range from 0.5 to 1. A value less than 0.5 is also possible. If more than one value for c-statistic was available, c-statistic for scoring-rule[Leary et al. 2019] or VC [Polites et al. 2016] was used.
**TRIPOD-score can range from 0% to 100%

Figure A2: Discriminative ability, application and TRIPOD-adherence of 30-day UHR predictive models in paediatrics (n=37)

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5 + Supplementary File
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Page 5 + Supplementary File
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5-7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5-6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6-7 + Table 1-4 + Supplementary File
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	NA
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Qualitative Synthesis, Page 6-7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Qualitative Synthesis, Page 6-7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Qualitative Synthesis, Page 6-7

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Qualitative Synthesis, Page 6-7& 18-19
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	NA
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6-7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	NA
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7-8 + Figure A1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Page 7-8 + Supplementary File
Study characteristics	17	Cite each included study and present its characteristics.	Page 8-17, Table 1+2+3
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 8 + Supplementary File
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1&2 + Supplementary File
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Qualitative Synthesis, Table 1-4, figure 1
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	NA
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 18-19
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 15-17, TRIPOD Assessment
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	NA
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Page 17-20

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
	23b	Discuss any limitations of the evidence included in the review.	Page 18-19
	23c	Discuss any limitations of the review processes used.	Page 18-19
	23d	Discuss implications of the results for practice, policy, and future research.	Page 18-19
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Page 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Page 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 20
Competing interests	26	Declare any competing interests of review authors.	Page 20
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 21 + Supplementary File

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Y
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Y
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Y
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	Y
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Y
Synthesis of results	6	Specify the methods used to present and synthesise results.	Y
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Y
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	Y
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	Y
Interpretation	10	Provide a general interpretation of the results and important implications.	Y
OTHER			
Funding	11	Specify the primary source of funding for the review.	Manuscript
Registration	12	Provide the register name and registration number.	NA

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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